

A counterexample to the Colin de Verdière chromatic conjecture

OpenAI

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Abstract

We disprove the Colin de Verdière chromatic conjecture by constructing graphs whose chromatic number exceeds their Colin de Verdière invariant by more than one. The examples have independence number at most two. In fact, their ordinary fractional chromatic number also exceeds their Colin de Verdière invariant by more than one.

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1 Introduction

The Colin de Verdière invariant connects graph colouring with the inertia and nullity of real symmetric matrices. We prove that the chromatic number need not be at most one more than this invariant.

1.1 The invariant and the main result

A real symmetric matrix M , indexed by the vertices of a finite nonempty simple graph $G = (V, E)$, is *well-signed for G* if, for distinct vertices u, v ,

$$M_{uv} < 0 \quad \text{when } \{u, v\} \in E, \quad M_{uv} = 0 \quad \text{when } \{u, v\} \notin E.$$

Its diagonal entries are unrestricted. Such a matrix has the *Strong Arnold Property* if the only real symmetric matrix X satisfying

$$MX = 0, \quad X_{uu} = 0 \ (u \in V), \quad X_{uv} = 0 \ (\{u, v\} \in E)$$

is $X = 0$. The Colin de Verdière invariant is

$$\mu(G) = \max \left\{ \dim \ker M : \begin{array}{l} M \text{ is well-signed for } G, \quad n_-(M) = 1, \\ M \text{ has the Strong Arnold Property} \end{array} \right\},$$

where $n_-(M)$ counts negative eigenvalues with multiplicity. This class is nonempty: perturb an invertible diagonal matrix with one negative entry by sufficiently small negative entries on the edges. Inertia and invertibility persist, and an invertible matrix has the Strong Arnold Property.

Write $\alpha(G)$ and $\chi(G)$ for the independence number and chromatic number. The chromatic conjecture for this invariant asserts that $\chi(G) \leq \mu(G) + 1$ for every finite nonempty simple graph. Our main result gives a negative resolution.

Theorem 1.1. *There are finite graphs G of arbitrarily large order for which*

$$\alpha(G) \leq 2, \quad \mu(G) + 1 < \frac{|V(G)|}{2} \leq \chi(G).$$

In particular, the Colin de Verdière chromatic conjecture is false.

We prove a stronger matrix obstruction than is necessary for the upper bound on μ : every well-signed matrix on the constructed graph with one negative eigenvalue has rank greater than $|V(G)|/2 + 1$. The Strong Arnold Property is not used in this bound. The maximum corank in this larger class is the related parameter κ considered by Lovász and Schrijver [21, pp. 572–574]. Thus the construction also satisfies $\kappa(G) + 1 < |V(G)|/2$. Unlike μ , this larger parameter is not minor-monotone in general; our argument uses it only through the stated matrix class.

1.2 History, significance, and the construction

Colin de Verdière introduced μ through the spectral study of Schrödinger operators on graphs [4, 27]. The Strong Arnold Property expresses a transversality condition from that spectral framework [3]. The small values of the invariant have concrete geometric meanings: $\mu \leq 2$ characterizes outerplanar graphs, $\mu \leq 3$ planar graphs, and $\mu \leq 4$ linklessly embeddable graphs [4, 19, 27]. The proposed chromatic bound holds throughout the range $\mu(G) \leq 4$ [27, p. 35]. Gram representations and nullspace embeddings have provided geometric ways to study these matrices [17, 20]. These connections make the proposed chromatic bound a natural test of how much colouring information the spectral invariant retains.

Write $h(G)$ for the largest t such that K_t is a minor of G . Minor monotonicity of μ , together with $\mu(K_t) = t - 1$, gives $h(G) \leq \mu(G) + 1$ [27, §1.3 and Theorem 2.4]. Thus Hadwiger’s conjecture $\chi(G) \leq h(G)$ [14] would imply the chromatic bound proposed by Colin de Verdière [4, 5, 27]. For completeness, the complete-graph identity follows directly from $-J_t$: it has one negative eigenvalue and nullity $t - 1$, and the Strong Arnold Property is automatic on a complete graph. No admissible matrix can have rank zero. The examples in Theorem 1.1 therefore also violate Hadwiger’s inequality. The matrix obstruction below is proved directly; its conclusion does not follow from a clique-minor obstruction.

A distinct connected-matching route to the failure of Hadwiger’s inequality is given in the companion manuscript [23, Theorem 1.1 and Corollary 1.2].

The construction starts with a finite-field relation of *holes*. Its vertices are injective frames. A hole is witnessed by a tensor shared by two frames, two compatible affine-gradient equations, and a parity condition. Summing those equations around a triangle gives a contradiction. The complement of the hole relation consequently has independence number at most two, and so needs at least half its order in colours. Our remaining task is to force matrix rank strictly above half the order. For this purpose, ordinary control of connected matchings is insufficient: the matrix argument needs conflicts between specified classes of matching edges and between a matching and a separately chosen vertex set.

The new probabilistic ingredient is a thinning of the allowed channel images that preserves the primal-frame distribution. It penalizes prescribed nonzero translates of channel combinations, turning the difficult phase alternative into one with a fixed positive mass of compatible pairs. This gives both a version of conflict supersaturation that distinguishes colours and an asymmetric edge–point version. We prove these extensions together with their common-limit and finite-sampling steps. The low-rank Boolean moment argument uses commuting multiplication operators, as in the flat-extension method for moments [18]; here the characteristic-two, partly truncated version is proved directly. The probabilistic comparison combines entropy control with the energy-increment method of weak regularity [7, 12]. For finite sampling we adapt the deterministic fingerprint procedure of Kleitman and Winston [16, 26], using a constrained entropy minimum as its potential [8].

The second ingredient is a matrix criterion for clique blow-ups. A finite-tree representation of hyperbolic flats locates large families of blocks that can be normalized simultaneously. A first spectral normalization produces an almost perfect matching of coordinates. A second normalization extracts additional positive directions from selected pair blocks. The resulting rank gain contradicts a kernel of approximately half the total dimension. The criterion needs only the graph properties stated next and may be used independently of the tensor construction.

1.3 The base graphs and the proof structure

A *hole* of a graph is a pair of distinct nonadjacent vertices. Two edges *conflict* if they are disjoint and all four cross pairs between their endpoints are holes. An edge and a vertex conflict if the vertex is distinct from both endpoints and both cross pairs are holes. Edges *touch* if they have a common endpoint or an edge joins their endpoint sets. Thus two disjoint edges fail to touch exactly when they conflict.

Theorem 1.2 (Base graphs). *There is a constant $a > 0$ such that, for every $\beta > 0$, there are arbitrarily large integers m and graphs H on m vertices with $\alpha(H) \leq 2$ and the following properties.*

- (G1) *Every matching of size at least $m/20$, partitioned into classes of size at most am , contains two conflicting edges in different classes.*
- (G2) *Every matching of size at least βm and every vertex set of size at least βm contain an edge and a vertex in conflict.*
- (G3) *Every two vertex sets of size at least βm contain a cross hole.*

The two vertex sets in G3 may overlap.

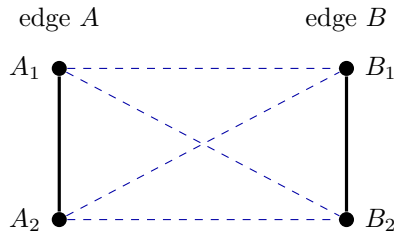


Figure 1: A conflict: solid segments are graph edges, and dashed segments are holes. The construction forces such a configuration between different classes of a large matching.

For a positive integer s , the *clique blow-up* $H[s]$ replaces each vertex of H by a clique of size s , and each edge by all edges between the two corresponding cliques. We also write $H[K_s]$ for

$H[s]$. An independent set in $H[s]$ uses at most one vertex from each clique and projects to an independent set in H . Hence $\alpha(H[s]) \leq 2$, and $\chi(H[s]) \geq \lceil ms/2 \rceil$.

Theorem 3.1 proves that, when a is fixed, β is sufficiently small, and m is sufficiently large, properties **G1–G3** force every well-signed one-negative matrix on $H[s]$ to have rank greater than $ms/2 + 1$, for every sufficiently large s . Choosing one base graph from **Theorem 1.2** and applying rank-nullity therefore proves **Theorem 1.1**. Notice the order: the base graph is fixed before the blow-up parameter tends to infinity.

The independence-number estimate also yields a stronger fractional-colouring consequence. For a finite nonempty simple graph G , let $\mathcal{I}(G)$ be the family of its independent vertex sets. A fractional colouring assigns nonnegative real weights w_I to these sets with $\sum_{I \ni v} w_I \geq 1$ at every vertex v . The ordinary fractional chromatic number $\chi_f(G)$ is the minimum of $\sum_I w_I$ over all such assignments. An ordinary colouring is a feasible assignment with unit weights on its colour classes, so $\chi_f(G) \leq \chi(G)$. On the other hand, summing the vertex constraints gives

$$|V(G)| \leq \sum_{v \in V(G)} \sum_{I \ni v} w_I = \sum_{I \in \mathcal{I}(G)} |I| w_I \leq \alpha(G) \sum_{I \in \mathcal{I}(G)} w_I,$$

and therefore $\chi_f(G) \geq |V(G)|/\alpha(G)$.

Corollary 1.3 (Fractional-colouring obstruction). *There are finite nonempty simple graphs G of arbitrarily large order for which*

$$\chi_f(G) \geq \frac{|V(G)|}{\alpha(G)} \geq \frac{|V(G)|}{2} > \kappa(G) + 1 \geq \mu(G) + 1 \geq h(G).$$

Proof. Take the constructed examples in **Theorem 1.1** and put $m = |V(G)|$. They have $\alpha(G) \leq 2$. Every well-signed one-negative matrix on these graphs has rank greater than $m/2 + 1$, so rank-nullity gives corank less than $m/2 - 1$. Maximizing in the larger matrix class yields $\kappa(G) + 1 < m/2$. Every matrix admitted in the definition of $\mu(G)$ is admitted for $\kappa(G)$, so $\mu(G) \leq \kappa(G)$. The inequality $h(G) \leq \mu(G) + 1$ was established above. Combining these bounds with the fractional-colouring estimate proves the chain. $\square$

The same graphs also violate the fractional-colouring weakening of Hadwiger’s conjecture discussed by Reed and Seymour [25, p. 148]: for every positive integer p , excluding a K_{p+1} minor was predicted to give a fractional p -colouring. Their definition uses nonnegative rational weights with coverage exactly one at each vertex and total weight at most p , which are admissible above; their result (1.3) proves the corresponding $2p$ -bound. Taking $p = h(G) \geq 1$ in the corollary rules out the coefficient-one prediction. The fractional relaxation here concerns colouring, while $h(G)$ remains the ordinary clique-minor number. The comparisons with κ and μ show that the obstruction survives both fractional colouring and omission of the Strong Arnold Property.

The list chromatic number $\chi_{\text{list}}(G)$ is the least integer k such that every assignment of finite colour lists of size at least k to the vertices admits a proper colouring from those lists.

Corollary 1.4 (A linear list-colouring bound). *Let C be the absolute integer in OpenAI [24, Theorem 1.1]. Every finite nonempty simple graph G satisfies*

$$\chi_{\text{list}}(G) \leq C(\mu(G) + 1).$$

Proof. That theorem gives $\chi_{\text{list}}(G) \leq Ch(G)$. Combining this with $h(G) \leq \mu(G) + 1$, proved above, gives the result. $\square$

1.4 From distribution tests to finite graphs

To prove [Theorem 1.2](#), we first work with probability laws on the finite set of frames. An ordered pair with no hole is called a *unit*. The two frames within a unit may be strongly dependent. The tests in [Theorem 5.4](#) give a positive lower bound for the probability that two independently sampled units conflict across distinct colours, and that a unit conflicts with an independently sampled point, under specified density bounds and, in the coloured test, an upper bound on each colour's mass.

The algebraic part gives sufficient certificates for four cross holes, using scalar conditions and equalities between ordered tuples of frame images, called *keys*. Once these equalities and scalar conditions hold, [Theorem 7.3](#) prescribes the remaining inner products between primal and channel frames, and [Theorem 8.1](#) bounds the probability of agreement with that prescription. The main probabilistic task is therefore to force overlap between the accepted key distributions despite dependence within each unit. Channel thinning supplies compatible scalar data. The histogram and product comparisons pass the actual query laws to a common limit, where finite-dimensional duality produces overlapping choices. Finally, the entropy fingerprint argument transfers the two distribution tests to one finite sample satisfying all three base-graph properties.

The paper first proves the matrix criterion. It then constructs the base graphs. [Sections 4](#) and [5](#) define the frame model and thinned law and formulate the distribution tests. [Sections 6](#) to [8](#) supply the finite algebraic certificates and the collision criterion. [Sections 9](#) and [10](#) establish compatibility on positive mass. [Sections 11](#) to [14](#) prove the comparison and overlap statements that establish the coloured distribution test. [Section 15](#) proves the independent-point distribution test. [Section 16](#) transfers both tests to finite graphs, proving [Theorem 1.2](#), and [Section A](#) records the parameter order and quantitative margins. [Section B](#) gives the compact measure constructions used in the limiting steps.

In the probabilistic construction, n is its unbounded input dimension, $N = M_0 n$ is the ambient frame dimension, and $m = 2^{1000gN}$ is the sampled graph order. The graph threshold β is distinct from the selector length b . The original raw probability law is denoted by P_0 , and its thinned version by π ; μ always denotes the graph invariant.

2 Finite trees for Lorentz blocks

The matrix argument needs a way to normalize several positive definite blocks at once, even when their dimensions tend to infinity. We first give a geometric construction whose constants depend only on the number of blocks. We then use the three conflict properties of [Theorem 1.2](#) to select the blocks that will be normalized together. All vector spaces and bilinear forms in this section are real. Logarithms in the hyperbolic estimates are natural logarithms.

2.1 Hyperbolic flats and block normalization

A *Lorentz space* is a finite-dimensional real vector space with bilinear form

$$[x, y] = -x_0 y_0 + \sum_{j=1}^d x_j y_j.$$

Its future unit hyperboloid is

$$\mathbb{H}^d = \{x : [x, x] = -1, x_0 > 0\}, \quad d_{\mathbb{H}}(x, y) = \operatorname{arcosh}(-[x, y]).$$

Unused positive coordinates may always be added, so we may take $d \geq 1$. The orthogonal complement of a future unit vector is positive definite. Indeed, if $x = (x_0, u)$ and $[x, (t, v)] = 0$, then $t = u \cdot v / x_0$, and

$$|v|^2 - t^2 \geq |v|^2 / x_0^2.$$

An orthonormal basis of this complement, together with x , therefore gives Lorentz coordinates in which $x = (1, 0)$. In these coordinates any other future unit vector has the form $(\cosh r, \sinh r u)$, with $|u| = 1$ and $r \geq 0$. In particular $-[x, y] \geq 1$. If y, z have radii r, s about x , the ordinary Cauchy–Schwarz inequality gives

$$-[y, z] = \cosh r \cosh s - \sinh r \sinh s (u \cdot v) \leq \cosh(r + s).$$

This proves the triangle inequality for $d_{\mathbb{H}}$; its other metric properties follow from the same coordinates.

The segment from x to y , of length $\ell > 0$, is

$$\gamma(t) = \frac{\sinh(\ell - t)}{\sinh \ell} x + \frac{\sinh t}{\sinh \ell} y, \quad 0 \leq t \leq \ell. \quad (2.1)$$

In coordinates about x it is $(\cosh t, \sinh t u)$, so $d_{\mathbb{H}}(\gamma(t), \gamma(t')) = |t - t'|$. When $x = y$ the segment is constant. A subspace P is *positive* if the Lorentz form restricted to it is positive definite. Its associated *flat* is

$$F_P = P^\perp \cap \mathbb{H}^d.$$

Orthogonal decomposition by a positive subspace leaves exactly one negative square, so F_P is nonempty. Formula (2.1) shows that it contains the segment between any two of its points.

Lemma 2.1 (Normalization near a flat). *Let P be positive and write $x = x_P + x_\perp$ for the orthogonal decomposition of $x \in \mathbb{H}^d$ relative to P . Then*

$$\sinh^2 d_{\mathbb{H}}(x, F_P) = [x_P, x_P]. \quad (2.2)$$

Suppose $P_1, \dots, P_k$ are positive subspaces whose flats have distance at most R from one common point. Choose any Lorentz-orthonormal basis in each P_i , concatenate these bases into a list A , and let $[A, A]$ be its Gram matrix. Then

$$\|[A, A]\|_{\text{op}} \leq k(\cosh^2 R + \sinh^2 R). \quad (2.3)$$

This bound does not depend on the dimensions of the subspaces. The subspaces are allowed to overlap. If $P_i \perp P_j$, then $F_{P_i} \cap F_{P_j} \neq \emptyset$.

Proof. Put $q = [x_P, x_P]$. Then $[x_\perp, x_\perp] = -1 - q$, and $y = x_\perp / \sqrt{1 + q}$ is future unit timelike. For $z \in F_P$, the hyperbolic coordinates above, now in $P^\perp$, give

$$-[x, z] = -[x_\perp, z] \geq \sqrt{1 + q},$$

with equality at $z = y$. This proves (2.2).

Use the common point as the time coordinate. For the basis matrix A_i , let t_i be its time-coordinate row and S_i its spatial-coordinate matrix. The projection formula gives $\|t_i\| \leq \sinh R$, and orthonormality gives

$$S_i^\top S_i = I + t_i^\top t_i, \quad \|S_i\|_{\text{op}} \leq \cosh R.$$

For the concatenated matrices S, t , Cauchy–Schwarz over the k blocks gives $\|S\|_{\text{op}}^2 \leq k \cosh^2 R$ and $\|t\|^2 \leq k \sinh^2 R$. Since $[A, A] = S^\top S - t^\top t$, this proves (2.3). Finally, if $P_i \perp P_j$, their sum is positive, and its orthogonal complement contains a future unit timelike vector perpendicular to both. $\square$

2.2 A finite configuration and its segment points

A *finite metric tree* is a finite tree whose edges are intervals of positive lengths, equipped with path distance; a single point is also allowed. Its segment between a, b is denoted $[a, b]_T$.

Finite hyperbolic configurations admit tree approximations with logarithmic additive error through Gromov's construction [13]. Its maximum–minimum chain closure is recorded explicitly by Cornet and Martínez-Pedroza [6, Section 2, equation (11)]. We give a direct hyperboloid argument with an error depending only on the number of marked points; the subsequent segment and flat lemmas supply the additional assertions needed for matrix normalization.

Lemma 2.2 (Finite tree approximation). *For $\ell \geq 1$, put*

$$K(\ell) = 4 + 2 \log(\max\{1, \ell - 1\}).$$

Given any ℓ points $z_1, \dots, z_\ell$ in a hyperboloid, in any dimension, there is a finite metric tree with marked points $z_1^T, \dots, z_\ell^T$ such that

$$|d_T(z_i^T, z_j^T) - d_{\mathbb{H}}(z_i, z_j)| \leq K(\ell). \quad (2.4)$$

The tree may be taken to be the hull of its marked points.

Proof. Take z_1 as origin and write $z_i = (\cosh r_i, \sinh r_i u_i)$. A direction with $r_i = 0$ can be chosen arbitrarily. For distinct indices set

$$q_{ij} = \frac{|u_i - u_j|}{2}, \quad a_{ij} = \min\{r_i, r_j, -\log q_{ij}\},$$

with $-\log 0 = +\infty$, and set $a_{ii} = r_i$. The exact identity

$$\cosh d_{\mathbb{H}}(z_i, z_j) = \cosh(r_i - r_j) + 2 \sinh r_i \sinh r_j q_{ij}^2 \quad (2.5)$$

implies

$$|d_{\mathbb{H}}(z_i, z_j) - (r_i + r_j - 2a_{ij})| \leq 4. \quad (2.6)$$

Here is a dimension-free verification. If $\min(r_i, r_j) < 1$, the triangle inequality and its reverse put both $(r_i + r_j - d_{\mathbb{H}}(z_i, z_j))/2$ and a_{ij} between zero and $\min(r_i, r_j)$, giving error at most two. Otherwise, writing

$$D = \max\{|r_i - r_j|, r_i + r_j + 2 \log q_{ij}\} = r_i + r_j - 2a_{ij},$$

the two terms in (2.5) give $e^D/4 \leq \cosh d_{\mathbb{H}}(z_i, z_j) \leq 2e^D$. Use $\log u \leq \operatorname{arcosh} u \leq \log(2u)$ for $u \geq 1$ to obtain an error at most $\log 4 < 4$.

For $i \neq j$, define A_{ij} as the largest bottleneck value $\min_{0 \leq t < k} a_{i_t i_{t+1}}$ over chains $i = i_0, \dots, i_k = j$, and put $A_{ii} = r_i$. Deleting loops never decreases a bottleneck, so only simple chains of at most $\ell - 1$ edges are needed. If a chain has bottleneck h , its endpoint radii are at least h , and the triangle inequality for the direction chords gives $q_{ij} \leq (\ell - 1)e^{-h}$. Consequently

$$a_{ij} \leq A_{ij} \leq a_{ij} + \log(\max\{1, \ell - 1\}), \quad A_{ij} \leq \min(r_i, r_j). \quad (2.7)$$

Concatenating chains also gives $A_{ij} \geq \min(A_{it}, A_{tj})$ for every t . Take intervals $[0, r_i]$, identifying their points of equal height t whenever $t \leq A_{ij}$. The last inequality makes this an equivalence relation at every height. These equivalence classes are nested as height decreases; their finitely many splitting heights therefore form a rooted finite tree. The distance between the two marked upper endpoints is $r_i + r_j - 2A_{ij}$. Equations (2.6)–(2.7) prove (2.4). Restricting to the hull of the marked points preserves their distances. $\square$

Endpoint approximation alone does not justify lifting a point of a tree hull back into a flat. The next lemma supplies this additional step, uniformly in the location of the point along its segment.

Lemma 2.3 (Comparison of segment points). *Let T and the points z_i^T satisfy (2.4). For $\theta \in [0, 1]$, associate the point at fraction θ of $[z_i^T, z_j^T]_T$ with the point at the same fraction of the hyperbolic segment $[z_i, z_j]$. If p, q are obtained in this way, from any two pairs of marked points, and $\hat{p}, \hat{q}$ are their associated hyperbolic points, then*

$$|d_{\mathbb{H}}(\hat{p}, \hat{q}) - d_T(p, q)| \leq 11K(\ell + 2). \quad (2.8)$$

For a degenerate segment any choice of its fractional parameter gives the same conclusion.

Proof. We first record an identity valid in any metric tree. If $p \in [a, b]_T$ and $q \in [c, d]_T$, then

$$d_T(p, q) = \max_{e \in \{a, b\}, f \in \{c, d\}} \{d_T(e, f) - d_T(e, p) - d_T(f, q)\}. \quad (2.9)$$

Every term is at most the left side by the triangle inequality. If $p \neq q$, choose an endpoint e whose path from p does not enter the component toward q , and an endpoint f whose path from q does not enter the component toward p . Such choices exist because p, q lie on their respective segments; an endpoint equal to the segment point is allowed. Then the path from e to f passes through p, q , giving equality. If $p = q$, either choose an endpoint equal to this common point, or choose endpoints in different components of its complement; again equality holds.

A second exact tree identity will locate the new marked points. If P_θ is at fraction θ of $[A, B]$ and X is any point of the tree, then

$$d(X, P_\theta) = \max\{d(A, X) - \theta d(A, B), d(B, X) - (1 - \theta)d(A, B)\}. \quad (2.10)$$

Indeed, if the projection of X onto $[A, B]$ is at distance u from A and at distance h from X , both sides equal $h + |u - \theta d(A, B)|$. This also holds for a degenerate segment.

Apply Theorem 2.2 to the original configuration together with $\hat{p}, \hat{q}$, obtaining a tree T' with error $K' = K(\ell + 2)$. Denote their marked points by p', q' . Suppose $\hat{p}$ is at fraction θ of the hyperbolic segment $[a, b]$, and let $\tilde{p}$ have that fraction on $[a', b']_{T'}$. The two exact hyperbolic distances from $\hat{p}$ to its endpoints are $\theta d_{\mathbb{H}}(a, b)$ and $(1 - \theta)d_{\mathbb{H}}(a, b)$. In (2.10), each corresponding difference is therefore at most $2K'$. Thus

$$d_{T'}(p', \tilde{p}) \leq 2K', \quad d_{T'}(q', \tilde{q}) \leq 2K'.$$

Together with the K' distance error for the pair $\hat{p}, \hat{q}$, this gives

$$|d_{\mathbb{H}}(\hat{p}, \hat{q}) - d_{T'}(\tilde{p}, \tilde{q})| \leq 5K'.$$

Between the original endpoint sets, distances in T and T' differ by at most $K(\ell) + K' \leq 2K'$. Distances from an endpoint to its fractional segment point are that fraction, or its complement, times the segment length. Each term in (2.9) therefore changes by at most $6K'$, and hence

$$|d_T(p, q) - d_{T'}(\tilde{p}, \tilde{q})| \leq 6K'.$$

The total error is at most $11K'$. All identities remain valid when an endpoint segment has length zero. $\square$

Proposition 2.4 (Finite tree principle). *For each index s , let $P_{1,s}, \dots, P_{k,s}$ be positive subspaces in a Lorentz space, whose dimension may depend on s . Fix a set $\mathcal{D}$ of pairs of distinct labels, and assume*

$$\sup_s d_{\mathbb{H}}(F_{P_{i,s}}, F_{P_{j,s}}) \leq D < \infty \quad (\{i, j\} \in \mathcal{D}).$$

After passing to a subsequence, there are a finite combinatorial tree T and a nonempty connected vertex set $S_i \subseteq V(T)$ for every label, with the following properties.

- (i) $S_i \cap S_j \neq \emptyset$ for every $\{i, j\} \in \mathcal{D}$.
- (ii) *For every vertex x , called a bag, choose any Lorentz-orthonormal basis in every $P_{i,s}$ with $x \in S_i$. The Gram matrix of the concatenated list has operator norm bounded uniformly in s . The bound depends only on k, D , and not on the dimensions or on the bases.*

In particular, mutually orthogonal blocks can always be required to have intersecting subtrees.

Proof. For every required pair, choose a point in each of its two flats at distance at most $D + 1$; an infimum need not be attained. Choose also one point in each flat. This gives $\ell = k + 2|\mathcal{D}| \leq k^2$ marked representatives. Apply [Theorem 2.2](#) and work in their tree hull K_s . Let $H_{i,s}$ be the hull of representatives chosen from the i -th flat, and set

$$R = \frac{D + 1 + K(\ell)}{2}, \quad U_{i,s} = \{x \in K_s : d_T(x, H_{i,s}) \leq R\}.$$

The R -neighborhood of a subtree in a tree is a closed subtree: the segment between two of its points lies in the union of their paths to the original subtree and the subtree itself, and stays within distance R of it. For a required pair, the midpoint of its two witness representatives belongs to both neighborhoods. Thus every required contact is preserved inside K_s .

Suppose $x \in U_{i,s} \cap U_{j,s}$. Choose $h_i \in H_{i,s}$, $h_j \in H_{j,s}$ at distance at most R from x . Every point of a finite tree hull lies on a segment between two of its defining representatives, allowing a repeated representative when the hull is a point. For example, after fixing one representative the union of its segments to all the others is already their hull. Lift h_i, h_j at their fractional parameters along the corresponding actual hyperbolic segments. Their lifts z_i, z_j belong to their respective flats by convexity, and [Theorem 2.3](#) gives

$$d_{\mathbb{H}}(z_i, z_j) \leq 2R + 11K(\ell + 2) =: R_*. \quad (2.11)$$

For any nonempty bag, fix one of these lifts. It has distance at most R_* from every flat in the bag. By [Theorem 2.1](#), their joint Gram norm is at most $k(\cosh^2 R_* + \sinh^2 R_*)$.

It remains to obtain a single finite combinatorial pattern. Suppress the degree-two vertices of K_s except marked representatives; there are at most 2ℓ vertices and at most 2ℓ edges. Each $U_{i,s}$ meets an edge in an interval or the empty set. Subdivide every edge at all endpoints of these intervals. This adds at most $2k$ vertices per edge, so the resulting tree has at most $2\ell(2k + 1)$ vertices. Each $U_{i,s}$ is now a subcomplex. There are only finitely many trees of this bounded size with k specified nonempty connected vertex sets. Along a subsequence the labelled pattern is constant. Take this pattern for T, S_i . Equation (2.11) proves the uniform bag assertion. The final assertion follows from [Theorem 2.1](#) with $D = 0$. $\square$

For a finite list of flat sequences, a preliminary subsequence can also arrange that every pairwise distance is either uniformly bounded or tends to infinity. For each pair in turn, take a bounded subsequence if one exists, and otherwise the distance tends to infinity; previous choices remain

valid after further subselection. Applying [Theorem 2.4](#) with all bounded pairs as required contacts preserves every such pair. No assertion that the remaining subtrees are disjoint is needed.

2.3 Weighted extraction from the conflict properties

We use a finite graph G on m vertices with the three properties in [Theorem 1.2](#), with parameters $a, \beta > 0$. For clarity, the properties used here are:

- (a) Every matching of size at least $m/20$, partitioned into classes of size at most am , contains conflicting edges from different classes.
- (b) Every matching of size at least βm and every vertex set of size at least βm contain an edge and a vertex in conflict.
- (c) Every two vertex sets of size at least βm have a cross hole.

A hole is a nonedge between distinct vertices. Edges conflict when they are disjoint and all four cross pairs are holes. An edge and a vertex conflict when the vertex is outside the edge and both cross pairs are holes. Thus a shared endpoint prevents a conflict. We may decrease a and assume $0 < a \leq 1$. Put

$$q = \lceil \beta m \rceil, \quad D_a = \lceil \log_2(5/a) \rceil + 1, \quad K_a = \lceil 5/a \rceil D_a, \quad c = c(a) = \frac{1}{40K_a}. \quad (2.12)$$

An edge family $\mathcal{E} \subseteq E(G)$ has *loads at most one* if its nonnegative weights satisfy

$$\sum_{e \in \mathcal{E}: v \in e} w_e \leq 1 \quad (v \in V(G)). \quad (2.13)$$

Then $\sum_e w_e \leq m/2$. We only need edges of positive weight.

Lemma 2.5 (Matching from bounded loads). *Every edge family of total weight W satisfying (2.13) has a matching of size at least $W/2$. If it contains no conflicting pair and $am \geq 1$, then $W < m/10$.*

Proof. Take a maximal matching of size k in the family. Its $2k$ endpoints cover every edge, so the load bound gives $W \leq 2k$. In the second assertion, $k \geq m/20$ would violate Property (a) with each matching edge given its own class. Hence $k < m/20$, proving the claim. $\square$

Lemma 2.6 (A bag with almost all vertex labels). *Let $V_0 \subseteq V(G)$, and assign a nonempty subtree S_v of a finite combinatorial tree to each $v \in V_0$. Assume $S_u \cap S_v \neq \emptyset$ whenever uv is a hole. Then some tree vertex belongs to all but fewer than $3q$ of the subtrees.*

Proof. At a tree vertex x , each subtree missing x is wholly contained in a unique component of $T - x$. No two components can each contain at least q whole subtrees: taking q labels from each would give two vertex sets with no cross hole, contrary to Property (c). If there is a component with at least q whole subtrees, orient the first edge from x toward that component. Each vertex has at most one outgoing edge. An edge cannot be oriented both ways, since its two orientations would again supply two separated sets of q labels. Following arrows in a finite tree must therefore end at a sink r .

Every component of $T - r$ contains at most $q - 1$ whole subtrees. If $q = 1$, no subtree misses r . If $q \geq 2$ and at least $3q - 2$ labels missed r , greedily unite components until their total first

reaches q . This total is at most $2q - 2$; the remaining components contain at least q labels. These two separated families contradict Property (c). The sink thus misses fewer than $3q - 2$, and in particular fewer than $3q$, labels. $\square$

Lemma 2.7 (A bag with positive edge weight). *Suppose $am \geq 2$. Let an edge family $\mathcal{E}$ satisfy (2.13) and have total weight greater than $3m/10$. Assign a nonempty subtree S_e of a finite tree to each edge, and suppose conflicting edges have intersecting subtrees. Then some vertex z satisfies*

$$\sum_{e: z \in S_e} w_e \geq cm.$$

Proof. Assign every edge's weight to one chosen vertex of its subtree. Every finite tree with an assigned nonnegative mass has a vertex whose removal leaves each component with at most half the total mass. To see this, minimize the sum of mass times graph distance from the vertex. Moving to a neighbor in a component of mass greater than half the total would strictly decrease that sum. Such a minimizing vertex is a weighted centroid.

Starting with T , cut at a weighted centroid in every current component whose assigned mass exceeds $am/10$, and repeat. At depth j each component has mass at most $m/2^{j+1}$, since the original mass is at most $m/2$. Consequently at most D_a rounds are needed. At each depth there are at most $\lceil 5/a \rceil$ components requiring a cut, because their masses sum to at most $m/2$. Thus at most K_a cut vertices are used. Mass located at a cut vertex is removed at that cut.

Suppose every bag has edge weight less than cm . Delete every edge whose subtree meets a cut vertex. The deleted weight is less than $K_a cm = m/40$, so the surviving weight W' satisfies $W' > m/4$. Each surviving subtree is wholly contained in one component of the cut tree; its assigned vertex lies there too. The total surviving weight in each such component is at most $am/10$. Use the components as classes of surviving edges. Subtrees of different classes are disjoint, so edges in different classes cannot conflict.

Put $Q = \lfloor am \rfloor$, so $Q \geq am/2$. Greedily choose a matching, with at most Q selected edges from any class, until no further edge can be added. If k edges have been selected, every surviving edge either meets one of their $2k$ endpoints or belongs to a class with Q selected edges. The first family has weight at most $2k$. There are at most k/Q saturated classes, each of weight at most $am/10$. Therefore

$$\frac{m}{4} < W' \leq 2k + \frac{k}{Q} \frac{am}{10} \leq \frac{11}{5}k.$$

It follows that $k > 5m/44 > m/20$. Its class sizes are at most $Q \leq am$, but it has no conflict between different classes. This contradicts Property (a). $\square$

Proposition 2.8 (A common bag). *Suppose the hypotheses of Theorems 2.6 and 2.7 hold in the same tree. Suppose additionally that $S_e \cap S_v \neq \emptyset$ whenever the edge e and the vertex $v \in V_0$ conflict. If $q \leq cm/4$, there is a vertex x such that*

$$|\{v \in V_0 : x \notin S_v\}| < 5q, \quad \sum_{e: x \in S_e} w_e \geq \frac{cm}{2}. \quad (2.14)$$

If all edges of $\mathcal{E}$ have both endpoints in V_0 and $q \leq cm/30$, deleting every edge incident with a missing vertex leaves bag weight at least $cm/3$.

Proof. Let r be the vertex bag from Theorem 2.6, let z be the edge bag from Theorem 2.7, and let $A = \{v \in V_0 : r \in S_v\}$. Along the path from r to z , each S_v with $v \in A$ is an initial interval.

Thus the number of absent labels from A is nondecreasing. If fewer than $2q$ are absent at z , use $x = z$. It misses fewer than $3q + 2q$ supplied vertex labels and has edge weight at least cm .

Otherwise let xy be the first edge of this path, in the direction from r to z , for which fewer than $2q$ labels of A are absent at x , and at least $2q$ are absent at y . Set

$$A_- = \{v \in A : y \notin S_v\}, \quad \mathcal{E}_+ = \{e : z \in S_e, x \notin S_e\}.$$

Every subtree indexed by A_- is contained in the x -side of $T - xy$: it contains r and avoids y . Every subtree indexed by $\mathcal{E}_+$ is contained in the y -side: it contains z and avoids x . In particular these two families are disjoint. This uses *all* root labels absent at y , including labels that ceased to occur earlier on the path; it does not require many labels to disappear on the single edge xy . See Figure 2.

If the bag weight at x were less than $cm/2$, then $\sum_{e \in \mathcal{E}_+} w_e > cm/2$. By Theorem 2.5, $\mathcal{E}_+$ has a matching of size greater than $cm/4$, hence at least q since $q \leq cm/4$. Also $|A_-| \geq 2q$. No selected edge conflicts with any label in A_- , because such a conflict would force intersecting subtrees. Taking any q of these labels violates Property (b). Thus the bag at x has weight at least $cm/2$; it misses fewer than $5q$ supplied labels.

Finally, the load bound shows that removing all edges incident with these missing vertices costs less than $5q$ weight. If $q \leq cm/30$, the surviving bag weight is at least $cm/2 - 5q \geq cm/3$, as asserted. $\square$

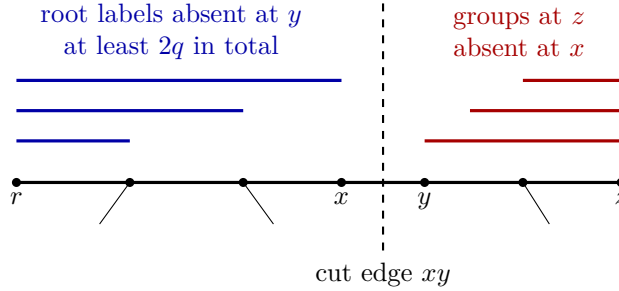


Figure 2: The path argument uses cumulative absence. The blue intervals show intersections with the path of root subtrees that avoid y ; some may end well before x . The red intervals come from group subtrees containing z and avoiding x . The two whole families lie on opposite sides of the cut, so no edge-vertex conflict can occur between them.

Remark 2.9 (Use with coordinate matchings). In the matrix application a base label indexes at most s coordinates, and the edge weight is the limit of the number of matched coordinate pairs of that type divided by s . A matching gives (2.13) automatically. Deleting all pairs incident with the fewer than $5q$ labels missing from the common bag removes fewer than $5qs$ pairs. It removes entire edge types, so any previously chosen normalization inside a retained type stays the same. The finite tree principle applies jointly to the original positive label blocks and the positive blocks of the candidate edge types. These subspaces may overlap. A common bag consequently bounds all their normalized Gram interactions at once. Passing to coordinate subsets is a compression by an isometry, so it preserves that bound.

3 The matrix obstruction

We now prove the conditional matrix statement. Only the three properties of the base graph enter this section; the Strong Arnold Property is not needed. Throughout, $v = ms$ denotes the order of the clique blow-up. Matrix traces and squared Frobenius norms will be divided by v , even when the matrix under consideration has fewer than v rows.

Theorem 3.1 (Matrix obstruction). *For every $a \in (0, 1]$ there are constants $\beta_0(a) > 0$ and $m_0(a) \in \mathbb{N}$ with the following property. Let $0 < \beta \leq \beta_0(a)$, let $m \geq m_0(a)$, and let G be an m -vertex graph satisfying the following conditions:*

- (i) *Every matching of size at least $m/20$, partitioned into classes of size at most am , contains two conflicting edges in different classes.*
- (ii) *Every matching of size at least βm and every vertex set of size at least βm contain a conflicting edge–point pair.*
- (iii) *Every two vertex sets of size at least βm have a cross hole.*

Then there is $s_0 = s_0(G, a, \beta)$ such that, for every integer $s \geq s_0$, every real symmetric well-signed matrix M for the clique blow-up $G[K_s]$ with exactly one negative eigenvalue satisfies

$$\text{rank } M > \frac{ms}{2} + 1.$$

Consequently $\mu(G[K_s]) + 1 < ms/2$ for all these s .

The proof uses two normalizations. A first normalization, separately inside the vertex cliques, produces almost disjoint coordinate pairs. A second normalization, inside groups of pairs having the same two base labels, supplies additional positive directions. The tree lemmas ensure that both normalizations can be used in one bounded Gram matrix. We first give the linear-algebra estimates that make the errors independent of the bounds furnished by the tree.

3.1 Positive blocks and bounded spectral limits

For a real symmetric matrix A , write $n_+(A)$ and $n_-(A)$ for its positive and negative indices, counting multiplicity. The notation $\|A\|$ means the Euclidean operator norm, and $\|A\|_F^2 = \text{Tr}(A^\top A)$. A symmetric matrix with at most one negative eigenvalue is a Gram matrix for the Lorentz form

$$\langle x, y \rangle = -x_0 y_0 + \sum_{i \geq 1} x_i y_i :$$

diagonalize the matrix and multiply each eigenvector coordinate by the square root of the absolute value of its eigenvalue. Unused positive coordinates can be added. Every restriction or coefficient pullback of this Gram matrix still has at most one negative eigenvalue. For lists X, Y of Lorentz vectors, $\langle X, Y \rangle$ denotes their rectangular Gram matrix; the same letter denotes the corresponding synthesis map.

The next facts are standard consequences of symmetric M -matrix theory and the Perron–Frobenius theorem; see Fiedler and Schneider [10, Theorem 5.2] and Debreu and Herstein [9, Theorem I]. Here an M -matrix has the form $cI - A$ with A entrywise nonnegative and c at least its spectral radius. We include the elementary proofs for the precise symmetric cases used below.

Lemma 3.2 (Normalization of a positive block). *Let D be real symmetric with nonpositive off-diagonal entries.*

- (i) *If D is positive definite, its positive inverse square root $D^{-1/2}$ is entrywise nonnegative and has positive diagonal entries.*
- (ii) *Suppose D is positive semidefinite and the graph of its strictly negative off-diagonal entries is connected. If D is singular, its kernel is one-dimensional and is spanned by a vector all of whose coordinates are positive. Every proper principal submatrix of D is positive definite.*

Proof. For (i), choose $c > \lambda_{\max}(D)$ and put $A = I - D/c$. All entries of A are nonnegative, and its eigenvalues lie in $[0, 1)$. The convergent binomial series gives

$$D^{-1/2} = c^{-1/2} \sum_{k=0}^{\infty} \frac{\binom{2k}{k}}{4^k} A^k. \quad (3.1)$$

Every term is entrywise nonnegative, and the constant term has positive diagonal.

For (ii), take $c > \lambda_{\max}(D)$ and consider $A = cI - D$. It is symmetric and entrywise nonnegative, with positive diagonal and connected off-diagonal support. A unit vector maximizing its Rayleigh quotient may be replaced by its coordinatewise absolute value, so a nonnegative maximizing eigenvector exists. Its eigenvalue is positive. The eigenvector equation and connectivity imply that it has no zero coordinate: a zero coordinate would force zeros at every adjacent coordinate and then everywhere. Moreover, every vector for the largest eigenvalue has one sign. Indeed, equality in the absolute-value Rayleigh inequality forces equal signs across each positive edge, while its absolute value is a nonnegative maximizing eigenvector and thus has full support. If the largest eigenspace had dimension at least two, it would contain a nonzero vector orthogonal to the positive eigenvector; that vector could not have one sign. The largest eigenvalue is therefore simple.

When D is singular and positive semidefinite, this largest eigenvalue of A is c , so the asserted kernel description follows. If a proper principal submatrix were singular, extend a nonzero vector of zero quadratic form by zero coordinates. Its quadratic form under D is zero, hence it belongs to $\ker D$ by the spectral theorem. This contradicts full support. $\square$

Lemma 3.3 (Spectral support from signs and inertia). *Let $v_j \rightarrow \infty$, and let B_j be symmetric entrywise nonnegative matrices of orders $d_j \leq v_j$, with zero diagonal and $\sup_j \|B_j\| < \infty$. Suppose $n_-(I - B_j) \leq 1$. Every weak subsequential limit of*

$$\nu_j = \frac{1}{v_j} \sum_{i=1}^{d_j} \delta_{\lambda_i(B_j)}$$

is supported on $[-1, 1]$ and has first moment zero. In particular, for each $\varepsilon > 0$, the number of eigenvalues of $I - B_j$ outside $[-\varepsilon, 2 + \varepsilon]$ is $o(v_j)$.

If also

$$\liminf_j \frac{d_j}{v_j} \geq 1 - \delta, \quad \text{rank}(I - B_j) \leq v_j/2 + 1,$$

then

$$\limsup_j \frac{\|B_j^2 - I\|_F^2}{v_j} \leq 2\delta. \quad (3.2)$$

The constant in this inequality is independent of the operator bound.

Proof. By [Theorem B.1](#), all measures have bounded mass and support in a common compact interval, so every subsequence has a weakly convergent further subsequence. Fix such a limit ν , and write $\theta = \nu(\mathbb{R}) \leq 1$. At most one eigenvalue of B_j exceeds 1, hence ν is supported on $(-\infty, 1]$. For every integer $k \geq 0$, entrywise nonnegativity gives $\text{Tr } B_j^{2k+1} \geq 0$, and therefore $\int x^{2k+1} d\nu(x) \geq 0$. If ν gave mass $h > 0$ to $(-\infty, -1 - \varepsilon]$, this integral would be at most $\theta - h(1 + \varepsilon)^{2k+1}$, which is negative for large k . This proves the support assertion. The zero diagonal gives $\int x d\nu(x) = 0$. The asserted eigenvalue count follows by applying these facts to any putative subsequence on which the normalized exceptional count stays positive.

For the additional assertion, pass to a subsequence realizing the limsup in (3.2) and then to a weak limit as above. The nullity assumption supplies at least $d_j - v_j/2 - 1$ eigenvalues equal to 1. Since singleton sets are closed, weak convergence gives $r := \nu(\{1\}) \geq \theta - 1/2$. Consequently

$$\int_{x \neq 1} (1+x) d\nu(x) = \theta - 2r \leq 1 - \theta \leq \delta.$$

For $-1 \leq x \leq 1$ we have $(1-x^2)^2 \leq 1-x^2 \leq 2(1+x)$, and the first expression vanishes at $x = 1$. Polynomial integrals converge on the common compact support, so

$$\lim_j \frac{\|B_j^2 - I\|_{\mathbb{F}}^2}{v_j} = \int (1-x^2)^2 d\nu(x) \leq 2\delta.$$

In this argument the bound on $\|B_j\|$ is fixed before j tends to infinity. It removes the contribution of any exceptional eigenvalue to each fixed moment; it does not enter the resulting bound. $\square$

Row-maximum rounding also appears in the study of nonnegative orthogonality constraints [2, 15]. Here symmetry and zero diagonal produce a partial matching, with bounds independent of the matrix order after normalization by v .

Lemma 3.4 (Extraction of coordinate pairs). *Let B be a symmetric entrywise nonnegative $d \times d$ matrix with zero diagonal, where $d \leq v$. Suppose*

$$\|B^2 - I\|_{\mathbb{F}}^2 \leq \varepsilon v, \quad 0 < \varepsilon \leq 8^{-4}, \quad \tau = \varepsilon^{1/4}.$$

There is a matching of coordinates with adjacency matrix W , extended by zero on the omitted coordinates, such that

$$\#\{\text{omitted coordinates}\} \leq 5\tau^2 v, \tag{3.3}$$

$$\|B - W\|_{\mathbb{F}}^2 \leq 8\tau^2 v, \tag{3.4}$$

$$|B_{ij} - 1| \leq \tau \quad \text{on every retained pair } \{i, j\}. \tag{3.5}$$

Every retained pair has $B_{ij} > 0$.

Proof. Put

$$d_i = \sum_j B_{ij}^2, \quad r_i = d_i^2 - \sum_j B_{ij}^4 \geq 0.$$

The diagonal entries of $B^2 - I$ show that $\sum_i (d_i - 1)^2 \leq \varepsilon v$. Nonnegativity also gives

$$\begin{aligned} \sum_i r_i &= \sum_{j \neq k} \sum_i B_{ij}^2 B_{ik}^2 \\ &\leq \sum_{j \neq k} \left(\sum_i B_{ij} B_{ik} \right)^2 \leq \varepsilon v. \end{aligned}$$

Call row i good when $|d_i - 1| \leq \tau$ and $r_i \leq \tau^2$. There are at most $2\tau^2 v$ bad rows. In a good row, let $a_i = \max_j B_{ij}^2$. Then

$$a_i \geq \frac{\sum_j B_{ij}^4}{d_i} = d_i - r_i/d_i, \quad d_i - a_i \leq \frac{\tau^2}{1 - \tau}.$$

Because $\tau \leq 1/8$, it follows that

$$a_i \geq \frac{1 - 2\tau}{1 - \tau} \geq \frac{6}{7}, \quad d_i - a_i \leq \frac{8}{7}\tau^2, \quad |\sqrt{a_i} - 1| \leq \tau.$$

For the last inequality, use $1 - \tau - \tau^2/(1 - \tau) \geq (1 - \tau)^2$ and $a_i \leq 1 + \tau \leq (1 + \tau)^2$. There is a unique entry attaining a_i . If its other endpoint is good, symmetry forces it to be the dominant entry there as well, since its square is at least $6/7$ whereas every nondominant squared entry in a good row is at most $8\tau^2/7$. These good-good pairs form the required matching.

For any set S of rows, Cauchy-Schwarz gives

$$\sum_{i \in S} d_i \leq |S| + \sqrt{|S|\varepsilon v}. \quad (3.6)$$

Thus the total squared row mass at bad rows is at most $(2 + \sqrt{2}\tau)\tau^2 v$. Every good row whose dominant endpoint is bad contributes at least $6/7$ to this mass, by symmetry. The total number of omitted coordinates is consequently at most

$$\left(2 + \frac{7}{6}(2 + \sqrt{2}\tau)\right)\tau^2 v \leq 5\tau^2 v.$$

For a retained row, its contribution to $\|B - W\|_F^2$ is at most $\tau^2 + 8\tau^2/7 = 15\tau^2/7$. By (3.6), the omitted rows contribute at most $(5 + \sqrt{5}\tau)\tau^2 v$. Since $d \leq v$ and $\tau \leq 1/8$, the sum is less than $8\tau^2 v$. The remaining assertions have already been proved. $\square$

Lemma 3.5 (A trace bound across the second normalization). *For each j , let H_j, Q_j be two lists of Lorentz vectors whose joint Gram matrix has operator norm at most a fixed K . Let $v_j \rightarrow \infty$. Suppose that for every $\varepsilon > 0$ the Gram matrix $\langle Q_j, Q_j \rangle$ has $o(v_j)$ eigenvalues above $2 + \varepsilon$. If*

$$\limsup_j \frac{\text{Tr}\langle H_j, H_j \rangle}{v_j} \leq \eta,$$

then

$$\limsup_j \frac{\|\langle H_j, Q_j \rangle\|_F^2}{v_j} \leq 2\eta. \quad (3.7)$$

The conclusion is independent of K .

Proof. Let J_j be the joint Gram matrix. Its negative part has rank at most one and norm at most K . Replacing J_j by its positive part changes any normalized trace by $O(K/v_j)$ and changes a Frobenius norm divided by $\sqrt{v_j}$ by $O(K/\sqrt{v_j})$. Write the resulting positive semidefinite matrix as

$$P_j = \begin{pmatrix} A_j & C_j \\ C_j^\top & D_j \end{pmatrix}.$$

A rank-one perturbation changes the number of eigenvalues above any fixed threshold by at most one, by the variational characterization of eigenvalues. Thus the spectral projector E_{hi} of D_j for $(2 + \varepsilon, \infty)$ has rank $o(v_j)$. Put $E_{\text{lo}} = I - E_{\text{hi}}$. Factor P_j as a Euclidean Gram matrix, say $A_j = U_j^\top U_j$, $D_j = V_j^\top V_j$ and $C_j = U_j^\top V_j$. Since $\|V_j E_{\text{lo}}\|^2 \leq 2 + \varepsilon$,

$$\|C_j E_{\text{lo}}\|_{\text{F}}^2 \leq (2 + \varepsilon) \text{Tr } A_j.$$

On the other spectral subspace, $\|C_j E_{\text{hi}}\|_{\text{F}}^2 \leq K^2 \text{rank } E_{\text{hi}} = o(v_j)$. Divide by v_j and take the limsup. This first bounds $\|C_j\|_{\text{F}}/\sqrt{v_j}$. The original cross block differs by Frobenius norm at most K , so it has the same squared normalized limsup bound, at most $(2 + \varepsilon)\eta$. Finally let $\varepsilon \downarrow 0$. In particular the hypotheses force $\eta \geq 0$, since the normalized trace of the original H_j block has nonnegative liminf. $\square$

3.2 The first normalization and the candidate groups

Proof of Theorem 3.1. Use the constant $c = c(a) > 0$ from Theorem 2.7; explicitly we may take

$$\begin{aligned} K(a) &= \left\lceil \frac{5}{a} \right\rceil \left(\left\lceil \log_2 \frac{5}{a} \right\rceil + 1 \right), & c(a) &= \frac{1}{40K(a)}, \\ \delta_0(a) &= \min \left\{ \frac{1}{3 \cdot 8^4}, \frac{c(a)}{12}, \frac{1}{3} \left(\frac{c(a)}{2340} \right)^4 \right\}, & \beta_0(a) &= \frac{\delta_0(a)}{10}, \\ m_0(a) &= \left\lceil \max \left\{ \frac{2}{a}, \frac{10}{\delta_0(a)}, 2 \right\} \right\rceil. \end{aligned} \quad (3.8)$$

Fix a, β, m, G as in the theorem with these choices, and put

$$q = \lceil \beta m \rceil, \quad \delta = 5(\beta + 1/m), \quad \tau = (3\delta)^{1/4}. \quad (3.9)$$

These choices imply

$$\delta \leq \delta_0, \quad \tau \leq \frac{1}{8}, \quad \tau \leq \frac{c}{2340}, \quad \frac{q}{m} \leq \frac{\delta}{5} \leq \frac{c}{60}, \quad am \geq 2. \quad (3.10)$$

Suppose the conclusion fails. There is then an unbounded sequence of integers s and corresponding well-signed matrices M_s with one negative eigenvalue such that, writing $v = ms$,

$$\text{rank } M_s \leq v/2 + 1. \quad (3.11)$$

All subsequences below keep a, β, m and the graph G fixed, and $s \rightarrow \infty$ is the only limiting operation. An expression $o(v)$ has ratio to v tending to zero in this limit; its constants may depend on the fixed graph and subsequence. The displayed constants in the errors involving δ or τ are absolute.

Represent M_s by Lorentz Gram vectors, one for each blow-up vertex. An original vertex label $u \in V(G)$ specifies its s -column clique block. By passing to a subsequence, the set of labels whose clique blocks have a negative eigenvalue can be fixed. Two such labels cannot be a hole: zero cross Gram would make their negative directions orthogonal and give two negative eigenvalues of M_s . These labels therefore form a clique in G . There are fewer than $2q$ of them, since two disjoint q -element subsets of a larger clique would violate condition (iii). Discard these labels.

The other clique blocks are positive semidefinite. They have strictly negative off-diagonal entries; Theorem 3.2 shows that deleting at most one coordinate makes each positive definite.

Work with $s \geq 2$, and apply [Theorem 2.4](#) to their positive spans. For every hole between two labels the blocks are orthogonal, so their tree subtrees intersect. By [Theorem 2.6](#), there is a bag missing fewer than $3q$ of the supplied single labels. Keep the labels in this bag and denote their set by U . There are fewer than $5q$ missing original labels. The retained coordinate dimension d_s satisfies

$$d_s \geq (m - 5q)(s - 1), \quad \liminf_{s \rightarrow \infty} \frac{d_s}{v} \geq 1 - \delta. \quad (3.12)$$

The tree and the chosen bag are fixed on a further subsequence.

Normalize each retained clique block by its positive inverse square root, and denote the resulting column lists by X_u , $u \in U$. Their diagonal Gram blocks are identities. By [Theorem 3.2](#), the normalizing coefficient matrices are entrywise nonnegative with positive diagonal. Thus cross blocks remain zero on holes and strictly negative on base edges. The full Gram matrix of the X_u is

$$A_s = I - B_s, \quad B_s \geq 0 \text{ entrywise}, \quad (B_s)_{uu} = 0 \text{ as a whole diagonal block.} \quad (3.13)$$

The bag bound from [Theorems 2.1](#) and [2.4](#) makes $\|B_s\|$ uniformly bounded in s . The rank is at most that of M_s , and its negative index is at most one. Consequently [Theorem 3.3](#) gives

$$\limsup_{s \rightarrow \infty} \frac{\|B_s^2 - I\|_F^2}{v} \leq 2\delta.$$

Since $\delta > 0$ is fixed, the left-hand quantity before taking a limsup is at most $3\delta = \tau^4$ for all sufficiently large s . Apply [Theorem 3.4](#) to obtain a coordinate matching with p_s^0 pairs and adjacency matrix W_s . Then

$$\begin{aligned} \liminf_s \frac{p_s^0}{v} &\geq \frac{1}{2} - \frac{\delta}{2} - \frac{5}{2}\tau^2, \\ \|B_s - W_s\|_F^2 &\leq 8\tau^2 v, \quad |(B_s)_{ij} - 1| \leq \tau \quad \text{on retained pairs.} \end{aligned} \quad (3.14)$$

Every pair connects two different adjacent base labels, because B_s vanishes within each original block and on base holes.

Group the pairs by their unordered label pair $e = \{u, w\} \in E(G)$. If group e contains $k_e(s)$ pairs, pass to a subsequence on which every $k_e(s)/s$ converges, and put

$$w_e = \lim_s k_e(s)/s, \quad \sum_{e \ni u} w_e \leq 1, \quad \sum_e w_e \leq m/2. \quad (3.15)$$

There are only finitely many types, so we may also fix whether each group is empty or its endpoint Gram block is positive definite, singular positive semidefinite, or has a negative eigenvalue.

Groups in the last category will not be candidates for the second normalization. Their types are pairwise touching: a conflict between two of them would make their cross Gram zero and give two orthogonal negative directions. Choose a maximal ordinary matching of these types in the base graph, of size k . If $k \geq m/20$, putting each edge in its own class contradicts (i), since $am \geq 1$. Thus $k < m/20$. The $2k$ endpoints of this maximal matching cover all the types under consideration. The load bound in [\(3.15\)](#) therefore gives

$$\sum_{e: \text{group has a negative eigenvalue}} w_e \leq 2k < m/10. \quad (3.16)$$

Their coordinate pairs are retained in the first matching.

For every group of positive weight with positive semidefinite endpoint Gram, that Gram has the form

$$\begin{pmatrix} I & -C \\ -C^\top & I \end{pmatrix}, \quad C \text{ entrywise strictly positive.}$$

The graph of negative entries is connected. If the group is singular, delete one whole matching pair. By [Theorem 3.2](#) its remaining Gram block is positive definite. Positive weight ensures that the group is eventually nonempty after this deletion. There are at most $\binom{m}{2}$ groups, so these deletions cost $O(m^2) = o(v)$ coordinates. Call the resulting positive definite groups of positive weight the *candidate groups*. Groups of zero limiting weight need not be candidates and keep their pairs in the first matching. By (3.14), (3.16) and $\tau \leq 1/8$, the total candidate weight satisfies

$$\sum_{e \text{ candidate}} w_e \geq m \left(\frac{2}{5} - \frac{\delta}{2} - \frac{5}{2} \tau^2 \right) > \frac{3m}{10}. \quad (3.17)$$

3.3 A common bag for both normalizations

Apply [Theorem 2.4](#) again, this time to the single-label positive spans for all $u \in U$ together with the positive spans of all candidate groups. These spans are allowed to overlap. The required subtree contacts hold in each of the following cases: two single labels form a hole; two group types conflict; or a group type and a single label conflict. In each case all cross Gram entries vanish, so the positive spans are orthogonal and their flats intersect.

The candidate weights satisfy [Theorem 2.7](#), so a bag has group weight at least cm . Apply [Theorem 2.8](#) to the supplied single labels U and these groups, using $q \leq cm/60$. It yields a bag containing all but fewer than $5q$ labels of U , and containing candidate group weight at least $cm/2$. The common-bag proof uses all root labels absent after its threshold edge, so it applies equally when the losses occur one label at a time.

Discard from the remaining coordinate matching every pair incident to one of the missing labels of U . This removes whole label-pair groups and at most $5qs$ pairs. Denote the surviving total number of pairs by $p = p_s$. Let $\mathcal{T}$ be the candidate groups in the common bag whose endpoints both remain, and let $t = t_s$ be their total number of pairs. Since the load at a missing label is at most one, the limiting weight lost from the bag is at most $5q$. Thus

$$\liminf_s \frac{p}{v} \geq \frac{1}{2} - \frac{\delta}{2} - \frac{5}{2} \tau^2 - \frac{5q}{m} \geq \frac{1}{2} - 3\tau^2, \quad (3.18)$$

$$\lim_s \frac{t}{v} \geq \frac{c}{2} - \frac{5q}{m} \geq \frac{c}{3}. \quad (3.19)$$

For the last inequality in (3.18), use $5q/m \leq \delta = \tau^4/3$ and $\tau \leq 1$. All group counts have limits after the earlier subselection, so t/v does as well.

Write X for the list of the $2p$ surviving matched endpoint columns, still in the first normalization. For $e \in \mathcal{T}$, let k_e denote its remaining number of pairs and write X_e for its $2k_e$ -column sublist, $D_e = \langle X_e, X_e \rangle$, and

$$Q_e = X_e D_e^{-1/2}, \quad Q = (Q_e)_{e \in \mathcal{T}}. \quad (3.20)$$

The common bag gives a uniform operator bound for the joint Gram of the first-normalized single blocks and all Q_e . Passing to the sublist X does not increase this bound. This point controls Q_e even when some eigenvalues of D_e tend to zero and the coefficients in $D_e^{-1/2}$ become large.

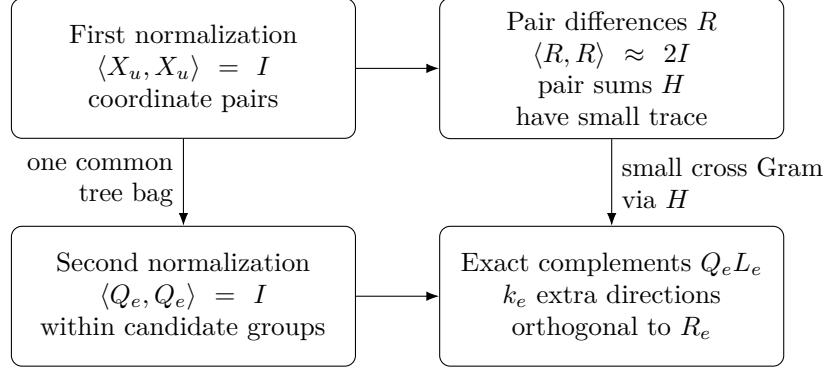


Figure 3: The two normalizations. The common bag bounds all indicated Gram matrices. The trace estimate for H , followed by a spectral cutoff at 2, controls the interaction with the second normalization using an absolute constant.

3.4 The additional positive directions

For each surviving pair $j = \{i, i'\}$ choose an order of its endpoints and form

$$R_j = \frac{X_i - X_{i'}}{\sqrt{2}}, \quad H_j = \frac{X_i + X_{i'}}{\sqrt{2}}.$$

The coefficient maps producing either list from X are Euclidean isometries. The restriction of $I - W_s$ to retained pair endpoints has Gram $2I$ on the difference coefficients. Hence (3.4) and contraction of Frobenius norm give

$$\|\langle R, R \rangle - 2I_p\|_F^2 \leq 8\tau^2 v. \quad (3.21)$$

Also $\langle H_j, H_j \rangle = 1 - (B_s)_{ii'}$, so

$$|\text{Tr}\langle H, H \rangle| \leq p\tau \leq \frac{\tau v}{2}. \quad (3.22)$$

These statements persist after all the pair deletions: they are restrictions of the original matching estimates.

The coefficients in each $D_e^{-1/2}$ are entrywise nonnegative. Distinct groups have disjoint endpoint coordinates, and all cross Gram entries between such coordinates are nonpositive. Coordinates with the same base label have zero interaction, since the first diagonal blocks are identities. Therefore

$$\langle Q, Q \rangle = I - C_s, \quad C_s \geq 0 \text{ entrywise}, \quad (C_s)_{ee} = 0 \text{ as a whole group block.} \quad (3.23)$$

It is bounded in operator norm and has at most one negative eigenvalue. By Theorem 3.3, its limiting spectral support lies in $[0, 2]$: more precisely, for every fixed $\varepsilon > 0$ there are $o(v)$ eigenvalues outside $[-\varepsilon, 2 + \varepsilon]$. The joint Gram of (H, Q) is bounded by the common bag. Applying Theorem 3.5 to (3.22) yields the absolute estimate

$$\limsup_s \frac{\|\langle H, Q \rangle\|_F^2}{v} \leq \tau. \quad (3.24)$$

In particular, no factor depending on the norm of the common bag appears here.

Fix a candidate group $e \in \mathcal{T}$. Let U_e be the $2k_e \times k_e$ Euclidean isometry whose columns are the coefficient vectors of its pair differences. Then

$$R_e = X_e U_e = Q_e T_e, \quad T_e = D_e^{1/2} U_e.$$

Because D_e is positive definite, T_e has rank exactly k_e . Choose a $2k_e \times k_e$ matrix L_e whose columns are an orthonormal basis of $(\text{im } T_e)^\perp$. Thus

$$L_e^\top L_e = I_{k_e}, \quad \langle R_e, Q_e L_e \rangle = T_e^\top L_e = 0. \quad (3.25)$$

This uses the full span of all the group's original difference directions; no small singular direction is discarded. Write $L = \text{diag}(L_e)_{e \in \mathcal{T}}$ and $Y = QL$. The list Y has exactly t columns.

For a pair j outside group e , each of the two numbers $x = \langle X_i, (Q_e)_k \rangle$ and $y = \langle X_{i'}, (Q_e)_k \rangle$ is nonpositive. Consequently

$$|\langle R_j, (Q_e)_k \rangle| = \frac{|x - y|}{\sqrt{2}} \leq -\frac{x + y}{\sqrt{2}} = -\langle H_j, (Q_e)_k \rangle. \quad (3.26)$$

This includes pairs in noncandidate groups. Let Z be $\langle R, Q \rangle$ with the entries between each R_e and its own Q_e replaced by zero. Then $\|Z\|_F^2 \leq \|\langle H, Q \rangle\|_F^2$, and (3.25) gives $\langle R, Y \rangle = ZL$. Since L is an isometry, Frobenius contraction and (3.24) imply

$$\limsup_s \frac{\|\langle R, Y \rangle\|_F^2}{v} \leq \tau. \quad (3.27)$$

Put $E_s = \langle Y, Y \rangle = L^\top \langle Q, Q \rangle L$. Every diagonal group block of E_s is an identity, so

$$\text{Tr } E_s = t. \quad (3.28)$$

This is a Euclidean coefficient compression. The variational characterization of eigenvalues shows that the number of eigenvalues of E_s above $2 + \varepsilon$ is at most that for $\langle Q, Q \rangle$, hence is $o(v)$. Its norm is bounded. If k_s denotes its number of eigenvalues greater than $1/4$, use the cutoff $2 + \varepsilon = 5/2$ to obtain

$$t = \text{Tr } E_s \leq \frac{t - k_s}{4} + \frac{5}{2}k_s + o(v).$$

Possible negative eigenvalues only decrease the left-hand trace contribution in this upper estimate. Rearranging gives

$$k_s \geq t/3 - o(v). \quad (3.29)$$

Finally consider the joint Gram matrix of (R, Y) . By (3.21), the number of eigenvalues of $\langle R, R \rangle$ at most $1/4$ is at most

$$\frac{16}{49} \|\langle R, R \rangle - 2I_p\|_F^2 \leq \frac{128}{49} \tau^2 v.$$

Restrict the two diagonal blocks to their spectral spaces above $1/4$. This leaves at least $p - (128/49)\tau^2 v$ dimensions on the R side and $t/3 - o(v)$ on the Y side. The restricted cross matrix has squared Frobenius norm at most $\|\langle R, Y \rangle\|_F^2$. Delete from the first side its left singular directions with singular values greater than $1/8$. The number deleted is at most $64\|\langle R, Y \rangle\|_F^2$. The remaining diagonal blocks are greater than $I/4$, and the cross operator norm is at most $1/8$. For their coefficient vectors x, y , the joint quadratic form is bounded below by

$$\frac{1}{4}(\|x\|^2 + \|y\|^2) - \frac{1}{4}\|x\|\|y\| \geq \frac{1}{8}(\|x\|^2 + \|y\|^2),$$

so this remaining space is positive definite. Using (3.27), the positive index of the joint Gram is at least

$$p + \frac{t}{3} - \frac{128}{49}\tau^2 v - 64\tau v - o(v). \quad (3.30)$$

Every column used here is a linear combination of the original Lorentz Gram columns for M_s . The joint Gram is therefore a coefficient pullback of M_s , and its positive index is at most rank M_s . This argument does not require the combined list (R, Y) to be linearly independent.

Combine (3.18), (3.19) and (3.30). Since $\tau \leq 1/8$,

$$\begin{aligned} \liminf_s \frac{\text{rank } M_s}{v} &\geq \frac{1}{2} + \frac{c}{9} - 64\tau - \frac{275}{49}\tau^2 \\ &\geq \frac{1}{2} + \frac{c}{9} - 65\tau \geq \frac{1}{2} + \frac{c}{12}, \end{aligned}$$

where the last inequality uses $\tau \leq c/2340$. This contradicts (3.11), whose normalized right-hand side tends to $1/2$. Hence only finitely many s can violate the stated rank bound. The positive gap also shows directly why the literal additive $+1$ is harmless for all sufficiently large s .

Every matrix in the definition of $\mu(G[K_s])$ is among the matrices just considered. Rank-nullity gives $\dim \ker M < ms/2 - 1$, and taking the maximum yields $\mu(G[K_s]) + 1 < ms/2$ as asserted. $\square$

For completeness, the matrix class defining $\mu(H)$ is nonempty for every finite nonempty graph H . Start from a diagonal matrix with one diagonal entry -1 and all others 1 , and put the same sufficiently small negative number in each edge position. The perturbation has operator norm less than $1/2$ when its size is sufficiently small, so the matrix is nonsingular with exactly one negative eigenvalue. Nonsingularity makes the Strong Arnold Property immediate.

If the base graph additionally satisfies $\alpha(G) \leq 2$, then $\alpha(G[K_s]) \leq 2$: an independent set uses at most one vertex of each clique, and its base labels are independent. Thus $\chi(G[K_s]) \geq \lceil ms/2 \rceil > \mu(G[K_s]) + 1$. Together with Theorem 1.2, the theorem above proves Theorem 1.1. One may also recover the chromatic number exactly as $\chi(G[K_s]) = ms - \nu(\overline{G[K_s]})$, where ν is the maximum matching size: every color class has one or two vertices, and the two-vertex classes are precisely disjoint hole pairs.

4 The tensor geometry and the hole relation

We construct a triangle-free relation of holes from tensor equations. The complement of this relation will be our base graph; the moment arguments in the next sections force its conflict properties. All vector spaces, linear maps, tensor products and ranks in the construction are over $\mathbb{F}_2 = \mathbb{F}_2$. We use the entrywise pairing of matrices:

$$\langle A, B \rangle = \sum_{r,c} A_{rc} B_{rc} = \text{Tr}(A^\top B).$$

Probabilities and entropy take real values.

4.1 The order of parameters

We first fix the parameter hierarchy, then define the coefficient space, its tensor-valued frame embeddings, and the hole equations. The fixed constants below bound the information retained about a unit and the number of vectors whose equality will certify a conflict. Their order ensures

that the colour-class scale precedes the graph threshold, and that the thinning rate precedes the ambient dimension multiplier.

The construction has one asymptotic parameter n . Every numerical parameter below, including the dimension multiplier M_0 , is fixed before n tends to infinity. Set

$$C_0 = 1000, \quad g = 10^9 + 1, \quad D = 4C_0g, \quad M = 2^{1000}, \quad k_{\max} = 56(g - 1). \quad (4.1)$$

In particular, g is odd. The first pin budget and the entropy slack are

$$\zeta = \frac{1}{1000k_{\max}}, \quad K_1 = \left\lceil \frac{4(D + 1)}{\zeta} \right\rceil. \quad (4.2)$$

For the phase estimates and a common upper bound on all pin dimensions, define

$$\begin{aligned} r &= 2K_1 + 2, & L_0 &= \lceil 100(D + 10) \rceil, & d_0 &= 4rL_0, \\ u_0 &= 10(K_1 + d_0 + 1), & K &= \left\lceil \frac{10(D + K_1 + u_0 + 10)}{\zeta} \right\rceil. \end{aligned} \quad (4.3)$$

The budget K is a conservative upper bound; the construction uses only the first peeling, whose pin dimension is at most K_1 . The additional probability thresholds in [Section 9](#) use only these early parameters and fixed numerical tolerances. They do not depend on the selector or channel dimensions.

The early probability margins determine a colour cap $\eta_{\text{col}} > 0$ and then the graph class scale $a > 0$. Given the graph threshold $\beta > 0$, choose the endpoint-density bound L required in [Section 16](#). Next choose the selector parameters j_*, b and the mixer parameters through J as in [Section 6](#). Choose r_0 sufficiently large after those choices and all fixed density bounds, and put $h = 1000r_0$. After h , choose a sufficiently small thinning rate $\rho > 0$ as in [Section 5](#). Finally choose a sufficiently large integer M_0 , also relative to $1/\rho$, and set

$$\begin{aligned} &(C_0, g, D, M, k_{\max}, \zeta, K_1; r, L_0, d_0, u_0, K) \\ &\longrightarrow (\eta_{\text{col}}, a) \longrightarrow (\beta, L) \longrightarrow (j_*, b, \text{mixer parameters}, J) \\ &\longrightarrow (r_0, h = 1000r_0) \longrightarrow \rho \longrightarrow M_0, \quad N = M_0n, \quad m = 2^{C_0gN}. \end{aligned} \quad (4.4)$$

We always take $n \geq 2r_0$ and subsequently increase its lower bound as needed. The matrices used below have dimensions depending on n ; for each n they are chosen to satisfy [Theorem 6.5](#), before any law of units is considered. The complete ledger of inequalities is in [Section A](#).

An $O(n)$ bit count below has a constant depending only on parameters chosen before M_0 . It can therefore be made a prescribed small fraction of N . This is different from an $o(N)$ bound: after M_0 is fixed, $n/N = 1/M_0$ is constant. Later fixed tests and tolerances may change how large n must be; they do not change any earlier construction parameter.

4.2 Moment matrices and cut profiles

Let

$$\mathcal{I} = \mathbb{Z}/g\mathbb{Z}, \quad \mathcal{E} = \binom{\mathcal{I}}{2}, \quad I_d = \{d + 1, \dots, d + (g - 1)/2\} \subset \mathcal{I}.$$

Elements of $\mathcal{I}$ are *tags*; elements of $\mathcal{E}$ are *components*. The ordinary, or base, variables comprise $g^2 + 3$ blocks of n bits:

$$O_{d,t} \quad (d, t \in \mathcal{I}), \quad S, \quad \#, \quad Z.$$

A separate selector has b bits. Its values $s \in \mathbb{F}_2^b$ are called *labels*, to distinguish them from tags.

The O -blocks carry testers whose availability depends on the tag, while S carries a tester available at every tag. The additional $\#$ -block lets the self-Gram form E , defined below, distinguish different selector labels. The Z -block is absent from E ; its free coordinates and change-of-basis symmetry are used in the later parameter tests.

Let

$$p_* = \sum_{j=0}^{j_*} \binom{b}{j}, \quad \mathcal{B} = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+(g^2+3)n}.$$

The first factor is indexed by squarefree selector monomials of degree at most j_* ; the second by the constant and the individual base variables. If p_s is the vector of selector-monomial evaluations and z is a base value, write

$$v(s, z) = p_s \otimes (1, z) \in \mathcal{B}. \quad (4.5)$$

Thus the row and column coordinates both represent monomials of base degree at most one and selector degree at most j_* .

Let $W \subset \mathcal{B} \otimes \mathcal{B}$ be the span of the point moments $v(s, z)v(s, z)^\top$. For $l \in \mathcal{I}$, let W_l be the span of those point moments for which

$$O_{d,t} = 0 \quad \text{whenever } l \notin I_d.$$

Let W_* be defined by requiring every O -block to be zero. The cut-profile space is

$$\mathcal{X} = \left\{ x = (x_{\{l,l'\}})_{\{l,l'\} \in \mathcal{E}} : x_{\{l,l'\}} = w_l + w_{l'}, \quad w_l \in W_l \right\}. \quad (4.6)$$

A tuple $w = (w_l)_{l \in \mathcal{I}}$ as in this display is a *representation* of x .

Lemma 4.1. *The intersection of W_l over any strict majority of tags is W_* . Two representations of the same cut profile differ by a constant tuple with value in W_* .*

Proof. Fix a base coordinate in $O_{d,t}$. It is allowed at exactly $(g-1)/2$ tags. A strict majority therefore contains a tag where that coordinate is disallowed. Every row and column involving it is zero in W_l for that tag. A matrix in the intersection consequently has zero rows and columns at all O -coordinates.

The coordinate projection that sets all O -variables to zero sends each point moment to a point moment defining W_* . Applied on both matrix factors, it fixes the matrix just described. That matrix is therefore in W_* , proving the nontrivial inclusion.

If w, w' give the same differences, then $w_l + w'_l = w_{l'} + w'_{l'}$ for every pair of tags. Their difference is a constant tuple. Its value belongs to every W_l , hence to W_* . Conversely every such constant tuple lies in the kernel of the cut map. $\square$

In each $O_{d,t}$ -block and in S , choose the first $2r_0$ coordinates in disjoint ordered pairs. The corresponding linear functionals on moments are

$$\eta_{d,t}(v(s, z)v(s, z)^\top) = \sum_{\rho=1}^{r_0} z_{O_{d,t}, 2\rho-1} z_{O_{d,t}, 2\rho}, \quad \eta_S(v(s, z)v(s, z)^\top) = \sum_{\rho=1}^{r_0} z_{S, 2\rho-1} z_{S, 2\rho}.$$

They are well defined by selecting the indicated ordered matrix entries. Put

$$\eta_O = \sum_{d,t} \eta_{d,t}, \quad \eta = \eta_O + \eta_S.$$

For a cut profile represented by w , define

$$\begin{aligned} a_t(x) &= \sum_{l,d} \eta_{d,t}(w_l), & a(x) &= \sum_t a_t(x) = \sum_l \eta_O(w_l), \\ b_t(x) &= \sum_{l \neq t} \eta_S(w_l), & \chi_*(w) &= \sum_l \eta(w_l). \end{aligned} \tag{4.7}$$

The functions a_t, a, b_t are well defined on $\mathcal{X}$. Indeed, a constant kernel value in W_* contributes zero to every O -tester, while it contributes $(g-1)\eta_S(w_*) = 0$ to b_t . The quantity $\chi_*(w)$ is used only when a representation has been specified.

Define a symmetric bilinear form T^0 on $\mathcal{X}$ by

$$T^0 = a \otimes a + \sum_{t \in \mathcal{I}} (a_t \otimes b_t + b_t \otimes a_t). \tag{4.8}$$

For $1 \leq j \leq J$ and ordered components e, f , choose endomorphisms L_j^{ef}, R_j^{ef} of $\mathcal{B}$. Their required properties will be proved in [Theorem 6.5](#). Set

$$\begin{aligned} B^1(x, y) &= \sum_{j=1}^J \sum_{e, f \in \mathcal{E}} \langle x_e, L_j^{ef} y_f (R_j^{ef})^\top \rangle, \\ T^1(x, y) &= B^1(x, y) + B^1(y, x), \quad T = T^0 + T^1, \quad r(x) = a + T(x, \cdot) \in \mathcal{X}^*. \end{aligned} \tag{4.9}$$

Here and later the letter r in $r(x)$ denotes an affine-gradient map; the early integer r in [Equation \(4.3\)](#) is its separately specified rank budget. Since the mixed terms in $T^0(x, x)$ cancel, T^1 is alternating, and $a(x)^2 = a(x)$, we have

$$T(x, x) = a(x) \quad (x \in \mathcal{X}). \tag{4.10}$$

The prescribed self-frame Gram matrix is denoted by E . It is the sum of the ordered tester matrices for η and a matrix $E^\#$ characterized on evaluation vectors by

$$v(s, z)^\top E^\# v(s', z') = \sum_{\rho=1}^b (s_\rho + s'_\rho) z_\#^\top M_\rho z'_\#, \tag{4.11}$$

where M_ρ are n -by- n matrices chosen in [Theorem 6.5](#). The expression is a bilinear form on $\mathcal{B}$ because $j_* \geq 1$. It vanishes when the two point inputs coincide. Consequently

$$\langle E, w \rangle = \eta(w) \quad (w \in W). \tag{4.12}$$

The Z -block does not occur in E . In particular, any invertible linear change of the Z -variables preserves E ; this symmetry will be used in unary probability estimates.

4.3 Raw vertices

For each $e \in \mathcal{E}$, take copies V_e^+, V_e^- of $\mathbb{F}_2^N$, paired by the usual dot product. We regard V_e^- as the dual of V_e^+ through that pairing. A raw vertex, also called an *orientation*, consists at each component of four maps

$$P_e, Q_e : \mathcal{B} \longrightarrow V_e^+, V_e^-, \quad X_e, Y_e : \mathbb{F}_2^h \longrightarrow V_e^+, V_e^-,$$

where the target is the first or second one as appropriate. They satisfy

$$[P_e, X_e] \text{ and } [Q_e, Y_e] \text{ are injective,} \quad P_e^\top Q_e = E, \quad X_e^\top Q_e = 0, \quad P_e^\top Y_e = 0. \quad (4.13)$$

No value is prescribed for $X_e^\top Y_e$.

Write $d = \dim \mathcal{B}$. For all large n , $N \geq 2(d+h)$, by the final choice of M_0 . The frame set is then nonempty. For example, take the plus columns to be the first $d+h$ coordinate vectors. Prescribe the full $(d+h)$ -square Gram matrix to be $\begin{pmatrix} E & 0 \\ 0 & 0 \end{pmatrix}$. Lift its columns to minus vectors using the first $d+h$ coordinates, and append distinct coordinate vectors in a complementary space of dimension $d+h$. These minus columns are independent and have the desired Gram values.

Take the uniform distribution on this finite frame set, independently over components. Let Ω_n be the raw vertex space and $P_0 = P_{0n}$ this distribution. We call P_e, Q_e the primal maps and X_e, Y_e the channel maps. The sampling law $\pi = \pi_n$, defined in [Section 5](#), will preserve the P_0 -marginal of all primal maps and restrict the conditional channel choices. All identities in this section hold for every raw vertex, and therefore under either law. There are at most $2^{2|\mathcal{E}|N(d+h)}$ raw vertices, so

$$\log_2 |\Omega_n| = O(N^2). \quad (4.14)$$

Use the common ambient tensor space

$$\mathcal{V} = \bigoplus_{e \in \mathcal{E}} V_e^+ \otimes V_e^-.$$

For a raw vertex o , define

$$U_o x = (P_{o,e} x_e Q_{o,e}^\top)_{e \in \mathcal{E}}, \quad u_o(\xi) = \sum_{e \in \mathcal{E}} \langle Y_{o,e} X_{o,e}^\top, \xi_e \rangle. \quad (4.15)$$

On a simple tensor $pq^\top$ in component e , the latter pairing is

$$u_o(pq^\top) = (Y_{o,e}^\top p) \cdot (X_{o,e}^\top q).$$

Thus the minus channel is paired with the first tensor factor and the plus channel with the second. All copies of coefficient spaces attached to different vertices are distinct, although the ambient $V_e^\pm$ are common.

Each U_o is injective: a left inverse of $P_{o,e}$ and the transpose of a left inverse of $Q_{o,e}$ recover x_e from $P_{o,e} x_e Q_{o,e}^\top$. The two annihilator conditions in [Equation \(4.13\)](#) also give

$$u_o U_o = 0. \quad (4.16)$$

Definition 4.2. Two raw vertices i, j have a *hole* between them if there exist $\lambda_i, \lambda_j \in \mathcal{X}$ such that

$$\begin{aligned} U_i \lambda_i &= U_j \lambda_j, & u_j U_i &= r(\lambda_i), & u_i U_j &= r(\lambda_j), \\ a(\lambda_i) + a(\lambda_j) &= 1. \end{aligned} \quad (4.17)$$

The middle two identities are identities of linear functionals on the whole cut-profile space.

Lemma 4.3. *The hole relation is symmetric, has no loops, and is triangle-free.*

Proof. Symmetry follows by interchanging the endpoints. A loop would have $U_i \lambda_i = U_i \lambda_j$, hence $\lambda_i = \lambda_j$ by injectivity, contradicting the last equation of Equation (4.17).

Suppose that three distinct raw vertices have holes on all three pairs. For a directed incidence ij , let x_{ij} be its witness at i . For distinct i, j, k , evaluate the gradient identity for ij on x_{ik} :

$$u_j(U_i x_{ik}) = a(x_{ik}) + T(x_{ij}, x_{ik}).$$

Sum over the six ordered choices of (i, j, k) . For fixed j , the two arguments on the left are equal by the sharing equation for the remaining edge ik , so they cancel. For fixed i , the two bilinear terms cancel by symmetry of T . The remaining sum is

$$\sum_{\{i,j\}} (a(x_{ij}) + a(x_{ji})) = 1 + 1 + 1 = 3,$$

a contradiction. $\square$

Sample m raw vertices independently from the law π defined in Section 5, and form G on their *positions*: two distinct positions are adjacent when their raw types have no hole. This gives a finite simple graph even when types repeat. Equal raw types are adjacent because holes have no loops. An independent triple of positions would therefore give three distinct raw types forming a hole triangle. By Theorem 4.3,

$$\alpha(G) \leq 2, \quad \chi(G) \geq \lceil m/2 \rceil. \quad (4.18)$$

5 The distribution problem and its preliminary tools

A *unit* is an ordered pair of raw vertices with no hole between its endpoints. The endpoints of a unit may be dependent. When two units are sampled independently, their four endpoints need not be independent within either unit. The event of interest is that all four cross pairs have holes, as in Figure 1.

The raw reference law P_0 is the uniform law of the exact frames in Equation (4.13). We will replace its conditional channel law by a thinned law π , while preserving its primal marginal exactly. The reference image and peeling arguments below use the absolute joint cap

$$\sigma \leq 2^{DN} P_0^2. \quad (5.1)$$

This condition alone imposes no constant bound on either endpoint marginal. The stronger marginal hypotheses required for the distribution tests will always be stated separately, relative to π .

5.1 Uniform frame laws

We record the linear algebra behind all reference image laws. For a subspace of a nominal coefficient space, its *image* is its image under the relevant frame map. Nominal independence is independence before applying these maps.

Lemma 5.1. *Let $A, A' : U \rightarrow V$ and $B, B' : W \rightarrow V^*$ be injective maps between finite-dimensional spaces over $\mathbb{F}_2$. If*

$$B^\top A = (B')^\top A',$$

there is $L \in \text{GL}(V)$ with $LA = A'$ and $L^{-\top}B = B'$. Thus based individually injective plus and minus frames with a prescribed Gram matrix form one orbit under the simultaneous primal and dual action.

Proof. Write $\Phi = B^\top : V \rightarrow W^*$ and $\Phi' = (B')^\top : V \rightarrow W^*$. Both are surjective. The based map $f : A(U) \rightarrow A'(U)$ given by $f(Au) = A'u$ satisfies $\Phi'f = \Phi$. It therefore maps $A(U) \cap \ker \Phi$ isomorphically onto $A'(U) \cap \ker \Phi'$. Extend this restriction to an isomorphism $L_0 : \ker \Phi \rightarrow \ker \Phi'$.

Choose a complement U_0 of $A(U) \cap \ker \Phi$ in $A(U)$. Its images under Φ are independent. Extend a basis of $\Phi(U_0)$ to a basis of W^* , and choose lifts of the added basis vectors under Φ and Φ' , respectively. For the basis vectors in $\Phi(U_0)$, choose the lifts in U_0 and their images under f . Define L to equal L_0 on $\ker \Phi$ and to carry each chosen lift to its primed lift. These are direct-sum decompositions of V , so L is an isomorphism. It extends f and satisfies $\Phi'L = \Phi$, which is equivalent to the two required frame identities. $\square$

Lemma 5.2. *The following statements hold for all sufficiently large n .*

- (a) *At independently sampled raw vertices and components, images of prescribed primal coefficient tuples that are independent separately at each vertex and sign are uniform subject to their individual injectivities and their prescribed same-vertex Gram matrices. For tuples using channels as well, the same assertion holds conditionally on the full self-channel Gram matrices $X_e^\top Y_e$; the relevant full Gram matrix is $\begin{pmatrix} E & 0 \\ 0 & X_e^\top Y_e \end{pmatrix}$.*
- (b) *Let a fixed tuple of nominal directions use arbitrary combinations of the two endpoints' primal and channel coefficients. Compute its rank separately in each component and sign in the direct sum over both endpoints, and let the total be t . For every preassigned $\epsilon > 0$, choosing M_0 sufficiently large gives, under P_0^2 ,*

$$\mathbb{P}\{\text{all those images equal prescribed vectors}\} \leq 2^{-(1-\epsilon)tN} \quad (t \geq 1). \quad (5.2)$$

This is valid for all possible ranks t , including $t = O(n)$, not only for bounded tuples.

- (c) *The plus frame $[P, X]$ is uniform among injective maps. Conditional on it, the minus columns are independent uniforms on the affine spaces prescribed in Equation (4.13), followed only by the conditioning that $[Q, Y]$ is injective. The failure probability of this last condition, before conditioning, is at most $2^{2(d+h)-N}$ per component.*

Given P, Q , the channels X, Y are independent uniform-column draws on $\ker Q^\top$ and $\ker P^\top$, respectively, conditioned on $[P, X]$ and $[Q, Y]$ being injective. The two rank-failure probabilities are exponentially small in N . Each channel frame alone is uniform among injective maps $\mathbb{F}_2^h \rightarrow \mathbb{F}_2^N$. For any fixed q independent vectors in its paired ambient space, the probability that transpose evaluation on their span is not injective is at most

$$2^{q-h+1}. \quad (5.3)$$

The unconditional joint channel law is bounded above by a constant, depending on g, h but not on n , times the completely independent uniform-column law.

- (d) *For p independent uniform plus vectors and q independent uniform minus vectors in paired copies of $\mathbb{F}_2^N$, and a prescribed p -by- q Gram matrix G , the probability of that Gram matrix together with individual injectivity of both frames is at most 2^{-pq} and at least*

$$2^{-pq}(1 - 2^{p-N})(1 - 2^{p+q-N}). \quad (5.4)$$

In particular, any consistent Gram specification on a bounded number of vectors, together with injectivity on each side, has probability bounded below by a positive constant for large N .

Proof. The raw law is invariant under L on each V_e^+ and $L^{-\top}$ on V_e^- , independently at vertices and components. For primal tuples, their restricted Gram is fixed by E . By [Theorem 5.1](#), all based realizations with that Gram and individual injectivity lie in one orbit. Projection of an invariant uniform raw law is uniform on this orbit: an element carrying one partial realization to another bijects their full-frame completions. There is at least one completion because these tuples are restrictions of a raw frame. After fixing $X_e^\top Y_e$, the same argument applies to arbitrary full-frame coefficient tuples. Independence over raw draws and components proves (a).

For (c), the action is transitive on injective plus frames, so their marginal is uniform. Given a plus frame, each column of Q has specified evaluations against P, X , while each column of Y has zero evaluations against P . There is no other constraint until full minus injectivity is imposed. The remaining allowed set is therefore the product of those affine column spaces restricted to full rank.

Each column space contains the common translation space $\text{ann}(\text{im}[P, X])$, of dimension $N - d - h$. For a nonzero coefficient vector $c \in \mathbb{F}_2^{d+h}$, the corresponding linear combination of minus columns is consequently uniform on an affine space containing that translation space. Its chance of being zero is at most $2^{-(N-d-h)}$. Union over the fewer than 2^{d+h} nonzero coefficient vectors gives the asserted bound $2^{2(d+h)-N}$.

Alternatively, condition first on P, Q . The allowed X -columns lie in $\ker Q^\top$, the allowed Y -columns in $\ker P^\top$, and their two full-rank restrictions are separate. This proves the stated conditional independence. For example, before imposing injectivity of $[P, X]$, any dependence with a nonzero X -coefficient has probability at most $2^{-(N-d)}$; union over its at most 2^{d+h} coefficient choices gives a bound 2^{2d+h-N} . The estimate for Y is identical.

The marginal of X alone is invariant under $\text{GL}(\mathbb{F}_2^N)$ and supported on injective maps; transitivity gives its uniformity, and likewise for Y . Before conditioning independent channel columns to be injective, their evaluations on a fixed independent q -tuple form a uniform h -by- q matrix. For each nonzero coefficient vector in $\mathbb{F}_2^q$, the chance that its evaluation is zero is 2^{-h} . Union over those coefficient vectors gives at most 2^{q-h} , and channel injectivity has probability at least $1/2$ for large N . This proves [Equation \(5.3\)](#). To compare their joint marginal with completely independent columns, condition on $G = X^\top Y$. The conditional marginal is uniform on the orbit of two individually injective h -frames having Gram matrix G . Under independent uniform columns that orbit has probability at least 2^{-h^2-2} for large N , by the argument for (d) below. Since the raw probability of any G is at most one, the density ratio on every such orbit is at most 2^{h^2+2} . Multiply over the fixed number of components to obtain the joint channel bound. Its possible dependence on h is harmless: h is fixed before n grows.

We next prove (b), keeping explicit the dependence on the rank. Replace each tested tuple by a basis of its nominal span; inconsistent dependent prescriptions have probability zero. For the plus directions at the two endpoints, start with completely independent uniform plus columns and then impose individual full-frame injectivity. The image of a rank- t_+ tuple in the joint nominal direct sum is exactly uniform on t_+ independent ambient vectors before this conditioning. Indeed, at each ambient coordinate the coefficient map is a surjective linear map onto $\mathbb{F}_2^{t_+}$, and different ambient coordinates use independent input bits. Full injectivity has probability $1 - 2^{-\Omega(N)}$, uniformly over the given construction, so the plus prescription costs at most $(1 - 2^{-\Omega(N)})^{-1} 2^{-t_+ N}$.

Now condition on all plus frames. At a component let S be the span of both endpoints' plus

columns; then $\dim S \leq 2(d + h)$. Before minus injectivity, every minus column has an affine distribution whose translation space contains $S^\perp$. Split off an independent uniform $S^\perp$ -part of each column and condition on the remaining parts. For a rank- t_- tuple of minus coefficient combinations, the $S^\perp$ -parts of its images are independent uniform vectors in $S^\perp$, because the coefficient map has rank t_- . Thus prescribed minus images have probability at most

$$2^{-(N-2(d+h))t_-}.$$

The additional conditioning on the two minus frames being injective changes this by a factor $1 + 2^{-\Omega(N)}$, by (c). Perform this argument componentwise. There is a fixed total conditioning factor bounded, for example, by $4^{|\mathcal{E}|}$, and the resulting bound is

$$4^{|\mathcal{E}|} 2^{-t_+N - (N-2(d+h))t_-}.$$

Choose M_0 so that $2(d + h)/N < \epsilon/2$ for large n , and then increase n so that the fixed prefactor is at most $2^{\epsilon N/2}$. Since $t = t_+ + t_- \geq 1$, this proves Equation (5.2) uniformly for every allowable t . In particular, mixing the two endpoints in a direction causes no loss of an entire N -bit image constraint.

Finally, to prove (d), the plus columns are independent with probability at least $1 - 2^{p-N}$: each nonzero coefficient dependence has probability 2^{-N} . Given their independence, each minus column takes its required evaluations with probability 2^{-p} , independently, so the Gram costs 2^{-pq} . Under this Gram conditioning, every nonzero combination of minus columns is uniform on an affine space of dimension at least $N - p$. Its probability of vanishing is at most $2^{-(N-p)}$. Union over fewer than 2^q combinations gives failure probability at most 2^{p+q-N} . This proves the lower bound. For the upper bound with plus injectivity required, condition on the plus columns and use the exact 2^{-pq} Gram probability whenever they are independent. A partially specified consistent Gram contains some full specification; for bounded p, q , the same positive lower bound for that full specification proves the last assertion. $\square$

5.2 Thinning the channel images

Write $e_* = |\mathcal{E}|$, $d = \dim \mathcal{B}$, and

$$q_h = 2^h - 1, \quad k_h = 2e_*q_h, \quad \varepsilon_{\text{ch}} = 10e_*2^h\rho.$$

After the channel dimension h has been fixed, choose $\rho > 0$ so small that $10^4e_*2^{2h}\rho < \zeta/100$ and that this quantity is smaller than every fixed rate slack used below. We then choose M_0 large enough also in terms of $1/\rho$. The channel frame in sign $+$ is X_e , and that in sign $-$ is Y_e .

Lemma 5.3 (Channel thinning). *Fix any bound q_{test} on the lengths of the ambient tuples to be tested; this bound is fixed before ρ, M_0, n . For all sufficiently large n there are subsets $A_e^\pm \subseteq V_e^\pm$, each of density $(1 + o(1))2^{-\rho N}$, with the following properties. Keep the $\mathbf{P}_0$ -marginal on all primal frames $(P_e, Q_e)_e$. Given those frames, condition the original channel law on*

$$X_e c \in A_e^+, \quad Y_e c \in A_e^- \quad (e \in \mathcal{E}, 0 \neq c \in \mathbb{F}_2^h). \quad (5.5)$$

This conditional event has positive probability for every primal frame. The resulting probability law π satisfies:

(a) *Its primal marginal is exactly that of $\mathbf{P}_0$, and*

$$\pi^2 \leq 2^{\varepsilon_{\text{ch}}N} \mathbf{P}_0^2. \quad (5.6)$$

(b) For every fixed independent ambient q -tuple, $q \leq q_{\text{test}}$, transpose evaluation by any specified channel frame has failure probability at most $2^{q-h+1} + o(1)$ under π , uniformly in the tuple. More generally every fixed finite union of these rank-failure tests has its P_0 -probability plus $o(1)$ as an upper bound.

(c) Given any primal frames, any component and sign, and any nonzero $v \in V_e^\pm$,

$$\pi\{\exists c \neq 0 : X_{ec} \in A_e^+ + v \mid (P, Q)\} \leq 2^{-\rho N/2}, \quad (5.7)$$

with the analogous statement for Y_e, A_e^- .

The error in (b) is exponentially small in N . The ambient tuple there is fixed before sampling the vertex; no uniform conditional rank bound for a tuple selected from the sampled primal frame is asserted.

Proof. Include every ambient vector in each $A_e^\pm$ independently with probability $p = 2^{-\rho N}$; subsets for different components and signs are independent. Fix a complete primal frame f , and let ν_f be the original conditional channel law. Define its survival fraction $Z_f = \nu_f(\text{Equation (5.5)})$, a random variable in the random subsets. Channel injectivity makes the q_h nonzero values in each component and sign distinct. Consequently

$$\mathbb{E}_A Z_f = p^{k_h}.$$

For any transpose-rank test T , or any union of at most N such tests, the numerator $Z_{f,T} = \nu_f(T \cap \text{Equation (5.5)})$ similarly has expectation $p^{k_h} \nu_f(T)$.

We establish simultaneous concentration of these fractions by the bounded-differences method [1, 22], with the needed exponential-moment estimate included below. By Theorem 5.2(c), a prescribed nonzero channel combination, conditional on f , has probability at most $2^{-(N-d)+1}$ of taking any prescribed value for all sufficiently large N . Indeed before full-frame injectivity it is uniform on the appropriate annihilator, and injectivity has probability $1 - 2^{-\Omega(N)}$, uniformly in f . Changing one membership bit of a subset can therefore change any such fraction by at most

$$c_N = C_h 2^{-N+d},$$

where C_h is fixed in n . There are $2e_* 2^N$ membership bits, so the sum of their squared influence bounds is at most

$$S_N = C_{g,h} 2^{-N+2d}.$$

For clarity, expose these independent bits and use the Doob martingale of any one fraction. A centered increment whose conditional range has length at most c has exponential moment at most $\exp(t^2 c^2 / 8)$. To see this, the second derivative of its log moment generating function is the variance under an exponential tilt, at most $c^2 / 4$; integrate twice using value and derivative zero at the origin. Multiplication of the conditional bounds and Markov's inequality give

$$\Pr_A(|Z - \mathbb{E}_A Z| > u) \leq 2 \exp(-2u^2 / S_N).$$

Take $u = \epsilon_N = 2^{-N/10}$. Since d/N can be made arbitrarily small by increasing M_0 , these tails are at most $2 \exp(-c_{g,h} 2^{N/2})$. There are $2^{O(N^2)}$ primal frames and $2^{O(N)}$ primitive transpose tests, since their tuple lengths are bounded by the fixed q_{test} . The unions of at most N primitive tests therefore number at most $2^{O(N^2)}$ as well. A union bound gives simultaneous error at most ϵ_N for

all these denominators and test numerators. The same argument for $2^{-N}|A_e^\pm|$, whose bit influences are 2^{-N} , gives $|A_e^\pm|/2^N = (1 + o(1))p$.

We include the translated numerators in this simultaneous estimate. Fix $f, e, \pm, v \neq 0$, and let $Z_{f,v}^{\pm,e}$ be the original conditional probability of survival and of the event in Equation (5.7). Consider, for example, sign $+$. If $X_e c + v$ coincides with another already required nonzero channel value $X_e c'$, then $v = X_e(c + c')$ is a nonzero channel value. The original probability of this possibility is at most $C_h 2^{-N+d}$, by the point bound and a union over the nonzero coefficients. On that exceptional event we bound the extra test by one; the survival probability over the random subsets is still p^{k_h} . In every other case $X_e c + v$ is a new membership bit and supplies an additional factor p . This includes the case $X_e c + v = 0$, since zero is not among the required nonzero channel values. Union over c yields

$$\mathbb{E}_A Z_{f,v}^{+,e} \leq p^{k_h} (C_h 2^{-N+d} + q_h p).$$

A changed membership bit can affect this numerator only by being an original channel value or a translated channel value. Both have point probability at most $C_h 2^{-N+d}$; hence the same influence and concentration estimates apply. The additional 2^N choices of translate do not affect the union bound. We may therefore fix one realization of all subsets for which every displayed concentration statement holds simultaneously.

Our choice of ρ gives $k_h \rho < 1/100$. Thus

$$Z_f = p^{k_h} (1 + O(\epsilon_N/p^{k_h})) > 0$$

uniformly, and the relative error tends to zero exponentially. The pointwise density of π with respect to P_0 is the survival indicator divided by Z_f . Consequently

$$\frac{d\pi^2}{dP_0^2} \leq 4p^{-2k_h} \leq 2^{\epsilon_{ch} N}$$

for large N ; the primal law was left unchanged by definition. Division of $Z_{f,T}$ by Z_f gives $\pi(T | f) = \nu_f(T) + O(\epsilon_N/p^{k_h})$, uniformly in every specified fixed test. Average over the original primal marginal and apply Equation (5.3) to prove (b). Every fixed finite union is among our tested unions once N is large enough; applying the same numerator estimate directly to that union gives its raw probability plus the stated error, without replacing the union probability by the sum of its constituent probabilities. Finally, dividing the translated numerator by Z_f bounds it by

$$C_h 2^{-N+d} + q_h 2^{-\rho N} + O(\epsilon_N/p^{k_h}) \leq 2^{-\rho N/2}$$

for all sufficiently large N . This proves (c) and the lemma. $\square$

5.3 The two distribution tests

For independent units $A = (A_1, A_2)$, $B = (B_1, B_2)$, write $A \bowtie B$ when all four pairs (A_i, B_j) are holes. For a point z , write $A \bowtie z$ when both (A_i, z) are holes. These definitions also cover repeated raw types; the absence of hole loops automatically forbids a conflict with a shared endpoint.

Theorem 5.4 (Distribution tests). *There is a constant $\eta_{\text{col}} > 0$, depending only on the early parameters and M , with the following property. Fix any $L < \infty$ after η_{col} and before the selector and channel parameters; choose all remaining parameters in the order of Equation (4.4) and Section A. For every sufficiently large n , the law π constructed above satisfies both conclusions.*

- (i) Let (A, c) have a probability law on units marked by a colour in any fixed finite palette. The colour may be randomized conditional on A . Write σ_1, σ_2 for the endpoint marginals of its projected unit law σ , and suppose

$$\sigma_1 \leq M\pi, \quad \sigma_2 \leq M\pi, \quad \sigma \leq 2^{3000gN}\pi^2, \quad \Pr(c = j) \leq \eta_{\text{col}} \quad \text{for every } j. \quad (5.8)$$

For two independent draws $(A, c), (B, c')$,

$$\Pr(A \bowtie B, c \neq c') \geq 2^{-100gN}.$$

- (ii) Let A have a unit law σ and let z be an independent point with law θ , where

$$\sigma_1 \leq L\pi, \quad \sigma_2 \leq L\pi, \quad \sigma \leq 2^{3000gN}\pi^2, \quad \theta \leq L\pi. \quad (5.9)$$

Then $\Pr(A \bowtie z) \geq 2^{-100gN}$.

The threshold for n may depend on the fixed palette and on L . All inequalities between measures are pointwise on their finite spaces.

The two conclusions are proved after the collision and overlap arguments, in [Theorem 14.8](#) and [Section 15](#), respectively. The finite graph transfer is given in [Section 16](#). Here we establish their leaf and image hypotheses. In particular, [Equation \(5.6\)](#) places every projected test law below $2^{(3000g+\varepsilon_{\text{ch}})N}\mathbf{P}_0^2$, leaving strict slack below the absolute budget $D = 4000g$.

5.4 Sparse profiles arising from true intersections

For a matrix profile, its *total component rank* means $\sum_{e \in \mathcal{E}} \text{rank } x_e$.

Lemma 5.5. *Let σ satisfy the joint cap in [Equation \(5.1\)](#). Outside a set of σ -probability $2^{-\Omega(N)}$,*

$$\sum_{e \in \mathcal{E}} \dim(\text{im } P_{1,e} \cap \text{im } P_{2,e}) \leq 2D.$$

For any unit outside that exceptional set and any $U_1\lambda_1 = U_2\lambda_2$, both λ_i have total component rank at most $2D$. Each has a unique representation $(w_{i,l})_l$ vanishing at a strict majority of tags, and

$$|\{l : w_{i,l} \neq 0\}| \leq 4D/g, \quad \chi_*(w_1) = \chi_*(w_2).$$

Proof. If the summed intersection dimension exceeds $2D$, choose $2D$ independent common-image vectors, componentwise. Write each as $P_{1,e}a = P_{2,e}b$. Their nominal relation directions (a, b) are independent in the direct sum of the two primal coefficient spaces at that component. Indeed a dependence between the pairs would, after applying $P_{1,e}$, be the corresponding dependence between the selected common-image vectors. Across components their ranks add.

There are at most $2^{O(n)}$ ways to choose this bounded number of coefficient pairs and their component indices. For each choice, [Equation \(5.2\)](#) bounds the probability of their zero images by $2^{-(1-\epsilon)2DN}$ under $\mathbf{P}_0^2$. After paying 2^{DN} for σ , and making the coefficient count a sufficiently small fraction of N by the choice of M_0 , the union bound is $2^{-\Omega(N)}$.

Suppose $U_1\lambda_1 = U_2\lambda_2 = \xi$. The column space of ξ_e lies in both primal plus spans. Since multiplication by the two injective frame maps preserves matrix rank,

$$\text{rank}(\lambda_{i,e}) = \text{rank}(\xi_e) \leq \dim(\text{im } P_{1,e} \cap \text{im } P_{2,e}).$$

This proves the total-rank assertion.

Take any representation $(w_l)_l$ of one of the profiles. A component $w_l + w_{l'}$ is nonzero whenever $w_l \neq w_{l'}$, so there are at most $2D$ unequal unordered pairs. If no value occurred at a strict majority of tags, write c_v for its multiplicities. Then

$$\#\{\text{unequal pairs}\} = \frac{1}{2} \left(g^2 - \sum_v c_v^2 \right) \geq \frac{1}{2} \left(g^2 - \frac{g}{2} \sum_v c_v \right) = \frac{g^2}{4} > 2D,$$

a contradiction. Let the majority have size $g - z$. There are at least $z(g - z) \geq zg/2$ unequal pairs, hence $z \leq 4D/g$. Its common value belongs to W_* by [Theorem 4.1](#), so subtract it from every coordinate. The resulting representation vanishes at a strict majority. It is unique: two such representations differ by a constant, and their two majority zero sets intersect.

Finally, equality of the component tensors gives equality of their ambient traces. By cyclicity of trace and [Equation \(4.12\)](#),

$$\text{Tr}(P_{i,e} \lambda_{i,e} Q_{i,e}^\top) = \langle P_{i,e}^\top Q_{i,e}, \lambda_{i,e} \rangle = \eta(\lambda_{i,e}).$$

For $e = \{l, l'\}$, it follows that

$$\eta(w_{1,l}) + \eta(w_{2,l}) = \eta(w_{1,l'}) + \eta(w_{2,l'}).$$

This value is independent of the tag. The two sparse representations together have at most $8D/g = 32000 < g$ nonzero coordinates, so some tag has both entries zero. The common value is therefore zero. Summing over tags proves equality of the two specified χ_* -values. $\square$

5.5 Peeling into leaves with exact pins

For a unit $O = (1, 2)$, introduce nominal coefficient spaces at each component

$$\mathcal{L}_{O,e} = \bigoplus_{i \in O} (\mathcal{B}_i \oplus H_i^+), \quad \mathcal{R}_{O,e} = \bigoplus_{i \in O} (\mathcal{B}_i \oplus H_i^-), \quad H_i^\pm = \mathbb{F}_2^h.$$

Their frame-image maps add the endpoint images in the common ambient $V_e^\pm$. A *pin space* is a specified subspace $D_{O,e}^\pm$ of this nominal direct sum together with specified exact images of every vector in a chosen basis. It may mix both endpoints, and it may mix primal and channel coordinates. Once the basis images are known, every image on the pin space is fixed.

For a tested tuple of directions, its *rank modulo the pins* is the sum, over components and signs, of the dimensions of the spans of its images in the quotients by $D_{O,e}^\pm$. The rank is nominal; it is not computed after applying the raw frames.

Lemma 5.6 (First peeling and restriction). *Take the constants in [Equations \(4.2\)](#) and [\(4.3\)](#), and choose the reference accuracy in [Equation \(5.2\)](#) with $\epsilon < \zeta/4$. For every law σ satisfying the absolute joint cap [Equation \(5.1\)](#), the following decompositions are available.*

- (a) *After discarding the exception in [Theorem 5.5](#) and at most $2^{-\zeta N}$ further mass, the law is partitioned into leaves. Each leaf has fixed pin spaces with total dimension at most K_1 , and its normalized law σ_λ satisfies*

$$\sigma_\lambda \{ \text{a specified exact-image constraint of rank } t \text{ modulo the pins} \} \leq 2^{-(1-\zeta)tN} \quad (t \geq 1). \quad (5.10)$$

The normalized leaf laws satisfy a common bound $2^{O(N)} \mathbf{P}_0^2$, with the implied constant determined by the early parameters.

(b) Suppose a restriction of the mixed law has relative mass $2^{-o(N)}$. One may discard an exponentially small fraction of that restricted law so that every remaining normalized leaf restriction satisfies

$$\sigma_\lambda\{a \text{ specified exact-image constraint of rank } t \text{ modulo the pins}\} \leq 2^{-(1-2\zeta)tN} \quad (t \geq 1), \quad (5.11)$$

and still has density $2^{O(N)}$ relative to $\mathbf{P}_0^2$. For several successive restrictions, the same statement applies to their combined relative cost.

In all parts the mixed law uses each leaf's actual mass, normalized only after the indicated discards. No equal reweighting of leaves is used.

Proof. We give the finite procedure and its quantitative bounds. Work first with the original, unnormalized restriction of σ obtained by removing the intersection exception. In particular its density is still at most 2^{DN} relative to $\mathbf{P}_0^2$. Let R be its residual support. As long as its residual mass exceeds $2^{-\zeta N}$, begin with its normalized restriction and no pins. If Equation (5.10) fails, condition on a violating image constraint and add its nominal directions to the pins. Given the existing pins, a consistent tuple constraint can be reduced to a basis of its span modulo the pins. The other prescriptions are either automatic or inconsistent; an inconsistent event cannot violate a positive probability bound. Thus every step adds positive rank.

If the cumulative added rank is u , the unnormalized mass of the current part is greater than

$$2^{-\zeta N - (1-\zeta)uN}.$$

The cumulative directions are independent after taking a basis, so the absolute density cap and Equation (5.2) give the upper bound

$$2^{DN - (1-\epsilon)uN}.$$

Comparing the bounds yields

$$(\zeta - \epsilon)u < D + \zeta. \quad (5.12)$$

This is strictly below the rank budget K_1 . The same comparison applies to a whole violating tuple at once, so a large tuple cannot jump beyond that budget. The rank therefore cannot increase indefinitely. The process reaches a part with no violating constraint; remove that entire part as a leaf and restart on the residual. The underlying raw unit space is finite, and every leaf is nonempty, so the procedure terminates with residual mass at most $2^{-\zeta N}$.

A leaf of rank $u \leq K_1$ has mass greater than $2^{-\zeta N - (1-\zeta)uN}$. Dividing the original density cap by this lower bound shows, for example,

$$\sigma_\lambda \leq 2^{(D+K_1+1)N} \mathbf{P}_0^2.$$

This proves (a), including uniformity of its density constant.

For (b), write the current normalized mixture as $\sum_\lambda w_\lambda \sigma_\lambda$. Let q_λ be the relative mass retained by the combined restriction in leaf λ , and let $q = \sum_\lambda w_\lambda q_\lambda = 2^{-o(N)}$. Discard leaves for which $q_\lambda < 2^{-\zeta N}$. Their mass in the restricted law is at most

$$q^{-1} \sum_{\lambda: q_\lambda < 2^{-\zeta N}} w_\lambda q_\lambda \leq 2^{-\zeta N + o(N)}.$$

On every other leaf, division by q_λ multiplies probabilities by at most $2^{\zeta N}$. For $t \geq 1$,

$$2^{\zeta N} 2^{-(1-\zeta)tN} \leq 2^{-(1-2\zeta)tN}.$$

This proves the claimed min-entropy and density bounds. It also explains why the small mass bound is needed for the mixed law, rather than separately asserted for every old leaf.

Throughout these operations, disintegration over the finite parts is exact. Multiplying each normalized part by its actual mass recovers the retained law. This proves the last assertion as well. $\square$

The per-leaf density bounds in [Theorem 5.6](#) are joint bounds relative to $\mathbf{P}_0^2$. They do not assert that each leaf has the original one-vertex marginal cap. If the initial mixed law has endpoint caps $C\pi$, restricting it by mass q gives endpoint caps $(C/q)\pi$. Its primal marginal caps are therefore $(C/q)\text{pr}_*\mathbf{P}_0$, where pr forgets all channel columns. For $q = 2^{-o(N)}$ these are $C2^{o(N)}$ times the original *primal* law. The corresponding full-frame bound is $C2^{o(N)}\pi$, not $C2^{o(N)}\mathbf{P}_0$. The absolute joint bound is $2^{(D+o(1))N}\mathbf{P}_0^2$. [Section A.5](#) records where these distinct controls are used.

Corollary 5.7 (Colours and actual leaf weights). *For a law satisfying [Equation \(5.8\)](#), discard colours whose mass is below $2^{-\zeta N}$, and apply [Theorem 5.6\(a\)](#) to each remaining conditional colour law. The total discarded mass tends to zero exponentially, the leaves have at most K_1 pins and satisfy [Equation \(5.10\)](#), and each leaf has an absolute density bound $2^{(D+K_1+1)N}\mathbf{P}_0^2$. With their actual retained masses these leaves form a mixed law whose endpoint marginals are bounded by $(1+o(1))M\pi$. Any subsequent combined restriction of mass $2^{-o(N)}$ has the first-pin freshness in [Equation \(5.11\)](#), after the trimming in [Theorem 5.6\(b\)](#).*

Proof. The palette size is fixed, so deleting light colours costs at most a fixed multiple of $2^{-\zeta N}$. Every retained conditional colour law has absolute density at most $2^{(3000g+\varepsilon_{\text{ch}}+\zeta)N}\mathbf{P}_0^2$, which is below $2^{DN}\mathbf{P}_0^2$. The intersection exception and the residual mass in [Theorem 5.6\(a\)](#) are exponentially small uniformly over these laws. Multiply each normalized leaf by its actual colour mass and its mass within that colour. Summing recovers exactly the retained restriction of the original law, so its endpoint marginals are bounded by the original ones divided by the total retained mass. This proves the asserted pooled bounds. The final assertion is precisely [Theorem 5.6\(b\)](#), applied to this mixture. In particular no inverse colour mass enters the pooled marginal constant. $\square$

Corollary 5.8 (An independent pair as an unpinned leaf). *Let $\theta \leq L\pi$, and draw two independent θ -points. The probability that they form an internal hole is $2^{-\Omega(N)}$. Their law conditioned on being a unit is a single leaf with no pins, with endpoint marginals at most $(1+o(1))L\pi$, and it satisfies [Equation \(5.10\)](#) for every rank $t \geq 1$. Its effective spaces C_1, C_2 , defined in [Section 6](#), are consequently zero.*

Proof. Under the raw primal law, each P_e is uniform injective. Two independent such images have nonzero intersection with probability at most $(2^d - 1)^2 / (2^N - 1)$, by a union bound over nonzero coefficient pairs. The primal marginals under θ have density at most L relative to those laws. Union over components therefore bounds the probability of any plus-span intersection by $O_g(L^2 2^{2d-N}) = 2^{-\Omega(N)}$. If the plus spans are disjoint in every component, a tensor common to the two endpoint primal tensor spaces has every component zero: each of its column spaces lies in the corresponding plus-span intersection. Injectivity of the primal frame maps then makes the sharing witnesses zero, contrary to the parity sum one in [Equation \(4.17\)](#). A hole is thus included in the exceptional event just bounded.

Before deleting internal holes the pair law is at most $L^2\pi^2$. For a rank- t exact-image event, [Equation \(5.2\)](#) and [Equation \(5.6\)](#) bound its probability by

$$L^2 2^{\varepsilon_{\text{ch}}N - (1-\epsilon)tN}.$$

Take $\epsilon < \zeta/4$. Since $\varepsilon_{\text{ch}} < \zeta/4$ and $t \geq 1$, division by the nonhole probability $1 - 2^{-\Omega(N)}$ leaves this at most $2^{-(1-\zeta)tN}$ for large N , uniformly for all ranks. The same normalization gives the endpoint caps. Empty pin spaces have zero primal projections, which gives the assertion about C_i . $\square$

5.6 Bilinear sign estimates

Lemma 5.9 (Walsh estimates). *Let all weights below have absolute value at most one.*

(a) *If x, y are independent uniform vectors and $B(x, y)$ is a bilinear form of rank d , then*

$$\left| \mathbb{E} f(x) g(y) (-1)^{B(x, y)} \right| \leq 2^{-d/2}.$$

Replacing either uniform law by a law of density at most C_i multiplies this bound by at most $C_1 C_2$.

(b) *If α, β are subprobability measures on $\mathbb{F}_2^d$, with point masses at most p_1, p_2 , respectively, then*

$$\left| \sum_{x, y} \alpha(x) \beta(y) f(x) g(y) (-1)^{x \cdot y} \right| \leq 2^{d/2} (p_1 p_2)^{1/2}.$$

(c) *For independent uniform global N -bit vector slots in two groups, a cross-pairing character with coefficient matrix C has bit rank $N \text{rank } C$. In particular, if its cross coefficient pattern is nonzero, its correlation against two bounded weights, one on each group, is at most $2^{-N/2}$. Arbitrary phases depending separately on the groups may be included in these weights.*

Proof. Let H be the 2^d -square matrix with entries $H_{xy} = (-1)^{x \cdot y}$. For $x, x' \in \mathbb{F}_2^d$,

$$(HH^\top)_{xx'} = \sum_y (-1)^{(x+x') \cdot y} = \begin{cases} 2^d, & x = x', \\ 0, & x \neq x'. \end{cases}$$

The second value is zero by pairing y with $y + e_j$ at any coordinate where $x + x'$ is nonzero. Thus the Euclidean operator norm of H is $2^{d/2}$.

For (b), set $a_x = \alpha(x)f(x)$ and $b_y = \beta(y)g(y)$. Then

$$|a^\top H b| \leq 2^{d/2} \|a\|_2 \|b\|_2, \quad \|a\|_2^2 \leq p_1 \sum_x \alpha(x) \leq p_1,$$

and similarly for b , proving the result.

For (a), invertible changes of coordinates reduce the form to the dot product on its d active coordinates. Average each bounded weight over its inactive coordinates. Apply (b) to the uniform probabilities $p_1 = p_2 = 2^{-d}$. A bounded-density change can be absorbed into the weights after dividing their bounds by C_1, C_2 .

Finally, the matrix of the full bit form in (c) is $C \otimes I_N$. Choose bases reducing C to $\text{diag}(I_{\text{rank } C}, 0)$; the tensor matrix then has rank $N \text{rank } C$. Apply (a), absorbing any within-group phases into the corresponding bounded weight. $\square$

6 Low-rank moments and mixing forms

This section supplies the finite algebra used in the queries. Low-rank moment representations bound the number of selector labels, protected labels keep attack atoms independent of the pins, and fixed mixing forms provide the rank estimates used in later parameter and status tests.

The pin budget K is fixed before the parameters chosen in this Section. The arguments in this section concern the coefficient spaces and exact frame constraints; they apply to the thinned law π as well as to P_0 . Set

$$R_* = 2(K + 20), \quad j_* \geq 10(K + 1)(R_* + 20). \quad (6.1)$$

All inequalities concerning ranks in this section are over $\mathbb{F}_2$. The selector size b will be chosen below. Put $p_* = \sum_{j=0}^{j_*} \binom{b}{j}$ and write $d_b = (g^2 + 3)n$ for the number of ordinary base coordinates. Thus $\mathcal{B} = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+d_b}$ and $v(s, z) = p_s \otimes (1, z)$, as in Section 4. The constant base coordinate has index 0.

The quotient by the radical and the commuting coordinate operators used below belong to the flat-extension approach to moment matrices; see Laurent and Mourrain [18]. We prove the required Boolean version with separate selector and base-degree bounds.

6.1 Boolean moment matrices

Lemma 6.1. *The span of $(1, z)(1, z)^\top$, $z \in \mathbb{F}_2^{d_b}$, consists precisely of symmetric $(1 + d_b)$ -by- $(1 + d_b)$ matrices Z satisfying $Z_{ii} = Z_{0i}$ for every i . For any u distinct selectors, their evaluation vectors p_s are linearly independent if $j_* \geq u - 1$.*

Proof. Every point matrix has the stated identities. Conversely the space specified by these identities has a basis consisting of the matrix supported at $(0, 0)$, the matrices supported at $(0, i)$, $(i, 0)$, (i, i) , and the matrices supported at (i, j) , (j, i) for $0 < i < j$. The first is the point matrix at zero. The second is the sum of the point matrices at zero and at the i th unit vector. The third is the sum of the point matrices at zero, at the i th and j th unit vectors, and at their sum. This proves the first assertion.

To isolate a selector s among u distinct selectors, for each other selector choose a coordinate on which it differs from s , and take the product of the corresponding affine bit functions that equal one at s and zero at the chosen other selector. This polynomial has degree at most $u - 1$ and evaluates to the indicator of s on the specified set. Applying these interpolating linear functionals to a relation among the evaluation vectors proves independence. $\square$

Lemma 6.2 (Label support). *Every $w \in W$ of rank at most R_* has a representation*

$$w = \sum_{s \in L} (p_s \otimes I) Z_s (p_s \otimes I)^\top, \quad |L| \leq R_*, \quad (6.2)$$

where each Z_s is symmetric, $(Z_s)_{ii} = (Z_s)_{0i}$, and $\sum_{s \in L} \text{rank } Z_s \leq R_*$.

Proof. Choose a linear combination of point matrices representing w . It defines a linear functional Λ on the Boolean functions of selector degree at most $2j_*$ and base degree at most two. The matrix w is the matrix of the pairing $B(f, g) = \Lambda(fg)$ on functions of base degree at most one and selector degree at most j_* . Products and degrees here are in the Boolean algebra, so each variable satisfies $x^2 = x$.

Let V_j denote the subspace with selector degree at most j , and let r_j be the rank of $B|_{V_j \times V_j}$. The sequence r_j is nondecreasing and bounded by R_* . Because $j_* \geq 6R_* + 4$, there is $j \geq 4R_* + 2$ such that

$$r_j = r_{j+1} = r_{j+2}. \quad (6.3)$$

Indeed, if no such three successive values occurred between levels $4R_* + 2$ and j_* , every two successive increments in that interval would contain a strict increase, giving more than R_* increases.

Let Q be the quotient of V_{j+2} by the radical of its restricted pairing. The induced pairing on Q is nondegenerate. Equality of the first and last ranks in (6.3) means that the image of V_j is all of Q : its dimension is at least $r_j = \dim Q$. Since the quotient pairing is nondegenerate, a vector of V_j pairing to zero with V_j consequently pairs to zero with V_{j+2} . For a selector coordinate s_a , define $M_a[f] = [s_a f]$ using $f \in V_j$. This is well defined: if $[f] = 0$, then for every $g \in V_j$,

$$B(s_a f, g) = B(f, s_a g) = 0,$$

and V_j spans Q . The same identity proves that M_a is self-adjoint. Replacing $s_b f$ by a representative in V_j and transferring s_a to the other argument gives

$$B(M_a M_b[f], [g]) = \Lambda(s_a s_b f g).$$

All representatives needed in this comparison lie in V_{j+2} . Thus $M_a M_b = M_b M_a$ and, since $s_a^2 = s_a$, $M_a^2 = M_a$.

Commuting idempotents over $\mathbb{F}_2$ have a simultaneous eigenspace splitting

$$Q = \bigoplus_{s \in L} Q_s, \quad M_a|_{Q_s} = s_a I.$$

For completeness, a single idempotent splits its space into kernel and image; all the other commuting maps preserve both summands, so iteration proves the assertion. Distinct joint eigenspaces are orthogonal, by self-adjointness of a coordinate on which their labels differ. Each nonzero summand has positive dimension, so $|L| \leq \dim Q \leq R_*$. Define Z_s by pairing the projections onto Q_s of the affine base coordinates. Then Z_s is symmetric and $\text{rank } Z_s \leq \dim Q_s$.

Repeated application of the M_a gives the class of a base coordinate times any selector monomial of degree at most j . It follows that (6.2) reproduces all moments of selector degree at most $2j$. To verify the diagonal identity separately on a label, use an interpolating selector polynomial e_s of degree at most $|L| - 1$. Its operator on Q is the projection onto Q_s . The Boolean identities give

$$B(z_i e_s, z_i e_s) = \Lambda(z_i e_s^2) = B(e_s, z_i e_s),$$

which is $(Z_s)_{ii} = (Z_s)_{0i}$. The interpolation degrees are within j .

Extend the right side of (6.2) to the full row and column indexing. Its rank is at most $\sum_s \dim Q_s \leq R_*$. By Theorem 6.1 it belongs to W . If its difference from w were nonzero, choose a nonzero difference moment of minimal selector degree $d' > 2j$. Write its selector monomial as the product on a set U of size d' , and keep the two affine base coordinates in that moment fixed. Index rows by the subsets $A \subset U$ of size $\lfloor d'/2 \rfloor$ and columns by the complementary monomials $U \setminus A'$ with the same indexing. The union $A \cup (U \setminus A')$ is U exactly when $A = A'$; otherwise its degree is smaller than d' . The difference on this square submatrix is therefore the identity. Both its row and column degrees are at most j_* , whereas its rank is $\binom{d'}{\lfloor d'/2 \rfloor} > 2R_*$. This contradicts the rank bound $2R_*$ for the difference and completes the proof. $\square$

6.2 Labels protected from the pins

For an endpoint i of a leaf, let $S_{i,e}^\pm \subset \mathcal{B}$ be the projection of $D_{O,e}^\pm$ onto its individual primal coordinate block. These are projections, not intersections; a stored pin may mix endpoints and channel coordinates.

Lemma 6.3. *Put $L_0^{\text{lab}} = R_* + 14$. At an endpoint, the union of the nonzero label supports of all vectors in all $S_{i,e}^\pm$ that can be written as $\sum_{s \in L} p_s \otimes z_s$ with $|L| \leq L_0^{\text{lab}}$ has size at most*

$$B_0 = 2|\mathcal{E}|K(R_* + 14). \quad (6.4)$$

Proof. For one projected space, choose a basis from its members admitting such sparse representations, for the span of all these members. There are at most K chosen members. Comparing any further sparse member with its expression in this basis uses at most $(K + 1)(R_* + 14)$ labels. Their evaluation vectors are independent by [Theorem 6.1](#) and (6.1). Consequently every label of the further member occurs in one of the chosen basis representations. There are at most $K(R_* + 14)$ such labels per projected space. Summing over signs and components proves the stated, deliberately generous, bound. $\square$

Exclude these labels whenever an attack atom is selected at that endpoint. For later finite exclusions set

$$B_* = 10(B_0 + 10000g + 1)$$

and choose b so large that

$$2^{b-2j_*} > 100g(B_* + 1). \quad (6.5)$$

The effective test space at endpoint i is

$$C_i = \{x \in \mathcal{X} : x_e \in S_{i,e}^+ \otimes S_{i,e}^- \text{ for every } e \in \mathcal{E}\}. \quad (6.6)$$

Fix a basis of this space on each leaf. Since the total stored pin dimension is at most K , the sum over components of the individual projection dimensions in either sign is at most K . Therefore

$$\dim C_i \leq K^2, \quad \sum_e \text{rank } x_e \leq K \quad (x \in C_i). \quad (6.7)$$

If pins are enlarged, retain the old effective spaces and their chosen basis data as well; they are subspaces of the new ones.

Definition 6.4 (Atoms and flavors). For a tag l , selector s , and base point z allowed at l , let $\mathbf{a}_l(s, z) \in \mathcal{X}$ be the cut profile represented by $w_l = v(s, z)v(s, z)^\top$ and $w_t = 0$ for $t \neq l$. Its diagonal bit is $q = \eta(w_l)$ and its role bit is $a(\mathbf{a}_l(s, z))$. A generic flavor retains all base coordinates allowed at l . A shared-only flavor sets all O blocks to zero. A pure flavor sets S to zero and retains a specified subset of the allowed O blocks, setting the others to zero. Every flavor retains both $\#$ and Z in their entirety. At a fixed label, a parameter test samples all retained bits independently and uniformly. Its tester summary is the list of all individual quadratic tester values. In addition to this summary we may record the basis values of $T(\mathbf{a}_l(s, z), \cdot)|_{C_i}$.

6.3 Choice of the mixers

For a quadratic function f on a binary vector space, its polar form is $(u, v) \mapsto f(u + v) + f(u) + f(v) + f(0)$; its polar rank is the rank of this bilinear form.

Fix, for example,

$$A = 10(K + 1)p_*(g^2 + 5), \quad s_0 = 4\lceil A \rceil, \quad J > 100(s_0 + K^2 + 1). \quad (6.8)$$

These constants precede r_0, h and M_0 .

Lemma 6.5 (Uniform mixing forms). *For all sufficiently large n one can choose the matrices L_j^{ef}, R_j^{ef} and M_a in [Section 4](#) so that:*

- (i) *For every nonzero $x \in \mathcal{X}$ of total component rank at most K , every tag, label and flavor, and every fixing of the non- Z base coordinates, the quadratic function $T(x, \mathbf{a}_l(s, z))$ of the n remaining Z bits has polar rank at least $2(J - s_0)$. It uses a space of linear forms on Z of bounded dimension, independent of n . This space can be chosen before the non- Z coordinates are fixed.*
- (ii) *For two distinct labels and any two flavors, every nonzero parity of the indexed L_j^{ef}, R_j^{ef} evaluations between the two point inputs, allowing each matrix in either order or both, and optionally either or both orders of E , has bilinear-part rank at least $n/5$. Here a nonzero parity involving an L or R evaluation means that at least one such indexed ordered evaluation occurs. The same rank bound holds for the three parities consisting of E alone, its reverse alone, or their sum.*

The choices are made before any unit law is selected.

Proof. A symmetric rank- r matrix x on a d_B -dimensional coordinate space can be written $ZHZ^\top$, where Z has r independent columns and H is a nonsingular symmetric r -by- r matrix. To verify the factorization, take a left inverse L of the chosen column basis Z and put $H = LxL^\top$. The projection ZL fixes x on the left and, by symmetry, on the right; hence $ZHZ^\top = x$. Its rank forces H to be nonsingular. Neither H nor the original matrix is required to have nonzero diagonal. For a fixed component rank tuple of total at most K , counting Z, H gives at most $2^{d_B K + K^2 + K}$ choices. There are at most $(K + 1)^{|\mathcal{E}|}$ such rank tuples. Therefore the number of profiles under consideration satisfies

$$\#\{x \in \mathcal{X} : \sum_e \text{rank } x_e \leq K\} \leq 2^{An} \quad (6.9)$$

for all sufficiently large n , since $d_B = p_*(1 + (g^2 + 3)n)$.

Choose all L, R matrices independently and uniformly. Fix a nonzero profile and write $x_e = Z_e H_e Z_e^\top$, with $r_e = \text{rank } x_e$. For a tag- l atom the contribution from a single indexed matrix uses the forward forms $Z_e^\top L v$ when $l \in f$, and the reverse forms $v^\top L Z_f$ when $l \in e$, and the corresponding forms for R . Regard these initially as separate formal variables. Across all indices their number is at most

$$m_0 = 4J(g - 1)K.$$

Each forward or reverse pair contributes a bilinear product with nonsingular middle matrix H_e or H_f . These products use disjoint formal variables. Since $x \neq 0$, for each j there is at least one such nonzero product, so the formal quadratic has polar rank at least $2J$.

We justify the distribution of their linear parts on Z even when forward and reverse restrictions overlap. For a fixed matrix L , specifying $Z_e^\top L$ and $L Z_f$ imposes at most $r_e r_f$ compatibility bits,

namely their common $Z_e^\top L Z_f$ block. After restriction to the free Z directions, the resulting list remains uniform on a linear subspace of codimension at most $r_e r_f$ in the space of all lists. Equivalently its law is dominated by $2^{r_e r_f}$ times the completely uniform law. Over all matrices the domination factor is at most 2^{2JK^2} ; only matrices with both relevant incidences contribute a compatibility term.

For a uniform m -by- n matrix, the probability of corank at least $s_0 + 1$ is at most $2^{m(s_0+1)-(s_0+1)n}$: choose $s_0 + 1$ independent row relations and require all their n evaluations to vanish. If $m < s_0 + 1$ the probability is zero. The preceding domination therefore bounds the bad probability for a fixed profile, tag and label by

$$2^{2JK^2+(s_0+1)m_0} 2^{-(s_0+1)n}.$$

There is no union over non- Z values or flavors in this estimate: they affect affine shifts, not these linear parts on the always retained Z block. A union over at most $2^{An}g^{2b}$ choices tends to zero. On the complementary event the image of the Z inputs in the formal variable space has codimension at most s_0 . Restricting a bilinear form to a subspace of codimension s_0 reduces rank by at most $2s_0$: in a basis extending that subspace one deletes s_0 rows and s_0 columns, each deletion decreasing rank by at most one. This proves (i). The formal list itself supplies the claimed bounded space of linear forms. The term T^0 has no Z variables and does not affect this argument.

For (ii), distinct labels have independent selector evaluation vectors. Their two Z direction spaces are therefore disjoint. Restricting a random matrix to these two spaces in the two orders reads independent off-diagonal blocks. Any nonempty parity involving ordered L, R evaluations is consequently a uniform n -by- n matrix on these directions, even after all other matrices and E have been fixed.

For the parities using only E , choose its matrices M_a independently and uniformly. On the two $\#$ direction spaces, E restricts to

$$A_{s,s'} = \sum_{a=1}^b (s_a + s'_a) M_a,$$

which is uniform because $s \neq s'$. The reverse gives $A_{s,s'}^\top$, and their sum is alternating. An $\lfloor n/2 \rfloor$ square block with disjoint row and column indices is uniform also in this last case. A uniform d_1 -by- d_2 matrix has probability at most $2^{(d_1+d_2)r-d_1d_2}$ of rank at most r , by representing such a matrix as a product of a d_1 -by- r and an r -by- d_2 matrix. Taking $r < n/5$ gives probability $2^{-\Omega(n^2)}$ in all the preceding cases, including the square block of side $\lfloor n/2 \rfloor$.

There are at most 2^{2b} ordered label pairs and $2^{4J|\mathcal{E}|^2+2}$ parities to check. These counts are fixed in n . The Z and $\#$ restrictions also show that the tester part of E and the choice of flavor cannot reduce the ranks just proved. A union bound establishes both properties simultaneously. Fix one successful choice for each sufficiently large n . $\square$

Remark 6.6. Both P_0 and π have an exact symmetry under a common $G \in \mathrm{GL}_n(2)$ change of the Z input coordinates at one vertex, using that change at every component and in both primal modes. To see this, let R_G be the induced automorphism of $\mathcal{B}$: it acts as G on the Z block, as the identity on the constant and other base blocks, and as the identity on the selector factor. Replace

$$(P_e, Q_e, X_e, Y_e) \quad \text{by} \quad (P_e R_G, Q_e R_G, X_e, Y_e).$$

Since E has no Z rows or columns, $R_G^\top E R_G = E$. All prescribed self-Grams, annihilator equations and full-frame injectivities are therefore preserved. The inverse transformation uses G^{-1} , so this is a bijection preserving P_0 and its primal marginal.

Moreover $\text{im}(P_e R_G) = \text{im } P_e$ and $\text{im}(Q_e R_G) = \text{im } Q_e$. Conditional on the primal maps, the admissible channel maps are consequently exactly the same before and after this transformation: their annihilator and injectivity conditions depend only on these primal spans. The ambient channel maps themselves and all prescribed ambient thinning sets are fixed. Thus the allowed channel sets and their conditional normalizing counts are unchanged. Explicitly, if $\mathcal{S}$ denotes the channel-thinning event and $c(P, Q) = \mathbb{P}_{P_0}(\mathcal{S} \mid P, Q)$, then $c(PR_G, QR_G) = c(P, Q)$. Both the retained primal marginal and the conditional density $\mathbf{1}_{\mathcal{S}}/c(P, Q)$ defining π in [Section 5](#) are therefore preserved.

Tester summaries are unchanged. The assertion concerns the two reference laws; T^1 need not be invariant. Under the inverse change of variables in a parameter test, each resulting key and its tester data are preserved, while the bounded space of linear forms on Z in [Theorem 6.5](#), part (i), is reparametrized. In particular the unfiltered empirical mass of every test depending only on that key and the unchanged tester data is preserved. The same equality holds after conditioning on any feasible tester summary: this event is independent of Z , and its input normalizing count is unchanged.

Choose r_0 only after the preceding parameters and the bounded rank requirements in [Sections 7](#) and [9](#), with all fixed marginal-density bounds already specified; retain $h = 1000r_0$. Choose the thinning rate $\rho > 0$ after h as in [Section 5](#). Finally choose M_0 , also relative to $1/\rho$, large enough for every stated nominal-coefficient and scalar-record cost to be the required small fraction of $N = M_0 n$. None of the finite-test accuracies used later changes these choices.

7 Recipes and realization of the gradients

This section is deterministic. We construct cross Gram entries that turn a finite collection of matched vector keys into four hole witnesses. The construction uses the exact frame identities, the stored nominal pin spaces and finite-dimensional linear algebra. It therefore applies to every frame in the support of the thinned law π , and to every restricted leaf law, without any conditional uniformity assumption on its channels.

We use the nominal coefficient spaces of [Section 5](#). For a unit O and component e , these are

$$\mathcal{L}_{O,e} = \bigoplus_{i \in O} (\mathcal{B}_i \oplus H_i^+), \quad \mathcal{R}_{O,e} = \bigoplus_{i \in O} (\mathcal{B}_i \oplus H_i^-), \quad H_i^\pm = \mathbb{F}_2^h.$$

Their images are obtained by summing the corresponding frame columns in the common ambient space. If i' is the other endpoint of O , put $p_i = u_{i'} U_i \in \mathcal{X}^*$.

7.1 Small tables and scalar recipes

Definition 7.1. For a leaf and component, put

$$\mathcal{J}_{O,e}^\pm = \bigoplus_{i \in O} (S_{i,e}^\pm \oplus H_i^\pm).$$

For two leaves belonging to units A, B , a small table specifies the two cross bilinear forms on $\mathcal{J}_{A,e}^+ \times \mathcal{J}_{B,e}^-$ and $\mathcal{J}_{B,e}^+ \times \mathcal{J}_{A,e}^-$, for every e . It is admissible for two orientations if it agrees with their actual cross Grams whenever at least one of the two table arguments is a stored pin direction.

Let $H_{i,0}^\pm$ be the projection of the appropriate pin space onto $H_i^\pm$, and fix a complement $H_i^\pm = H_{i,0}^\pm \oplus H_{i,1}^\pm$. The complement is called private. We identify the two channel coordinate

copies with $\mathbb{F}_2^h$ in their common column order. The table is injecting if, for each component, each orientation of the roles O, O' , and $i \in O, z \in O'$, the following maps are injective:

$$\begin{aligned} D_{O,e}^+ \cap \mathcal{B}_i &\longrightarrow \mathbb{F}_2^h / H_{z,0}^+, & d &\longmapsto \text{table evaluation of } d \text{ against } Y_z, \\ D_{O,e}^- \cap \mathcal{B}_i &\longrightarrow \mathbb{F}_2^h / H_{z,0}^-, & d &\longmapsto \text{table evaluation of } d \text{ against } X_z. \end{aligned} \quad (7.1)$$

These quotient conditions use the standard coordinate dot pairing on the channels. For instance, the first image is zero in the quotient precisely when it pairs to zero with all vectors annihilating $H_{z,0}^+$.

Contractions on C_i computed from the table carry a star. Thus $(u_z U_i)^*(x)$, for $x \in C_i$, is computed by writing x_e in $S_{i,e}^+ \otimes S_{i,e}^-$ and taking the dot product of its two table evaluations against Y_z, X_z . This is linear in x . All coordinates involved have bounded dimension, although that bound may depend on h .

For every cross pair i, z , choose between one and seven atoms at i and the same number at z , with a bijection preserving the tag and the diagonal bit q of each position. At any one endpoint all labels in its two lists must be distinct and must avoid the exclusions of [Theorem 6.3](#). Write v_{iz} for the sum of its list toward z . A scalar recipe requires, in both orientations of the two unit roles,

$$\begin{aligned} \sum_{\text{list toward } z} q &= 1, & a(v_{iz}) + a(v_{zi}) &= 1, \\ T(v_{iz}, \cdot)|_{C_i} &= (a + (u_z U_i)^*)|_{C_i}, \\ (p_i + a)(v_{iz}) &= (p_{i'} + a)(v_{i'z}). \end{aligned} \quad (7.2)$$

An atom $\mathbf{a}_l(s, y)$ at i has vector key

$$(P_{ie}v(s, y), Q_{ie}v(s, y))_{e \ni l},$$

with a fixed order of its $2(g-1)$ entries. Matching keys means equality in this order at every matched atom position.

Lemma 7.2 (Table flags and binary prescriptions). *The following statements hold.*

- (i) *For a fixed pair of leaves and a numerical small table, admissibility is the intersection of two unary orientation filters, one on each unit. Its numerical coordinate data, including injection and the bases of the effective spaces, have bounded description length apart from the nominal coefficient bases, which require $O(n)$ bits.*
- (ii) *The key directions of the lists at either unit are jointly independent modulo its table-coordinate spaces, component by component and sign by sign.*
- (iii) *Given (7.2), there is a prescription of the binary L, R evaluations between distinct atoms at each endpoint, depending only on their tags, tester summaries and p bits, which makes*

$$r(v_{iz})(x) = \begin{cases} 0, & x \text{ is an atom of the list toward } z, \\ p_{z'}(\tilde{x}), & x \text{ is an atom toward } z', \end{cases} \quad (7.3)$$

where z' is the other endpoint of the opposite unit and $\tilde{x}$ is the matched atom there.

Proof. A pin image is fixed on its leaf. A cross entry with a pinned first argument therefore depends only on the other unit's orientation, and conversely for a pinned second argument. Entries

with both arguments pinned are numerical tests on the leaf pair. This proves the first assertion about filters. Use bases of the projected pin spaces to encode tensors of C_i : their coordinates lie in spaces of dimension at most K^2 , and there are at most K^2 basis tensors. The pin, projection and channel relationships and the table are bounded-dimensional matrices. The nominal bases themselves have $O(K \dim \mathcal{B}) = O(n)$ coefficients.

For the second assertion, project a purported dependence onto an individual primal coordinate block. It expresses a sum of at most fourteen attack rays as an element of $S_{i,e}^\pm$. Any nonzero such vector would have a sparse representation on attack labels and is forbidden by [Theorem 6.3](#). Independence of those label factors and the nonzero constant coordinate of each point then makes all coefficients zero. The channels introduce no relation because the table space is the direct sum of its individual primal and channel blocks.

For the last assertion, at one endpoint let I_1, I_2 be its two disjoint nonempty lists. The matrix of T on its atoms must be symmetric, with diagonal a on each atom. Prescribing the two list sums of its rows is possible exactly when their self-pairings equal the corresponding diagonal sums and their mutual pairings agree. To see sufficiency, subtract any symmetric matrix with the required diagonal. The remaining task prescribes an alternating form on the span of the two independent list indicators and its pairings with the whole coordinate space. Extend those indicators to a basis, retain the prescribed entries in their rows and columns, and assign the remaining alternating entries arbitrarily.

For (7.3), the self-pairing requirements hold because $T(v, v) = a(v)$. Mutual consistency is

$$p_z(v_{zi}) + p_{z'}(v_{z'i}) = a(v_{iz}) + a(v_{iz'}).$$

The coupled equation at the opposite unit gives the left side as $a(v_{zi}) + a(v_{z'i})$, which equals the displayed right side by the two role-sum equations in (7.2). Thus the symmetric matrix exists.

Every off-diagonal entry can be implemented by formal L, R bit prescriptions on that pair of atoms. The value of T^0 is already determined by their tester summaries. Choose one indexed product term in T^1 linking a component of the first star to a component of the second star; set its two factors to give the desired correction and set all other product factors to zero. Forward and reverse evaluations are distinguished. The two labels at this endpoint differ, so these are the indexed ordered evaluations covered by [Theorem 6.5](#). This argument specifies the bits; their occurrence in parameter tests will be supplied by [Theorem 11.9](#). It makes no assumption about independence of arbitrary actual evaluations at fixed points. $\square$

7.2 The deterministic realization theorem

Theorem 7.3 (Gradient realization). *Suppose two units in fixed leaves have lists satisfying (7.2), the binary prescriptions yielding (7.3), equal matched keys, and an admissible injecting small table. Freeze their actual cross Gram entries whenever either argument is a pin or a key direction. Then there are abstract cross bilinear forms on the full nominal spaces, extending all these frozen entries, whose primal-versus-channel entries satisfy*

$$u_z U_i = r(v_{iz}) \quad \text{on } \mathcal{X} \quad (i \text{ and } z \text{ in opposite units}). \quad (7.4)$$

Only the primal-versus-channel entries need be prescribed in the final test. Their number is at most $16|\mathcal{E}|h \dim \mathcal{B} = O(n)$, with its coefficient fixed before M_0 . Agreement of the actual cross Grams with these entries produces all four cross holes, with witnesses (v_{iz}, v_{zi}) .

Proof. Equal keys imply $U_i v_{iz} = U_z v_{zi}$, and the role sums in (7.2) supply the witness parity. It remains to construct the gradients. We distinguish abstract Gram entries from the random event that the actual entries agree with them.

A baseline of bounded primal rank. At each component and sign choose a splitting of the table space as $\mathcal{J} = D \oplus \mathcal{J}'$, append the key space, and append an additional primal complement. The independence in Theorem 7.2 makes these splittings possible. Retain the small table on the product of the two table spaces and the actual frozen entries in any pin or key row or column. These prescriptions agree on their overlaps by admissibility. Set the free blocks from an additional primal complement to the opposite $\mathcal{J}'$ to zero, in both orientations, and set the additional-primal by additional-primal block to zero. This defines baseline cross bilinear forms.

Fix an opposite endpoint z . Let

$$F_{z,e}^0 : \mathcal{L}_{O,e} \longrightarrow \mathbb{F}_2^h, \quad G_{z,e}^0 : \mathcal{R}_{O,e} \longrightarrow \mathbb{F}_2^h$$

be evaluation against its Y_z, X_z channels. On additional primal inputs, evaluation of an opposite channel factors through its projection from $\mathcal{J}$ onto D ; hence the rank there is at most $\dim D$. The remaining primal inputs are in bounded-dimensional projected pin and key spaces. It follows that the ranks of both baseline maps on all primal inputs are bounded independently of r_0, h, n . Arbitrary channel-channel entries of the small table do not affect this bound. A uniform bound for either map on the combined primal inputs of the two endpoints is

$$B_{\text{lin}} = 3K + 28 : \quad (7.5)$$

the opposite pin projection costs at most K dimensions, the two individual projected primal spaces cost at most $2K$, and the two endpoints have at most twenty-eight key rays.

Let $u_z^0 U_i$ denote the resulting contractions. They give the table values on C_i . On an attack atom toward z they are zero, since its tensor is the equal-key tensor at z and $u_z U_z = 0$. On an attack toward z' they equal $p_{z'}$ of its matched atom, since that tensor is the corresponding tensor at z' . These are precisely the values in (7.3).

Allowed changes. Let $\mathcal{A}_{O,e}^\pm$ be the span of all key directions on the indicated side of the unit, and put

$$\mathcal{F}_{O,e}^\pm = D_{O,e}^\pm + \mathcal{A}_{O,e}^\pm.$$

Write bars for the quotients

$$\overline{\mathcal{L}}_{O,e} = \mathcal{L}_{O,e} / (\mathcal{F}_{O,e}^+ + \bigoplus_i H_{i,1}^+), \quad \overline{\mathcal{R}}_{O,e} = \mathcal{R}_{O,e} / (\mathcal{F}_{O,e}^- + \bigoplus_i H_{i,1}^-).$$

We may change F_z^0, G_z^0 by any maps

$$\delta F_{z,e} : \overline{\mathcal{L}}_{O,e} \longrightarrow (H_{z,0}^-)^\perp, \quad \delta G_{z,e} : \overline{\mathcal{R}}_{O,e} \longrightarrow (H_{z,0}^+)^\perp. \quad (7.6)$$

Here each map is extended to a bilinear change supported on its opposite channel block and zero on the opposite primal blocks. It vanishes on all frozen entries: the domain quotient kills frozen directions on one side, while on the other side a pin's channel projection lies in $H_{z,0}$ and is annihilated by the allowed values. A reciprocal-role change in the same Gram block contributes nothing on this map's primal inputs, because it is supported on a channel argument there. On private-by-private blocks both changes are zero. They may add further channel-channel values, which are irrelevant to the primal contractions. Thus all choices below may be made independently for opposite endpoints and for the reciprocal unit roles as far as primal-input values are concerned.

The unrestricted linear response. Fix z and solve simultaneously for the two $i \in O$. Consider the linear space of responses formed by

$$F_z^0 \cdot \delta G_z + \delta F_z \cdot G_z^0$$

and all bilinear forms on $\overline{\mathcal{L}}_{O,e} \times \overline{\mathcal{R}}_{O,e}$, with their contractions summed over components on each individual cut profile. We claim that the pair of target forms

$$(r(v_{iz}) - u_z^0 U_i)_{i \in O} \tag{7.7}$$

belongs to this response space.

Let $(x_i)_{i \in O} \in \mathcal{X}^O$ annihilate all responses. Annihilating the arbitrary pure bilinear forms says that at each component the sum of the two nominal tensors has zero image in the product of barred quotients. Project further onto the individual primal spaces modulo $S_{i,e}^\pm$ and the individual key spans. This projection is well defined on the barred spaces, since every pin projects into $S_{i,e}^\pm$. It shows that $x_{i,e}$ vanishes after both such quotients. A matrix killed after quotienting its two factors by spaces of dimensions d_+, d_- has rank at most $d_+ + d_-$: in bases extending those spaces all entries outside their rows and columns vanish. Consequently

$$\text{rank } x_{i,e} \leq 2K + 28 \leq R_*.$$

Apply [Theorem 6.2](#) to every component.

For an attack label at i , compare its label summand with all key directions in that component. Their combined label set has at most $R_* + 14$ elements, whose direction spaces $p_s \otimes \mathbb{F}_2^{1+d_b}$ form a direct sum. Any vector of $S_{i,e}^\pm$ in this sum has zero attack-label component, by [Theorem 6.3](#). Thus the attack summand projects to zero modulo just its individual key ray if the atom uses this component, and is zero if there is no such ray. In the former case, writing its affine base vector as $v = (1, y)$, a symmetric matrix killed in the two quotients by v has the form

$$Z = cvv^\top + vd^\top + dv^\top.$$

Indeed choose v as the first basis vector: the surviving matrix can be supported only on its first row and column. The moment identities in the original affine coordinates now give

$$Z_{ii} = cv_i, \quad Z_{0i} = cv_i + d_i + d_0 v_i,$$

so $d_i = d_0 v_i$ and $Z = cvv^\top$.

The cut relation on any triangle of tags separates label by label: the union of the three supports has at most $3R_*$ labels, within the interpolation bound. At this attack label the nonzero components lie on its tag star; the triangle relation forces their scalar coefficients to be equal. Subtract that scalar multiple of the attack atom from x_i . Do this for every attack label. Each subtracted atom annihilates the response space, because both its factor directions are keys. The remaining profiles still annihilate the response and have only nonattack label support. Removing these separated label blocks does not increase their component ranks or label support sizes.

At one component contract a remaining $x_{i,e}$ on the minus side by a primal functional ϕ at i annihilating $S_{i,e}^-$ and all individual key directions. Extend ϕ by zero on other endpoint and channel blocks; it is a functional on $\overline{\mathcal{R}}_{O,e}$. Pure annihilation shows that the resulting individual plus vector d is a sum of a pin, key directions, and private channel directions. In this representation all key coefficients vanish. To see this, project onto each individual primal block. At the other endpoint

the projected pin would be a sparse sum of attack rays. At endpoint i it is the difference of the nonattack vector d , supported on at most R_* labels, and a sum of at most fourteen attack rays. In both cases the label exclusion prevents any attack coefficient. The channel components of the pin lie in the protected spaces H_0 , whereas the remaining channel term lies in their chosen complements H_1 . They must both vanish. Therefore

$$d \in D_{O,e}^+ \cap \mathcal{B}_i.$$

Test the derivative response using $\delta G_{z,e} = \phi \otimes c$ for arbitrary $c \in (H_{z,0}^+)^\perp$. It gives $F_{z,e}^0(d) \cdot c = 0$. Hence the table evaluation of d is zero modulo $H_{z,0}^+$, and injection in (7.1) yields $d = 0$. The reciprocal contraction proves the same assertion in the minus mode.

It follows that the remaining component tensors lie in the products of the individual projected pin plus key spaces. Their nonattack support removes the key directions by the same sparse-label argument. Thus the remaining profiles belong to C_i . We have proved that every annihilator is a sum of effective-space profiles and attack atoms. The target (7.7) vanishes on the former by (7.2) and on the latter by (7.3) and the baseline values. In a finite-dimensional vector space the image of a linear map is the annihilator of the kernel of its dual: this follows by extending a basis of the image to a basis of the target space. It proves the claimed linear solvability.

A low-rank representation of the target. For one i , the form $r(v_{iz})$ has a matrix representation on the components of rank at most

$$15r_0 + 14J|\mathcal{E}| \tag{7.8}$$

per component. Here is an explicit routing for its tester part. Let $w = (w_t)$ be the representation given by its atom list. The list condition gives $\chi_*(w) = \sum q = 1$. On an allowed $O_{d,t}$ block at any tag the coefficient in $a + T^0(v_{iz}, \cdot)$ is

$$1 + a(v_{iz}) + b_t(v_{iz}) = 1 + \chi_*(w) + \eta_S(w_t) = \eta_S(w_t).$$

There are at most seven tags t with nonzero $\eta_S(w_t)$. Route this tester from each allowed tag $l \in I_d$ to d through the component $\{l, d\}$. It contributes nothing at d , where $O_{d,t}$ is disallowed. At a component this uses at most fourteen ordinary testers, allowing either endpoint as center. For the shared tester, route a star centered at t with coefficient $a_t(v_{iz})$. Its contribution at the center is $(g-1)a_t(v_{iz}) = 0$, and at any other tag it gives the desired term in $a_t(v_{iz})b_t$. Consolidating gives at most one shared tester per component. Each ordered tester matrix has rank r_0 , proving the $15r_0$ bound.

For $T^1(v_{iz}, \cdot)$, each of the two orientations of an indexed term supplies a matrix of rank at most the component rank of v_{iz} . This profile is a sum of at most seven atoms, so its total component rank is at most $7(g-1)$. The total contribution per component has rank at most $14J(g-1) \leq 14J|\mathcal{E}|$. This proves (7.8).

Removing the bounded-dimensional obstructions. Let $M_{i,e}$ be the preceding representing matrices. On each individual primal factor choose a projection $A_{i,e}^\pm$ whose kernel is the sum of its projected pin and key spaces. The matrix $(A_{i,e}^+)^\top M_{i,e} A_{i,e}^-$, extended by zero outside that individual block, is a permitted pure quotient form. Its sum over i has rank at most twice (7.8). Moreover

$$\begin{aligned} M_{i,e} - (A_{i,e}^+)^\top M_{i,e} A_{i,e}^- &= (I - A_{i,e}^+)^\top M_{i,e} + (A_{i,e}^+)^\top M_{i,e} (I - A_{i,e}^-), \\ \text{rank}(M_{i,e} - (A_{i,e}^+)^\top M_{i,e} A_{i,e}^-) &\leq \text{rank}(I - A_{i,e}^+) + \text{rank}(I - A_{i,e}^-) \leq 2K + 28. \end{aligned} \tag{7.9}$$

Subtract these pure forms from the linear task. Its residual, including the baseline contraction, has a per-individual matrix representation of bounded rank independent of r_0, h, n , and remains linearly soluble.

In any unrestricted solution, replace the derivative output maps by bounded-rank maps in the same allowed value spaces, preserving all their dot products with the opposite baseline values on primal inputs. Such replacement is possible as follows. The map from the allowed value space to the dual of the span of those baseline values has bounded rank. Choose a linear section of its image and compose with that map. This has bounded-dimensional image and preserves every observed pairing. Postcomposing the original derivative map with this value projection leaves its primal derivative response unchanged and preserves its domain quotient. The two new derivative maps have bounded rank. Their extra product $\delta F_z \cdot \delta G_z$ is also a bounded-rank pure form, which we include with opposite sign in the remaining pure task. That task still has an unrestricted pure solution. Its per-individual target matrices have rank at most

$$R_{\text{res}} = 2K + 28 + 4B_{\text{lin}} = 14K + 140. \quad (7.10)$$

Here the first term bounds the projection difference, one B_{lin} bounds the baseline contraction, two bound the derivative terms, and one bounds their extra product.

Compression of the remaining pure task. We show that the latter task has a pure solution of bounded rank. At endpoint i , expand all projected pin and key vectors, and all row and column forms in its bounded-rank target representation, along selector coordinates and base blocks. Their counts depend on earlier construction parameters but not on r_0, h, n . On each ordinary base block let U be the span of the vectors to fix and K_0 the common kernel of the forms to preserve. Choose a complement in K_0 to $U \cap K_0$, and project along that complement onto a subspace containing U . The resulting endomorphism fixes U , preserves all the forms, and has rank at most $\dim U + \text{codim } K_0$. There are at most $2|\mathcal{E}|(K + 14)$ vectors to fix and at most $2|\mathcal{E}|R_{\text{res}}$ forms to preserve at one endpoint before selector expansion. Thus an explicit bound for the rank on an ordinary block is

$$D_{\text{blk}} = 2p_*|\mathcal{E}|(K + 14 + R_{\text{res}}). \quad (7.11)$$

Use these endomorphisms blockwise, fix the constant base coordinate, and act trivially on the selector factor. Use the same resulting map at every component and in both modes at this endpoint. It sends each allowed point to an allowed point and therefore preserves all W_l and the cut domain. It fixes the individual projected pin and key vectors. Taking identity on channels consequently fixes every mixed pin itself and preserves private channel subspaces. The map descends to the barred spaces and has bounded rank there: only the bounded protected channel dimensions survive their quotients. On either barred space a uniform bound is

$$D_{\text{quo}} = 2p_*(1 + (g^2 + 3)D_{\text{blk}}) + 2K. \quad (7.12)$$

The first term counts the two compressed primal blocks and the second bounds their surviving channel projections.

Precompose an unrestricted pure solution on its two arguments with these maps. Its rank per component becomes bounded. On individual cut profiles its value is unchanged, because the compressed profile remains in that cut domain and every form in the target representation is preserved. It is therefore the required bounded-rank pure solution.

Factoring the pure forms through channels. Restore the earlier projected pure forms. Their sum with the compressed correction has rank at most

$$30r_0 + 28J|\mathcal{E}| + D_{\text{quo}} \quad (7.13)$$

per component for the fixed opposite endpoint. In particular the additive constant is independent of r_0, h, n . In the two allowed channel value spaces impose, in addition, orthogonality to the opposite baseline and bounded derivative values on primal inputs. These are only boundedly many linear restrictions. Each remaining value space has codimension at most $K + 2B_{\text{lin}}$, so their dot pairing has rank at least $h - 2K - 4B_{\text{lin}}$: restricting a nondegenerate pairing by codimensions c_1, c_2 loses at most $c_1 + c_2$ in rank. With $h = 1000r_0$, the sufficient requirement is

$$970r_0 > 28J|\mathcal{E}| + D_{\text{quo}} + 2K + 4B_{\text{lin}}. \quad (7.14)$$

Every quantity on its right is fixed before r_0 , so this is one of the permitted requirements on that choice.

Factor a rank- r desired pure bilinear form as $\sum_{a=1}^r \ell_a \otimes m_a$. Choose vectors f_a, g_a in these remaining channel spaces with $f_a \cdot g_b$ equal to the Kronecker delta; they exist by the pairing rank. The changes $\sum_a \ell_a f_a$ and $\sum_a m_a g_a$ realize that form as their dot product. Their cross terms with the baseline and the previous small derivative changes vanish on primal inputs by the additional orthogonality conditions. They obey (7.6) and hence preserve every frozen entry. This realizes the target exactly for both endpoints $i \in O$.

Repeat for both opposite endpoints and both unit roles. The independence of primal-input changes already proved ensures that the resulting two cross Gram blocks realize all equations (7.4) simultaneously. Only primal-channel entries have been used. In each of the two blocks there are at most $8h \dim \mathcal{B}$ such entries per component, proving the asserted count. Actual agreement with them gives all gradient equations; equal keys and the role sums give the other witness conditions. All four cross pairs are therefore holes. $\square$

8 From overlapping key distributions to four holes

The deterministic construction becomes useful when its keys collide. The principal cost is equality of ambient vectors; the remaining primal-channel scalar entries have a smaller cost. We give a criterion that retains this distinction, including for cells selected using the parameters on the opposite side. The two unit laws may be different. We also keep colour and type tests in the successful event throughout the argument.

8.1 Queries and their reference measure

A numerical template fixes the shape of all lists in a scalar recipe: positions, endpoints, tags, labels, flavors and desired q bits, with matching tags and q bits across the two sides. It also fixes each atom's *numerical unary status*: its tester summary, role $a(\text{atom})$, basis values of $T(\text{atom}, \cdot)|_{C_i}$, and p_i bit. It fixes the within-endpoint L, R bit prescriptions of Theorem 7.2. Its labels are distinct at every endpoint. For a fixed pair of leaves it may specify a small table and two separate orientation filters, including its admissibility filters. Only choices for which equal accepted keys satisfy the hypotheses of Theorem 7.3 will be used.

The query at a unit draws fresh independent point parameters at all its positions and evaluates their keys. Parameters may be uniform on the retained flavor coordinates, or normalized conditional

on the prescribed, possible q bit at each position. This latter conditioning depends only on point parameters and has a fixed positive probability whenever it is possible. In particular the two units' parameter draws remain independent. No normalization is made for the other status, binary, Gram or orientation acceptance conditions. Every resulting accepted measure is a subprobability measure.

Let k be the number of ambient vector slots on one template side, counting components and signs. There are at most four lists of seven atoms, each with $2(g-1)$ slots, so

$$k \leq 56(g-1) = k_{\max}. \quad (8.1)$$

Let $\mathcal{K}_n$ be the set of global tuples of these k vectors satisfying the prescribed paired diagonal bit at every position and component, and having every off-position plus/minus inner product within a component equal to zero. Write ν_n for the uniform probability measure on $\mathcal{K}_n$.

The set $\mathcal{K}_n$ has density bounded below by a positive constant in the space of all k independent uniform vectors, and

$$|\mathcal{K}_n| \leq 2^{kN}. \quad (8.2)$$

Indeed the number of slots is fixed. With probability tending to one the plus vectors in each component are independent; given them, each minus vector fulfills a fixed list of independent linear equations with the prescribed probability 2^{-r} , where r is the number of equations on it. Their total number is fixed. This supplies a constant lower bound, including all diagonal and off-position requirements. Queries are accepted only when their keys belong to $\mathcal{K}_n$.

For a pair of leaves, let d_A, d_B be the densities relative to ν_n of the separate accepted key measures, integrating over their normalized leaf orientation laws and parameter draws. The densities include their orientation filters and all query conditions but are not renormalized for success. Thus $\int d_A d\nu_n, \int d_B d\nu_n \leq 1$. Both filters may depend on the ordered pair of leaves, while each is a predicate on its own orientation and query parameters. A test on the leaf pair, such as distinct colours or prescribed law types, is written as an external indicator $J_n(\ell_A, \ell_B) \in \{0, 1\}$. Equivalently, one can set both accepted measures to zero when that test fails.

Proposition 8.1 (Filtered collision criterion). *Let σ_A, σ_B be two laws on units, possibly carrying finitely many colour or type marks. After forgetting these marks, suppose each law has density at most 2^{3000gN} relative to $\pi \otimes \pi$. For $O \in \{A, B\}$, let $\tilde{\sigma}_O$ be a normalized restriction of σ_O of relative mass at least $2^{-o(N)}$, decomposed into leaves satisfying (5.11). On each side mix the leaves with their actual masses in $\tilde{\sigma}_O$.*

Fix a required predicate $\Psi(A, B)$ on the marked units and an allowed leaf-pair indicator J_n . Use only numerical query options whose joint acceptance on an allowed leaf pair, including their separate orientation filters, implies $\Psi(A, B)$. Suppose these accepted queries satisfy

$$\mathbb{E}_{\text{independent leaf pair}} J_n(\ell_A, \ell_B) \int d_A d_B d\nu_n \geq 2^{-o(N)}. \quad (8.3)$$

Then independent draws $A \sim \sigma_A, B \sim \sigma_B$ satisfy $\Psi(A, B)$ and have all four cross holes with probability at least

$$2^{-(k_{\max} + .03)N - o(N)}. \quad (8.4)$$

It suffices to obtain (8.3) after skipping some leaf pairs, or after summing over a bounded number of numerical options. In particular, a positive subsequential overlap lower bound yields (8.4) along that subsequence. With $\sigma_A = \sigma_B$ and Ψ the distinct-colour test, the conclusion retains that test. The statement also applies when the two sides have different prescribed law types.

Proof. First note where thinning enters the hypotheses. By [Theorem 5.3](#), with $\varepsilon_{\text{ch}} = 10|\mathcal{E}|2^h\rho$, the projected unit laws satisfy

$$\sigma_O \leq 2^{3000gN} \pi^{\otimes 2} \leq 2^{(3000g+\varepsilon_{\text{ch}})N} \mathbf{P}_0^{\otimes 2}.$$

The choice of ρ leaves this exponent strictly below $D = 4000g$, also after a restriction of mass $2^{-o(N)}$. Thus the reference image estimates and first peeling are charged against the original law $\mathbf{P}_0$; they supply the all-rank leaf hypothesis used here. No uniform conditional channel law under σ_O or π is used in this proof.

We first work with one shape of vector slots and one option per leaf pair. A bounded number of shapes or options will be handled at the end. We count only leaf pairs with $J_n = 1$, and every accepted measure already contains its filters enforcing Ψ on these pairs. Leaves are drawn independently from their two actual mass distributions; none of the following estimates requires those distributions to be equal.

The numerical records. Fix two leaves, a key $q \in \mathcal{K}_n$, and point parameter values a, b on the two sides. Freeze the images of all pin and key directions. The pin images are known from the leaves and the key images are the entries of q . At each unit record the pairings of all its nominal columns with these opposite frozen images. Record any numerical statuses in [Equations \(7.2\)](#) and [\(7.3\)](#) not already fixed by the query conditions. For fixed parameters these records, together with acceptance, partition the unit's orientations into unary cells. A partition is allowed to depend on the opposite parameters; it does not depend on the opposite orientation once its frozen images have been specified.

There are at most

$$L_n = 2^{Cn} \tag{8.5}$$

cells on either side, for a constant C fixed before M_0 . To verify the count, there are $O(\dim \mathcal{B} + h) = O(n)$ columns and only boundedly many pin and key vectors to pair against. Nominal projected-pin bases use $O(n)$ coefficients. The basis tensors of C_i are encoded in their bounded projected pin bases, rather than as full matrices on $\mathcal{B}$; this uses boundedly many further coordinates. Parameter values require $O(n)$ bits and the remaining table and status information has bounded size. The ambient pin and key images are already fixed and are not charged again as new records.

The numerical records determine every frozen cross entry. Together with the fixed algebraic data, leaves, parameters, and table, they determine a choice of the abstract solution in [Theorem 7.3](#). One can make this choice deterministic by taking the first solution in fixed coordinate orders on the finite spaces. Its unknown entries need not be encoded as additional records. In particular, a global signature cylinder or a complicated orientation filter is only an acceptance predicate, not an input requiring a new coordinate description for this solver.

Pruning at a fixed parameter pair. Put

$$\tau_n = 2^{-(k+.01)N}. \tag{8.6}$$

For the fixed parameter pair (a, b) and key q , let $\alpha_a^b(c; q)$ be the mass of a first-side accepted orientation in record cell c , measured in its normalized leaf law. Define $\beta_b^a(d; q)$ similarly. Denote their sums over records by $\alpha_a(q), \beta_b(q)$; these sums do not depend on the opposite parameter, since it only refines the record partition. Their sums over all keys are at most one.

Discard for this fixed pair only cells of mass less than τ_n . The product mass lost from small first-side cells at key q is at most

$$L_n \tau_n \beta_b(q).$$

Sum over keys and average the independent parameter weights. The loss in collision probability is at most $L_n \tau_n$, since $\sum_q \beta_b(q) \leq 1$ for every b . Discarding small second-side cells loses at most the same amount. Because ν_n is uniform, the overlap integral is $|\mathcal{K}_n|$ times the probability of an accepted key collision. Thus the total overlap loss, for any fixed leaf pair and also after averaging leaf pairs, is at most

$$2|\mathcal{K}_n|L_n\tau_n \leq 2^{1+Cn-.01N} \leq 2^{-.005N} \quad (8.7)$$

for the chosen sufficiently large M_0 and all sufficiently large n .

Every retained cell is its original unary cell and has mass at least τ_n . We have not intersected the retained sets over different opposite parameter values; such an intersection could reduce a cell below its threshold. Instead all subsequent estimates are made for this fixed parameter pair and its unaltered cells, and then averaged. This preserves both their mass bounds and the product of the two conditional orientation laws.

Image entropy inside a retained cell. By [Theorem 7.2](#), the k nominal key directions are jointly independent modulo the pins, with rank computed separately by component and sign and jointly across the two endpoints. Suppose a further tuple has rank $t \geq 1$ modulo pins and these key directions. Apply (5.11) to the combined tuple of rank $k+t$, prescribe the known key images and any images of the further tuple, and divide by the cell mass at least τ_n . Its largest conditional point mass is at most

$$2^{-(1-2\zeta)(k+t)N+(k+.01)N} = 2^{[-(1-2\zeta)t+.01+2\zeta k]N} \leq 2^{-.95tN}. \quad (8.8)$$

Here $k \leq k_{\max}$ and $\zeta = 1/(1000k_{\max})$, so $.01 + 2\zeta k \leq .012$. The inequality holds for every $t \geq 1$, including tuples whose rank grows with n . Further restrictions defining the cell cause no additional loss: the numerator was bounded by an event containing the entire cell and the prescribed images, and only the original cell mass has been divided out.

Testing the remaining scalar Gram entries. For a retained cell pair the abstract target is fixed. Let s_n be the number of its tested primal-channel entries in the two cross Gram blocks. By [Theorem 7.3](#) and the final choice of M_0 ,

$$s_n \leq 16|\mathcal{E}|h \dim \mathcal{B} < .01N.$$

Expand the indicator that all these entries agree with the target as a sum of 2^{s_n} binary characters, with the factor 2^{-s_n} .

A character has a coefficient tensor in the direct sum of the two cross nominal tensor products, over all components. Take its image after quotienting every nominal factor by pins and keys. If this image is zero, its value depends only on frozen cross entries. More explicitly the kernel of a tensor-product quotient is the sum of tensors with a frozen factor on at least one side; all their pairings have been recorded. The actual and target Grams agree there, so the disagreement character is identically one.

Otherwise let $t \geq 1$ be the sum of the ranks of the quotient coefficient matrices, over components and the two cross orientations. A rank factorization expresses its phase, up to a constant and two

separate phases involving frozen images, as

$$\sum_{a=1}^t X_a \cdot Y_a.$$

On each unit the corresponding coefficient directions have joint rank t modulo its frozen space. The two different cross orientations use opposite nominal signs, so their ranks add. By (8.8), each tuple of actual images has largest point mass at most $2^{-.95tN}$ under its conditional cell law. The two laws are independent. Absorb the separated phases into bounded weights. [Theorem 5.9](#), in dimension tN , bounds the absolute character mean by

$$2^{tN/2} (2^{-.95tN} 2^{-.95tN})^{1/2} = 2^{-.45tN} \leq 2^{-.45N}. \quad (8.9)$$

There is at least the trivial character with value one, and all other zero-quotient characters also have value one. Consequently the conditional agreement probability on every retained cell pair is at least

$$2^{-s_n} (1 - 2^{s_n} 2^{-.45N}) \geq 2^{-s_n-1} \geq 2^{-.02N}$$

for all sufficiently large n . No independence of the individual tested Gram bits is asserted or required.

Averaging and the exponent. The assumed overlap is subexponentially large. The error in (8.7) is exponentially small, so the retained collision probability, after averaging leaves and independent parameters, is at least $2^{-o(N)}/|\mathcal{K}_n|$. Integrating the preceding scalar agreement bound gives four holes with probability at least

$$2^{-(k+.02)N-o(N)}$$

under $\tilde{\sigma}_A \otimes \tilde{\sigma}_B$, with Ψ satisfied. This implication is valid even if many successful queries certify the same orientation pair: the fresh query experiment is a probability space, and its successful event is contained in the event that the underlying orientations satisfy Ψ and have all four holes. Each original law dominates its restricted normalized law by its own restriction mass. The product of those two masses is at least $2^{-o(N)}$, which gives the same additional cost even when the laws and restrictions are different.

If there are boundedly many options, sum the corresponding overlap quantities and choose one whose expectation is at least the sum divided by their number, or partition into these fixed cases. Shapes of $\mathcal{K}_n$ and varying finite numerical formats can likewise be separated into boundedly many cases. These constant factors are absorbed by the slack from .02 to .03 in (8.4). Finally

$$k_{\max} + .03 = 56(g-1) + .03 < 100g,$$

so (8.4) contradicts a sequence whose probability of four holes satisfying the required predicate Ψ is less than 2^{-100gN} . In particular, failure of a filtered distribution test forces vanishing overlap only for its permitted options; the argument makes no assertion about the overlaps of excluded colour or type pairs. $\square$

9 Phase estimates before key collisions

The estimates in this section supply the compatible small tables used in [Theorem 10.5](#). Channel thinning makes the retained mass and the compatibility margin positive constants chosen before the channel size. The original frame law is $\mathbf{P}_0$, whereas all marginal bounds below are against the thinned law π .

Throughout this section, different units are independent, but the two endpoints of one unit may have arbitrary dependence. A marked unit is a unit together with choices $\lambda_i \in C_i^{\text{old}}$, $i = 1, 2$, where C_i^{old} denotes its effective space after the first peeling. Marks are permitted to depend on the entire unit. Put

$$\Delta_O = U_1 \lambda_1 + U_2 \lambda_2, \quad u_{O,S} = \sum_{i \in S} u_i \quad (S \subseteq \{1, 2\}). \quad (9.1)$$

The total component rank of Δ_O is at most $r = 2K_1 + 2$. We use L_0, d_0, u_0, K from [Equations \(4.2\)](#) and [\(4.3\)](#); in particular,

$$L_0 = \lceil 100(D + 10) \rceil, \quad d_0 = 4rL_0, \quad u_0 = 10(K_1 + d_0 + 1).$$

Write $\varepsilon_{\text{ch}} = 10|\mathcal{E}|2^h \rho$, as in [Theorem 5.3](#). A marked law with endpoint marginals at most $\Lambda \pi$ has unchanged primal marginal bound $\Lambda \text{pr}_* \mathbf{P}_0$, but its full-frame bound is only $\Lambda 2^{\varepsilon_{\text{ch}} N} \mathbf{P}_0$. Here Λ is a fixed constant specified before h . In the coloured test it comes from the pooled law and the total mass retained, and does not depend on the probability of an individual colour. The starting absolute pair cap is $2^{3000gN} \pi^2$. After the first peeling and restrictions of fixed positive mass it is at most $2^{(D+.01)N} \mathbf{P}_0^2$, by [Theorem 5.3](#) and $D = 4000g$. Every use of a full-frame image bound below explicitly pays the loss $\varepsilon_{\text{ch}} N$. We choose $\varepsilon_{\text{ch}} < \min\{\zeta/100, 10^{-4}\}$, after h , and choose M_0 afterward to make each primal ratio loss as small as required. A fixed multiplicative constant is absorbed by taking n sufficiently large.

Definition 9.1. A componentwise cover is a collection of subspaces $T_e^\pm \subseteq V_e^\pm$. A tensor $\xi = (\xi_e)_e$ is covered by it if

$$\xi_e \in T_e^+ \otimes V_e^- + V_e^+ \otimes T_e^- \quad \text{for every } e.$$

The dimensions of the two modes of the cover are $\sum_e \dim T_e^+$ and $\sum_e \dim T_e^-$.

Lemma 9.2 (Rank growth without a cover). *Let $r \geq 1$ be an integer, set $m_r = 100r + 10$ and $q_r = 2^{-m_r}$, and let s be a power of two with $s \geq 2^{m_r}$. Suppose $T_1, \dots, T_s$ are independent copies of a random tensor of total component rank at most r . If every fixed componentwise cover with dimension at most rs in each mode has probability less than p , then*

$$\mathbb{P} \left\{ \text{rank} \left(\sum_{j=1}^s T_j \right) < q_r s / 4 \right\} \leq 4^s p^{q_r s / 2}. \quad (9.2)$$

Here rank means the sum of component ranks.

Proof. Regard all components as blocks in the direct sums of the two mode spaces. Their column and row spaces are direct sums of componentwise spaces, so that every cover obtained in this proof is componentwise. We first make a deterministic observation about any realized list of tensors.

Expose a subset E of the indices and quotient the two modes by the column and row spans of the exposed tensors. If $b = s - |E|$ indices remain, let C_j, R_j be their projected original column and row spaces. Define their span deficits by

$$\delta_C = \sum_{j \notin E} \dim C_j - \dim \sum_{j \notin E} C_j, \quad \delta_R = \sum_{j \notin E} \dim R_j - \dim \sum_{j \notin E} R_j.$$

The projected sum has rank at least

$$\#\{j \notin E : \bar{T}_j \neq 0\} - \delta_C - \delta_R. \quad (9.3)$$

Indeed, before adding the mode spaces, place the projected tensors on the separate formal summands $C_j \otimes R_j$. Their block diagonal sum has rank at least the displayed number of nonzero tensors. Applying the two addition maps can decrease rank by at most their kernel dimensions, which are δ_C, δ_R .

There is an exposed set leaving $b = s/2^j$ indices for some $0 \leq j < m_r$ such that $\delta_C + \delta_R \leq b/4$. To prove this, order the realized direction spaces by a uniform random permutation. In either mode, let a_t be the expected increase of their span dimension at the t -th step. Submodularity and exchangeability give

$$r \geq a_1 \geq a_2 \geq \dots \geq a_s \geq 0.$$

For $b_j = s/2^j$, set

$$f_j = a_{s-b_j+1}, \quad v_j = f_j - \frac{1}{b_j} \sum_{t=s-b_j+1}^s a_t.$$

Each remaining index has expected projected dimension f_j immediately after the first $s - b_j$ exposures. Consequently its mode's expected deficit divided by b_j is v_j . The average increment on the first half of this tail is at least f_{j+1} . It follows that

$$v_j \leq f_j - f_{j+1} + \frac{1}{2}v_{j+1}.$$

Summing for $0 \leq j < m_r$ yields

$$\sum_{j=0}^{m_r-1} v_j \leq 2(f_0 - f_{m_r}) - v_0 + v_{m_r} \leq 2r.$$

The two modes therefore have total expected deficit ratios at most $4r$ across these m_r scales. Some scale has expected ratio less than $1/4$, and some permutation realizes a ratio at most $1/4$, as claimed.

If the original sum has rank less than $q_r s/4$, its projection at this exposed set has no larger rank. By Equation (9.3), at most $q_r s/4 + b/4 \leq b/2$ remaining tensors have nonzero projections. At least $q_r s/2$ of them thus belong to the cover formed by the exposed column and row spans. Both cover dimensions are at most rs .

For fixed disjoint index sets E, J , with $|J| = \lceil q_r s/2 \rceil$, condition only on the tensors indexed by E . The cover is then fixed, and the tensors indexed by J are still independent. The probability that all are covered is at most $p^{|J|}$. There are at most 2^s choices for each index set. Taking their union proves Equation (9.2). $\square$

Theorem 9.3 (Phase alternative). *Suppose a marked-unit law σ satisfies*

$$\sigma_i \leq \Lambda\pi \quad (i = 1, 2), \quad \sigma \leq 2^{(D+.01)N} \mathbf{P}_0^2,$$

where Λ is fixed before the channel size, and suppose $\Delta_O \neq 0$ throughout. For all sufficiently large n , one can retain a sublaw of relative mass at least $\delta_{\text{ph}} > 0$, depending only on the early parameters, such that one of the following holds for independent draws A, B :

(i) *every nontrivial phase mean*

$$\mathbb{E}(-1)^{u_{B,S}(\Delta_A) + u_{A,R}(\Delta_B)}, \quad (S, R) \neq (\emptyset, \emptyset),$$

has absolute value less than .005;

(ii) *writing $F(A, B) = u_{B,\{1,2\}}(\Delta_A)$, one has $F(A, B) = 0$ for every pair of retained units, and every displayed mean for which $|S| = 1$ or $|R| = 1$ is $2^{-\Omega(N)}$.*

All marks remain in their first effective spaces. No additional image directions are pinned. Neither the retained-mass bound nor the tiny-cover dimension used in the proof depends on selector or channel sizes; the lower bound required of h may depend on Λ .

Proof. We first choose the parameters and treat the case with no large cover. In the cover case we pass to offset records. A tiny cover either makes F identically zero or permits a gain from thinning; in the absence of a tiny cover, a fresh-direction count gives mixing.

Write

$$c_s = 10^{-12}(1+r)^{-4}, \quad q_0 = 2^{-100r-10}, \quad \theta = 10^{-4}, \quad \Gamma = D + 2r + 10,$$

and let s be the largest power of two not exceeding $c_s N$. For large n , it satisfies the size requirement of [Theorem 9.2](#) and $s \geq c_s N/2$. Choose $p_0 > 0$, using only these early parameters, so small that

$$\frac{q_0}{2} \log_2(1/p_0) \geq 2 + \log_2(1/\theta) + \frac{2\Gamma}{c_s}. \quad (9.4)$$

When the channel size is chosen, require also

$$\frac{hq_0}{4} \geq \log_2(1/\theta) + \frac{2\Gamma}{c_s}. \quad (9.5)$$

Further lower bounds for h arising below will again involve only early parameters.

No large cover. Suppose no fixed cover of dimension at most rs in each mode has mass at least p_0 . Fix a candidate value ξ of Δ_B , a nonempty S , and a set R . Define the unit-side sign $\psi(A) = (-1)^{u_{A,R}(\xi)}$. Against completely iid channel columns at the endpoints selected by S , expansion of the even s -th moment gives

$$\begin{aligned} \mathbb{E}_{\text{iid channels}} \left| \mathbb{E}_A \psi(A) (-1)^{u_{B,S}(\Delta_A)} \right|^s \\ \leq \mathbb{E}_{A_1, \dots, A_s} 2^{-h \text{rank}(\Delta_{A_1} + \dots + \Delta_{A_s})}. \end{aligned} \quad (9.6)$$

For one channel pair the expectation of its bilinear character on a tensor ξ is $2^{-\text{rank} \xi}$. Independence over channels and components proves this assertion; selecting both endpoints only squares the

factor. The signs $\psi(A_j)$ have absolute value one and can be removed for an upper bound. By [Theorem 9.2](#), the last expression is at most

$$2^{2s} p_0^{q_0 s/2} + 2^{-h q_0 s/4}.$$

The channel marginal comparison in [Theorem 5.2](#) multiplies this bound by a constant depending on h and the number of components, but independent of n . Markov's inequality and [Equations \(9.4\)](#) and [\(9.5\)](#) bound the probability that the inner mean has magnitude at least θ by $C_h 2^{-\Gamma N}$.

There are at most $C_r 2^{2rN}$ tensors of total component rank at most r : choose their component ranks, then factor each matrix through that rank. The component-rank choices and the overcounting factors are bounded independently of n . Take a union over these candidate values and the bounded choices of S, R , and then pay the joint-law density $2^{(D+.01)N}$. The exceptional B -mass is exponentially small. On all other B 's the conditional A -mean is at most θ , even after inserting the actual value Δ_B . Every required phase mean is therefore at most $\theta + 2^{-\Omega(N)} < .005$. Interchanging A, B deals with $S = \emptyset$. This proves the first alternative without a further restriction.

A large cover and its offsets. Otherwise retain mass at least p_0 in a fixed cover (T_e^+, T_e^-) of dimension at most rs in each mode. Each individual primal frame span avoids these fixed subspaces except on exponentially small mass. Indeed its dimension is $\dim \mathcal{B}$, its location is uniform under P_0 , and

$$\mathbb{P}_{P_0}\{\text{im } P_{ie} \cap T_e^+ \neq \emptyset\} \leq 2^{\dim \mathcal{B} + \dim T_e^+ - N + 1},$$

the same estimate holds for the minus mode. Here $rs \leq rc_s N$ and $\dim \mathcal{B}/N$ is sufficiently small. The primal marginal bound $\Lambda \text{pr}_* P_0$ permits deletion of these exceptions; this step does not use a full-frame bound against P_0 . Normalize the retained law, still denoted σ . Its marginal bound $\sigma_i \leq L\pi$ has $L \leq 2\Lambda/p_0$ for all sufficiently large n . Thus this fixed bound may also depend on the given Λ , whereas the retained-mass bound remains independent of Λ .

Project both modes of each component modulo $T_e^\pm$. Because Δ is covered, the two endpoint tensors have the same quotient. The projections are injective on the individual primal spans, so they preserve each tensor rank. Choose a shortest outer-product factorization of the common quotient tensor and lift each factor to the two endpoint primal spans. Componentwise, the difference has the form

$$\Delta = \sum_{\alpha} (p_{\alpha} b_{\alpha}^{\top} + a_{\alpha} q_{\alpha}^{\top} + a_{\alpha} b_{\alpha}^{\top}), \quad a_{\alpha} \in T^+, \quad b_{\alpha} \in T^-. \quad (9.7)$$

The first endpoint is the reference here; its factors are p_{α}, q_{α} , and those at the second endpoint are $p_{\alpha} + a_{\alpha}, q_{\alpha} + b_{\alpha}$. The factor lists at either endpoint are independent in their respective modes.

Choose bases for the spans of the offsets a_{α}, b_{α} in each component. Their combined ordered list is denoted ω_O , and its size is t_O , with

$$1 \leq t_O \leq 2r. \quad (9.8)$$

The lower bound follows from $\Delta \neq 0$. Collect the companions of these offset basis vectors in a tuple Z_O : minus companions pair with plus offsets and vice versa. The companions are independent within each component and mode. For example, writing $b_{\alpha} = \sum_k c_{k\alpha} \omega_k^-$ gives a full-row-rank matrix $(c_{k\alpha})$, so the vectors $\sum_{\alpha} c_{k\alpha} p_{\alpha}$ are independent. Using the second endpoint as reference changes each companion by a known combination of offsets, and preserves independence.

An offset record consists of these basis vectors, their component and sign assignments, and the bounded coefficient arrays in [Equation \(9.7\)](#). Its logarithmic count is at most

$$2r^2 s + O_{r,g}(1) < .002N$$

for sufficiently large n . Choosing a nominal representation of a companion tuple in one endpoint's primal frame costs $2^{O(r \dim \mathcal{B})} = 2^{O(n)}$ possibilities. This last cost has an arbitrarily small ratio exponent after choosing M_0 . Fix deterministic conventions for all these records and factorizations; their choices can be arbitrary functions of the marked unit.

For a linear tensor functional u , let $\Phi_u(\omega)$ be the tuple obtained by contracting u with the offset basis vectors. In particular, on a plus offset a its corresponding plus output is $X(Y^\top a)$; on a minus offset b the minus output is $Y(X^\top b)$. At fixed two offset records the phase in the theorem is, up to a sum of two unary phases,

$$Z_A \cdot \Phi_{u_{B,S}}(\omega_A) + Z_B \cdot \Phi_{u_{A,R}}(\omega_B). \quad (9.9)$$

The unary phases are evaluations of the last term of Equation (9.7); fixed offset records make them separate functions of A and B .

Singleton phase estimates. All record laws in the following calculation are unnormalized subprobabilities of σ . For any record, the companion tuple alone has point masses at most

$$p_{\max}(Z_O) \leq 2^{-t_O N + .01N}. \quad (9.10)$$

To obtain this, enumerate its nominal primal representations, apply the individual primal frame-image estimate and the unchanged primal marginal bound $L \operatorname{pr}_* \mathbf{P}_0$, and choose M_0 large enough to absorb the coefficient count and lost primal codimensions.

Suppose $S = \{j\}$, and use endpoint j as the reference for the B -companions. For fixed ω_A , discard pairs on which the coefficient lists $Y_j^\top a$ or $X_j^\top b$, for its independent offsets, fail full rank in any component. Under the original law, comparison with iid columns bounds the transpose-rank failure on d independent ambient vectors by 2^{d-h+1} for large n . The fixed-list assertion of Theorem 5.3 bounds the same event under π by $2^{d-h+1} + o(1)$, uniformly over the fixed list. For each fixed A , the list being tested is fixed before drawing B . Consequently the discarded pair mass is at most

$$L O(r) 2^{2r-h} + o(1) < 10^{-4},$$

on making h sufficiently large in terms of the fixed marginal cap and the early rank bound.

On the remaining pairs, the tuple $(Z_B, \Phi_{u_{B,j}}(\omega_A))$ has point masses at most

$$2^{-(t_B + t_A)N + (.02 + \varepsilon_{\text{ch}})N}. \quad (9.11)$$

Here is a direct conditional count. Enumerate a nominal representation of Z_B , first test its values, and condition on the primal frames at endpoint j . The two channel maps are independent uniform-column draws in the appropriate annihilators, apart from a bounded injectivity conditioning. Freeze the values of $Y^\top a$ and $X^\top b$. Their number of possibilities is bounded in n , and their lists have full rank on the retained event. Ignore the equations that originally defined these coefficient values. Prescribing their images under X and Y then costs at least $(N - \dim \mathcal{B})t_A$ bits, up to a bounded factor. This is a count under $\mathbf{P}_0$, conditional on its primal frames. Replacing its channel law by that of π costs at most $2^{\varepsilon_{\text{ch}} N}$; applying the endpoint marginal bound costs the fixed factor L . Together with the primal companion test this gives Equation (9.11). The ignored equations only enlarge the event, so no independence between a coefficient list and the map defining it has been assumed.

Apply Theorem 5.9 to Equation (9.9), on $(t_A + t_B)N$ bits. On the A -side the point-mass bound for the entire tuple is no larger than Equation (9.10); on the B -side use Equation (9.11). Separate

unary phases have absolute value one. The contribution for two fixed records is at most

$$2^{(t_A+t_B)N/2} \left(2^{-t_A N + .01N} 2^{-(t_A+t_B)N + (.02+\varepsilon_{\text{ch}})N} \right)^{1/2} = 2^{(-t_A/2 + .015 + \varepsilon_{\text{ch}}/2)N}.$$

There are at most $2^{.004N}$ record pairs. By Equation (9.8), their total is at most $2^{-.480N}$. Thus every singleton phase mean is bounded by $10^{-4} + 2^{-.480N}$. The same proof applies when $|R| = 1$.

A tiny cover or mixing of the full endpoint sum. Put $\delta = .0005$. If a cover of total dimension at most d_0 contains the whole offset list with probability at least $\delta/4$, restrict to those units. Intersect its spaces componentwise with the preceding large cover. This preserves every captured offset list and does not increase its dimension. Each individual primal span therefore still avoids it. Delete units whose channel transpose at either endpoint is not injective on a fixed basis of any component and sign of this tiny cover. By Theorem 5.3 and the endpoint bound, the deleted relative mass is at most

$$L' O(d_0 + |\mathcal{E}|) 2^{d_0-h} + o(1), \quad L' \leq 4L/\delta.$$

Take h so this is less than $1/2$. This leaves relative mass at least $\delta/8$ in the large-cover law. All future offset lists in this tiny cover now pass the singleton rank tests.

Repeat Equation (9.7) with the tiny cover as the quotient directions. Fix an offset record: the actual offset basis, its component and sign assignments, and its bounded binary coefficient arrays. These data have at most

$$R_0 = 2^{100(r+1)^2(d_0+|\mathcal{E}|+1)}$$

possibilities. Indeed at most $2r$ vectors are chosen from a fixed space of dimension d_0 , at most $2r$ component/sign assignments are made, and all coefficient arrays have bounded sizes in terms of $r, |\mathcal{E}|$. The deliberately generous exponent bounds the product of these finite counts. Retain a record of mass at least $1/R_0$. This fixes a common ordered tuple ω , of length $1 \leq t \leq 2r$, but fixes neither the companions nor any frame image. In particular this restriction is not an additional pinning step.

Split once more, retaining at least half the mass, according as $\Phi_{u_{O,\{1,2\}}}(\omega)$ is zero or nonzero. All normalizations in this paragraph are bounded by early constants. In the zero case each term in Equation (9.7) has one factor from ω , so

$$u_{B,\{1,2\}}(\Delta_A) = 0 \quad \text{for every retained } A, B.$$

This includes the last, offset–offset term of the lift. The singleton count already proved, with no rank discard because the whole tiny cover passes the transpose tests, gives exponentially small singleton means under this fixed restriction. Thus alternative (ii) holds.

In the nonzero case we claim the stronger full-sum point bound

$$\mathbb{P}_\sigma\{Z_O = z, \Phi_{u_{O,\{1,2\}}}(\omega) = \phi\} \leq 2^{-tN - \rho N/3} \quad (\phi \neq 0). \quad (9.12)$$

Here σ denotes the normalized retained law; its endpoint marginals are at most $L''\pi$ for an early constant L'' . To verify the claim, choose endpoint one as reference in the lift. Enumerate all nominal primal descriptions of its t independent companions; their count is $2^{O(n)}$. For each description, prescribing the companion images z has probability at most $2^{-tN + O(n)}$ under the primal marginal of π , which is exactly the primal marginal of P_0 . Choose a nonzero coordinate d of the prescribed output ϕ . The corresponding contraction of each endpoint's channel form is a nonzero channel combination, because its transpose is injective on the fixed cover basis. Call these

combinations v_1, v_2 . They lie in the same prescribed component/sign subset, and $v_1 + v_2 = d \neq 0$. Necessarily, therefore, endpoint one has a nonzero channel combination in the translate by d of its allowed subset. Conditional on all its primal frames, this endpoint-one event has π -probability at most $2^{-\rho N/2}$, by [Theorem 5.3](#).

This is a necessary event on endpoint one alone. Consequently its probability under the dependent two-endpoint law is bounded using only the endpoint marginal $L''\pi$: no independence of the endpoints, and no conditional density bound for endpoint two, is used. Sum the nominal descriptions to obtain

$$\mathbb{P}_\sigma\{Z_O = z, \Phi = \phi\} \leq L'' 2^{-tN + Cn} 2^{-\rho N/2}$$

for a constant C fixed before M_0 . Choose M_0 so $C/M_0 < \rho/12$, and then n so $\log_2 L''/N < \rho/12$. This proves [Equation \(9.12\)](#).

For the other side of the Walsh estimate the companion-only bound can now be sharpened to

$$p_{\max}(Z_O) \leq 2^{-tN + \rho N/12}. \quad (9.13)$$

Its errors are solely $O(n)$ primal image and nominal-description costs plus a fixed normalization, so increasing M_0 after ρ proves this bound. There is no full-frame thinning loss in [Equation \(9.13\)](#). For a phase containing a full-sum term, apply [Theorem 5.9](#) to [Equation \(9.9\)](#), using [Equation \(9.12\)](#) on that term's side and [Equation \(9.13\)](#) on the other. Both records are already fixed, so there is no exponential record union. The absolute value is at most

$$2^{2tN/2} \left(2^{-tN + \rho N/12} 2^{-tN - \rho N/3} \right)^{1/2} = 2^{-\rho N/8}.$$

When the other phase term is absent, its output coordinates are zero; the companion bound still bounds the full tuple. Phases containing a singleton have the previous exponential estimate. Every nontrivial phase is of one of these types, so the nonzero case gives alternative (i).

Assume instead that no such cover captures mass $\delta/4$. Fix an A -offset list $v = \omega_A$, and use endpoint 1 as reference throughout the next calculation. Call B frequent for this list if its offset record m , its companion value z , and its output $\phi = \Phi_{u_{B,\{1,2\}}}(v)$ have joint point probability greater than $2^{-(t_B + .5)N}$. For fixed m, z , define the deterministic list

$$\mathcal{F}_v(m, z) = \left\{ \phi : \sigma(m_B = m, Z_B = z, \Phi_{u_{B,\{1,2\}}}(v) = \phi) > 2^{-(t_B + .5)N} \right\}. \quad (9.14)$$

By [Equation \(9.10\)](#), $|\mathcal{F}_v(m, z)| \leq 2^{.51N}$.

We claim that the frequent-pair mass is at most δ . Otherwise a set of B 's of mass at least $\delta/2$ has frequent A -mass at least $\delta/2$. Draw L_0 independent A -probes. At each stage the spans of all past offsets have total dimension at most $2rL_0 < d_0$. The mass of a probe whose whole offset list lies in those spans is less than $\delta/4$. Thus each stage has conditional probability at least $\delta/4$ of being frequent and having a fresh offset vector. Select one fresh vector by a fixed rule using only the probe lists. Averaging fixes a probe collection with independent selected witnesses in each component and mode, and with successful B -mass at least

$$q_* = (\delta/2)(\delta/4)^{L_0} > 0. \quad (9.15)$$

This lower bound is independent of selector and channel sizes. Require

$$LO(L_0) 2^{L_0 - h} < q_*/2. \quad (9.16)$$

By [Theorem 5.3](#), the successful B 's can then be trimmed so that the channel transpose at endpoint 1 has full rank on all these fixed witness vectors, while leaving mass at least $q_*/2$, after absorbing the uniform $o(1)$ error in a strict version of [Equation \(9.16\)](#).

Now estimate this event under the raw law P_0^2 . Condition on all of endpoint 2 and on the primal frames at endpoint 1. Enumerate the offset record m , and enumerate a nominal primal tuple representing Z_B at endpoint 1. There are at most $2^{.002N+O(n)}$ such choices. For each choice the ambient value z , all the lists in [Equation \(9.14\)](#), and their endpoint-2 translations are fixed. The event defined by the adaptive marks is contained in the union of these raw events. This uses no conditional density assertion about the marked law, and does not enumerate $2^{t_B N}$ arbitrary ambient companion values.

Project each of the L_0 lists onto the coordinate of its chosen fresh witness. Their total product size is at most $2^{.51L_0 N}$. Freeze the bounded lists of channel coefficient values on the witnesses. On their full-rank event, prescribing the corresponding endpoint-1 output tuple costs at least $.99L_0 N$ bits, by the same annihilator count as in [Equation \(9.11\)](#). The upper probability under σ , after paying its joint density, is therefore at most

$$2^{(D+.01)N+.002N+O(n)-.48L_0 N+O(1)}.$$

Choose M_0 to make the $O(n)/N$ term small. Because $L_0 \geq 100(D+10)$, this is exponentially small, contradicting the mass $q_*/2$. The claim follows.

Discard the frequent events in a full-sum phase test. For fixed records, this is a restriction on B alone once the A -offset record is fixed. The B -tuple point bound is now $2^{-(t_B+.5)N}$; the A -tuple bound is $2^{-(t_A-.01)N}$. Walsh on $(t_A+t_B)N$ bits gives $2^{-.245N}$, before the $2^{.004N}$ record count. Thus every phase with $S = \{1, 2\}$ has magnitude at most $\delta + 2^{-\Omega(N)}$. Interchanging the units covers the cases with $R = \{1, 2\}$. Together with the singleton estimates, all nontrivial means are less than .005.

All retained laws have relative mass at least

$$\delta_{\text{ph}} = \frac{p_0 \delta}{64R_0} > 0. \quad (9.17)$$

The large-cover span deletion leaves at least $p_0/2$; the tiny-cover restriction, transpose deletion, record restriction and zero/nonzero split retain at least $p_0 \delta / (32R_0)$ in total. This proves the stated, slightly smaller uniform bound. The lower bounds on h depend on the fixed marginal caps after these early restrictions. Only after h do we choose the thinning rate ρ ; the new full-sum estimate then requires M_0 large relative to $1/\rho$. No inverse mass depending on n or on a numerical table is paid. $\square$

Corollary 9.4 (A uniform phase compatibility margin). *On either retained law of [Theorem 9.3](#), for each fixed $c \in \mathbb{F}_2$ and all sufficiently large n ,*

$$\mathbb{P}\{u_{B,1}(\Delta_A) = u_{B,2}(\Delta_A) = u_{A,1}(\Delta_B) = u_{A,2}(\Delta_B) = c\} \geq 1/32. \quad (9.18)$$

Also $\mathbb{P}\{F(A, B) + F(B, A) = 0\} \geq .49$, and each single prescribed phase bit has probability at least .49.

Proof. In alternative (i), Fourier inversion of the four bits gives every prescribed pattern probability at least $(1 - 15 \cdot .005)/16 > 1/32$. The even-sum condition and each single-bit condition have probability at least $(1 - .005)/2 > .49$. In alternative (ii), both full endpoint sums are zero

identically. Among the sixteen Fourier characters, the four with even support in each endpoint pair therefore equal one. Their coefficients for the common pattern (c, c, c, c) are all one, for either value of c . Each of the other twelve characters contains a singleton and has mean $2^{-\Omega(N)}$. Thus the probability in Equation (9.18) is $1/4 + O(2^{-\Omega(N)})$. The remaining assertions follow from the same zero sums and singleton means. $\square$

Lemma 9.5 (Injection on independent pairs). *Let σ_A, σ_B be two possibly different marked-unit laws, each with pins of total dimension at most K . Suppose all four mixed endpoint marginals are at most $\Lambda\pi$, where Λ is fixed before h . Draw $A \sim \sigma_A$ and $B \sim \sigma_B$ independently; the endpoints within either unit may be dependent. For every fixed $\varepsilon > 0$, choosing h sufficiently large in terms of K, ε, g makes the probability that the pair fails a small-table injection condition at most ε , for all sufficiently large n . This is an unconditional estimate, so its error can be subtracted from any independently established compatibility mass.*

Proof. It suffices to treat one component, endpoint pair and sign, and then sum over at most $16|\mathcal{E}|$ tests. All copies of A used below are drawn from σ_A independently of $B \sim \sigma_B$. The argument uses only this independence and the separate marginal bounds, and does not require the two unit laws to coincide. Let V_A be the image of the individual pinned primal space $D_{A,e}^+ \cap \mathcal{B}_i$, and let $H_B \subseteq \mathbb{F}_2^h$ be the protected coefficient space at the opposite tested endpoint. Both have dimension at most K . Failure means that some $0 \neq v \in V_A$ has $Y_B^\top v \in H_B$. Both the pins and H_B may depend on their entire units.

Put $\ell = \lfloor N/(10(K+1)) \rfloor$. The probability that an independent new individual primal span meets a fixed space of dimension at most $K\ell$ is at most

$$\Lambda 2^{\dim \mathcal{B} + K\ell - N + 1} = 2^{-\Omega(N)}.$$

This uses only the unchanged primal marginal, and tests the whole primal span, so it is uniform over the adaptive pin choices. If a fixed B has failure probability greater than $\eta > 0$, successive independent draws $A_1, \dots, A_\ell$ can each witness failure while the spaces V_{A_j} remain mutually independent, with conditional probability at least $c = \eta/2$ at every step. The probability of this complete test is at least c^ℓ .

Conversely fix any such independent spaces. There are at most $2^{K\ell}$ choices of one nonzero witness in each, and the chosen witnesses are independent. For iid Y_B columns their transpose images are independent uniform elements of $\mathbb{F}_2^h$. For a fixed H_B of dimension at most K , the probability that all these images belong to it is at most $2^{-(h-K)\ell}$. There are at most $(K+1)2^{Kh}$ possible subspaces H_B , independently of n . The original channel marginal comparison in Theorem 5.2 therefore bounds this event under $\mathbf{P}_0$ by $C_{K,h} 2^{-(h-2K)\ell}$. Here the tested list has length proportional to N ; we use this absolute reference-law count, rather than the bounded-list rank assertion of Theorem 5.3.

The endpoint bound against π and the absolute thinning density loss multiply this upper bound by $\Lambda 2^{\varepsilon_{\text{ch}} N}$. Fubini and the lower bound c^ℓ give

$$\mathbb{P}_B\{\mathbb{P}_A(\text{failure} \mid B) > \eta\} \leq C_{K,h} \Lambda 2^{-(h-2K-\log_2(1/c))\ell + \varepsilon_{\text{ch}} N}.$$

Take $h > 2K + \log_2(1/c) + 30(K+1)$, before choosing $\varepsilon_{\text{ch}} < 10^{-4}$. The right side is exponentially small. The unconditional failure probability for this test is at most $\eta + 2^{-\Omega(N)}$. Choose $\eta < \varepsilon/(32|\mathcal{E}|)$, sum the tests, and then take n sufficiently large. This proves the assertion. $\square$

10 Preparation of the two-endpoint statuses

We work along a sequence of unit laws with endpoint marginals at most $M\pi$ and projected pair density at most $2^{3000gN}\pi^2$, as in the first part of [Theorem 5.4](#). A colour is an additional mark and may be randomized conditional on the unit. Passing to subsequences is allowed. By [Theorem 5.3](#), the absolute joint density relative to P_0^2 has ample slack below 2^{DN} . Apply [Theorem 5.7](#), or [Theorem 5.6](#) when there is no colour, and discard the exception in [Theorem 5.5](#). This removes only $o(1)$ total mass.

All restrictions in this section act on the resulting *pooled law*, with the actual leaf weights. Although a colour is fixed on each leaf, no prediction or phase argument is applied separately to its normalized colour class. In particular the endpoint marginal bound depends on the total retained mass, not on the smallest colour mass or the number of colours. All effective spaces in the following definition are the first spaces C_i^{old} , with total pin dimension at most K_1 . They will not be enlarged.

Let pr forget the channel columns and retain the primal frames. The marginal fact used below is exactly

$$\text{pr}_* \pi = \text{pr}_* P_0.$$

Thus a restriction of fixed positive pooled mass has endpoint law at most a fixed multiple of π , and its primal projection is at most that same multiple of $\text{pr}_* P_0$. This does not assert a constant full-frame density relative to P_0 .

For a tag l and a diagonal bit q , let $\Omega_{l,q,n}$ be the probability space of global plus and minus vector slots on the star of l , with the uniform law subject to paired diagonal q in every component. Requiring its individual vectors to be nonzero changes this law by exponentially small total variation, which will be harmless.

Definition 10.1 (Common prediction). A prediction along a subsequence consists of common binary functions

$$f_{i,n}(l, q, \cdot) : \Omega_{l,q,n} \longrightarrow \mathbb{F}_2, \quad i = 1, 2,$$

a set of units of mass at least 10^{-6} , and, on every unit in this set, choices

$$\lambda_i \in C_i^{\text{old}}, \quad \alpha_i \in \mathbb{F}_2, \quad \mathcal{E}_i \subseteq \mathbb{F}_2^b, \quad |\mathcal{E}_i| \leq B_*,$$

such that the following uniform statement holds. There is $\varepsilon_n \rightarrow 0$ for which, at both endpoints, at every label outside $\mathcal{E}_i$, at every tag, and in every permitted flavor, a uniform parameter atom x satisfies

$$p_i(x) = T(\lambda_i, x) + \alpha_i a(x) + f_{i,n}(l, q, \text{key}(x)) \quad (10.1)$$

except with probability at most ε_n . Here $p_i = u_{i'} U_i$, where i' is the other endpoint. The functions $f_{i,n}$ are the same for all retained units. The marks $\lambda_i, \alpha_i, \mathcal{E}_i$ may depend on the whole unit.

The errors in [Theorem 10.1](#) refer to uniform parameters in each flavor, before conditioning on a tester bit. Conditioning on a possible fixed tester value only multiplies the error by a fixed constant. There are finitely many labels, tags and flavors, with their numbers fixed independently of n .

If a prediction exists, retain its marked population and pigeonhole the values of the two α_i 's, of

$$c = a(\lambda_1) + a(\lambda_2),$$

and of the indicator $\Delta = 0$. This loses a factor at most 16, leaving mass at least $6.25 \cdot 10^{-8} > 10^{-12}$. Pass to a subsequence on which every reference error probability of every binary combination of f_1, f_2, q has a limit, for each tag and bit. This requires only finitely many comparisons.

For sequences of common functions, write $f \simeq g$ if their disagreement probability tends to zero on every $\Omega_{l,q,n}$, and identify sequences related in this way. Then quotient their binary vector space by the span of the common function with value q . Denote the resulting class by $[f]$, and put

$$d_i = 1 + \alpha_i, \quad e_i = (d_i, [f_i]). \quad (10.2)$$

The pairs e_i are *prediction classes*; a query's numerical unary status records the scalar outcomes specified in [Section 8](#). The support of this pair of statuses is $\{i : e_i \neq 0\}$. These statuses are common to all retained units, although the marked λ_i 's can vary.

Lemma 10.2 (Exactification of empty status). *If $e_1 = e_2 = 0$, deleting $o(1)$ unit mass makes*

$$p_i = T(\lambda_i, \cdot) + a \quad \text{on all of } \mathcal{X}, \quad i = 1, 2, \quad (10.3)$$

an exact identity on every retained unit. The same conclusion holds on any marked population of fixed positive mass satisfying [Equation \(10.1\)](#) with $\alpha_i = 1$ and $f_i \simeq \gamma_i q$; the threshold 10^{-6} in [Theorem 10.1](#) is not needed here.

Proof. Empty status gives $\alpha_i = 1$ and $f_i \simeq \gamma_i q$ for a fixed $\gamma_i \in \mathbb{F}_2$. Restriction to any fixed positive population only changes the primal marginal bound by a fixed factor, which is all the proof uses. At any fixed tag, label and parameter point, the key depends only on the primal frames. Under π its law is therefore the same as under P_0 : the single-key reference up to an exponentially small injectivity error, by [Theorem 5.2](#). The bounded *primal* marginal density transfers the reference disagreement of f_i and $\gamma_i q$ to an average $o(1)$ parameter error on the marked population. No channel coordinate is tested in this comparison. There are only finitely many endpoint, tag and label choices. Markov's inequality therefore allows deletion of $o(1)$ unit mass so that this error tends to zero uniformly on the remaining units and choices. Combining it with [Equation \(10.1\)](#) gives, at every nonexceptional label,

$$p_i(x) + T(\lambda_i, x) + a(x) + \gamma_i q(x) = 0$$

with parameter error tending to zero in the generic flavor.

For fixed label and tag the left side is a Boolean polynomial of base degree at most two. A nonzero Boolean polynomial of degree at most k has nonzero probability at least 2^{-k} under uniform bits. For completeness, this follows by induction: choose a coordinate with a nonzero derivative; the derivative has degree at most $k - 1$, and whenever it is nonzero at least one of the two values on that coordinate is nonzero. The constant nonzero polynomial is the initial case. Thus the displayed relation, once its error is below $1/4$, is exact for every allowed base value.

For any fixed allowed base value its dependence on the selector has degree at most $2j_*$. It vanishes at every selector outside a set of size at most B_* . The choice of b gives $B_* < 2^{b-2j_*}$, so the same polynomial bound forces it to vanish at the excluded labels also.

Use one shared-only base point with $\eta_S = 1$, the same at every tag. The sum of its g star atoms is zero in the cut space. Summing the exact relation over these atoms gives $0 = \gamma_i g = \gamma_i$, because g is odd and the first three terms are linear on $\mathcal{X}$. Consequently $\gamma_i = 0$. The point atoms span the cut space by its definition, proving [Equation \(10.3\)](#). $\square$

10.1 A comparison for affine parameter slices

The next elementary comparison is stated separately because the global predictors in [Theorem 10.1](#) need not be polynomial. All tests in it are common functions of the displayed image columns; they are not arbitrary orientation-dependent tests.

Lemma 10.3 (Affine-slice comparison). *Fix a selector, a flavor, and a fixed number d of affine parameter variables. Write a random affine coefficient map as*

$$C(t, u) = p_s \otimes (t, z_0 t + Lu), \quad (t, u) \in \mathbb{F}_2 \oplus \mathbb{F}_2^d.$$

Assume the flavor retains an n -bit Z -block absent from E . Suppose we condition the affine coefficients on $C^\top EC = G$, where G is fixed, this event has probability at least a fixed $\beta > 0$, and it depends on no Z coefficient. Let $\mathcal{T}_n$ be any common test of absolute value at most one on the image column arrays $(P_e C, Q_e C)_e$, using a fixed set of components. Its empirical average over the conditioned affine coefficients, with the primal frame sampled from $\text{pr}_ P_0 = \text{pr}_* \pi$, has variance*

$$O(2^{-N/2} + 2^{2d+1-n})$$

around its mean under uniform injective image column arrays with internal Gram G . The constants may depend on the fixed column counts and β , but the estimate is uniform over the tests $\mathcal{T}_n$.

Proof. The Z -coefficients remain independent uniform bits after the conditioning. The columns of C are independent except with probability $O(2^{d-n})$. For two independent conditioned maps C, C' , a relation

$$C(a, u) + C'(a', u') = 0$$

has $a = a'$, by its constant coordinate. Its Z -part is

$$a(z_0 + z'_0) + L_Z u + L'_Z u' = 0.$$

The matrix $[z_0 + z'_0, L_Z, L'_Z]$ is a uniform $n \times (2d + 1)$ bit matrix. A union over its nonzero coefficient vectors bounds the failure of combined independence by $(2^{2d+1} - 1)2^{-n}$. In particular, the two translation columns do not share a forced nominal constant direction.

For fixed independent nominal columns, the restricted-image transitivity in [Theorem 5.2](#) gives the uniform injective law with its specified Gram. One batch consequently has the claimed reference law, independently of the remaining nominal coefficients. For two batches, compared with two independent copies of this reference law, the actual images have finitely many additional mutual Gram entries prescribed. Joint injectivity contributes only $O(2^{-N+O(d)})$ to the comparison: under independent uniform vector slots, a dependence among a fixed number of columns has this probability, and the fixed Gram conditioning has bounded reciprocal probability.

Expand the mutual Gram indicator in binary characters, collapsing any repeated identical entries. A nonzero character has a nonzero coefficient matrix on at least one interaction between the two batches. More explicitly the coefficient blocks pairing first-batch plus vectors with second-batch minus vectors and first-batch minus vectors with second-batch plus vectors give cross-batch bit rank $N(\text{rank } A + \text{rank } B) \geq N$. The internal Gram constraints, individual injectivity indicators, and the two arbitrary tests are separate bounded weights on the two batches; their normalization factors are fixed. [Theorem 5.9](#) bounds every nontrivial character contribution by $O(2^{-N/2})$. The same computation without the tests gives the normalization of the mutual Gram conditioning. There are only a fixed number of characters. Thus the conditional two-batch product expectation

differs from the product of the one-batch expectations by $O(2^{-N/2})$, uniformly over the nominal mutual Gram values.

Averaging the coefficients, and including the probability of nominal dependence, proves the asserted second-moment estimate. The one-batch mean differs from the reference mean only by the already bounded nominal-dependence error. $\square$

Lemma 10.4 (Equal coefficients when the tensor difference vanishes). *On a predicted branch with $\Delta = 0$ throughout, $\alpha_1 = \alpha_2$.*

Proof. Suppose otherwise. By [Theorem 5.5](#), the two canonical sparse representations $w^{(i)} = (w_l^{(i)})_l$ of the marked λ_i 's have the same value of χ_* . Each has at most $4D/g$ nonzero tag entries. As $\alpha_1 \neq \alpha_2$, on every unit exactly one endpoint has $\chi_*(w^{(i)}) + \alpha_i = 1$. Fix an endpoint i on a positive fraction of the units. Pigeonhole the set

$$T_0 = \{t : \eta_S(w_t^{(i)}) = 1\}, \quad |T_0| \leq 4D/g,$$

and choose a single selector outside the stored exceptions on a positive fraction of this remaining population. The latter is possible by averaging over selectors, since $B_* < 2^{b-1}$. All these restrictions have fixed positive mass in n ; their endpoint primal marginal is at most $\Lambda_0 \text{pr}_* \mathbf{P}_0$ for a fixed Λ_0 . No restriction made solely for this contradiction will be used later to choose a channel-size threshold.

At each tag l , use the pure flavor retaining exactly the allowed $O_{d,t}$ blocks with $t \notin T_0$, setting all S bits to zero and retaining the $\#, Z$ blocks. For a pure tag atom, $a(x) = q(x)$ and every $b_t(x) = 0$. The coefficient of its tester $\eta_{d,t}$ in $T^0(\lambda_i, x) + \alpha_i a(x)$ is

$$a(\lambda_i) + b_t(\lambda_i) + \alpha_i = \chi_*(w^{(i)}) + \eta_S(w_t^{(i)}) + \alpha_i = 1$$

on the retained t 's. Hence [Equation \(10.1\)](#) gives

$$f_i(l, q, \text{key}(x)) + q = p_i(x) + T^1(\lambda_i, x) \tag{10.4}$$

with uniform $o(1)$ parameter error.

The right side of [Equation \(10.4\)](#) is a sum of at most

$$s_1 = (g-1)(h + 2JK_1) \tag{10.5}$$

products of affine functions of the base variables. Indeed p_i contributes at most h products in each of the $g-1$ star components. Factoring the old λ_i 's, whose total component rank is at most K_1 , gives at most $2J(g-1)K_1$ products for T^1 . The selected pure tester retains at least

$$\frac{g-1}{2}(g - 4D/g)r_0 > 2s_1 + 1 \tag{10.6}$$

disjoint ordered bit pairs. To check the last inequality, use $D = 4000g$ and $h = 1000r_0$. After division by $(g-1)r_0$, the left side is $(g-16000)/2$ and the right side is at most $2000 + 4JK_1/r_0 + o(1)$. The fixed value $g = 10^9 + 1$ and the later choice of sufficiently large r_0 make the inequality hold.

Put $m_1 = 2gs_1 + 1$, and introduce synthetic variables $\xi, v \in \mathbb{F}_2^{m_1}$ and

$$q_* = \sum_{j=1}^{m_1} \xi_j v_j.$$

For every component independently, choose uniform injective maps of the affine coefficient columns $(1, \xi, v)$ into the plus and minus ambient spaces with full paired Gram

$$\langle \text{plus}(1, \xi, v), \text{minus}(1, \xi', v') \rangle = \sum_j \xi_j v'_j.$$

Such maps exist for large n , since the number of columns is fixed. Their point evaluations form one synthetic array of global component keys; all tag stars use the corresponding entries of this same array.

We next prove two properties of this array with probability tending to one.

Shared-only parity. In a local shared-only test use the fixed selector and the same point at every tag. The sum of the tag atoms is zero in $\mathcal{X}$. Summing Equation (10.1) therefore gives

$$\sum_l f_i(l, q, \text{key}_l) = 0$$

except on $o(1)$ parameter mass, uniformly on the retained units. Both tester bits have probability at least $1/4$: the bias of the sum of r_0 independent bit-pair products is 2^{-r_0} .

Apply Theorem 10.3 with no affine variables ($d = 0$), the shared-only flavor, and internal Gram (q). The Gram event depends only on the fixed tester coordinates and has probability at least $1/4$. The common test is the displayed parity predicate of the global keys in all components. Its empirical frequency under the primal reference law concentrates at its global reference probability. The endpoint primal marginal bound $\Lambda_0 \text{pr}_* \mathbf{P}_0$, and its frequency $1 - o(1)$ on the retained population, force that reference probability to tend to one. At every point of the synthetic array the point-key marginal is this reference, up to the exponentially small nonzero-vector conditioning. There are a fixed finite number of points. Thus, with probability tending to one,

$$\sum_l f_i(l, q_*, \text{key}_l) = 0 \quad \text{at every synthetic point.} \quad (10.7)$$

Affine-product representations on every small slice. Fix sets $I, J \subseteq \{1, \dots, m_1\}$, each of size at most $2s_1 + 1$, and set all ξ and v coordinates outside I, J , respectively, to zero. There are $d = |I| + |J| \leq 4s_1 + 2$ remaining variables. At a given tag, map this slice into its selected local pure flavor by independent uniform affine functions for every retained base bit, including independent translations. Condition their full oriented Gram to equal the synthetic slice Gram.

This conditioning has a fixed positive probability. At the fixed selector $E^\#$ vanishes on pairs. There are at most $2s_1 + 1$ active terms $\xi_j v'_j$ in the target Gram, so Equation (10.6) lets us route them into distinct retained ordered tester pairs, and set all other tester affine coefficients to zero. If t_0 scalar tester coordinates are prescribed, this assignment alone has probability $2^{-t_0(d+1)} > 0$, independent of n . The Z -affine coefficients remain iid. At every fixed slice point the unconditioned base input is uniform; after conditioning, its density is bounded by the fixed reciprocal Gram probability.

Consequently Equation (10.4) holds simultaneously at all points of the slice except on $o(1)$ of these parametrizations, uniformly on the marked population. Its right side pulls back to a sum of at most s_1 affine products. Consider only the event

the array of values of the common function $f_i + q_*$ on this specified slice admits such an affine-product representation.

This is a common test of the image column arrays. It contains neither the mark λ_i nor any coefficients of the unit-specific representation. Its empirical success probability is $1 - o(1)$ on the retained population. [Theorem 10.3](#) and the bounded primal marginal therefore force its probability under the synthetic slice law to tend to one. That law is also the marginal of the corresponding slice in the full synthetic array, by restricted frame transitivity.

There are only finitely many choices of tag and slice, with the number fixed independently of n . We may therefore require all these events and [Equation \(10.7\)](#) simultaneously. Notice that we embedded only the small slices locally; no local realization of the entire synthetic tester was required.

Choose one synthetic array on which all these properties hold, and define

$$F_l(\xi, v) = f_i(l, q_*, \text{key}_l) + q_*.$$

For each tag define its Boolean cross-coefficient matrix by

$$C_l[j, k] = F_l(e_j, e_k) + F_l(e_j, 0) + F_l(0, e_k) + F_l(0, 0).$$

On a coordinate slice where F_l is a sum of s_1 affine products, this is the matrix of $\xi_j v_k$ coefficients. One affine product contributes a sum of two rank-one matrices to this cross matrix. Every $(2s_1 + 1)$ -square minor of C_l is therefore singular, so $\text{rank } C_l \leq 2s_1$. On the other hand [Equation \(10.7\)](#), together with odd g , gives $\sum_l F_l = q_*$ and hence

$$\sum_l C_l = I_{m_1}.$$

Rank subadditivity would imply $m_1 \leq \sum_l \text{rank } C_l \leq 2gs_1$, contradicting $m_1 = 2gs_1 + 1$. This proves $\alpha_1 = \alpha_2$. $\square$

10.2 Compatible tables and the prepared alternatives

For an admissible small table, a contraction on a marked λ_i means its table contraction. This is well defined because $\lambda_i \in C_i^{\text{old}}$ and all first effective spaces are retained. We use the following phase requirements:

$$\begin{aligned} u_{B, S^\circ}(\Delta_A) + u_{A, S^\circ}(\Delta_B) &= |S^\circ|d, & \text{if the nonzero } e_i \text{ all coincide,} \\ & S^\circ = \{i : e_i \neq 0\} \neq \emptyset, \quad d = \text{their first bit;} \\ u_{B, j}(\Delta_A) = c, \quad u_{A, i}(\Delta_B) = c \quad (i, j = 1, 2), & \text{if } e_1 = e_2 = 0. \end{aligned} \tag{10.8}$$

In all other status cases there is no additional phase condition. The four equations in the second line all use the *same* previously fixed bit $c = a(\lambda_1) + a(\lambda_2)$. For a candidate table these contractions are computed from its entries and the numerical coordinates of the marks.

Proposition 10.5 (Status preparation). *There are constants $\delta_*, c_* > 0$, fixed before the selector and channel sizes, with the following property. Along a subsequence, the original pooled unit law has a restriction of mass at least δ_* whose first pin dimension is at most K_1 and whose normalized leaves satisfy [Equation \(5.11\)](#). One of the following descriptions applies.*

- (i) *There is no prediction as in [Theorem 10.1](#) on the first pooled law before the final negligible leaf trimming. For independent draws from the prepared law, the probability of an admissible injecting actual small table is greater than .99.*

- (ii) A prediction with fixed α_i, c and common classes e_i is retained. Its marks and exceptions are preserved, and Equation (10.1) has uniform $o(1)$ error on every retained unit. If both classes vanish, Equation (10.3) is exact on every unit. For independent draws from the prepared law, the probability of an admissible injecting actual table satisfying Equation (10.8) is at least $1/64$.

In particular the corresponding successful pair event, measured in two draws from the original pooled law, has mass at least c_* . One may take

$$\delta_* = \frac{10^{-6}\delta_{\text{ph}}}{64}, \quad c_* = \frac{\delta_*^2}{64}, \quad 0 < \eta_{\text{col}} < \frac{c_*}{4}, \quad (10.9)$$

where $0 < \delta_{\text{ph}} \leq 1$ is the early retained-mass constant in Theorem 9.3. The colour bound in Equation (10.9) can be chosen using the initial M and early geometric constants only; the marginal constant L of the second distribution test does not enter it.

After forgetting colour and other marks, the prepared law σ' satisfies

$$\sigma'_i \leq \frac{M}{\delta_*} \pi \quad (i = 1, 2), \quad \sigma' \leq 2^{(D+o(1))N} \mathbf{P}_0^2. \quad (10.10)$$

The endpoint primal marginals consequently have the same constant bound relative to $\text{pr}_* \mathbf{P}_0$. Every normalized leaf has joint density at most $2^{O(N)} \mathbf{P}_0^2$. The successful table event is detected by a finite list of numerical tables and separate unary orientation flags, together with leaf-pair tests. The size of this list is fixed in n , but need not be early.

Proof. Consider first the pooled law after the first peeling and intersection deletion. If no prediction exists on any subsequence, apply Theorem 9.5 to independent draws with no additional acceptance condition. Its failure tolerance can be chosen below $1/128$, so the success probability is greater than .99 for large n . The actual table is automatically admissible. Later leaf trimming removes only $o(1)$ mass. A prediction on the trimmed law with mass bounded strictly above 10^{-6} would give one on the preceding law, by multiplying its mass by the retained proportion and preserving its common functions and unitwise error. This is the form of the no-prediction conclusion used below.

Otherwise retain a prediction population and pigeonhole α_1, α_2, c and the indicator $\Delta = 0$, as above. Its relative mass is at least $10^{-6}/16$. If the statuses are empty, perform the $o(1)$ deletion in Theorem 10.2. All subsequent restrictions then preserve the exact identities Equation (10.3).

Suppose $\Delta = 0$. By Theorem 10.4, $\alpha_1 = \alpha_2$. If precisely one status is active, its first bit cannot be 1: that would give $\alpha_i = 0$ there and $\alpha_{i'} = 1$ at the inactive endpoint. Thus the right side of the first line of Equation (10.8) is zero. With two equal active statuses it is also zero, because $2d = 0$ over $\mathbb{F}_2$. Every left side vanishes when $\Delta = 0$, so all these conditions hold automatically.

If both statuses are empty and $c = 1$, then $U_1\lambda_1 = U_2\lambda_2$, while the two identities Equation (10.3) give the affine-gradient equations for an internal hole and $c = 1$ gives its unequal a -values. This contradicts the assumption that the pair is a unit. Consequently $c = 0$, and all four required phase bits vanish. Thus compatibility is automatic also in this case.

Suppose next that $\Delta \neq 0$. Apply Theorem 9.3 to the pooled marked law. It retains at least the early fraction δ_{ph} , and gives either full mixing or the identity

$$F(A, B) = u_{B, \{1, 2\}}(\Delta_A) = 0 \quad \text{for every two retained units } A, B,$$

together with exponentially small singleton character means. For clarity, write the four phase bits as

$$X_j = u_{B,j}(\Delta_A), \quad Y_i = u_{A,i}(\Delta_B), \quad i, j \in \{1, 2\}.$$

Full mixing bounds the absolute mean of every nonconstant character of (X_1, X_2, Y_1, Y_2) by .005. Binary Fourier inversion therefore gives every prescribed four-bit pattern probability at least

$$\frac{1 - 15(.005)}{16} > \frac{1}{32}.$$

Each condition in [Equation \(10.8\)](#) admits such a pattern. In particular the empty-status pattern is exactly (c, c, c, c) .

In the other alternative, $X_1 = X_2$ and $Y_1 = Y_2$ identically. If there is one active status, the required bit is $X_i + Y_i = d$; its nontrivial character has singleton support in both units, so its probability is $1/2 + o(1)$. If two active statuses coincide, the condition is $F(A, B) + F(B, A) = 0$, which holds identically. With empty statuses put $X = X_1 = X_2$ and $Y = Y_1 = Y_2$. The characters of $X, Y, X + Y$ all have singleton representatives. Hence

$$\begin{aligned} \mathbb{P}\{X_1 = X_2 = Y_1 = Y_2 = c\} &= \frac{1}{4} \left(1 + (-1)^c \mathbb{E}(-1)^X + (-1)^c \mathbb{E}(-1)^Y + \mathbb{E}(-1)^{X+Y} \right) \\ &= \frac{1}{4} + o(1) > \frac{1}{32}. \end{aligned}$$

This explicitly requires the same fixed c on both sides. The remaining status cases impose no condition. Thus all branches have conditional compatibility probability at least $1/32$ for large n , in agreement with [Theorem 9.4](#).

Apply the unconditional form of [Theorem 9.5](#) to this retained law, with failure below $1/128$. The probability of a compatible injecting actual table is then at least

$$\frac{1}{32} - \frac{1}{128} - o(1) > \frac{1}{64}.$$

This is subtraction of exceptional mass; it makes no independence assumption about injection and compatibility. The injection estimate uses the endpoint bound relative to π , together with the unchanged primal marginal and the allowed thinning loss in the full-frame estimates. Its tolerance is fixed before h .

We now justify the claimed leaf estimates without adding any pins. Let w_λ be the original actual weights of the first leaves, and let q_λ be the fraction of leaf λ retained by all restrictions just made. Discard the retained parts with $q_\lambda < 2^{-\zeta N}$. Their total unnormalized mass is at most

$$\sum_{\lambda: q_\lambda < 2^{-\zeta N}} w_\lambda q_\lambda \leq 2^{-\zeta N}.$$

On each remaining leaf, division by q_λ weakens [Equation \(5.10\)](#) to

$$2^{\zeta N} 2^{-(1-\zeta)tN} \leq 2^{-(1-2\zeta)tN} \quad (t \geq 1),$$

as required. Its joint density gains at most $2^{\zeta N}$ and remains $2^{O(N)} \mathbf{P}_0^2$. The pin spaces and all old marks are unchanged. This exponentially small deletion changes the successful pair probability by $o(1)$, already allowed above.

The first peeling loses $o(1)$ mass. On a predicted branch, the finite initial selection costs at most $16 \cdot 10^6$, and the phase restriction costs at most $1/\delta_{\text{ph}}$. All further losses tend to zero; for

large n their combined retained factor, including first peeling, is at least $1/4$. This proves the unit-mass lower bound δ_* in Equation (10.9). The no-prediction branch retains $1 - o(1)$ mass and satisfies the same lower bound. Multiplying the conditional pair-success bound by δ_*^2 proves the original-scale bound c_* . The density statements follow by dividing the initial bounds by the retained unit mass and using Equation (5.6). All marginal constants required by the phase and injection steps are fixed in advance, for example below $2M/\delta_*$, so their channel-size thresholds can be imposed after these early margins have been selected.

Finally, every numerical table has bounded coordinate dimensions. The marked tensors are recorded by their coordinates in the first effective bases, and the exception sets have bounded descriptions. For a fixed leaf pair and a fixed numerical option, the two admissibility conditions are separate unary orientation flags, since the pin images are fixed on each leaf. Fixing numerical mark coordinates is also a unary flag. The phase requirements then depend only on the table entries and those coordinates. An actual successful table supplies at least one successful option in this finite list. This proves the asserted finite-data form without selecting one table during preparation or conditioning the two draws jointly on table success. $\square$

Corollary 10.6 (Removing equal colours). *Suppose the original colour masses are $p_\gamma \leq \eta_{\text{col}}$. The successful prepared pair event of Theorem 10.5, with the further requirement that its two colours differ, has mass at least $c_* - \eta_{\text{col}} > 3c_*/4$ in the original pooled law.*

Proof. Two independent original draws have equal colours with probability

$$\sum_{\gamma} p_{\gamma}^2 \leq \eta_{\text{col}} \sum_{\gamma} p_{\gamma} = \eta_{\text{col}}.$$

Subtract this upper bound from the original-scale successful mass c_* . Equivalently, in a restriction of mass δ , the equal-colour probability is at most $\eta_{\text{col}}/\delta^2$. This calculation applies also to randomized colours, which are simply part of the original marked probability space. It uses actual colour probabilities and imposes no lower bound on them. $\square$

Remark 10.7 (No feedback from later accuracies). The margins δ_*, c_* , the choice of η_{col} , and the injection tolerance are fixed before h . At this stage success means that some numerical table passes. The later choice of one option from a finite list may have a much smaller mass depending on h , which is irrelevant to the earlier injection estimate. The errors in the affine-slice comparison and in subsequent fixed-test limits are handled by increasing n ; their slice counts, partitions and flag thresholds do not change the channel size or pin budget. Only the first peeling has been used throughout.

11 Histogram and parameter estimates

We prepare the estimates needed to pass successful queries to a common limit. Histogram control gives uniform integrability and finite nets within each leaf. Typicality compares prescribed common tests with their raw reference averages, binary mixing controls the specified parameter constraints, and unary positivity supplies the base-coordinate choices used in Section 14.

All construction parameters, including M_0 , are fixed throughout this section, and $N = M_0 n$. Constants in estimates may depend on those parameters. A bounded number always means a number bounded independently of n . In particular, the numbers of tags, labels, coordinates in a small table, and positions in a query are bounded in this sense.

11.1 Query measures and the distinction between tests and flags

A position specifies an endpoint of a unit, a tag l , a label s , and one of the flavors from [Section 6](#). Its parameter is a uniform allowed base input. We may condition this input on a possible value of q , or on a feasible tester summary. The probability of any such fixed conditioning is bounded below by a positive constant. The Z block remains uniform and independent of the other input coordinates. The parameters of different positions are independent, even when some positions have the same label. At a fixed orientation O , the key at position j is denoted by $K_j(O, z_j)$. A query consists of a fixed bounded list of positions and their fresh parameters. The key uses only the primal maps P_e, Q_e . Write $\text{pr}(O)$ for the tuple of those maps, with both endpoints included when O is a unit. Forgetting the channel maps and any auxiliary marks in a law will always mean this projection.

For a position of tag l and diagonal value q , let $\Omega_{l,q,n}$ be the uniform law on its global vector slots, subject to the prescribed plus-minus diagonal q at each component of its star. For a query, write

$$\bar{\Omega}_n = \bigotimes_j \Omega_{l_j, q_j, n}.$$

Let ω_n instead be uniform on all its vector slots without any Gram conditions. Finally, ν_n is uniform on $\mathcal{K}_n$, the set obtained by imposing all the individual diagonals and all the off-position Gram zeros required in [Section 8](#). Each of these bounded lists of Gram conditions has probability bounded below under ω_n .

We will use three different kinds of assertions.

- (i) A *common key test* is a prescribed function of global keys. It may depend on n and on the law under consideration, but it is not chosen using the particular orientation being queried.
- (ii) A *common input-and-key test* may also use the nominal parameters. The concentration result below applies to these more general tests. The positivity result for basis statuses applies only to common key tests.
- (iii) A histogram *flag* is any one of a fixed bounded list of bits depending on an orientation and its query parameters. The histogram compactness result permits arbitrary filters of the orientation and these flags. In particular a flag may use all channel maps; the bits p_i used later are such flags. Later successful queries use a narrower class: products of unary status requirements, the specified Gram and L, R constraints, and an additional orientation-only filter. The product comparison in [Section 12](#) is stated for that narrower class.

All filtered measures in this section are subprobability measures. We do not divide them by the probabilities of their filters. Colours and law types may be retained as finite marks on the orientations and on the leaves. The number of their possible values is fixed before n grows. Neither a mark nor a channel-dependent flag becomes a common key test merely by being recorded.

11.2 Finite Gram comparisons

We first record the counting and Fourier facts used several times below.

Lemma 11.1 (Gram normalization). *Let a plus slots and b minus slots in one ambient component be independent uniform vectors, and prescribe their entire a by b Gram matrix. If $a + b \leq N/2$, the probability of that prescription together with injectivity in each mode is*

$$2^{-ab}(1 + O(2^{-N+a+b})),$$

uniformly in the prescribed matrix. The same statement holds for a fixed collection of independent endpoint/component groups, with the sum of their Gram-bit counts in place of ab and a fixed factor in the error bound. In particular, for a bounded number of slots all relevant Gram normalizations and their ratios are bounded above and below by positive constants, uniformly in the nominal inputs.

Proof. The plus slots are linearly independent with probability $\prod_{j=0}^{a-1} (1 - 2^{j-N}) = 1 - O(2^{-N+a})$. Given such plus slots, each minus slot has probability 2^{-a} of satisfying its prescribed values. Conditional on those values the minus slots are independent uniform vectors in specified affine cosets of one common annihilator of dimension $N - a$. For any nonzero combination of the b slots, the resulting vector is uniform in an affine coset of that annihilator. Its probability of being zero is at most $2^{-(N-a)}$. A union bound over fewer than 2^b combinations proves the claimed injectivity error. Multiplication over independent groups gives the last assertion. $\square$

When the number of slots is bounded, dropping an injectivity condition in a normalized Gram law changes a bounded integral by $O(2^{-N+O(1)})$. For the growing batches used below we will instead retain injectivity in the denominator and drop it only from a nonnegative numerator; this avoids paying an inappropriate additive error before a density bound.

Lemma 11.2 (Comparison of two bounded batches). *Consider two bounded batches of prescribed nominal columns in one raw endpoint, or in two independent raw endpoints. Suppose their union is nominally independent in every endpoint, component, and mode. Let F and G be bounded functions of the separate image batches, respectively. The difference between their joint raw expectation and the product of their separate raw expectations is*

$$O(\|F\|_\infty \|G\|_\infty 2^{-N/2}).$$

The bound is uniform in the independent nominal columns and all their prescribed same-endpoint Gram entries. The functions may also depend on the fixed nominal inputs determining their own batch.

Proof. By [Theorem 5.2](#), the image laws are uniform laws on the indicated independent vectors with their same-endpoint Gram entries prescribed. Relative to the product of the two separate batch laws, the union law adds only the mutual Gram entries and joint injectivity. The latter may be omitted at an exponentially small error because the number of columns is bounded.

Expand the indicators of the mutual Gram entries into binary characters. Their values, which are determined by the nominal inputs, enter this expansion only as signs. In a nontrivial character, after identical descriptions of an unordered slot pair have been collapsed, the coefficient matrix between the two batches is nonzero. Its bit rank is at least N . The individual batch Gram densities can be included in the two separate bounded weights. [Theorem 5.9](#) bounds this character by a constant times $\|F\|_\infty \|G\|_\infty 2^{-N/2}$. There are only boundedly many characters. Applying the same expansion with $F = G = 1$ controls the normalization, while [Theorem 11.1](#) bounds all internal normalization factors. The zero character gives the product of separate expectations. $\square$

Lemma 11.3 (Product tests under the key reference). *For a fixed query shape and arbitrary global single-key functions $f_{j,n}$ with $|f_{j,n}| \leq 1$,*

$$\int \prod_j f_{j,n}(k_j) d\nu_n = \prod_j \int f_{j,n} d\Omega_{l_j, q_j, n} + O(2^{-N/2}).$$

The bound is uniform over the functions.

Proof. Start with $\bar{\Omega}_n$ and expand the off-position Gram conditions. For any nonzero character, choose an interacting pair of positions and fix all other positions. The remaining phase has rank at least N between the two selected position groups. The prescribed single-position diagonal densities and the unary functions are separately bounded weights. [Theorem 5.9](#) gives the error above. The identical expansion with all functions one gives the normalization of the off-position Gram event. Its zero-character contribution is a fixed positive constant, and all other contributions are exponentially small. Dividing proves the assertion. $\square$

11.3 A many-query Fourier bound

Here and in the histogram lemma, a leaf is normalized to have mass one. Its law may be very far from having bounded individual marginals.

Lemma 11.4 (Many queries and prescribed cells). *Fix a bounded query shape. There are constants $c_1, c_2 > 0$, independent of any key-space partition, with the following property. Put $\ell = \lfloor c_1 n \rfloor$ and repeat the query independently ℓ times at a common orientation. Let $A_1, \dots, A_\ell$ be cells of a partition of the full slot space, with weights $w_j = \omega_n(A_j)$. Suppose the partition has bounded size and its nonempty weights have a positive lower bound along the sequence under consideration. Conditional on any nominal inputs whose queried columns are jointly independent in every endpoint, component, and mode, the raw image probability that each query j lies in A_j is at most*

$$2^{c_2 \ell} \prod_{j=1}^{\ell} w_j \quad (11.1)$$

for all sufficiently large n . How large n must be may depend on the partition and its weight bound. The constants c_1, c_2 do not.

The probability, over the repeated fresh parameters, that the required nominal independence fails tends to zero exponentially in n , uniformly over the common orientation.

Proof. Let k bound the number of vector slots in one draw, enlarging it by a fixed factor when convenient. At any endpoint and component, a new point column contains the component $p_s \otimes Z$ with a fresh uniform n -bit vector Z . Its probability of lying in a previously exposed space of dimension v is at most 2^{-n+v} : the projection of that space onto its $p_s \otimes Z$ direction space has dimension at most v . This remains valid with repeated labels, and also after the input is conditioned on q or a tester summary. A union bound, with $k\ell$ a small fraction of n , proves the independence assertion.

Fix independent nominal inputs. The raw image law is an iid global slot law conditioned on all required same-endpoint Gram entries and on injectivity. Divide the Gram entries into those within a draw and those between draws, after collapsing duplicate entries. Write their numbers as b_{in} and b_{cross} , respectively. We have $b_{\text{in}} = O(\ell)$. The normalization in the denominator is at least a fixed constant times $2^{-b_{\text{in}} - b_{\text{cross}}}$ by [Theorem 11.1](#). In the nonnegative numerator, drop injectivity and the within-draw conditions, and expand only the cross-draw conditions. It remains to bound

$$2^{O(\ell)} \sum_{\chi} |\mathbb{E}_{\omega_n^{\otimes \ell} \chi} \prod_{j=1}^{\ell} \mathbf{1}_{A_j}|,$$

where the zero character has contribution $\prod_j w_j$.

Represent a nonzero character by a symmetric binary matrix on the distinguished slot indices, grouped into ℓ draw blocks of size at most k . Its diagonal draw blocks are zero. A coefficient records one unordered interacting slot pair; in particular there is no nontrivial character represented by the zero matrix. Let $t \geq 1$ be the maximum rank of a coefficient rectangle whose row draw-block set and column draw-block set are disjoint.

We claim that the number of such patterns with this value of t is at most $2^{Ck^2\ell t}$ for an absolute constant C . Choose a nonsingular t by t minor witnessing the rank, with row-block set I and column-block set J . Each of I, J has at most t blocks. Specify all entries incident with those blocks. If u, v lie in two distinct unselected blocks, border the chosen minor by row u and column v . The enlarged row and column block sets are still disjoint. Its rank cannot exceed t , so its bottom-right entry is uniquely determined by the other entries and the invertible minor. Entries in a single unselected diagonal block are already zero. Thus the specified entries determine the whole pattern. Their number is $O(k^2\ell t)$; the choices of the minor add at most $2t \log_2(k\ell)$ bits, which is absorbed in the asserted bound.

For one pattern, fix slots outside the draw blocks $I \cup J$. Across I and J the phase has bit rank at least tN . Phases internal to each group or involving the fixed slots are separate weights, and the cell indicators within the two groups are bounded by one. The Walsh estimate gives $2^{-tN/2}$. Integrating the other cell indicators gives

$$|\mathbb{E}\chi \prod_j \mathbf{1}_{A_j}| \leq 2^{-tN/2} \prod_{j \notin I \cup J} w_j \leq 2^{-tN/2} w_{\min}^{-2t} \prod_j w_j, \quad (11.2)$$

where $w_{\min} = \min_j w_j$. This argument treats components and modes as distinguished slots; their only effect is to constrain which coefficient entries may be nonzero.

The sum of nonzero characters, divided by $\prod_j w_j$, is therefore at most

$$\sum_{t \geq 1} 2^{Ck^2\ell t - tN/2 + 2t \log_2(1/w_{\min})}.$$

Choose c_1 small enough both for nominal independence and for $Ck^2c_1 < M_0/8$. For any fixed positive weight bound, the displayed sum is at most $\sum_{t \geq 1} 2^{-tN/4}$ for all sufficiently large n . It can consequently be absorbed in the within-draw factor $2^{O(\ell)}$. This proves (11.1) with a constant independent of the partition and its positive weight bound. $\square$

11.4 Uniform integrability and per-leaf nets

For a finite partition $\mathcal{P}$ and reference weights w_C , use relative entropy in bits,

$$D_2(p||w) = \sum_{C \in \mathcal{P}} p_C \log_2 \frac{p_C}{w_C}, \quad 0 \log_2 0 = 0.$$

Only nonempty reference cells are used.

For the finite entropy chain rule and Pinsker inequality in these units, see Cover and Thomas [7, Theorem 2.5.3 and Lemma 11.6.1]. We include the short derivation of the inequality and its constants in bits. For probability vectors p, q , put $A = \{j : p_j > q_j\}$. Convexity of $t \log t$ (the log-sum inequality) collapses their relative entropy to that of the two probabilities $p(A), q(A)$. Binary relative entropy in natural units has second derivative $1/[u(1-u)] \geq 4$ in its first argument and has value and first derivative zero at $u = q(A)$. Since $p(A) - q(A) = \|p - q\|_1/2$, Taylor's integral formula gives

$$D_2(p||q) \geq \frac{\|p - q\|_1^2}{2 \ln 2} \geq \frac{1}{2} \|p - q\|_1^2. \quad (11.3)$$

Zero probabilities follow by continuity, with infinite relative entropy when appropriate.

Lemma 11.5 (Uniform histogram control). *Fix a query shape, a constant C_L , and an integer F . Consider any collection of normalized leaf laws ν_L satisfying*

$$\nu_L \leq 2^{C_L N} \mathbf{P}_0^2. \quad (11.4)$$

The following assertions are uniform over those leaves and over the bounded numerical choices of a query.

- (i) *If A_n is a sequence of slot sets with $\omega_n(A_n) \rightarrow 0$, then the probability that the unfiltered query key belongs to A_n , averaged on any choice of leaves, tends to zero. Equivalently, the averaged key laws are uniformly absolutely continuous in this asymptotic small-set sense.*
- (ii) *On each leaf, choose any F binary flags of the orientation and query parameters. Let $\mathcal{H}_{L,n}$ be the family of unnormalized key densities obtained by filtering by any event of the orientation and the flag values. For every $\varepsilon > 0$, this family has an $L^1(\omega_n)$ ε -net of cardinality bounded independently of n and of the leaf, for all sufficiently large n . The net itself may depend on the leaf.*
- (iii) *Both conclusions persist after restricting the query to K_n and expressing the subdensities relative to ν_n . In particular these subdensities are uniformly integrable: for every $\eta > 0$ there is $R < \infty$ such that, for all sufficiently large n , all their tails above R have integral at most η .*

The flag functions may vary with the leaf and with n . The number of flags is fixed. No bound on the normalized marginal of an individual leaf is assumed here. The displayed cap concerns the orientation law after any marks are forgotten. Marks can be fixed on a leaf or included among the bounded flag data.

Proof. We first obtain a uniform bound for coarse conditional information. Fix a partition with a bounded number r of cells and with every nonempty reference weight bounded below by a positive constant. At an orientation O , let $H_O(C)$ be its input probability of cell C . Repeated queries are independent with this cell law. Uniformly over probability vectors on the fixed r -simplex, the empirical histogram of $\ell = \lfloor c_1 n \rfloor$ samples converges to H_O . Relative entropy against the fixed positive reference weights is uniformly continuous on that simplex: each cell frequency has variance at most $1/(4\ell)$, and $u \log u$ is continuous at zero. Consequently, on orientations with $D_2(H_O \| w) > T + 1$, the empirical histogram has entropy greater than T with probability $1 - o(1)$, uniformly over those orientations. The nominal independence event of [Theorem 11.4](#) can be imposed simultaneously at a further $o(1)$ loss, also uniformly over orientations.

There are at most $(\ell + 1)^r$ empirical types. For a type v , the multinomial bound gives

$$\frac{\ell!}{\prod_C (\ell v_C)!} \prod_C w_C^{\ell v_C} \leq 2^{-\ell D_2(v \| w)}.$$

Indeed the probability of that same type under cell probabilities v is at most one, which bounds its multinomial coefficient by $\prod_C v_C^{-\ell v_C}$. Apply (11.1) for each cell sequence of a type, and then pay (11.4). We obtain

$$\begin{aligned} (1 - o(1)) \nu_L \{D_2(H_O \| w) > T + 1\} \\ \leq 2^{C_L M_0 n + (c_2 - T)\ell + r \log_2(\ell + 1)}. \end{aligned} \quad (11.5)$$

Choose $T > c_2 + C_L M_0 / c_1 + 2$. This bound is exponentially small. The density cap has been applied only on the event of nominal independence. Its failure was removed in the uniform per-orientation lower bound, not multiplied by $2^{C_L N}$ as an additive raw exceptional probability.

Since $D_2(H_O \| w) \leq \log_2(1/w_{\min})$, it follows that

$$\limsup_{n \rightarrow \infty} \mathbb{E}_{O \sim \nu_L} D_2(H_O \| w) \leq C_H \quad (11.6)$$

for a constant C_H independent of the fixed partition size and its positive weight bound. The onset of the bound may depend on both. The constant may depend on C_L, M_0 and the query shape.

Let V denote the flag vector, let $C_{\mathcal{P}}(K)$ denote the cell of $\mathcal{P}$ containing the key K , and let $p_{O,V}$ be its cell law conditional on O, V . Average this law with the actual joint distribution of (O, V) ; flag values of probability zero may be given any conditional law. The entropy chain rule gives

$$\mathbb{E} D_2(p_{O,V} \| w) = \mathbb{E} D_2(H_O \| w) + I(C_{\mathcal{P}}(K); V | O) \leq \mathbb{E} D_2(H_O \| w) + F.$$

Indeed, split each logarithm $\log_2(p_{O,V}(C)/w_C)$ through $H_O(C)$. The first term averages to $I(C_{\mathcal{P}}(K); V | O)$; the second averages to $\mathbb{E} D_2(H_O \| w)$ because averaging $p_{O,V}$ over the conditional flag law gives H_O . The information term is at most the entropy of the F -bit flag vector, hence at most F . Thus (11.6) holds for these conditional histograms with $C_H + F$ in place of C_H .

For the small-set assertion, suppose by contradiction that $\omega_n(A_n) \rightarrow 0$ but the averaged query probability of A_n is bounded below by $a > 0$ on a sequence of leaves. Fix $0 < \delta < 1/2$ and enlarge A_n to a set of reference weight tending to δ ; the uniform slot atoms tend to zero, so this is possible. On the resulting two-cell partition, the binary entropy inequality gives

$$D_2((p, 1-p) \| (\delta, 1-\delta)) \geq p \log_2(1/\delta) - 1.$$

Taking expectations and limits in (11.6) yields $a \log_2(1/\delta) \leq C_H + 1$. Sending δ to zero is a contradiction. This proves (i), and the same assertion holds for every filtered submeasure by domination.

We next prove the net assertion. For a partition $\mathcal{P}$, write $\Pi_{\mathcal{P}} d$ for its conditional-average density relative to ω_n . If some filtered density d satisfies $\|d - \Pi_{\mathcal{P}} d\|_1 > \varepsilon$, split every cell by the set where $d - \Pi_{\mathcal{P}} d$ is positive. On the refined partition the filtered cell masses differ in ℓ^1 by more than ε from their parent masses subdivided in reference proportions. Since the filter is a function of (O, V) with values in $[0, 1]$, the triangle inequality implies that the *expected* corresponding discrepancy of the unfiltered conditional vectors $p_{O,V}$ is at least ε .

For any refinement, the relative entropy chain rule and (11.3) give

$$\begin{aligned} & \mathbb{E} D_2(p_{\text{child}} \| w_{\text{child}}) - \mathbb{E} D_2(p_{\text{parent}} \| w_{\text{parent}}) \\ & \geq \frac{1}{2} (\mathbb{E} \|p_{\text{child}} - \text{split}_w(p_{\text{parent}})\|_1)^2. \end{aligned} \quad (11.7)$$

Here split_w divides each parent mass among its children in their reference proportions. To see the last step, apply the usual conditional relative entropy identity inside each parent, then Pinsker, and then Jensen first with the parent masses and then over (O, V) .

A bounded number of refinements must suffice uniformly asymptotically. For otherwise choose any large fixed depth d and a sequence of leaves with d successive witnessing refinements. Their final partitions have at most 2^d cells. Pass to a subsequence on which the reference weights of the entire finite tree converge and the laws of the conditional histogram vectors converge. A

final cell of zero limiting reference weight has zero limiting expected conditional mass by (i), and hence zero conditional mass almost surely. Delete these zero-weight cells in the limit. For a positive-weight parent, reference subdivision ratios converge. For a zero-weight parent, both the actual and subdivided total masses tend to zero in expectation. The positive discrepancy at every refinement consequently persists in the limit, at least as $\varepsilon/2$ if needed.

The limiting final conditional entropy is at most $C_H + F$. Indeed, before taking limits merge all zero-limiting-weight final cells into one positive cell, and use the conditional version of (11.6) on that partition. All its limiting weights are positive, so entropy is continuous. This merging is used only for the final upper bound. The remaining positive limit cells still form a genuine refinement tree, on which (11.7) telescopes. That identity forces the final entropy to be at least $d\varepsilon^2/8$, a contradiction for large fixed d .

Apply the preceding argument with $\varepsilon/2$. On a partition of bounded size r'_0 that approximates every family member, quantize each cell mass downwards in increments $\varepsilon/(2r'_0)$. The L^1 error in the piecewise-constant density is the sum of the cell-mass errors, at most $\varepsilon/2$. There are at most $(1 + 2r'_0/\varepsilon)^{r'_0}$ possible quantized mass lists. They form the required net, proving (ii).

For (iii), the restriction of a density d relative to ω_n has density $\omega_n(\mathcal{K}_n)d$ on $\mathcal{K}_n$ relative to ν_n . Restriction therefore cannot increase the L^1 distance. A ν_n -small set is also ω_n -small, so (i) transfers as well. Every resulting subdensity has integral at most one; its superlevel set above R has reference measure at most $1/R$. Assertion (i), applied along any contrary sequence, makes the integrals on those superlevel sets tend to zero uniformly as $R \rightarrow \infty$. This is uniform integrability. $\square$

Remark 11.6. The preceding lemma allows a leaf to have strongly nonuniform histograms. For example, forcing a whole primal frame map into a fixed ambient subspace of constant codimension costs $2^{O(n)}$ and can be consistent with (11.4). It puts its keys on a fixed small reference set. The uniform entropy constant is therefore allowed to depend on the leaf density exponent and on M_0 . The next results concern typical orientations under a mixed law and use a different, stronger marginal hypothesis.

11.5 Typical empirical integrals under a paired law

Write $P_0^{\text{pr}} = \text{pr}_* P_0$ for the raw single-endpoint primal law. In the rest of the section let ρ_n be a law on units satisfying, after forgetting any finite marks, for a fixed C and some $a_n \rightarrow 0$,

$$\rho_n \leq 2^{CN} P_0^2, \quad \text{pr}_*(\rho_n)_i \leq M 2^{a_n N} P_0^{\text{pr}} \quad (i = 1, 2). \quad (11.8)$$

Changing the fixed factor M does not matter. Since M_0 is fixed, $a_n N = o(n)$. The law may have arbitrary dependence between its two endpoints. A decomposition into leaves does not change (11.8) for its mixture. The second inequality is solely a primal marginal bound. No $M 2^{o(N)} P_0$ bound on the full endpoint law is required or asserted. In particular $\text{pr}_* \pi = P_0^{\text{pr}}$ allows endpoint caps relative to the thinned law π to supply this hypothesis. Projecting the first inequality also gives $\text{pr}_* \rho_n \leq 2^{CN} (P_0^{\text{pr}})^2$.

Lemma 11.7 (Typicality of common input-and-key tests). *Let F_n be a uniformly bounded common function of the inputs and primal keys of a fixed bounded query, with no other orientation dependence. In particular it does not inspect channel maps or the orientation's marks. Put*

$$A_n(O) = \mathbb{E}_z F_n(z, K(O, z)), \quad m_n = \mathbb{E}_{O \sim P_0^2, z} F_n(z, K(O, z)).$$

For every fixed $\varepsilon > 0$, under (11.8),

$$\rho_n\{|A_n - m_n| > \varepsilon\} = 2^{-\Omega(n)}.$$

Repeated labels in the query are allowed. The constants may depend on the query and ε , but the estimate is uniform over tests with the given sup bound. The exceptional set may depend on the test.

For a query at just one endpoint, its raw variance about its raw mean is $2^{-\Omega(n)}$. This variance bound is also uniform when an arbitrary external input-and-image datum is fixed as an argument of the test. This datum may include a channel frame from a separately fixed vertex; uniformity refers to each fixed datum.

Proof. Every empirical integral and exceptional event used in this proof is measurable with respect to the primal maps. We can therefore work throughout with the projected law, paying only the projected marginal bound in (11.8). Raw expectations are unchanged by this projection. Rescale the test to have absolute value at most one. First consider one raw endpoint and duplicate its parameter batch. The two batches have jointly independent nominal columns except with $2^{-\Omega(n)}$ probability, because their Z inputs are independent and their total number of columns is bounded. With the nominal inputs fixed on this event, Theorem 11.2 compares the product of the two tests with the product of their separate raw expectations, at error $O(2^{-N/2})$. Averaging the two independent input batches and bounding the dependent-input event trivially proves

$$\text{Var}_{O \sim P_0}(\mathbb{E}_z F_n(z, K(O, z))) \leq C_1 2^{-c'_1 n}. \quad (11.9)$$

All these estimates are uniform in a fixed external datum, since it only changes the separate bounded test functions. Chebyshev's inequality proves the one-endpoint assertion at every fixed tolerance.

For the paired assertion, assume that both endpoints are used; the other case follows at once from the marginal bound. Let D_2 denote the second endpoint's input-and-image data under its raw model, and for a first primal orientation o define

$$h_o(D_2) = \mathbb{E}_{z_1} F_n(z_1, K_1(o, z_1), D_2), \quad \bar{h}(D_2) = \mathbb{E}_{o \sim P_0} h_o(D_2).$$

Fix a small constant $\eta > 0$. By (11.9), uniformly in D_2 , $\mathbb{P}_{P_0^{\text{pr}}}\{|h_o(D_2) - \bar{h}(D_2)| > \eta\}$ is exponentially small. Fubini and Markov show that, outside an exponentially small set of first primal orientations, the raw-model probability of the bad data set

$$B_o = \{D_2 : |h_o(D_2) - \bar{h}(D_2)| > \eta\}$$

is at most 2^{-cn} for some $c > 0$. The projected marginal cap in (11.8) still makes the discarded first-endpoint mass exponentially small.

Fix any of the remaining first orientations. We claim that a raw second orientation o' has empirical input probability at least η of B_o only with probability $2^{-\Omega(n^2)}$. Draw $\ell' = \lfloor c'_2 n \rfloor$ independent second-endpoint input batches. For an orientation with empirical bad probability at least η , at each step success and nominal independence of the new columns modulo the previous ones have conditional probability at least

$$\eta - O(2^{-n+O(\ell')}) \geq \eta/2.$$

The failure bound depends only on fresh uniform Z coordinates and holds for any past inputs. Thus all successes together with joint nominal independence have probability at least $(\eta/2)^{\ell'}$.

For independent nominal columns, the joint raw image law of these batches has density at most $2^{C_2(\ell')^2}$ relative to the product of the individual raw batch laws. Indeed it adds only $O((\ell')^2)$ mutual Gram conditions and injectivity; count their normalizations by [Theorem 11.1](#) and bound the indicator in the numerator by one. After averaging the inputs, the all-success probability on the independence event is consequently at most

$$2^{C_2(\ell')^2} \mathbb{P}(D_2 \in B_o)^{\ell'} \leq 2^{C_2(\ell')^2 - cn\ell'}.$$

Choose $c'_2 > 0$ small enough that $C_2(c'_2)^2 < cc'_2/2$ and that nominal independence is available. Dividing by $(\eta/2)^{\ell'}$ proves the claim.

This quadratic-exponential estimate is uniform in the fixed good first primal orientation. Hence the set of paired primal orientations with a good first endpoint but a bad oversampling second endpoint has $(P_0^{\text{pr}})^2$ -mass $2^{-\Omega(n^2)}$. Multiplying by the projected joint density $2^{CN} = 2^{CM_0n}$ still leaves quadratic-exponentially small mass. On every remaining pair,

$$|\mathbb{E}_{z_2} h_o(D_2) - \mathbb{E}_{z_2} \bar{h}(D_2)| \leq 3\eta.$$

Finally, $\bar{h}$ is a common input-and-key test of the second endpoint. Apply the one-endpoint estimate again and its projected marginal cap to compare its empirical integral with its raw mean to within η . Taking $4\eta < \varepsilon$ proves the lemma. $\square$

Remark 11.8. Uniformity in the test means that each prescribed test has a small exceptional set with the stated bound. It does not mean that one orientation is simultaneously good for every possible global test. The later arguments use only finitely many common tests at a time. For two distinct laws, apply these results on a mixture with fixed positive type weights and retain the type as a mark. A test prescribed for each of the finitely many types is handled by taking the union of its exceptional sets. The same argument permits colours to be recorded on leaves while using the original, mass-weighted mixture for every marginal estimate. It does not require the normalized individual colour classes to satisfy a common marginal cap. Restriction to a sublaw of inverse-polynomial mass, or more generally of reciprocal mass $2^{o(n)}$, preserves (11.8). Such restriction need not preserve a uniform cap on every individually normalized old leaf. The mixed-law lemmas do not require that additional conclusion.

11.6 Mixing of the binary parameter constraints

Lemma 11.9 (Binary mixing). *Fix a query template whose labels are distinct at each used endpoint. Impose the off-position Gram zeros of K_n and a consistent fixed specification of the within-endpoint ordered L, R evaluations used in [Section 7](#). Let $\mathcal{A}_n(O, z)$ be their joint indicator. There is a template constant $\kappa > 0$ such that, outside an exponentially small set of orientations under (11.8),*

$$\mathbb{E}_z \mathcal{A}_n(O, z) \prod_j w_j(O, z_j) = \kappa \prod_j \mathbb{E}_{z_j} w_j(O, z_j) + O(2^{-cn}) \quad (11.10)$$

uniformly over all unary weights with $|w_j| \leq 1$. These weights may depend on the entire fixed orientation. The input laws may be conditioned separately on feasible diagonal values or tester summaries. The exceptional set is independent of the unary weights.

Proof. Collapse all identical Gram entries, including the repeated same-endpoint entries that equal the same E evaluation, and impose only one copy of each. Consistency means that their prescribed values respect these repetitions. Expand the resulting binary indicators into characters. The zero character has coefficient $\kappa = 2^{-b}$, where b is the number of the resulting specified bits. We show that every nonzero character has an exponentially small integral against the product of unary weights.

If an interacting pair of positions belongs to the same endpoint, their labels are distinct. The second property of [Theorem 6.5](#) says that every nonzero parity of its ordered L, R bits, with its possible E bits in either order, has bilinear rank at least $n/5$ on the retained free variables. This includes characters involving only E bits.

For positions from opposite endpoints, we establish the analogous claim for their global Gram bits except on a $2^{-\Omega(n^2)}$ raw exceptional set. Write $d = \dim \mathcal{B} + h$ for the full plus-frame dimension. At one component, an intersection of dimension at least $v = \lfloor n/10 \rfloor$ between the two full plus spans gives v independent image relations. Their coefficient choices number at most 2^{2dv} , and the frame image estimate bounds each event by $2^{-(1-\epsilon)vN}$, for a fixed small ϵ . The choice of M_0 , hence d/N small, makes this $2^{-\Omega(n^2)}$. There are only boundedly many components.

Condition on plus frames off this exception. Choose a nonzero pair pattern and a component where, after exchanging the endpoints if necessary, it uses a plus slot at endpoint 1 paired with a minus slot at endpoint 2. At least $n - v$ of the first position's Z directions are independent modulo the second endpoint's full plus frame. The second position's nominal minus Z directions are independent. Conditional on the plus frames, the latter minus columns have independent uniform values on the appropriate affine annihilator spaces, before their full-rank conditioning. Their evaluations on these $n - v$ directions therefore form a uniform rectangular matrix, up to a fixed translation. Terms in the reverse order and other components may be frozen first; they use separate minus-frame randomness. The final injectivity conditioning costs only a bounded multiplicative factor.

A uniform a by b binary matrix has rank at most r with probability at most $2^{(a+b)r-ab}$, by writing it as a product of an a by r and an r by b matrix and counting. With $a \geq n - v$ and $b = n$, this shows that the bilinear rank is greater than $n/5$ except with probability $2^{-\Omega(n^2)}$. Union over the boundedly many nonzero pair patterns. The joint density in (11.8) preserves this exceptional scale.

For any nonzero full character, choose a pair with nonzero coefficient pattern and fix the other position inputs. The remaining terms become separate unary phases. The selected pair has rank at least $n/5$ by one of the preceding two arguments. [Theorem 5.9](#) gives an error $O(2^{-n/10})$ against the two remaining bounded unary weights. Input conditioning costs only the fixed reciprocal probabilities of the diagonal values or tester summaries; those events can be absorbed into the respective weights. Summing the bounded number of characters proves (11.10). All estimates were uniform in the unary weights, so the exceptional set does not depend on them. $\square$

11.7 Positive basis fractions on common key cells

We give the small-cell argument explicitly, because its uniform lower fraction will be used on a countable collection of signature cells.

Lemma 11.10 (Quadratic signs on affine subspaces). *Let Q be a quadratic function on a binary vector space and let its polar form have rank $2r$. Its uniform sign bias has magnitude at most 2^{-r} . On an affine subspace of codimension c , the restricted polar rank is at least $2r - 2c$, and its sign*

bias is at most 2^{-r+c} whenever this bound is less than one.

Proof. Squaring the mean sign and translating one variable gives

$$|\mathbb{E}_x(-1)^{Q(x)}|^2 = \mathbb{E}_h(-1)^{Q(h)+Q(0)} \mathbf{1}_{\{B(h,\cdot)=0\}},$$

where B is the polar form. Its absolute value is at most the kernel proportion 2^{-2r} . Restricting a bilinear form to a codimension- c subspace in each of its two factors can decrease rank by at most $2c$. Apply the first assertion on that affine subspace, where translation affects only the linear and constant terms. $\square$

Lemma 11.11 (Unary reference law and base-status positivity). *Fix an endpoint, tag, label, flavor, and feasible tester summary with diagonal q . For a bounded number of common global key tests, its empirical conditional key frequencies differ from $\Omega_{l,q,n}$ by any prescribed fixed tolerance except on exponentially small mixed mass under (11.8).*

At each unit, let $C_i \subset \mathcal{X}$ be any orientation-dependent space of dimension at most K^2 all of whose elements have total component rank at most K , and choose any basis. Fix a sequence of global single-key sets A_n with $\Omega_{l,q,n}(A_n) \geq \delta > 0$. Except on exponentially small mixed mass, every basis assignment has, conditional on the tester summary and on $K_i \in A_n$, fraction at least $(7/8)2^{-\dim C_i}$. For the fixed common key sets, the exceptional event depends only on the primal frames and is chosen uniformly over all nominal profiles of the indicated rank before C_i or its basis is selected. Thus these selections may depend on the channels and all orientation marks. We use the first effective spaces from (6.6); the larger generality in this statement requires no further peeling.

For the full generic flavor and inputs conditional just on q , the following conservative constant is valid:

$$\beta_{\text{base}} = 2^{-K^2-1} 4^{-(g^2+1)} > 0. \quad (11.11)$$

For every feasible specified tester summary and basis assignment, outside an exponentially small exceptional set,

$$\mathbb{P}_z\{K_i \in A_n, \text{ the specified summary and basis bits} \mid q\} \geq \beta_{\text{base}} \Omega_{l,q,n}(A_n). \quad (11.12)$$

In particular each joint choice of the role $a(\text{atom})$ and the basis bits has this lower bound, after choosing a compatible summary. Both roles are available for either possible generic q . No prescribed value of the channel flag p_i is included in these lower bounds. If that bit is recorded together with the role and basis bits, the bound holds only after summing over its two values.

The lower constants do not depend on δ , on the key tests, or on the orientation-dependent choice of C_i . The onset of the asymptotics and the exceptional-set constants may depend on a fixed positive lower bound δ . The exceptional set is allowed to depend on the prescribed tests: there is no simultaneous assertion over all measurable key sets at one orientation.

Proof. For each fixed point parameter, the raw key law is the single-key reference up to an exponentially small injectivity error, by Theorem 5.2. The first conclusion therefore follows from the one-endpoint part of Theorem 11.7, also after conditioning on a feasible summary.

Fix a nonzero profile $x \in \mathcal{X}$ of total component rank at most K , and consider the sign

$$s_x(z) = (-1)^{T(x, \text{atom}(z))}.$$

By the first property of [Theorem 6.5](#), its dependence on Z factors through a bounded-dimensional space of linear forms, independent of the other input values, and its Z -polar rank is at least $2(J - s_0)$. Put

$$\gamma = 2^{-K^2-3}, \quad b_* = 2^{-J+2s_0} < \gamma.$$

The strict inequality follows from the chosen lower bound on J .

Apply a uniform $G \in \text{GL}_n(\mathbb{F}_2)$ simultaneously to the Z input coordinates in all primal frames at the queried endpoint. This preserves the raw primal law because E has no Z rows or columns. We perform this averaging on P_0^{pr} , not on the thinned conditional channel law. It also preserves the empirical mass of a global key cell: change variables in the uniform Z input and leave the other coordinates unchanged. After this change of variables the cell event is fixed and the sign's bounded row-bit space is moved by the inverse transformation.

Fix a starting orientation for which the cell, conditional on the summary, has input mass at least $\delta/2$. We claim that at most 2^{-2An} of the transformations can make its conditional absolute sign bias exceed γ , for sufficiently large n . Suppose the contrary and retain one bias sign, losing at most a factor two. For any fixed integer u , choose u such transformations successively so that each new row-bit space meets the span of the previous ones in dimension at most s_0 . This is possible for large n . Indeed, for a uniformly moved space of bounded dimension and a fixed space of bounded dimension, the probability of an intersection of dimension greater than s_0 is $O_u(2^{-s_0n})$: choose s_0 independent vectors in the former and bound the probability that their uniformly independent images all lie in the latter. The number of such choices and target vectors is bounded independently of n . Since $s_0 = 4\lceil A \rceil > 2A$, this excluded fraction is smaller than the retained fraction at each of the fixed number of choices.

Let E_0 be the fixed cell event in the conditional input probability space and let $\delta' = \mathbb{P}(E_0) \geq \delta/2$. Denote the chosen signs, with their common bias sign adjusted to be positive, by $s_1, \dots, s_u$. Let $\mathcal{F}_{j-1}$ contain all non- Z inputs and the row bits of the preceding chosen spaces, and put

$$m_j = \mathbb{E}(s_j \mid \mathcal{F}_{j-1}), \quad D_j = s_j - m_j.$$

Conditioning the current row bits on the past imposes at most s_0 linear conditions on their distribution. The polar rank can therefore lose at most $2s_0$ further dimensions. By [Theorem 11.10](#),

$$|m_j| \leq b_*$$

pointwise, including after the non- Z inputs are fixed. The D_j are martingale differences with second moments at most one, so $\|u^{-1} \sum_j D_j\|_2 \leq u^{-1/2}$. On the other hand every selected sign has conditional mean greater than γ on E_0 . Consequently

$$(\gamma - b_*)\delta' \leq |\mathbb{E} \mathbf{1}_{E_0} u^{-1} \sum_j D_j| \leq \sqrt{\delta'/u}. \quad (11.13)$$

Here the intrinsic bias costs $b_*\delta'$, because its conditional mean is bounded pointwise; it does not cost an absolute b_* before division by the cell mass. Taking $u > 2/[\delta(\gamma - b_*)^2]$ contradicts (11.13). This proves the transformation claim with J fixed independently of the cell mass.

Integrate that claim over the starting raw primal orientation, using frame-law invariance. The first part of the lemma makes the cell mass at least $\delta/2$ outside an exponentially small set. More precisely, the transformation estimate is used only on orientations with this mass; that property is itself invariant under the transformations. This cell-mass exceptional set is common to all profiles and is discarded only once. For a fixed x , the event that the cell mass is at least $\delta/2$ and the

conditional sign bias is bad has raw probability at most $2^{-2^{An}}$. There are at most 2^{An} nonzero profiles of the required rank for all sufficiently large n , by [Theorem 6.5](#). Union over them, and over the bounded numerical position and summary choices. The resulting event is still exponentially small, also after the projected marginal inflation in [\(11.8\)](#). This union is taken before making any channel-dependent choice of effective space or basis: its exceptional event is a function of the primal frames alone.

On its complement every nonzero linear combination of a basis of C_i has conditional sign bias at most γ , regardless of how the space and basis were selected from the unit. Fourier inversion, for $d_i = \dim C_i$, gives for every assignment

$$\mathbb{P}(\text{assignment} \mid E_0, \text{summary}) \geq 2^{-d_i} (1 - (2^{d_i} - 1)\gamma) \geq (7/8)2^{-d_i}.$$

This also covers $d_i = 0$.

It remains to check the uniform generic constant. For one retained tester block, the sum of $r_0 \geq 1$ independent bit-pair products has probabilities $(1 \pm 2^{-r_0})/2$, both at least $1/4$. Distinct tester blocks use disjoint input bits. There are at most $g^2 + 1$ such blocks, so every feasible summary has probability at least $4^{-(g^2+1)}$. Conditional on its compatible value of q , its probability is no smaller. Its empirical key-cell mass, conditional on that summary, is at least $(3/4)\Omega_{l,q,n}(A_n)$ outside an exponentially small set by the first part. Combining this with the basis lower fraction gives [\(11.12\)](#), since $(3/4)(7/8) > 1/2$. In the generic flavor there is at least one retained O tester and the independent shared tester. Prescribing their parities realizes either a role at either fixed q , and a feasible summary for each has the same lower probability bound. $\square$

Remark 11.12. The key-only hypothesis cannot be omitted. For a fixed nonzero profile x_0 , the input event $\{z : T(x_0, \text{atom}(z)) = 0\}$ has zero conditional mass for the opposite basis bit. This is an allowed input-and-key test for concentration, but is not a common global key cell for positivity. The constants in [Theorem 11.11](#) apply on every fixed positive limiting key cell, however small. Countably many resulting cell inequalities can be passed through a diagonal subsequence; they do not assert simultaneous finite- n control of all possible cells. An orientation-only subfilter preserves an almost-sure limiting positivity statement, whereas an arbitrary additional input filter need not.

12 Weak product testing

The preceding histogram lemma gives compact families within each leaf. We now prove a different statement: a fixed global test on a tuple of keys can be replaced, for the integrations that occur here, by a bounded sum of products of global single-key functions. Both its reference and its actual-query conclusions are needed.

Fix a query template with r positions and with labels distinct at each used endpoint. Its input laws and reference measures are those of [Section 11](#). In particular $\bar{\Omega}_n = \bigotimes_{j=1}^r \Omega_{l_j, q_j, n}$ and ν_n is its off-position-Gram-conditioned reference on $\mathcal{K}_n$. The additional prescribed binary tests are the within-endpoint ordered L, R evaluations from [Section 7](#). Let $\mathcal{A}_n(O, z)$ be the successful binary indicator from [Theorem 11.9](#), including the off-position Gram zeros. All common functions below inspect primal keys only. Channel maps may enter the unary input weights and the orientation filters.

Theorem 12.1 (Weak product comparison). *Let W_n be any sequence of real functions on $\mathcal{K}_n$ with $\|W_n\|_\infty \leq B$, where B is fixed. Extend it to the product of the single-key spaces with the same*

bound. For every $\varepsilon > 0$ there are constants $L, C < \infty$, independent of n , and functions

$$S_n(k_1, \dots, k_r) = \sum_{a=1}^L c_{a,n} \prod_{j=1}^r f_{a,j,n}(k_j), \quad |c_{a,n}| \leq C, \quad |f_{a,j,n}| \leq 1, \quad (12.1)$$

with the following properties.

(i) For arbitrary global unary functions $g_{j,n}$ of bound one,

$$\left| \int (W_n - S_n) \prod_j g_{j,n}(k_j) d\nu_n \right| \leq \varepsilon \quad (12.2)$$

for all sufficiently large n .

(ii) Under any mixed unit law satisfying (11.8), outside an exponentially small exceptional set of orientations,

$$\sup_{|w_j| \leq 1} |\mathbb{E}_z \mathcal{A}_n(O, z) (W_n - S_n)(K(O, z)) \prod_j w_j(O, z_j)| \leq \varepsilon. \quad (12.3)$$

The unary input weights may depend on the entire fixed orientation and on any additional marks fixed with it. The exceptional set is independent of these weights. Thus one may prescribe channel flags such as p_i in a unary weight without making them common key tests or asserting positivity for their individual values.

The functions $f_{a,j,n}$ are global key functions; they can be chosen to be indicators of sets, or finite linear combinations of such indicators with uniformly bounded complexity. They do not depend on the sampled orientation. The constants may depend on B, ε and the fixed template and construction parameters, but not on n .

An additional orientation-only filter of bound one may multiply (12.3) without changing it. In particular, after averaging normalized leaves, the same error control applies to unnormalized admissibility-filtered query measures, even when the filter is chosen using the opposite leaf. No assertion is made for an arbitrary additional joint input filter. For distinct marked laws, use their fixed-weight mixture, with caps checked after forgetting the marks. A retained ordered type pair or different-colour condition is an orientation/leaf-pair filter, so the same conclusion applies before normalization. All leaf averages use their actual weights; no marginal hypothesis on a separately normalized colour class is needed.

The proof uses a finite family of twists. If I, J partition a set of positions into two nonempty groups, let $\mathcal{T}_{I,J}$ consist of all products of their available inter-position global Gram signs: the dot products of a plus slot at one position with a minus slot at another position in the same component. These signs are defined on the product reference whether or not that Gram is prescribed by a raw same-endpoint frame law. Every such twist equals one on $\mathcal{K}_n$. The number of twists is a constant depending only on the query shape.

For a real function D on the keys of $I \cup J$, define its twisted cut norm by

$$\|D\|_{\square, \tau} = \max_{\tau \in \mathcal{T}_{I,J}} \sup_{|f|, |g| \leq 1} \left| \int D(k_I, k_J) \tau(k_I, k_J) f(k_I) g(k_J) d\bar{\Omega}_{I \cup J, n} \right|. \quad (12.4)$$

Here f, g may be arbitrary functions on their whole respective groups. Only after the recursion below will they become unary.

We use the finite-partition energy-increment method underlying weak regularity; see Frieze and Kannan [12]. The additional Gram conditioning needed here is retained explicitly in the following self-contained proof.

Lemma 12.2 (Finite twisted regularity). *Let $|W| \leq B$ on the product key space of $I \cup J$, and let $\delta > 0$. There is a decomposition*

$$W = \sum_{a=1}^{L'} c_a f_a(k_I) g_a(k_J) \tau_a(k_I, k_J) + D, \quad (12.5)$$

where $|f_a|, |g_a| \leq 1$, $\tau_a \in \mathcal{T}_{I,J}$, $|c_a| \leq B$, $\|D\|_\infty \leq 2B$, and $\|D\|_{\square, \tau} \leq \delta$. The number L' is bounded in terms of B, δ and the twist count only. The functions f_a, g_a may be chosen as indicators of cells in finite group partitions.

Proof. Let Γ be the finite list of crossing Gram bits generating the twists. Begin with the trivial partitions of the two groups. For any current pair of finite partitions, condition W on their cells and on Γ ; call the resulting conditional expectation S . Then $\|S\|_\infty \leq B$ and $\|W - S\|_\infty \leq 2B$.

If $D = W - S$ has twisted cut norm greater than δ , choose a twist and witnesses f, g giving correlation greater than δ . Quantize f, g uniformly within their bounds finely enough, with a mesh depending only on B, δ , that their quantized product times the twist still correlates with D by more than $\delta/2$. Refine the group partitions by the values of these quantized functions. The witness product is measurable for the new conditioning algebra and has L^2 norm at most one. Thus the new conditional expectation of D has L^2 norm at least $\delta/2$. Since D is orthogonal to the old conditioning algebra, the squared L^2 norm of the new conditional expectation of W increases by at least $\delta^2/4$. It is bounded above by B^2 . At most $4B^2/\delta^2 + 1$ refinements are possible. Each increases the partition sizes by a factor bounded in terms of B, δ , so the final partitions have bounded size.

On each product of group cells, the final conditional expectation is a bounded function of Γ . Assign value zero to any impossible combination of a cell pair and Gram bits. Its finite Fourier expansion on the Gram bits has coefficients of absolute value at most B . Multiplying by the two cell indicators gives (12.5). The decomposition is exact everywhere on the product key space, regardless of which Gram bit values occur in a particular cell pair. $\square$

We next prove that a small twisted-cut remainder is negligible on actual queries. This is the step that is not supplied by unary histograms alone.

Lemma 12.3 (A terminal remainder on actual inputs). *Let I, J be a nontrivial partition of some of the template positions. Let D_n be a common function of their global keys with $\|D_n\|_\infty \leq B_1$ and $\|D_n\|_{\square, \tau} \leq \delta$. Let $\psi_n(I, J)$ be any product of the specified nominal L, R signs and global Gram signs crossing from I to J , evaluated on their actual inputs and keys. For each fixed $\varepsilon_{\text{emp}} > 0$, except on exponentially small mixed mass,*

$$\sup_{|f|, |g| \leq 1} |\mathbb{E}_{z_I, z_J} D_n(K_I, K_J) \psi_n(I, J) f(O, z_I) g(O, z_J)| \leq (C_3 \delta + \varepsilon_{\text{emp}} + o(1))^{1/4}. \quad (12.6)$$

The constant C_3 and the vanishing error depend only on B_1 and the fixed template. The weights may depend arbitrarily on the fixed orientation. The exceptional set does not depend on those weights.

Proof. For one fixed orientation, put $U(I, J) = D_n(K_I, K_J) \psi_n(I, J)$ and use independent copies of the parameter groups I, J . Cauchy-Schwarz in I gives

$$|\mathbb{E} U(I, J) f(I) g(J)|^2 \leq \mathbb{E}_I (\mathbb{E}_J U(I, J) g(J))^2.$$

Expand the square, and apply Cauchy–Schwarz in the two J variables. The factors $g(J_0)g(J_1)$ have bound one. We obtain

$$|\mathbb{E}U(I, J)f(I)g(J)|^4 \leq Q_n(O) := \mathbb{E}_{I_0, I_1, J_0, J_1} \prod_{a, b \in \{0, 1\}} U(I_a, J_b). \quad (12.7)$$

In particular $Q_n(O) \geq 0$: it is the expectation over J_0, J_1 of the square of $\mathbb{E}_I U(I, J_0)U(I, J_1)$. All separate weights, including their orientation dependence, have disappeared.

The right-hand side is a bounded common input-and-key test, involving two fresh copies of each original position. It depends only on the primal maps: the channel-dependent weights and all marks have disappeared in (12.7), and the nominal L, R signs involve only the displayed fresh parameters. Repeated labels across these copies are permitted by [Theorem 11.7](#). It follows that, outside an exponentially small mixed exceptional set,

$$Q_n(O) \leq \mathbb{E}_{P_0^2} Q_n + \varepsilon_{\text{emp}}. \quad (12.8)$$

We bound the raw mean by $C_3\delta + o(1)$.

Fix the nominal inputs of the four groups. Their columns are jointly independent within each endpoint, component, and mode except with $2^{-\Omega(n)}$ probability, since all copies have fresh Z inputs. The dependent-input contribution is $o(1)$ by the fixed sup bound. On the independence event, [Theorem 5.2](#) gives the uniform image law with prescribed same-endpoint Grams and injectivity. Ignore injectivity at an exponentially small error and express this law relative to the product of all the individual single-key references. There are only boundedly many additional Gram bits, and their normalization is bounded uniformly in the nominal inputs by [Theorem 11.1](#). Expand their indicators into characters. The nominal L, R phases are now constants. The global Gram phases already in the four copies of ψ_n can be combined with the expansion characters.

For completeness, distinguish a slot by its group I or J , its copy, its position, component, and sign. Remove the diagonals already in the reference and collapse repeated descriptions of the same unordered slot dot. Every remaining Gram coefficient belongs to exactly one of the following classes:

- (a) both slots lie in one of I_0, I_1, J_0, J_1 ;
- (b) the slots join I_0 to I_1 , or J_0 to J_1 ;
- (c) the slots lie in one of the four I_a – J_b quadrants.

Class (a) contributes bounded separate group phases. Suppose a nonzero class-(b) pattern between I_0, I_1 remains. Fix both J groups. The four D_n factors split into two bounded functions of I_0 and I_1 . The class-(c) phases become separate functions of those two groups, as do all class-(a) phases. Their only remaining interaction is the nonzero coefficient rectangle between I_0 and I_1 , whose bit rank is at least N . The individual diagonal densities may also be absorbed into the respective bounded weights. [Theorem 5.9](#) bounds this term by $O(2^{-N/2})$. The same argument applies to a nonzero J_0 – J_1 pattern.

If neither class-(b) pattern remains, attach each class-(c) pattern as an available twist to its corresponding $D_n(K_{I_a}, K_{J_b})$. Fix I_1, J_1 . The remaining integral in I_0, J_0 has one factor D_n times an available twist, tested against bounded separate functions. The two other unfixed D_n factors have bound B_1 and the fixed factor has bound B_1 . The twisted cut estimate therefore bounds this term by $B_1^3\delta$, with the harmless fixed normalization factors included in the constant.

These cases exhaust the character expansion. The groups I and J may each contain positions from both endpoints of the unit: this only removes some same-endpoint Gram bits from the

expansion and does not create a new class. A nonzero coefficient rectangle has rank at least one in distinguished global slots, hence bit rank at least N . The two dots $P_a \cdot Q_b$ and $P_b \cdot Q_a$, for example, use different slots and are not duplicates. The prescribed nominal Gram values affect character coefficients only by signs.

Summing the bounded number of terms and averaging the nominal inputs gives $\mathbb{E}_{\mathbf{P}_0^2} Q_n \leq C_3 \delta + o(1)$. Combine this with (12.7) and (12.8). The same $Q_n(O)$ bounds every choice of separate weights, so the exceptional set is independent of them. $\square$

Proof of Theorem 12.1. If $r = 1$, quantize the values of W_n in $[-B, B]$ with mesh at most $\varepsilon/2$, choosing representatives in the same interval. The resulting S_n satisfies $\|W_n - S_n\|_\infty \leq \varepsilon/2$ and is a linear combination of at most $1 + \lceil 4B/\varepsilon \rceil$ global level-set indicators, with coefficients bounded by B . Both required errors are at most $\varepsilon/2$, since the reference measure is a probability measure and the actual-query indicator and weights have bound one. This includes $B = 0$, when $S_n = 0$. For $r \geq 2$, apply Theorem 12.2 to a nontrivial split of the full set of positions. Recursively apply it to group factors in the structured terms until every factor is unary. Whenever a remainder is produced, leave that whole term terminal; do not expand its other factors. We obtain the exact decomposition, on the unconditioned product reference,

$$W_n = S_n^{\text{tw}} + \sum_{v \in \mathcal{V}_n} R_{v,n}. \quad (12.9)$$

Each structured term is a product of unary global functions times available Gram twists. Each terminal term has the form

$$R_{v,n}(k) = c_{v,n} D_{v,n}(k_I, k_J) \prod_H f_{v,H,n}(k_H) \tau_{v,n}(k), \quad (12.10)$$

where I, J partition one group, the groups H are disjoint outside $I \cup J$, all outside functions are bounded, $\tau_{v,n}$ is a product of available Gram signs, and $D_{v,n}$ has a specified small twisted cut norm on $I|J$. The sup bounds, numbers of terms, and partition complexities are all bounded independently of n .

Here is an explicit way to organize the tolerance choices. Along any structured branch a split replaces one group by two, so at most $r - 1$ splits are needed before all groups are unary. At any fixed depth the number of pending terms and all their coefficient and factor bounds have already been bounded using previous choices. Allocate at most $\varepsilon/(2r)$ of each desired final error to that depth, dividing this budget among its pending terms and their known outside bounds. Choose the next cut tolerances small enough for the reference estimates below and for the fourth roots in Theorem 12.3. These choices determine a finite bound on the next level's number of structured terms. Continue for at most $r - 1$ levels. The empirical tolerance ε_{emp} for each terminal test is chosen at the same stage, sufficiently small for its allotted fourth-root error. All choices are constants; none requires a change in the construction parameters or in M_0 .

On $\mathcal{K}_n$, every Gram twist in S_n^{tw} equals one. Removing them gives the function S_n in (12.1). Its unary factors are indicators of the finite partitions used in the recursion; coefficients and term counts have the asserted bounds.

To prove the reference assertion, multiply a terminal term (12.10) by a product of arbitrary bounded unary tests and expand the off-position Gram event defining ν_n . Fix all positions outside its remainder. Their group factors are bounded constants. Gram phases internal to I or J and phases involving a fixed outside position are separate bounded group weights. The only other phases are available crossing twists. Thus the integral is bounded by the twisted cut norm of $D_{v,n}$

times a constant depending only on its known bounds and the fixed Gram normalization. The tolerance allocation makes the sum of these errors at most ε , proving (12.2). This estimate is uniform over all unary reference test functions.

For the actual-query assertion, expand $\mathcal{A}_n$ into its Gram and nominal L, R characters. The number of terms is bounded. Consider a terminal term and fix the outside position parameters. All phases internal to I or J , or involving an outside position, can be absorbed into separate group weights. The remaining crossing phases are a product $\psi_n(I, J)$ of precisely the form in Theorem 12.3. It is independent of the outside parameter values: a binary bit involving an outside position has already been put into a separate weight. All original unary weights, even when they depend on the orientation, can also be included in these two weights.

Apply Theorem 12.3, scaling the known outside bounds as necessary. Its common four-copy test is independent of the fixed outside parameters, so one exceptional orientation set controls every such fixation. Integrate the outside parameters, sum the bounded binary-character expansion, and sum the terminal terms. The chosen cut and empirical tolerances make the total at most ε for sufficiently large n . The union of the finitely many exceptional sets is exponentially small. Since each four-copy estimate is uniform over its separate weights, the final exceptional set is independent of all unary input weights and of orientation-only filters. This proves (12.3). $\square$

Corollary 12.4 (Factorized integration of the approximant). *For the structured sum in (12.1) and arbitrary bounded unary input weights, outside exponentially small mixed mass,*

$$\begin{aligned} \mathbb{E}_z \mathcal{A}_n(O, z) S_n(K(O, z)) \prod_j w_j(O, z_j) \\ = \kappa \sum_{a=1}^L c_{a,n} \prod_j \mathbb{E}_{z_j} [f_{a,j,n}(K_j(O, z_j)) w_j(O, z_j)] + o(1). \end{aligned} \quad (12.11)$$

The error is uniform over the unary weights. An orientation-only filter may be included before averaging this identity over leaves. In particular the same statement applies to marked mixtures and retained type/colour pairs as in Theorem 12.1.

Proof. Apply Theorem 11.9 to each structured term, with $f_{a,j,n}(K_j)w_j$ as its unary weights. The number of terms and their bounds are fixed at the chosen approximation accuracy, so the sum of the exponentially small errors tends to zero. The exceptional set in that lemma is independent of these weights. $\square$

Remark 12.5 (What the two comparisons do not assert). The reference error in (12.2) is weak testing error, not L^1 error. The actual comparison allows arbitrary orientation-dependent *unary* parameter weights, and the specified binary Gram/ L, R conditions. It does not allow an unprescribed joint input filter to be inserted without proof. For instance, two complementary bounded tuple densities determined by a high-rank global bilinear bit can have nearly uniform unary marginals and zero overlap. A two-element histogram net would not exclude that example. The additional actual-query conclusion of Theorem 12.1, proved by the four-copy expansion, is what rules it out for the successful-query format used here. All factors added to the common signature lists by this theorem are global key functions. Nominal input status bits remain weights, not common key cells in Theorem 11.11.

13 A common limit for successful queries

We work with two independently sampled unit laws, which may be different. Each has a family of first-peeling leaves satisfying the per-leaf image bounds of [Theorem 5.6](#); their mixtures satisfy the absolute joint and projected primal marginal bounds of [Equation \(11.8\)](#). The restrictions supplied by [Theorem 10.5](#) are one application; an independent pair of points will be another. All arguments in this section concern a sequence with $n \rightarrow \infty$, with the construction parameters, including M_0 , fixed. We freely pass to subsequences, using the compactness and measurable-density constructions in [Section B](#). The leaves retain their actual mass weights. Conditional on a leaf, its orientation law is normalized; a restriction imposed on a query is never normalized away. More explicitly, if $w_{A,n}(L)$ and $w_{B,n}(L')$ are the respective leaf probabilities, every average below uses $w_{A,n}(L)w_{B,n}(L')$. Neither a uniform law on leaves nor a new normalization of an individual colour is permitted. The marginal hypotheses apply to these mixtures, not to their normalized leaves.

We also fix a permitted-pair indicator $J_n(L_A, L_B) \in \{0, 1\}$. It can test a finite colour mark or the ordered types of the two laws. Randomized colours are sampled as part of the original unit and then included in the leaf record. In particular, different-colour acceptance is a leaf-pair test. All statements about vanishing overlap and product zero below are restricted to $J_n = 1$; no assertion about the complementary pairs is needed.

The limit has two tasks. First, own-leaf nets and global projections pass truncated overlaps to a common space; uniform integrability then removes the truncation. Second, the actual-query and reference conclusions of [Theorem 12.1](#) identify each limiting density for the successful-query format [\(13.1\)](#) from its unary status measures. [Theorems 13.5](#) and [13.6](#) combine these tasks in [Theorem 13.7](#). A limit of unary marginals alone would not justify this conversion.

13.1 Options and the permitted filters

Definition 13.1 (Numerical option). A numerical option specifies a recipe shape with between one and seven positions on each cross edge, together with the following data:

1. the endpoint, tag, label, flavor, and diagonal bit at every position;
2. the full numerical unary statuses required at those positions, including the tester summary, the role $a(\text{atom})$, the basis coordinates of $T(\text{atom}, \cdot)|_{C_i}$, and the p_i bit;
3. the required within-endpoint binary evaluations, and the off-position Gram conditions defining the reference key space;
4. the small tables and the finite coordinate formats in which they are written.

The two sides have matching tags and diagonal bits position by position. All labels used at one endpoint are distinct. Coordinate formats of smaller dimension are padded by zeros.

Once the construction parameters have been chosen, the collection of options is finite. In particular, a large selector length does not make this collection grow with n . A basis of each C_i was chosen on its own leaf. Its tensor coordinates in the projected pin bases, its a values, the pin/projection relations, and the mandatory support exclusions all have bounded finite descriptions. No arbitrary ambient tensor is included in this metadata.

Fix an option ω and a pair of leaves (L_A, L_B) . Call the option *usable* on this pair if its finite metadata satisfy the label and injection requirements and its numerical statuses imply [\(7.2\)](#), with its binary prescriptions supplying [\(7.3\)](#). Denote this indicator by $U_\omega(L_A, L_B)$. Usability is

determined by bounded discrete metadata. The phase conditions in [Theorem 10.5](#) will help us find usable options; they need not be added to the definition of usability. Likewise, prediction exceptions may remain orientation metadata. They restrict our later choice of an option, rather than its definition of usability on an entire leaf pair.

For a specified table, admissibility against pinned images is checked separately on the two orientations. Write these flags as $F_{A,\omega}$ and $F_{B,\omega}$. They may depend on both leaves, but each is a function of only one orientation once the leaves are fixed. The successful query on one side therefore consists exactly of

$$\begin{aligned} & \text{an orientation-only flag} \times \prod_{\ell} \text{a unary status indicator at } \ell \\ & \times \text{ the specified binary and Gram indicators.} \end{aligned} \tag{13.1}$$

The parameter input at each position is independent and uniform in its specified flavor, conditional on the requested feasible diagonal bit. Only the diagonal conditioning is included in this input normalization. The other successful-query factors remain unnormalized.

Remark 13.2 (Two different classes of flags). The density nets in [Theorem 11.5](#) cover arbitrary orientation filters and a bounded family of additional numerical flags. The factorization argument below uses the narrower format (13.1). It does not introduce an arbitrary additional joint predicate on the point parameters. In particular, membership in a nominal basis-bit cell is a status requirement, not a global key test on which all basis assignments are asserted to remain possible.

For the key shape of ω , let $(\mathcal{K}_n, \nu_n)$ be the uniform reference space from [Section 8](#). Define $d_{A,n}^\omega$ and $d_{B,n}^\omega$ to be the densities of the two successful queries with respect to ν_n , after averaging orientations in their normalized leaves. Their integrals are at most one. Set

$$I_n(\omega) = \mathbb{E}_{L_A, L_B} J_n(L_A, L_B) U_\omega(L_A, L_B) \int_{\mathcal{K}_n} d_{A,n}^\omega d_{B,n}^\omega d\nu_n. \tag{13.2}$$

Here the leaves are independent before the permitted-pair test, usability, or flag passage is tested. All restrictions in the integrand have already been charged to the subdensities. By [Theorem 8.1](#), a positive lower bound for $I_n(\omega)$ along a subsequence suffices for the desired collision. We shall analyze the contrary possibility

$$I_n(\omega) \longrightarrow 0 \quad \text{for every numerical option } \omega. \tag{13.3}$$

13.2 A global projection for independent leaf families

The nets supplied by [Theorem 11.5](#) are chosen separately for each leaf. The next elementary Hilbert-space argument produces the common partitions that will be needed in the limit.

Lemma 13.3 (Global projections with adaptive filter choices). *Let (K_n, ν_n) be finite probability spaces. On each first-side leaf L and each second-side leaf L' let there be a family of subdensities. Assume that, for each fixed $R < \infty$ and $\varepsilon > 0$, their truncations at height R have $L^1(\nu_n)$ nets of uniformly bounded size, with the nets chosen using their own leaf only. The bounds may depend on R, ε but not on n or the leaf.*

For every $R < \infty$ and $\eta > 0$, there are deterministic partitions $\mathcal{P}_n$ of uniformly bounded size such that the following holds. These partitions may depend on the leaf laws, but are chosen before the leaves are sampled. For independent leaves, choose a member f and a member g of the two

truncated families, allowing either choice to depend on both leaves. If P_n is conditional expectation onto $\mathcal{P}_n$, then

$$\mathbb{E}|\langle f, g \rangle - \langle P_n f, P_n g \rangle| \leq \eta. \quad (13.4)$$

The same conclusion holds for every deterministic refinement of $\mathcal{P}_n$ chosen independently of the sampled leaves. Multiplication of the discrepancy by any indicator determined by the leaf pair does not increase the bound.

Proof. We may truncate net representatives to $[0, R]$. Since $\|f - g\|_2^2 \leq R\|f - g\|_1$ for functions in this interval, we obtain L^2 nets with uniformly bounded size, say at most m . Pad shorter nets by zero functions. Write their members as $f_{L,a}$ and $g_{L',b}$, and form the positive operators on $L^2(\nu_n)$

$$C_A = \mathbb{E}_L \sum_{a=1}^m f_{L,a} \otimes f_{L,a}, \quad C_B = \mathbb{E}_{L'} \sum_{b=1}^m g_{L',b} \otimes g_{L',b}.$$

Here $(f \otimes f)v = f\langle f, v \rangle$. Both traces are at most $T = mR^2$.

Fix $\delta > 0$. There are at most T/δ eigenvalues of C_A larger than δ . A unit eigenfunction e for such an eigenvalue λ satisfies, pointwise,

$$|e(x)| \leq \lambda^{-1} \mathbb{E}_L \sum_a |f_{L,a}(x)| |\langle f_{L,a}, e \rangle| \leq T/\lambda \leq T/\delta.$$

Quantize these finitely many bounded eigenfunctions with a common finite partition. Its conditional expectation P can be made to satisfy $\|(1 - P)e\|_2 \leq \varepsilon$ for every one of them. With $Q = 1 - P$, the spectral decomposition gives

$$\|QC_AQ\|_{\text{op}} \leq \delta + T\varepsilon^2.$$

The number of partition cells is bounded in terms of $m, R, \delta, \varepsilon$. If P' is a refinement and $Q' = 1 - P'$, then $Q' = Q'Q = QQ'$, so the same operator bound holds for $Q'C_AQ'$.

For net representatives, independence of the leaves gives

$$\begin{aligned} \mathbb{E}_{L,L'} \sum_{a,b} |\langle f_{L,a}, g_{L',b} \rangle - \langle P' f_{L,a}, P' g_{L',b} \rangle|^2 &= \text{Tr}(Q'C_AQ'C_B) \\ &\leq (\delta + T\varepsilon^2)T. \end{aligned}$$

This estimate permits pairwise-adaptive choices: the squared error for any chosen pair of representatives is pointwise at most the displayed sum. Cauchy–Schwarz bounds its expected absolute error. Approximating arbitrary family members by their net representatives costs at most a constant times R times the L^2 net error, since orthogonal projections are contractions. Choose the net accuracy, then δ and ε , sufficiently small in terms of η . This proves (13.4), also for refinements. The final indicator assertion follows from the absolute-value bound before the indicator is applied. $\square$

For a fixed numerical option, all parameter-dependent predicates needed by one side can be put in a bounded own-leaf flag family: its own basis bits, tester outcomes, and prescribed binary tests are already fixed by the option. All remaining dependence on the opposite leaf is an orientation filter. Consequently Theorem 11.5 supplies the net hypothesis of Theorem 13.3 for all such filters simultaneously. No net need be chosen anew using the opposite leaf.

13.3 Common signatures and stored conditional measures

Lemma 13.4 (Common signature model). *There is a subsequence and a common limiting experiment with the following properties.*

- (i) *For each tag and diagonal bit (l, q) there is a compact bit-sequence space $\Xi_{l,q}$ with reference law $m_{l,q}$. For each key shape there is a compact tuple-test space Υ with reference law ρ . Here ρ denotes this measure, distinct from the scalar thinning parameter in [Theorem 5.3](#). Its single-position projections ξ_ℓ have the independent laws m_{l_ℓ, q_ℓ} .*
- (ii) *A base orientation array θ contains its bounded discrete metadata and its single-atom status measures. For a unary type $t = (i, l, s, \varphi, q)$ and a full numerical status v , these measures have jointly measurable densities*

$$h_{t,v}^\theta : \Xi_{l,q} \longrightarrow [0, 1], \quad \sum_v h_{t,v}^\theta = 1 \quad m_{l,q}\text{-almost everywhere.} \quad (13.5)$$

For the full generic flavor, the sum of these densities over statuses with any specified role and basis vector in the recorded actual dimension is bounded below by a positive construction constant, almost everywhere. Compatible finer tester requirements have the lower fractions supplied by [Theorem 11.11](#).

- (iii) *There is first a random limiting leaf-pair record Z , which includes two conditional laws Λ_A, Λ_B of the arrays and their finite flag vectors. Conditional on Z , the two orientation records are drawn independently from these laws. After flags are dropped and Z is averaged out, θ_A, θ_B are independent with their respective base laws; they are identically distributed when the two original laws coincide. The record also includes the limiting permitted-pair bit $J(Z)$.*
- (iv) *Every discrete assertion of table usability, injection, and any imposed phase compatibility is retained, jointly with J and with its probability. Thus the preparation guarantees from [Theorem 10.5](#), when imposed, remain valid in this experiment: the no-prediction branch has a passed injecting table on mass at least .99, and the predicted branch has the stated positive mass of passed compatible injecting tables. Each imposed prediction whose error tends to zero is exact in the limiting status measures outside its stored exceptions. No prediction is required for the construction of the model itself.*

All single-key tests used to define $\Xi_{l,q}$ are functions of global keys only.

Proof. Choose countable lists of global binary tests on each $\Omega_{l,q,n}$. Include every imposed global predictor bit, if there is one. A key's values under this list form its unary signature. For each key shape also choose countably many binary tests on $\mathcal{K}_n$, including the pullbacks of all unary tests at its positions. Their values form a tuple signature.

The lists are constructed by countable closure. For every numerical option, positive integer truncation height, and successively improving fixed accuracy, include the cells of a partition provided by [Theorem 13.3](#). For every tuple cylinder test and successively improving accuracy, include all unary indicator factors needed in [Theorem 12.1](#); close under the finite Boolean operations needed to record them. Repeat these operations. Every finite stage has bounded complexity independent of n , so it defines sequences of tests for all sufficiently large n ; finitely many missing initial values may be assigned arbitrarily. Bounded coefficients in the approximations are also included among the scalar coordinates that will converge. Crucially, regularity in [Theorem 12.1](#)

is performed on key spaces. Its new unary factors remain global key functions. Nominal input phases handled in the proof of that theorem are not added to the signature lists.

At an orientation, and for every unary type t , record the empirical mass of each finite signature cylinder together with each numerical status outcome. Include the finite metadata needed for the chosen C_i bases and all numerical options; in predicted cases include the old marked profiles in these bases, the fixed role coefficients, and their exception sets. Call this array θ_n . The cylinder masses lie in $[0, 1]$ and obey consistent finite additivity identities. Their closed consistency space, together with the finite metadata, is compact and metrizable.

For each pair of leaves retain its permitted-pair bit J_n and, as part of the same random record, the *conditional measures* of $(\theta_n, \text{flag vector})$ on each side. Also retain the full and truncated successful-query measures on all tuple cylinders, for each option. These are measures of mass at most one on compact bit-sequence spaces; their spaces are compact for weak convergence. Take a common subsequence of the laws on the resulting countable product, and simultaneously of the deterministic reference laws and the bounded approximation coefficients. Consistent limiting cylinder masses define countably additive measures on the bit-sequence spaces, for example by first defining the measure on finite prefixes and taking the unique measure with those finite-dimensional laws.

Cylinder evaluation is continuous because bit cylinders are clopen. For each fixed unary cylinder, [Theorem 11.7](#) says that the sum of its empirical status masses differs in probability from its reference mass by a quantity tending to zero. This test uses only primal images. The projected primal marginal assumption on the actual mixture is the one needed here; no original full-frame marginal bound and no assertion about every individually normalized leaf is used. The status bits may depend arbitrarily on channels: summing all statuses removes those bits. Countably many limiting cylinder equalities imply that the total status measure is $m_{l,q}$ almost surely. Each status measure is therefore dominated by $m_{l,q}$, giving (13.5). Versions jointly measurable in (θ, ξ) are obtained by taking the limits of their conditional density averages on increasing finite cylinder partitions, as in [Theorem B.2](#). An exceptional null set for the base law also has Λ_A - and Λ_B -mass zero for almost every stored record on the respective side, because averaging these conditional laws recovers that side's base law. The same is true after any flag restriction.

For a generic role/basis assignment, apply [Theorem 11.11](#) to each fixed cylinder of positive limiting reference weight. The lower fraction is independent of the cell; the onset of the estimate may depend on its positive weight. All profiles of the relevant rank are covered by that lemma before the profile is selected, so the chosen C_i may vary with the entire orientation, including its channels. The p_i flag is summed over in this positivity statement; no value of p_i is prescribed as a positive base coordinate. Its lower inequalities pass to the limiting cylinder masses. Zero-reference cylinders already have zero total status mass. The cylinder inequalities extend by finite unions and monotone approximation to all measurable sets, giving the claimed almost-everywhere lower densities. This argument does not condition on arbitrary nominal input cells.

By [Theorem 11.3](#), every finite collection of unary signature tests has its product reference law in the tuple limit. Therefore the unary projections under ρ are independent with the stated marginal laws.

There is no need to interchange disintegration and weak convergence. The conditional laws themselves were retained as random coordinates. For bounded continuous tests, taking their product is continuous in these two laws. Consequently sampling from $\Lambda_A \otimes \Lambda_B$ conditional on the limiting record Z reproduces the limiting joint orientation tests. Before flags are applied, the two base orientations were independent under their respective mixture laws at every n , so

independence with those respective laws remains true after averaging Z and dropping flags. If the laws agree, their limits agree as well. Finite metadata are discrete coordinates, hence all their incidence and compatibility identities are retained. This proves the table assertions and their mass statements. Finally the prediction errors tend uniformly to zero on the retained orientations. Since the predictor bits are included as unary tests, each such error is a finite status-cylinder mass and becomes zero in the limit. $\square$

13.4 Truncation and the zero-product implication

Fix the common model from [Theorem 13.4](#). For an option and a side, let $\eta_R(Z)$ and $\eta(Z)$ denote the limiting successful-query measures obtained respectively from $\min(d_n, R)\nu_n$ and $d_n\nu_n$. All are measures on the tuple space for that shape.

Lemma 13.5 (Passage of vanishing overlaps). *Assume (13.3). For almost every Z , the measures $\eta_A(Z), \eta_B(Z)$ are absolutely continuous with respect to ρ . For each usable option their densities have product zero ρ -almost everywhere on the event $J(Z) = 1$.*

Proof. For each integer R , cylinder inequalities give $0 \leq \eta_R \leq R\rho$, and the measures increase with R . The uniform small-set conclusion of [Theorem 11.5](#) also gives uniform integrability of the successful-query densities. Indeed, since their masses are at most one, $\nu_n\{d_n > R\} \leq 1/R$; uniform small-set control bounds the mass on this set by a quantity tending to zero as $R \rightarrow \infty$, uniformly asymptotically in the leaf and in its orientation filter. Thus there is a deterministic function $\varepsilon_R \rightarrow 0$ such that, in the limit,

$$0 \leq \eta(Z)(Y) - \eta_R(Z)(Y) \leq \varepsilon_R \quad \text{almost surely.}$$

It follows that $\eta = \sup_R \eta_R$. In particular it has a density which is the increasing supremum of the truncated limiting densities.

Fix R . The prelimit inner product of the two truncated densities, multiplied by J_n and usability, tends to zero because it is bounded above by (13.2). Choose a global partition with projection error at most ε in [Theorem 13.3](#). For that finite partition the inner product of the projected densities is

$$\sum_{C: \nu_n(C) > 0} \frac{\eta_{A,R,n}(C)\eta_{B,R,n}(C)}{\nu_n(C)}.$$

This expression passes to its limiting counterpart in expectation, jointly with the discrete permitted-pair and usability indicators. A cell whose reference weight tends to zero contributes at most $R^2\nu_n(C)$ and may be omitted. On the other cells ordinary continuity applies.

Take increasing cylinder partitions which generate all tuple tests and include global projection partitions at accuracies tending to zero. The refinement assertion in [Theorem 13.3](#) preserves the error estimate. Conditional expectations of the limiting bounded densities on these partitions converge in $L^2(\rho)$: this follows, for instance, by approximating an L^2 function by cylinder-simple functions and using that conditional expectation is an L^2 contraction. Their inner products therefore converge in expectation, being bounded by R^2 . We obtain

$$\mathbb{E}_Z J(Z) U_\omega(Z) \int \frac{d\eta_{A,R}}{d\rho} \frac{d\eta_{B,R}}{d\rho} d\rho = 0.$$

All terms are nonnegative. Taking the countable intersection of the resulting almost-sure assertions, then increasing R , proves the same product-zero conclusion for the full densities on $J = 1$.

Multiplication by J is legitimate at the projection step because the projection error bound is in absolute value and permits every leaf-pair indicator. There are only finitely many options. $\square$

13.5 Identification of the full limiting density

For a side of an option, let α_Z be the subprobability law of θ obtained from its conditional law Λ by retaining the appropriate orientation flag. Let t_ℓ, v_ℓ be its required unary type and status at position ℓ . Write $\kappa > 0$ for the binary mixing constant for this side and numerical prescription. Constants on the two sides need not be denoted by the same letter.

Lemma 13.6 (Successful-query density). *For almost every Z , the full limiting successful-query measure on this side has density*

$$G_Z(u) = \kappa \int \prod_{\ell} h_{t_\ell, v_\ell}^{\theta}(\xi_\ell(u)) d\alpha_Z(\theta) \quad (u \in \Upsilon) \quad (13.6)$$

with respect to ρ .

Proof. The displayed expression is jointly measurable and nonnegative, by [Theorem 13.4](#). We prove equality against every tuple cylinder indicator W .

At a fixed accuracy ε , apply [Theorem 12.1](#) to its prelimit global key test W_n . On the reference key space its structured approximation has the form

$$W_n^{(\varepsilon)}(k) = \sum_{a=1}^{m_\varepsilon} c_{a,n} \prod_{\ell} \phi_{a,\ell,n}(k_\ell),$$

where the number of terms and all bounds are fixed at this accuracy. The unary factors may be taken to be finite linear combinations of indicator factors already in the common test lists. In the actual-query assertion of that theorem, take as unary weights the numerical status indicators of this option. These may depend on the orientation. The extra admissibility condition is orientation-only. Thus, after averaging over leaves, the successful mass tested by W_n differs from that tested by $W_n^{(\varepsilon)}$ by at most the prescribed accuracy and a vanishing error.

For clarity, this last use does not require typicality separately on every leaf. At a typical orientation the product-testing error is uniform over the allowed bounded unary weights. Multiplying by any orientation flag only decreases its bound. After both leaves are averaged, the mass of exceptional orientations is bounded by the mixed orientation error in [Theorem 11.7](#). An opposite-leaf-dependent choice of the flag cannot increase that unconditional error.

Apply [Theorem 11.9](#) to each structured term. Again its unary weights are the status indicators times bounded global unary factors. The resulting expression is

$$\kappa \int \sum_a c_{a,n} \prod_{\ell} \left(\int \phi_{a,\ell,n} d\eta_{t_\ell, v_\ell, n}^{\theta_n} \right) d\alpha_{Z_n}(\theta_n), \quad (13.7)$$

up to a vanishing error at this fixed complexity. Here the measures inside the product are the empirical single-atom status measures at the given orientation. Every integral in parentheses is a finite linear combination of status-cylinder coordinates. The coefficients and conditional measures were stored in the compactification. Consequently (13.7) converges jointly with the successful cylinder mass to the analogous expression in the limit. The preceding errors are controlled in mean absolute value over Z .

It remains to compare that limiting structured expression with the integral of WG_Z . This is where the *reference* conclusion of [Theorem 12.1](#) is also needed. Its error bound, uniform against products of bounded unary functions, passes to the limiting reference law first for cylinder-simple unary functions, since all the relevant finite test probabilities converge. It then holds for all bounded measurable unary functions on the signature spaces: approximate each one in $L^1(m_{l,q})$ by cylinder-simple functions, and telescope the product. At this fixed approximation accuracy the structured test has a fixed finite bound, so this passage costs a quantity tending to zero.

We may therefore use the unary functions $h_{t_\ell, v_\ell}^\theta$ in the limiting reference comparison, for each θ , and integrate it under the subprobability law α_Z . Independence of the unary projections under ρ gives

$$\int W^{(\varepsilon)}(u)G_Z(u) d\rho(u) = \kappa \int \sum_a c_a \prod_\ell \left(\int \phi_{a,\ell}(\xi) h_{t_\ell, v_\ell}^\theta(\xi) dm_{l_\ell, q_\ell}(\xi) \right) d\alpha_Z(\theta).$$

This is exactly the limit of (13.7). Replacing $W^{(\varepsilon)}$ by W in this last integral costs at most κ times the reference approximation accuracy. Combining the empirical and reference estimates and sending $\varepsilon \rightarrow 0$ yields

$$\mathbb{E}_Z \left| \eta(Z)(W) - \int WG_Z d\rho \right| = 0.$$

There are countably many cylinder indicators. Equality on all of them determines the measures and proves the lemma. $\square$

Proposition 13.7 (The limiting overlap obstruction). *Assume (13.3). In the common experiment of [Theorem 13.4](#), almost every sampled pair of orientation records with $J(Z) = 1$ has the following property: there is no usable numerical option whose two orientation flags pass and for which, at every matched position,*

$$\int_{\Xi_{l,q}} h_{t_A, v_A}^{\theta_A}(\xi) h_{t_B, v_B}^{\theta_B}(\xi) dm_{l,q}(\xi) > 0. \quad (13.8)$$

Proof. For a usable option, [Theorems 13.5](#) and [13.6](#) give $\int G_{A,Z} G_{B,Z} d\rho = 0$ for almost every Z with $J(Z) = 1$. Fubini's theorem and independence of the unary projections express this integral as

$$\kappa_A \kappa_B \int \int \prod_\ell \left(\int h_{t_{A,\ell}, v_{A,\ell}}^{\theta_A}(\xi) h_{t_{B,\ell}, v_{B,\ell}}^{\theta_B}(\xi) dm_{l_\ell, q_\ell}(\xi) \right) d\alpha_{A,Z}(\theta_A) d\alpha_{B,Z}(\theta_B).$$

All factors are nonnegative and both κ constants are positive. Hence the product inside is zero for almost every passed pair from the conditional laws. Averaging Z and taking the finite union over options proves the assertion. $\square$

14 Solving the status constraints

We now construct the options forbidden by [Theorem 13.7](#). The argument takes place in the common signature model, but the no-prediction alternative will require a transfer back to global functions on the finite key spaces. We show that a positive mass of permitted pairs with soluble scalar constraints forces positive key overlap. Applied to the prepared coloured law, the collision criterion then supplies a four-hole event between different colours.

Write the endpoints of the two units as $(A, 1), (A, 2)$ and $(B, 1), (B, 2)$. In this section $h_{A,i}$ or $h_{B,j}$ denotes an atom's role bit $a(\text{atom})$, not the channel dimension. The basis status vector at an

endpoint is denoted by $\mathbf{x}$, and its dimension is the recorded dimension of that endpoint's effective space C_i .

14.1 Generators and the coupled linear system

Fix two orientation arrays θ_A, θ_B satisfying the almost-sure conclusions of [Theorem 13.4](#). Some labels at each endpoint may be excluded. In particular, the mandatory support exclusions from [Equation \(6.4\)](#) are always excluded.

Definition 14.1 (Available generator). On the cross edge (i, j) , a vector

$$g = (q, h_{A,i}, h_{B,j}, \mathbf{x}_{A,i}, \mathbf{x}_{B,j}, p_{A,i}, p_{B,j}) \quad (14.1)$$

is an available generator if some common tag, surviving label at each end, and allowed flavors give positive overlap as in [\(13.8\)](#) for these entries. Entries of the tester summary not displayed in [\(14.1\)](#) are summed over. A generator is retained with one such certificate of its tags, labels, and flavors.

Let V_{ij} be the binary span of these generators. Its *base projection* forgets the final two p entries and has target

$$B_{ij} = \mathbb{F}_2 \oplus \mathbb{F}_2 \oplus \mathbb{F}_2 \oplus \mathbb{F}_2^{\dim C_{A,i}} \oplus \mathbb{F}_2^{\dim C_{B,j}}.$$

For any surviving pair of labels and any tag, the full generic flavor gives at least one available generator above every base vector. Indeed, at fixed q the sum of the status densities with any requested role/basis assignment is bounded below almost everywhere by [Theorem 13.4](#). The product of the two such sums has positive integral on the common signature space. Splitting its finite sum over the two p bits yields an available generator. Both diagonal bits are feasible in the full generic flavor. Thus

$$V_{ij} \longrightarrow B_{ij} \text{ is surjective,} \quad \dim \ker(V_{ij} \longrightarrow B_{ij}) \leq 2. \quad (14.2)$$

This remains true after any exclusions leaving surviving generic labels.

Fix a small table. Its local target basis vectors are the evaluations of

$$(a + (u_{B,j}U_{A,i})^*)|_{C_{A,i}}, \quad (a + (u_{A,i}U_{B,j})^*)|_{C_{B,j}},$$

in the recorded bases. Denote these vectors by $\mathbf{t}_{A,i;j}$ and $\mathbf{t}_{B,j;i}$, respectively. We seek one sum vector $v_{ij} \in V_{ij}$ for each cross edge, subject to

$$\begin{aligned} q_{ij} &= 1, & h_{A,i;ij} + h_{B,j;ij} &= 1, \\ \mathbf{x}_{A,i;ij} &= \mathbf{t}_{A,i;j}, & \mathbf{x}_{B,j;ij} &= \mathbf{t}_{B,j;i}, \end{aligned} \quad (14.3)$$

$$\begin{aligned} \sum_{i=1}^2 (p_{A,i;ij} + h_{A,i;ij}) &= 0 \quad (j = 1, 2), \\ \sum_{j=1}^2 (p_{B,j;ij} + h_{B,j;ij}) &= 0 \quad (i = 1, 2). \end{aligned} \quad (14.4)$$

These are precisely the scalar recipe equations [\(7.2\)](#). The occurrence of a given endpoint in two different sums is distinguished by its edge subscript. By [Theorem 7.3](#), these are the effective-space conditions needed for the full gradients; the attack-atom conditions are supplied by [\(7.3\)](#). This use does not require pins to be supported at individual endpoints. The realization theorem already treats the mixed pin spaces through their projections and the injection hypothesis.

Lemma 14.2 (Dual relations). *Any obstruction to the scalar system has a nonzero vector of weights $(s_1, s_2, r_1, r_2) \in \mathbb{F}_2^4$. On every edge span V_{ij} these give a relation*

$$s_j(p_{A,i} + h_{A,i}) + r_i(p_{B,j} + h_{B,j}) = \mathbf{c}_{ij} \cdot \mathbf{x}_{A,i} + \mathbf{c}'_{ij} \cdot \mathbf{x}_{B,j} + \beta_{ij}(h_{A,i} + h_{B,j}) + \gamma_{ij}q \quad (14.5)$$

for suitable coefficients. The scalar system is soluble if, for every such relation, the sum of the right sides evaluated at the local targets is zero.

Proof. Apply finite-dimensional duality to the linear constraint map on $\bigoplus_{i,j} V_{ij}$. The coefficients of the four coupled equations are s_j, r_i , giving (14.5) separately on each edge. If they were all zero, the local projection onto $(q, h_{A,i} + h_{B,j}, \mathbf{x}_{A,i}, \mathbf{x}_{B,j})$ would be surjective by (14.2), forcing all other dual coefficients to vanish. Hence a nonzero obstruction uses some coupled equation. Conversely the constraint target lies in the image exactly when it annihilates every dual relation. The coupled targets are zero, so their evaluated contribution is the sum stated in the lemma. $\square$

14.2 A dual pattern on positive mass gives a global prediction

Call a pair of arrays *obstructable* if there are total exclusion sets of size at most B_* at each endpoint, containing the mandatory exclusions, and some nonzero weights for which (14.5) holds throughout. We do not require the relation to take a nonzero value on a particular table target. This stronger event is useful because its complement works for every table.

Lemma 14.3 (Extraction of a global prediction). *On the no-prediction alternative of Theorem 10.5, the probability that the iid base arrays are obstructable is less than .1.*

Proof. Suppose otherwise. Let E_s be the event that a witnessing relation has some $s_j = 1$, and define E_r similarly. Interchanging the two iid units interchanges these events, so they have the same probability. Their union is the obstructable event. Therefore $\mathbb{P}(E_s) \geq .05$.

Fix a witness with $s_j = 1$. For each first-side endpoint i , rewrite its edge relation as

$$p_{A,i} + \mathbf{c}_{ij} \cdot \mathbf{x}_{A,i} + (1 + \beta_{ij})h_{A,i} = r_i p_{B,j} + \mathbf{c}'_{ij} \cdot \mathbf{x}_{B,j} + (r_i + \beta_{ij})h_{B,j} + \gamma_{ij}q. \quad (14.6)$$

For any surviving label/flavor choices and any common feasible (l, q) , this equality holds almost surely under the product of the two status distributions conditional on the same signature. Otherwise a violating pair of numerical statuses would have positive overlap and would be a generator violating the relation. The two bits in (14.6) are conditionally independent at a fixed signature. If two independent bits agree almost surely, both are constant almost surely with their common value: writing their success probabilities as a, b , the mismatch probability $a(1-b) + (1-a)b$ is zero only for $(a, b) = (0, 0)$ or $(1, 1)$.

Compare all surviving first-side choices to one fixed generic surviving choice at (B, j) . Its total conditional status mass is one on every signature space, and its generic flavor admits either q . It follows that, simultaneously over all tags, feasible bits, labels and flavors remaining at (A, i) ,

$$p_{A,i} = T(\lambda_{A,i}, \text{atom}) + \alpha_{A,i} h_{A,i} + f_i(l, q, \xi), \quad (14.7)$$

where $\lambda_{A,i}$ is the basis combination specified by $\mathbf{c}_{ij}$ and $\alpha_{A,i} = 1 + \beta_{ij}$. The coefficients are common to these choices. The offset is also common across them; one does not choose a new offset for every label or flavor.

There are very few possible offsets after the second-side array is fixed. If $r_i = 0$, generic positivity of the role and basis at (B, j) forces $\mathbf{c}'_{ij} = 0$ and $\beta_{ij} = 0$ in (14.6). The offset is then either zero or q . If $r_i = 1$, it is, up to a common multiple of q , the deterministic value of

$$p_{B,j} + \mathbf{c}' \cdot \mathbf{x}_{B,j} + \alpha' h_{B,j}. \quad (14.8)$$

For this fixed array and endpoint there is at most one possible deterministic expression (14.8), including its coefficients, even if the permitted exclusions vary. Indeed two candidate exclusion sets of size at most B_* have a common surviving generic label, since the label universe is larger than $2B_*$. At that label every role/basis assignment has positive density conditional on the signature. Subtracting the two claimed deterministic expressions therefore forces their coefficient differences to be zero, and then their offset functions agree almost everywhere on every (l, q) space. The same generic label can be used at every tag and diagonal bit.

Thus, for a fixed second-side array, each choice of j supplies at most four offsets for each first-side endpoint: the two multiples of q , and the unique function (14.8) plus those two multiples, if that function exists. Across both choices of j there are at most $2 \cdot 4^2 = 32$ pairs (f_1, f_2) .

All events used here are measurable. There are finitely many types, status vectors, coefficients, and exclusion sets, and generator availability is a positivity test on the integral of jointly measurable densities. By Fubini, choose a fixed second-side array for which the first-side mass of E_s exceeds .04. One of the at most 32 pairs of offset functions then gives (14.7) on first-side mass at least $\delta_0 = .04/32$, allowing its coefficients and exclusion sets to vary with the first-side array. This is much larger than 10^{-6} .

We now transfer these fixed signature functions to a prediction in the finite problem. This step is necessary because the definition of prediction asks for global functions of actual keys. For any $\varepsilon > 0$, approximate each fixed binary function f_i by a binary cylinder function $f_i^{(\varepsilon)}$, with disagreement measure at most ε on each of the finitely many (l, q) reference spaces. Such approximation follows by approximating the measurable event where $f_i = 1$ in measure by the algebra of finite cylinders. The total unary measure of every type is its reference measure, so replacing f_i increases every relation error in (14.7) by at most ε .

For fixed cylinder functions and a fixed choice of coefficient bits and exclusions, all these error probabilities are finite linear combinations of status-cylinder coordinates. Their maximum over the remaining unary types is continuous. Minimizing over the finitely many permitted coefficient/exclusion choices is also continuous, separately on each discrete coordinate format. The strict event that this minimum is below 2ε consequently has limiting probability at least δ_0 . Weak convergence implies that its prelimit probability exceeds $\delta_0/2$ for all sufficiently large indices. One may see this directly by approximating the indicator of an open strict-error event from below by continuous functions.

Choose decreasing $\varepsilon_m \rightarrow 0$ and increasing prelimit indices at which the last mass bound holds. At index m use the two global functions obtained by evaluating $f_i^{(\varepsilon_m)}$ on the actual global key signature. Keep the orientations in the strict-error event and choose one of their successful coefficient/exclusion records. The kept mass is at least $\delta_0/2 > 10^{-6}$, and the maximum relation error over all remaining unary types tends uniformly to zero. Conditional-on- q errors give the same conclusion for the unconditioned flavor tests. These are precisely the requirements of [Theorem 10.1](#). The no-prediction branch has only the first effective spaces and only negligible trimming, so this mass margin also gives a prediction before that trimming. This contradiction proves the lemma. $\square$

14.3 The predicted dual relations

In the predicted alternatives, exclude the stored prediction exceptions as well as the mandatory support exclusions. Write the marked profiles as $\lambda_{A,i}$ and $\lambda_{B,i}$. The constants α_i and the functions f_i are common by endpoint index across units. Put $d_i = 1 + \alpha_i$. The prepared class $e_i = (d_i, [f_i])$ is as in (10.2): functions are first identified almost everywhere on all tag/bit signature spaces, and then quotiented by the span of the one function q , with a common coefficient on those spaces.

In a limiting table computation, use the notation

$$\mathbf{u}_{B,j}(\Delta_A) = \sum_{i=1}^2 (u_{B,j} U_{A,i})^*(\lambda_{A,i}), \quad \mathbf{u}_{A,i}(\Delta_B) = \sum_{j=1}^2 (u_{A,i} U_{B,j})^*(\lambda_{B,j}). \quad (14.9)$$

These are finite contractions in the recorded table coordinates. The notation does not require retaining an ambient tensor Δ in the compactification.

Lemma 14.4 (Predicted scalar compatibility). *For a predicted pair with a table satisfying the compatibility conditions of Theorem 10.5, the system (14.3)–(14.4) is soluble. The assertion survives any further bounded label exclusions leaving generic labels.*

Proof. The exact limiting prediction is

$$p_{A,i} = \boldsymbol{\lambda}_{A,i} \cdot \mathbf{x}_{A,i} + \alpha_i h_{A,i} + f_i, \quad p_{B,j} = \boldsymbol{\lambda}_{B,j} \cdot \mathbf{x}_{B,j} + \alpha_j h_{B,j} + f_j,$$

where bold $\boldsymbol{\lambda}$ denotes coordinates in the effective basis. Substitute this into a dual relation (14.5). At a fixed signature the two generic status distributions have every pair of role/basis assignments available. Comparing their coefficients forces

$$\mathbf{c}_{ij} = s_j \boldsymbol{\lambda}_{A,i}, \quad \mathbf{c}'_{ij} = r_i \boldsymbol{\lambda}_{B,j}, \quad \beta_{ij} = s_j d_i = r_i d_j, \quad (14.10)$$

$$s_j f_i + r_i f_j = \gamma_{ij} q. \quad (14.11)$$

Thus

$$s_j e_i = r_i e_j \quad (i, j \in \{1, 2\}). \quad (14.12)$$

Equality of functions here is exactly the almost-everywhere equality recorded in the prepared subsequential class.

If both e_i vanish, Theorem 10.2 supplies the exact identity $p_i = T(\lambda_i, \text{atom}) + h_i$ on all atoms. Hence on each span $p_i + h_i = \boldsymbol{\lambda}_i \cdot \mathbf{x}_i$. At the local targets, the first coupled sum for a fixed j is

$$\sum_i a(\lambda_{A,i}) + \mathbf{u}_{B,j}(\Delta_A) = c + \mathbf{u}_{B,j}(\Delta_A) = 0$$

by the empty-status compatibility condition. The reciprocal coupled sums vanish in the same way. Local surjectivity therefore solves all constraints in this case.

Suppose that some e_i is nonzero. A nonzero solution of (14.12) is possible only when all nonzero e_i are equal, and then both weight vectors are the indicator of their support S . To check this without any assumption on the dimension of the function space, first consider a support with one index, say $e_1 \neq 0, e_2 = 0$. The off-diagonal equations force $r_2 = s_2 = 0$, and the diagonal gives $s_1 = r_1 = 1$ for a nonzero solution. If both e_i are nonzero, the diagonal equations give $s_i = r_i$. The off-diagonal equation $s_2 e_1 = s_1 e_2$ either makes both weights zero, or makes both one and $e_1 = e_2$. These cases exhaust the possibilities.

If the two nonzero classes are unequal, no dual obstruction exists. Otherwise let d be the common first bit on S . With $s_i = r_i = \mathbf{1}_{i \in S}$, (14.11) is symmetric in i, j . Its diagonal is zero. Since the function q is not zero on the family of reference spaces, this gives $\gamma_{ij} = \gamma_{ji}$ and $\gamma_{ii} = 0$. The sum of all four q -contributions to a target is therefore zero. The total role contribution is

$$\sum_{i,j} \beta_{ij} = |S|d \quad \text{in } \mathbb{F}_2.$$

Evaluate the basis terms in (14.10) at their local targets. The a -terms sum to $|S|c + |S|c = 0$, because Theorem 10.5 fixed the same value $c = \sum_i a(\lambda_i)$ in both units. The remaining terms are

$$\sum_{j \in S} u_{B,j}(\Delta_A) + \sum_{i \in S} u_{A,i}(\Delta_B).$$

By (10.8), this equals $|S|d$. Together with the role contribution it is zero. Thus every dual relation vanishes on the target, and Theorem 14.2 proves solvability. The argument used only surviving generic labels and the prediction on those labels, so additional exclusions of the indicated kind do not change it. $\square$

Lemma 14.5 (Scalar solvability on positive mass). *In each positive-mass alternative of Theorem 10.5, the common experiment has positive mass of passed injecting tables for which the scalar system is soluble, robustly under the exclusions needed below. On the no-prediction branch this holds whenever the total exclusions stay within B_* per endpoint. On a predicted branch it holds after the stored exceptions and any further bounded exclusions leaving generic labels.*

Proof. Without prediction, the probability of an obstructable base pair is less than .1 by Theorem 14.3. This statement uses the unflagged iid base law. In the full conditional-law experiment, a passed injecting table exists on mass at least .99 by Theorem 13.4. Their intersection therefore has mass greater than .89. On it no nonzero dual pattern exists for any exclusions within the stated budget, so Theorem 14.2 solves the scalar system for every such table. No independence of the two events is being assumed.

With prediction, a passed compatible injecting table exists on positive mass in the same experiment. The exact prediction and all almost-sure unary properties continue to hold on that event, and Theorem 14.4 gives the assertion. $\square$

14.4 Stable spans and fresh short lists

It remains to replace a vector in an edge span by a short list of actual generators, using distinct labels across both lists at an endpoint. The small dimension in (14.2) is decisive.

Lemma 14.6 (Fresh lists of at most seven generators). *On any pair for which scalar solvability survives the bounded exclusions specified in Theorem 14.5, further bounded exclusions can be made so that every scalar solution on the resulting four edge spans can be realized by between one and seven available generators per edge. The two endpoint labels of each generator are retained with its certificate, and all labels used at an endpoint are distinct across its two incident lists. Every realized matched position has positive status overlap.*

Proof. Begin with the mandatory support exclusions and, in the predicted case, the stored exceptions. More generally, start with the initial exceptions allowed by the scalar-solvability

hypothesis. Whenever deleting at most 100 additional labels at each endpoint can shrink any of the four edge spans, make such a deletion. Each span continues to project onto its full base space. Its dimension exceeds that base dimension by at most two, so there are at most eight dimension-reducing steps in total. At the end, every edge span is unchanged by deleting up to 100 further labels at each endpoint. There remain generic labels, since (6.5) makes the label universe much larger than all these bounded exclusions. On the no-prediction branch the total budget, including later fresh-label removals, is at most $B_0 + 900 < B_*$. On a predicted branch it is at most the stored exceptions plus this quantity, still leaving many generic labels. Scalar solvability is retained by the scalar-solvability hypothesis. Fix one solution on these stabilized spans.

We prove a statement for one edge after any labels used on earlier edges have also been reserved. Write its stabilized span as V , its base space as B , and its projection as pr_B . Let G_L be the generators avoiding the current reserved sets L . Their span is still V . Let H_L be the span of sums of one, two, or three members of G_L whose total base projection is zero and whose endpoint labels are mutually disjoint within that word. We claim

$$H_L = \ker \text{pr}_B. \quad (14.13)$$

To prove the claim, take two generator lifts of the same $b \in B$. There is a third generic lift of b at labels avoiding both of them and L . Comparing each original lift to this third one gives an allowed two-generator kernel word. Hence all lifts of b have the same class modulo H_L . Call that class $\ell(b)$. A generator above base zero is itself an allowed one-generator kernel word, so $\ell(0) = 0$. Choose mutually label-disjoint generic lifts of $b, c, b + c$. Their sum is an allowed three-generator kernel word, and therefore $\ell(b) + \ell(c) = \ell(b + c)$. The generators span V , so this additive lift identifies V/H_L with B : the induced projection is inverse to ℓ . This proves (14.13), including the case of zero base projection.

For the desired vector in V , first choose a generic singleton lift of its base projection. Reserve its labels. The remaining discrepancy belongs to $\ker \text{pr}_B$, whose dimension is at most two. On the remaining labels (14.13) still holds because stability has preserved V . If the kernel is zero no correction is needed. If it is one-dimensional, choose a nonzero short kernel word and use it if needed. If it is two-dimensional, choose a nonzero short word, reserve its at most three labels at each end, and apply (14.13) again. The new short words still span the full kernel, so one lies outside the first word's direction. These two words form a basis and the discrepancy is the sum of an appropriate subset of them. This uses at most $1 + 3 + 3 = 7$ generators. The initial lift is always retained, so the list is nonempty; reserving labels before each choice ensures disjointness even when two generators have the same numerical bit vector.

Carry out this procedure on the four edges in turn. Each endpoint is incident to only two lists and uses at most fourteen labels in total. All removals and auxiliary freshness comparisons in the procedure are within the 100-label stability allowance. The generators keep their certificates of positive overlap, proving the lemma. $\square$

14.5 Positive overlap on permitted pairs

Proposition 14.7 (Positive-mass overlap). *Consider two independent unit laws with actual leaf weights, the mixed caps (11.8), the final per-leaf image bounds, and a permitted-pair indicator J_n as in Section 13. Suppose their common model has positive mass of pairs with $J = 1$ admitting a passed injecting table whose scalar system is soluble after every bounded exclusion used by Theorem 14.6. Then some fixed numerical option has expected unnormalized, permitted-pair overlap*

bounded below by a positive constant along a subsequence. Consequently it satisfies (8.3), with the same permitted-pair predicate. The two laws need not coincide. If the original laws additionally satisfy the hypotheses of Theorem 8.1, its probability bound holds along the same subsequence, with any initial restrictions of reciprocal mass $2^{o(N)}$ charged to that probability.

Proof. Work on the subsequence defining the common model in the hypothesis. If the conclusion failed, finiteness of the option collection would give (13.3) along this same subsequence. Apply Theorem 13.7 in this model.

By hypothesis, there is positive model mass of permitted pairs with a table whose flags pass and whose scalar solution survives the needed exclusions. Apply Theorem 14.6. This gives between one and seven matched generators per edge with fresh allowed labels. Their sum vectors obey (14.3) and (14.4); equivalently, the lists obey all scalar recipe equations. At each generator the unspecified tester bits were only aggregated. Expanding the positive overlap integral as a finite sum over the two full tester summaries yields at least one pair of summaries for which the overlap is still positive. Make that refinement independently at each position.

The scalar consistency argument in Section 7, in particular (7.3), now supplies a consistent numerical prescription of the required within-endpoint binary bits from these summaries and p bits. Together with the already passed table, these data define a usable numerical option. The only parameter filters added are precisely the unary statuses and the prescribed binary conditions in (13.1). At every matched position the refined status overlap is positive. This contradicts Theorem 13.7 on a positive-mass event.

No pointwise uniform lower bound for these individual positive integrals is needed. The options form a finite collection and every product has finitely many nonnegative factors. A positive-mass event on which some such product is positive gives positive expectation for at least one option. Its lower bound is a constant on the chosen subsequence, however small that constant may be. The preceding contradiction therefore proves the stated subsequential lower bound.

For this probability consequence, assume additionally that the original laws satisfy the hypotheses of Theorem 8.1. The successful densities are unnormalized with respect to the retained law, and Theorem 8.1 explicitly charges any law restriction of reciprocal mass $2^{o(N)}$. A positive constant, or its product with such restriction costs, is $2^{-o(N)}$. The resulting four-hole probability, including the permitted-pair predicate, has exponent at most $k_{\max} + .03 + o(1)$, whereas

$$100g - (k_{\max} + .03) = 44g + 55.97 > 0.$$

Thus these overlaps are on the scale required for Theorem 5.4. $\square$

14.6 The coloured unit distribution test

Proposition 14.8 (Different-colour conflicts). *Let the early constant $\eta_{\text{col}} > 0$ satisfy*

$$\eta_{\text{col}} < \frac{1}{4}c_*, \quad c_* := \frac{\delta_*^2}{64},$$

where δ_* is the retained-mass margin in (10.9). This choice depends on the original endpoint bound M and the early geometric constants, and precedes the selector and channel dimensions. For every fixed finite palette and all sufficiently large n , a unit law with both endpoint marginals bounded by $M\pi$, pair law after forgetting colour bounded by $2^{3000gN}\pi^2$, and each colour of mass at most η_{col} satisfies

$$\Pr\{A \text{ and } B \text{ have four cross holes and different colours}\} \geq 2^{-100gN},$$

where A, B are independent draws from that law. In particular, [Theorem 5.4\(i\)](#) holds.

Proof. Suppose that the assertion fails along an unbounded sequence of n . The laws and their randomized colour marks may vary with n ; the palette is fixed. Refine by colour and discard the exponentially light colour classes as in [Theorem 5.6](#). Peel each remaining conditional class, but retain the original mass of every leaf. Their mixture is a restriction of the original law with total mass $1 - o(1)$ before the fixed preparation restrictions. Hence its endpoint marginals remain bounded by a fixed multiple of $M\pi$ and its primal marginals by the same multiple of the original primal law. The conditioning of a small colour may change a leaf's joint density bound; this is already included in its absolute density and image budgets. It never changes the mixed marginal constant used next.

Apply [Theorem 10.5](#) to this pooled law, carrying its colour as a mark. In the predicted case this yields a retained law of original mass at least δ_* , and an event of passed compatible injecting tables of conditional pair mass at least $1/64$. Its scalar systems are soluble by [Theorem 14.4](#), robustly under the fresh-label exclusions. In the no-prediction case the retained mass is also at least δ_* and [Theorem 14.5](#) supplies conditional pair mass greater than .89. In both cases the successful scalar-table event has mass at least c_* in two draws from the original law.

If the original colour probabilities are p_c , their equal-colour pair mass is

$$\sum_c p_c^2 \leq (\max_c p_c) \sum_c p_c \leq \eta_{\text{col}}.$$

The equal-colour part of any subevent, including a prepared subevent, has no larger original mass. Subtracting it therefore leaves at least $c_* - \eta_{\text{col}} > 0$ of successful scalar-table pairs of different colours. Equivalently, if preparation retained mass δ , its normalized equal-colour pair mass is at most $\eta_{\text{col}}/\delta^2$. This subtraction takes place before any pigeonholing of a numerical table. Thus neither the number of tables nor the palette size enters the early choice of η_{col} .

Use the permitted-pair indicator $J_n(L_A, L_B) = \mathbf{1}_{\{\text{col}(L_A) \neq \text{col}(L_B)\}}$. Store this bit in the common model. The preceding original-mass lower bound, together with the retained prediction and injection assertions of [Theorem 13.4](#), gives positive limiting mass on $J = 1$ satisfying the hypothesis of [Theorem 14.7](#). That proposition yields a fixed option with positive subsequential overlap on different-colour pairs. The filtered form of [Theorem 8.1](#) now gives probability at least $2^{-(k_{\max} + .03)N - o(N)}$ for a four-hole event on those same pairs. Because $k_{\max} + .03 < 100g$, this is larger than 2^{-100gN} for all sufficiently large indices, contradicting the chosen violating sequence.

The overlap constant is allowed to depend on the subsequential model. This causes no loss in the uniform eventual assertion: any sequence violating it has a further subsequence with one fixed positive constant, and the fixed exponential gap just used defeats that sequence. $\square$

15 Conflicts between a unit and an independent point

We prove the second distribution test. Its two endpoint bounds are separate assumptions: throughout this section, σ is a law on units, ν is a law on points, and

$$\sigma \leq 2^{3000gN} \pi^2, \quad \sigma_1, \sigma_2 \leq L\pi, \quad \nu \leq L\pi. \quad (15.1)$$

Here π is the thinned law and P_0 is the original frame law. In particular only their primal marginals agree. We use $\varepsilon_{\text{ch}} = 10|\mathcal{E}|2^h\rho$ as an upper bound on the exponent lost in comparing π^2 with P_0^2 .

All restrictions made below have fixed positive mass, so all their constant marginal factors can be specified before the channel size is chosen.

Draw $A \sim \sigma$ and, independently, two points $B_1, B_2 \sim \nu$. We use this independent experiment for all calculations involving the two points. By [Theorem 5.8](#),

$$\mathbb{P}\{B_1 B_2 \text{ is a hole}\} \leq O_g(L^2 2^{2 \dim \mathcal{B} - N}) = o(1). \quad (15.2)$$

Indeed the two primal spans in each component are disjoint except on the displayed exceptional mass. On their complement, the first hole equation forces both profiles to be zero, contradicting the last hole equation. Only after the independent-point calculations will we delete these internal holes. The resulting unit law has a single unpinned leaf and

$$C_{B,1} = C_{B,2} = 0. \quad (15.3)$$

The full-frame image bounds needed for that leaf pay $\varepsilon_{\text{ch}} N$; they do not assert an unchanged full-frame marginal. Conditioning on a nonhole costs $1 + o(1)$.

15.1 No common prediction for the point side

Write $\Omega_{l,q,n}$ for the single-key reference space from [Section 10](#). For a channel frame v and a key $k = ((x_e, y_e))_{e \ni l}$ in this space, put

$$\chi_v(k) = (-1)^{u_v(k)}, \quad u_v(k) = \sum_{e \ni l} (Y_{v,e}^\top x_e) \cdot (X_{v,e}^\top y_e).$$

Every component has $x_e \cdot y_e = q$. This notation agrees with evaluating u_v on the tensor represented by the key.

Lemma 15.1 (Channel correlations on a common key). *Let v, w be two frames for which the concatenated channel columns $[X_{v,e}, X_{w,e}]$ and $[Y_{v,e}, Y_{w,e}]$ have rank $2h$, in every component of the star of l . Uniformly in these frames and in q ,*

$$\mathbb{E}_{\Omega_{l,q,n}} \chi_v = 2^{-h(g-1)} + O_g(2^{-N+4h}), \quad (15.4)$$

$$\mathbb{E}_{\Omega_{l,q,n}} \chi_v \chi_w = 2^{-2h(g-1)} + O_g(2^{-N+4h}). \quad (15.5)$$

For independent frames of law $\nu \leq L\pi$, these rank conditions fail with probability $o(1)$.

Proof. Before imposing $x \cdot y = q$ in one component, the four channel evaluation vectors

$$(Y_v^\top x, Y_w^\top x, X_v^\top y, X_w^\top y)$$

are uniform on $\mathbb{F}_2^{4h}$. Conditional on their values, x, y are independent uniforms on affine subspaces of codimension $2h$. The pairing between their translation spaces has rank at least $N - 4h$. For a bilinear form of rank R , its sign average on an affine product is either zero or has absolute value 2^{-R} : average first in one variable, leaving at most a 2^{-R} fraction of the other variable. Thus every fixed evaluation tuple has diagonal probability $\frac{1}{2} + O(2^{-N+4h})$. Conditioning on the diagonal changes its uniform law by $O(2^{-N+4h})$ in total variation. Deleting zero key vectors incurs an additional $O(2^{-N})$ error.

For independent uniform bits a, b , $\mathbb{E}(-1)^{ab} = 1/2$. There are h such products for one frame and $2h$ independent products for the pair of frames in each of the $g - 1$ components. Independence across components proves the two formulas.

Under P_0^2 , two independent channel frames have concatenation rank less than $2h$ with probability at most $O_g(2^{2h-N})$, by exposing their uniform injective columns. The domination $\nu^2 \leq L^2 2^{\varepsilon_{\text{ch}} N} P_0^2$ still makes this probability exponentially small, since $\varepsilon_{\text{ch}} < 1/2$. This proves the last assertion. $\square$

Lemma 15.2 (Absence of a point-side prediction). *In the common limit of the independent two-point law, there is no positive mass on which, at either endpoint, a prediction*

$$p_{B,j} = \alpha h_{B,j} + f(l, q, \xi) \quad (15.6)$$

holds for a fixed measurable common-signature function f , a fixed bit α , and all surviving unary types after bounded label exclusions. The assertion holds for every fixed positive mass, however small, and after deletion of internal holes.

Proof. Suppose the limiting mass is positive. Approximate f , on each of the finitely many tag/diagonal reference spaces, by binary cylinder functions. For a fixed approximation, the maximum prediction error over the surviving unary types is a continuous function of the finitely many stored status-cylinder coordinates. Minimizing over the finite allowed exclusion choices preserves continuity. The strict-error transfer in the proof of [Theorem 14.3](#) therefore gives a subsequence and global key functions f_n for which the finite prediction error tends to zero on a set of pairs of mass at least some $\delta > 0$. The functions are fixed before sampling these pairs. We use no supremum over functions chosen by a sampled orientation.

There is a fixed surviving generic label on positive pair mass: average over the finite label set and retain one of its positive terms. Fix also a tag, a feasible diagonal, and a feasible generic tester summary τ . This summary has a fixed positive nominal probability and fixes the role $a(\tau)$. Conditioning the vanishing error on it still leaves vanishing error. Absorb the term $\alpha a(\tau)$ into the global binary test $\phi_n(k)$. Write the two points in the relevant order as U, V , so that $p_{B,j}$ evaluates the channel form of V on the primal key supplied by U . Set

$$\begin{aligned} D_n(U, V) &= \mathbb{P}_z\{u_V(K_U(z)) \neq \phi_n(K_U(z)) \mid \tau\}, \\ d_n(V) &= \mathbb{P}_{k \sim \Omega_{l,q,n}}\{u_V(k) \neq \phi_n(k)\}. \end{aligned}$$

For each fixed V , the one-endpoint assertion of [Theorem 11.7](#) applies with that entire frame as external data. Its exceptional set depends on this prescribed test, and its bound is uniform in the external data. The raw mean is $d_n(V) + o(1)$ by the single-key frame law. The marginal of the primal frame of U is at most L times its P_0 -marginal. Consequently, for every fixed $t > 0$,

$$\mathbb{P}_{U,V \sim \nu}\{|D_n(U, V) - d_n(V)| > t\} = o(1).$$

For example the variance proof bounds the left side, apart from the negligible reference error, by $LC2^{-cn}/t^2$. Choose $t_n \downarrow 0$ slowly enough for the exceptional probability still to vanish. If $D_n \leq \epsilon_n \rightarrow 0$ on the successful pair set, its projection to the V -coordinate therefore yields

$$G_n = \{v : d_n(v) \leq \epsilon_n + t_n\}, \quad \nu(G_n) \geq \delta - o(1).$$

Any adaptive choice of the successful pair set is harmless here: the typicality exception was bounded on the independent product law before restricting it.

Take two independent frames V, W from ν conditioned on G_n . The probability of channel rank failure is at most its unconditioned $o(1)$ probability divided by $\nu(G_n)^2$, and hence still tends to zero. On a pair with full ranks, agreement of both channel signs with the same test gives

$$\mathbb{E}_{\Omega_{l,q,n}} \chi_V \chi_W \geq 1 - 4(\epsilon_n + t_n).$$

This contradicts (15.5), whose right side tends to $2^{-2h(g-1)} < 1$. This use of two successful frames is essential: it introduces no required lower bound on δ besides positivity. Finally, (15.2) changes all bounded error or event probabilities by $o(1)$, so deleting internal holes cannot create the asserted limiting prediction. $\square$

15.2 A singleton phase estimate

The point side needs only one phase calculation. In the following lemma the marks on A belong to its first effective spaces, and $\Delta_A = U_{A,1}\lambda_{A,1} + U_{A,2}\lambda_{A,2}$.

Lemma 15.3 (One point, then two independent points). *Let a marked unit law have a fixed marginal bound relative to π and an absolute joint bound $2^{(D+o(1))N}$ relative to P_0^2 . Suppose $\Delta_A \neq 0$ throughout, with its component rank bounded by r , and fix $c \in \mathbb{F}_2$. One can retain a fixed positive fraction of the unit law such that, for an independent point $Z \sim \nu$,*

$$\mathbb{P}\{u_Z(\Delta_A) = c \text{ and } R(A, Z)\} > .49.$$

Here R is either the whole space or the transpose-rank condition on the offset list used below. The retained fraction can be fixed before h ; the sufficiently large choice of h may depend on the fixed marginal constants. With two independent points, under this same retained unit law,

$$\mathbb{P}\{u_{B_1}(\Delta_A) = u_{B_2}(\Delta_A) = c, R(A, B_1), R(A, B_2)\} > .49^2. \quad (15.7)$$

Proof. Use the cover threshold p_0 , even moment s , and rank-growth constants in Theorem 9.2 and the proof of Theorem 9.3. Explicitly, s is the largest power of two at most $c_s N$, where $c_s = 10^{-12}(1+r)^{-4}$, so $s \geq c_s N/2$ for large n . Put $\theta = 10^{-4}$ and $\Gamma = D + 2r + 10$. If no cover of the specified size has unit mass at least p_0 , expand the moment against independent uniform channel columns of the one point:

$$\mathbb{E}_{\text{iid channels}} \left| \mathbb{E}_A (-1)^{u_Z(\Delta_A)} \right|^s \leq \mathbb{E}_{A_1, \dots, A_s} 2^{-h \text{rank}(\Delta_{A_1} + \dots + \Delta_{A_s})} \leq 4^s p_0^{q_0 s/2} + 2^{-h q_0 s/4}.$$

This is (9.6) with no reciprocal phase and no candidate value of Δ_Z . The inequalities (9.4) and (9.5), followed by Markov's inequality at threshold θ , bound the raw channel exception by $C_h 2^{-\Gamma N}$. The channel comparison in Theorem 5.2 is included in the fixed factor C_h . Paying $L 2^{\varepsilon_{\text{ch}} N}$ once leaves an exponentially small point-law exception. Thus $\left| \mathbb{E}_{A,Z} (-1)^{u_Z(\Delta_A)} \right| < .01$ for large n . Take R to be the whole space; each value of the bit has probability greater than .495.

Otherwise retain the p_0 mass in a fixed cover and delete the negligible set whose individual primal spans meet that cover. The deletion uses the unchanged primal marginal. The lifting identity (9.7) gives offset records with at most $2^{0.02N}$ possibilities and offset basis size $1 \leq t \leq 2r$. At one fixed record, let Z_A be its companion tuple. Its unnormalized subprobability law satisfies

$$p_{\max}(Z_A) \leq 2^{-tN + .01N}$$

by (9.10). For that record let $R(A, Z)$ require the channel transposes of Z to have full rank on the plus and minus offset lists. The transpose estimate in Theorem 5.3, multiplied by the fixed point marginal factor, gives

$$\mathbb{P}\{R(A, Z) \text{ fails}\} \leq L O(r) 2^{2r-h} + o(1) < .01$$

after increasing h . We may, and do, make this bound less than .005.

On the retained rank event the output tuple $\Phi_{u_Z}(\omega_A)$ has point masses at most $2^{-tN+.01N}$. To check this directly, condition on the point's primal frame and freeze the bounded lists of transpose coefficients on the offsets. Each list has full rank. Ignoring the equations defining those coefficients, their prescribed images under the opposite channel maps cost $t(N - \dim \mathcal{B})$ bits. The number of coefficient lists is fixed in n . The density loss from thinning and the ratio $\dim \mathcal{B}/N$ are absorbed into $.01N$. This is the singleton image calculation of (9.11) with no point-side companion tuple.

The phase is $Z_A \cdot \Phi_{u_Z}(\omega_A)$, plus a function of the point alone at this fixed record. The Walsh bound Theorem 5.9 therefore bounds its unnormalized signed mass, including the rank filter, by

$$2^{tN/2} \left(2^{-tN+.01N} 2^{-tN+.01N} \right)^{1/2} = 2^{(-t/2+.01)N}.$$

Summing over records gives at most $2^{-.488N}$. Since $\mathbb{P}(R) > .995 - o(1)$, the mass where R holds and the phase bit has either prescribed value exceeds $.49$. This proves the singleton assertion.

For the last assertion put

$$f(A) = \mathbb{P}_{Z \sim \nu} \{u_Z(\Delta_A) = c, R(A, Z) \mid A\}.$$

The two points are conditionally independent given A , with the identical separate rank filter, so the left side of (15.7) is

$$\mathbb{E}_A f(A)^2 \geq (\mathbb{E}_A f(A))^2 > .49^2.$$

This argument asserts an averaged singleton estimate; it does not require a pointwise estimate for each unit. Internal holes have not yet been deleted. $\square$

15.3 The mixed dual relations and the conflict bound

Proposition 15.4 (The asymmetric distribution test). *For every fixed L , with the late parameters chosen as above, all sufficiently large n and all laws satisfying (15.1) obey*

$$\mathbb{P}_{A \sim \sigma, Z \sim \nu} \{A \text{ and } Z \text{ conflict}\} \geq 2^{-100gN}.$$

Thus Theorem 5.4(ii) holds.

Proof. Suppose a violating sequence exists. Peel the A law once by Theorem 5.6(a); retain the actual leaf weights and the resulting first effective spaces. After making all restrictions on this law, we will apply part (b) once to restore the required normalized-leaf image bounds without adding pins. The margins between $3000g$ and $D = 4000g$ absorb thinning and fixed normalizations. Throughout, the mixed marginal bounds used in Sections 11 and 13 concern the primal frames.

Preparing the unit and the point pair. Make the following subsequential choice on the A law. If an empty prediction exists on mass at least 10^{-6} , retain it. Here empty means $\alpha_1 = \alpha_2 = 1$ and the offsets are $\gamma_1 q, \gamma_2 q$ with fixed bits γ_i , modulo errors tending to zero in the reference spaces. Theorem 10.2 exactifies it, after negligible deletion, to

$$p_{A,i} = T(\lambda_{A,i}, \cdot) + a \quad \text{on all of } \mathcal{X}, \quad i = 1, 2. \quad (15.8)$$

In particular its proof eliminates the multiples of q , using the odd number of star atoms. Fix $c = a(\lambda_{A,1}) + a(\lambda_{A,2})$ and the indicator of $\Delta_A = 0$, losing a factor at most four. If no such empty

prediction exists, retain the first law without prediction preparation. Only this specialized empty prediction is tested; arbitrary common predictions need not be excluded.

In the predicted branch with $\Delta_A \neq 0$, apply [Theorem 15.3](#). In the branch with $\Delta_A = 0$, necessarily $c = 0$. Indeed the identities (15.8) give $(u_{A,1} + u_{A,2})U_{A,i} = T(\lambda_{A,i}, \cdot) + a$, since $u_{A,i}U_{A,i} = 0$; if $c = 1$, these profiles and $\Delta_A = 0$ satisfy all hole equations for the unit A , a contradiction. Thus in either predicted branch the independent experiment has fixed positive mass on

$$u_{B,j}(\Delta_A) = c \quad (j = 1, 2). \quad (15.9)$$

In the nonzero case this mass exceeds $.49^2$ under the normalized retained A law. Before constructing either common model, apply [Theorem 5.6\(b\)](#) to the combined restriction made on A . More explicitly, let w_λ be the original first-leaf weights and let q_λ be the fraction retained in leaf λ by all the A -only restrictions just made. Their total mass $q = \sum_\lambda w_\lambda q_\lambda$ is bounded below by a fixed positive constant. Discard the leaves with $q_\lambda < 2^{-\zeta N}$. They have mass at most $2^{-\zeta N}/q = o(1)$ in the restricted law; every remaining normalized leaf has the image bound (5.11) and an absolute density $2^{O(N)}$ relative to P_0^2 . The pins and first effective spaces have not changed, and the actual remaining leaf weights are retained. This operation depends only on A , so the two points remain independent conditional on A . On the predicted branch the two-bit compatibility mass is still at least $.49^2 - o(1)$.

Now delete internal B holes, costing $o(1)$ by (15.2), and take the single unpinned B leaf from [Theorem 5.8](#). Apply the independent-law version of [Theorem 9.5](#) to these two laws, with failure probability less than .01. On the predicted branch compatibility and injection therefore coexist on mass at least $.49^2 - .01 - o(1) > .22$, for large n . On the branch without prediction a passed injecting actual table has mass at least .99. These are subtractions of exceptional mass, and require no independence between the exceptional events and compatibility.

Separate flags for candidate tables. We record this mass in the permitted finite-table format. For each leaf pair enumerate all numerical candidate small tables T , as in [Theorem 13.1](#); there are finitely many, with a bound independent of n . A fixed T has separate admissibility flags $F_{A,T}(A)$ and $F_{B,T}(B)$ by [Theorem 7.2](#). In the predicted branch define

$$H_{A,T}(A) = \mathbf{1} \left\{ \sum_{i=1}^2 (u_{B,j}U_{A,i})_T^*(\lambda_{A,i}) = c \quad (j = 1, 2) \right\}. \quad (15.10)$$

The star denotes contraction in the numerical table T , as in (14.9). It is calculated from the entries of T , the recorded tensor coordinates of the $C_{A,i}$ bases, and the bounded coordinate vectors of the marks $\lambda_{A,i}$. Thus, for a fixed leaf pair and T , $H_{A,T}$ is a flag of A alone. Include the mark coordinates among the finite orientation metadata and replace $F_{A,T}$ by $F_{A,T}H_{A,T}$. In the branch without prediction set $H_{A,T} = 1$. Injection is a finite matrix-rank test on the candidate table and the leaf metadata.

Each pair counted by the ambient compatibility and injection estimate has at least one passed candidate: take T to be its actual small table. Therefore the union of these permitted candidate-table events has at least the mass just established. We store their entire finite flag vector in the common model. This argument does not impose equality of a candidate table with the actual cross entries between private directions; such an equality would be a joint condition. The actual table is used only to witness the existence of a passed candidate. Nor do we condition the two unit laws on the ambient phase event (15.9). They remain independent laws with the separate flags just described.

Solving the mixed scalar system. Construct the common model for these two different unit laws, retaining the ordered A – B type filter. The dual relations (14.5) become

$$s_j(p_{A,i} + h_{A,i}) + r_i(p_{B,j} + h_{B,j}) = \mathbf{c}_{ij} \cdot \mathbf{x}_{A,i} + \beta_{ij}(h_{A,i} + h_{B,j}) + \gamma_{ij}q, \quad (15.11)$$

because $C_{B,j} = 0$. We first show that no relation with some $r_i = 1$ can occur on positive model mass, for any of the permitted bounded exclusions. If it did, Fubini fixes an A array with a positive-mass set of such B arrays. Pigeonhole its finite coefficient and exclusion data and fix a surviving generic label on the A side. At a fixed common signature, the two status distributions are independent. The two expressions in (15.11) must agree almost surely, because a mismatching positive-density pair would be an available generator violating the relation. Two independent bits which agree almost surely are both deterministic. The generic A type has total status density one on every signature space, so its deterministic value gives a fixed common function on all surviving B types. Solving (15.11) for $p_{B,j}$ gives

$$p_{B,j} = (1 + \beta_{ij})h_{B,j} + f(l, q, \xi).$$

The strict-error cylinder transfer used in Theorem 15.2 applies, and that lemma rules this out. The finite number of possible dual and exclusion records implies that, almost surely, every remaining dual relation has $r_1 = r_2 = 0$.

In the branch without an empty prediction, suppose obstructable pairs had probability at least .1. A nonzero weight now has some $s_j = 1$. Generic positivity of the two possible B -role values in (15.11) forces $\beta_{ij} = 0$. The same relation, for both i , is then

$$p_{A,i} = T(\lambda_{A,i}, \text{atom}) + h_{A,i} + \gamma_{ij}q. \quad (15.12)$$

It holds on all surviving unary types, by comparison to a fixed generic B type of total density one. These are empty predictions. There are only two choices of j and four offset pairs $(\gamma_{1j}q, \gamma_{2j}q)$; therefore one choice occurs on A mass at least .1/8. The marked profiles and exclusions may still vary with A , as allowed in Theorem 10.1; their number is not a further pigeonhole cost. Cylinder transfer with strict error events, as in Theorem 14.3, loses at most a factor two in this mass and supplies a finite empty prediction of mass much larger than 10^{-6} . This is a contradiction. Thus obstructable mass is less than .1, whereas passed injecting tables have mass at least .99. Their intersection with nonobstructable pairs has mass greater than .89. On it Theorem 14.2 solves the scalar system for every such table and every allowed exclusion.

On the predicted branch, substitute (15.8) into (15.11). The full generic base projection forces

$$\mathbf{c}_{ij} = s_j \boldsymbol{\lambda}_{A,i}, \quad \beta_{ij} = 0, \quad \gamma_{ij} = 0.$$

For $s_j = 0$ all these coefficients are zero. Fix any passed injecting candidate table T , including the flag $H_{A,T} = 1$. At its local targets the dual evaluation is

$$\sum_{i,j} \mathbf{c}_{ij} \cdot \mathbf{t}_{A,i,j} = \sum_j s_j \left(\sum_i a(\lambda_{A,i}) + \sum_i (u_{B,j} U_{A,i})_T^*(\lambda_{A,i}) \right) = 0$$

by (15.10) and $\sum_i a(\lambda_{A,i}) = c$. This calculation uses the selected candidate's starred contractions throughout; it does not require that candidate to equal the actual cross table of the sampled pair. Every dual relation vanishes on its target. By Theorem 14.2, the scalar system is soluble on the positive mass having a passed compatible injecting candidate. The reasoning uses only surviving generic labels and hence persists through the further bounded exclusions for fresh lists.

Positive overlap and collision. In either branch we now have the hypothesis of the mixed, filtered [Theorem 14.7](#). Equivalently, [Theorem 14.6](#) supplies at most seven generators per cross edge, with distinct labels at each endpoint and positive status overlap at every matched position. The scalar recipe and gradient realization produce a usable numerical option; the filtered zero-product conclusion of [Theorem 13.7](#) cannot hold on this positive mass. Consequently some option has a positive subsequential expected unnormalized overlap. This positive constant need not be uniform over all possible limit models.

The collision criterion [Theorem 8.1](#) therefore gives four cross holes between A and B with probability

$$2^{-(k_{\max} + .03 + o(1))N}.$$

All retained-law normalizations have fixed positive mass and are included in the $o(1)N$ loss. In particular these four holes imply that A conflicts with B_1 . Viewed in the original independent experiment, the latter has exactly the probability of a conflict with one point of law ν . Since

$$k_{\max} + .03 = 56(g - 1) + .03 < 100g,$$

this contradicts the assumed bound below 2^{-100gN} for all sufficiently large indices of the violating sequence. The asserted uniform eventual distribution test follows. $\square$

16 From distribution tests to finite graphs

We now prove [Theorem 1.2](#). The only probabilistic input to this section is [Theorem 5.4](#), whose two parts were proved in [Theorems 14.8](#) and [15.4](#). The sample uses the thinned law π . Write

$$Q_N = 2^{3000gN}, \quad \epsilon_N = 2^{-100gN}, \quad m = 2^{1000gN}, \quad \Omega = \text{supp } \pi. \quad (16.1)$$

All spaces in this section are finite. In particular, π is strictly positive on Ω , and $\log |\Omega| = O(N^2)$: specifying all frame entries uses $O(N^2)$ bits, and restricting their support by thinning only decreases its size. We regard the fixed construction parameters as constants in this estimate.

16.1 Entropy and terminal covers

For probabilities p, q on a finite space with $q > 0$, define

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}, \quad 0 \log 0 = 0.$$

Logarithms in entropy expressions are natural logarithms.

This is the finite convex information-projection inequality underlying the entropy method of Csiszár [\[8\]](#).

Lemma 16.1 (Entropy increase after a deletion). *Let $\mathcal{P}$ be a nonempty compact convex family of probabilities and let p minimize $D(\cdot||q)$ on $\mathcal{P}$. Then p is positive on the union of the feasible supports. For every $p' \in \mathcal{P}$ supported on S ,*

$$D(p'||q) - D(p||q) \geq D(p'||p) \geq -\log p(S).$$

Proof. The minimum exists by continuity. If $p(x) = 0 < p'(x)$, the right derivative of entropy along $(1-t)p + tp'$ is $-\infty$ at zero; all terms on the support of p have finite derivative. This contradicts minimality. Differentiating now on the common feasible support gives

$$\sum_x (p'(x) - p(x)) \log \frac{p(x)}{q(x)} \geq 0.$$

Subtracting the two entropies proves the first inequality. If p' is supported on S , then

$$D(p' \| p) = D(p' \| p(\cdot | S)) - \log p(S).$$

Relative entropy is nonnegative: apply $\log t \leq t - 1$ with $t = q(x)/p'(x)$ and sum on the positive support of p' . This proves the second inequality as well. $\square$

We require a terminal cover for coloured units that accounts for four different capacities. The following elementary separation argument supplies the finite linear programming duality needed here.

Lemma 16.2 (Resource prices). *Let $\mathcal{R}$ be a nonempty finite collection of objects, each having a nonnegative load vector $v_r \in \mathbb{R}^d$, and let $c \in \mathbb{R}_{\geq 0}^d$ be the vector of capacities. If no probability $(p_r)_{r \in \mathcal{R}}$ has $\sum_r p_r v_r \leq c$ coordinatewise, there are nonnegative prices $w \in \mathbb{R}^d$ with*

$$w \cdot c < 1, \quad w \cdot v_r \geq 1 \quad (r \in \mathcal{R}).$$

Proof. The compact convex sets $C = \text{conv}\{v_r : r \in \mathcal{R}\}$ and $B = \prod_{j=1}^d [0, c_j]$ are disjoint. Choose a closest pair $z \in C, b \in B$, and put $u = z - b \neq 0$. For fixed z , the nearest point of B is its coordinatewise truncation, so $b_j = \min(z_j, c_j)$, $u \geq 0$, and $u \cdot b = u \cdot c$. For every $v \in C$, comparison with $(1-t)z + tv$ at $t = 0$ gives $u \cdot (v - z) \geq 0$. Consequently

$$u \cdot v \geq u \cdot z = u \cdot c + \|u\|^2 > u \cdot c \geq 0.$$

Take $w = u/(u \cdot z)$. This proves both assertions using only minimization on compact subsets of a Euclidean space. $\square$

Lemma 16.3 (The coloured terminal cover). *Fix a palette $[q] = \{1, \dots, q\}$. Suppose that a set $R \subseteq \Omega^2 \times [q]$ supports no probability σ whose two endpoint marginals are at most $M\pi$, whose projection onto Ω^2 is at most $Q_N\pi^2$, and whose mass on each colour is at most η_{col} . There are sets $S \subseteq \Omega$, $E \subseteq \Omega^2$ and $C \subseteq [q]$ such that*

$$\pi(S) < \frac{8}{M}, \quad \pi^2(E) < \frac{4}{Q_N}, \quad |C| < \frac{4}{\eta_{\text{col}}}, \quad (16.2)$$

and each $(x, y, c) \in R$ has $x \in S$, $y \in S$, $(x, y) \in E$, or $c \in C$.

Proof. The assertion is immediate for empty R . Otherwise introduce separate resources for each left vertex, right vertex, ordered pair, and colour, with capacities

$$M\pi(x), \quad M\pi(y), \quad Q_N\pi(x)\pi(y), \quad \eta_{\text{col}},$$

respectively. An object (x, y, c) uses one unit of each of its four resources. Apply [Theorem 16.2](#). Write the resulting prices as ℓ_x, r_y, e_{xy}, d_c ; their capacity-weighted sum is less than one, and

$\ell_x + r_y + e_{xy} + d_c \geq 1$ on R . Take all resources of price at least $1/4$. They cover R . Let S be the union of the high-price left and right vertex sets. In fact

$$\pi(S) \leq 4 \sum_x \pi(x)(\ell_x + r_x) < 4/M < 8/M.$$

The corresponding pair and colour inequalities are $\pi^2(E) < 4/Q_N$ and $|C| < 4/\eta_{\text{col}}$. This uses the capacity on the pair projection, so all colours of one pair share a single pair resource. $\square$

The next cover is a capacitated max-flow/min-cut argument [11]; we include the finite proof for the real capacities used here.

Lemma 16.4 (The uncoloured terminal cover). *If $R \subseteq \Omega^2$ supports no probability with both marginals at most $L\pi$ and joint law at most $Q_N\pi^2$, then there are $S \subseteq \Omega$ and $E \subseteq \Omega^2$ with*

$$\pi(S) < 1/L, \quad \pi^2(E) < 1/Q_N, \quad (16.3)$$

such that each pair in R meets S or belongs to E . A set $T \subseteq \Omega$ supports no probability at most $L\pi$ precisely when $\pi(T) < 1/L$.

Proof. Make a directed network from a source to a left copy of Ω , then to a right copy, then to a sink. The source and sink arcs have capacities $L\pi(x)$ and $L\pi(y)$. The middle arc $x \rightarrow y$, present for $(x, y) \in R$, has capacity $Q_N\pi(x)\pi(y)$. A flow of value one is exactly a feasible unit law, and a larger flow can be scaled to value one.

A maximum flow exists by compactness. Its value is less than one. In its residual network there is no source-to-sink path, since augmenting by the smallest residual capacity on such a path would increase the value. Let Z be the vertices reachable from the source. Every original arc leaving Z is saturated; every original arc entering Z carries zero flow. Summing flow conservation over Z therefore shows that its cut capacity is the maximum flow value, hence is less than one.

Let A be the left types outside Z , and B the right types inside Z . The cut inequality reads

$$L\pi(A) + L\pi(B) + Q_N\pi^2(R \cap ((\Omega \setminus A) \times (\Omega \setminus B))) < 1.$$

Take $S = A \cup B$ and take E to be the pair set in the last summand. This proves the cover and both bounds. For the point assertion, a law of total mass one on T bounded by $L\pi$ exists exactly when $L\pi(T) \geq 1$: necessity follows by summing, and sufficiency follows by normalizing $\pi|_T$. $\square$

16.2 Finite fingerprints

Call two coloured units adjacent when their colours differ and their four cross pairs are holes. This is a finite simple graph: a unit cannot conflict with itself, because a vertex has no hole to itself. For a fixed palette, consider its independent sets. Also consider the bipartite graph with raw units on one side and points of Ω on the other, with adjacency meaning an edge–point conflict. In this graph the relevant objects are pairs (I, J) having no adjacency between I and J .

The recording-and-replay construction follows the graph fingerprint method of Kleitman and Winston [16]; see also Samotij [26]. Its potential here is the constrained entropy minimum, whose increase makes the terminal count small.

Lemma 16.5 (Fingerprint bound). *There are families of terminal sets, respectively terminal pairs of sets, containing every such independent set, respectively every such pair (I, J) , with the following properties. Each coloured terminal has an infeasible coloured region as in [Theorem 16.3](#); in each bipartite terminal at least one of the two feasible regions in [Theorem 16.4](#) is empty. The number of terminals of either kind is*

$$\exp(O(N^3 2^{100gN})) = \exp(o(m)). \quad (16.4)$$

The constants may depend on the fixed palette and on L .

Proof. Start with the entire relevant graph as residual set. On a nonempty feasible region choose the entropy minimizer, breaking all choices deterministically. For coloured units the reference probability is π^2 times the uniform law on $[q]$. The projected pair cap gives density at most qQ_N against this reference. The minimum entropy is therefore in $[0, \log(qQ_N)]$. By [Theorem 5.4\(i\)](#), the average minimizer mass of a residual neighbourhood is at least ϵ_N . Select a vertex of maximal neighbourhood mass. If it belongs to the independent set being contained, record it and delete its neighbours. Otherwise delete only that vertex. These operations preserve containment of the target independent set.

For the bipartite problem, choose separate entropy minimizers on the two residual sides, against π^2 and π . Their sum lies in $[0, \log Q_N + \log L]$. By [Theorem 5.4\(ii\)](#), some vertex on either side has opposite minimizer neighbourhood mass at least ϵ_N . Choose deterministically a vertex of maximal such mass, including its side in the choice. If it belongs to the corresponding target set, record it and delete its neighbours on the opposite side; otherwise delete only the chosen vertex. The absence of edges between I and J preserves both containments.

In either procedure every step removes a vertex: a recorded vertex has a nonempty neighbourhood. Stop when feasibility fails (on either side in the bipartite case). Deleting vertices can only increase each minimum entropy. If a recording leaves the relevant feasible region nonempty, [Theorem 16.1](#) increases its minimum by at least $-\log(1 - \epsilon_N) \geq \epsilon_N$. Consequently the number of recordings is at most

$$1 + \left\lceil \frac{\max\{\log(qQ_N), \log Q_N + \log L\}}{\epsilon_N} \right\rceil = O(N 2^{100gN}). \quad (16.5)$$

The extra one permits a last recording that destroys feasibility.

After recording a vertex, its residual neighbourhood is empty forever. Since a selected vertex always has positive neighbourhood mass, it cannot be selected again. The set of recorded vertices, with side information in the bipartite case, thus determines the entire procedure: replay its deterministic choices, performing a recording exactly when the selected vertex is in that set, and otherwise deleting it. In particular it determines the terminal. The total number of available vertices has logarithm $O(N^2)$. Bounding the number of fingerprints of length at most (16.5) by the corresponding power of one plus that number proves (16.4). $\square$

16.3 One sample satisfies all three properties

Choose the colour threshold $\eta_{\text{col}} > 0$ from [Theorem 5.4\(i\)](#), and fix

$$0 < a < \min\{1/100, \eta_{\text{col}}/800\}, \quad L > 10^6/\beta. \quad (16.6)$$

Here we may assume $0 < \beta \leq 1$, since increasing β only weakens the required graph properties. The order is η_{col}, a , then β, L , followed by the remaining construction parameters as in [Section A](#).

Let $X_1, \dots, X_m$ be independent samples from π . On positions $[m]$, put an edge between distinct positions exactly when their types have no hole. Repeated types are allowed. By [Theorem 4.3](#), this graph has independence number at most two for every sample.

First consider any matching whose edges have been partitioned into classes of size at most am . Merge successive classes until a merged class first has size at least am , and repeat, leaving a possible final smaller class. Every merged class has size at most $2am$. As a matching has at most $m/2$ edges, there are at most

$$q = \left\lceil \frac{1}{2a} \right\rceil + 1$$

merged classes. Assign these the fixed palette $[q]$. Different merged colours imply different original classes.

For every coloured terminal, fix a cover (S, E, C) from [Theorem 16.3](#). Put $k = \lceil m/100 \rceil$. The probability that at least k sampled positions lie in S is at most

$$\binom{m}{k} (8/M)^k \leq (800e/M)^k = \exp(-\Omega(m)).$$

A fixed collection of k disjoint ordered position pairs gives independent π^2 -draws, and is entirely in E with probability less than $(4/Q_N)^k$. There are at most m^{2k} such collections, so its union bound is

$$m^{2k} (4/Q_N)^k = 2^{-(1000gN-2)k}. \quad (16.7)$$

Union over (16.4). With probability $1 - \exp(-\Omega(m))$, simultaneously every coloured terminal has fewer than $m/100$ exceptional positions and fewer than $m/100$ disjoint exceptional ordered pairs. The strict bounds use $k - 1 < m/100$.

Suppose there were a matching of size at least $m/20$ with no conflict between original classes. Orient its edges arbitrarily, merge its classes as above, and take the set of resulting coloured raw types. This is an independent set in the coloured conflict graph, so is contained in a terminal. Every matching edge is covered by an exceptional position, an exceptional ordered pair, or an exceptional colour. The first two covers each account for fewer than $m/100$ edges. The last accounts for fewer than

$$\frac{4}{\eta_{\text{col}}} 2am < m/100$$

edges. Their sum is less than $3m/100 < m/20$, a contradiction. This proves the first graph property. Multiple sampled edges may have the same raw type; the counting uses their disjoint sample positions and does not assume distinct types.

For the bipartite terminals put $k_\beta = \lceil \beta m/100 \rceil$. At a terminal with infeasible unit region, fix (S, E) from [Theorem 16.4](#). At a terminal with infeasible point region, its entire point region has π -mass less than $1/L$. For every one of these vertex sets, the position tail is at most

$$\binom{m}{k_\beta} L^{-k_\beta} \leq \left(\frac{100e}{\beta L} \right)^{k_\beta} = \exp(-\Omega_\beta(m)).$$

The exceptional ordered-pair tail is at most

$$m^{2k_\beta} Q_N^{-k_\beta} = 2^{-1000gNk_\beta}.$$

Another union over (16.4) shows that all these sets simultaneously contain fewer than $\beta m/100$ sample positions, and all pair exceptions contain fewer than that many disjoint ordered pairs, with probability tending to one.

If a matching and vertex set, each of size at least $\beta m/10$, had no edge–point conflict, their raw type sets would give a pair (I, J) contained in a bipartite terminal. An infeasible point region cannot contain that vertex set. An infeasible unit region covers fewer than $2\beta m/100$ matching edges, also a contradiction. We have proved the second property even at the stronger threshold $\beta m/10$. There is no disjointness requirement between the original vertex set and matching; a conflict itself ensures that its point differs from both edge endpoints.

Finally let $A, B \subseteq [m]$ each have size at least βm . A maximal matching in the induced graph on A leaves at most two vertices unmatched: the unmatched vertices are independent, and the whole graph has independence number at most two. Its size is at least $(|A| - 2)/2 \geq \beta m/10$ for all sufficiently large m . Apply the second property to this matching and B . The resulting edge–point conflict supplies a hole from A to B , proving the third property.

All the required events hold together with probability tending to one. Their intersection therefore contains a finite sample for every sufficiently large allowed n ; since $m = 2^{1000gM_0n}$ tends to infinity, this proves [Theorem 1.2](#).

A Parameters, ranks, and probability scales

This Appendix collects the choices made throughout the proof and checks their dependencies. All quantities chosen before n are constants. They can be very large; no upper bound on their size is needed, and none of these choices depends on an adversarial unit law. The fixed mixers may be chosen separately at each sufficiently large n , by [Theorem 6.5](#).

A.1 The acyclic order of construction

The early constants in (4.1)–(4.4) are

$$\begin{aligned} C_0 &= 1000, & g &= 10^9 + 1, & D &= 4C_0g, & M &= 2^{1000}, \\ k_{\max} &= 56(g - 1), & \zeta &= \frac{1}{1000k_{\max}}, & K_1 &= \left\lceil \frac{4(D + 1)}{\zeta} \right\rceil. \end{aligned} \quad (\text{A.1})$$

In particular g is odd and $D = 4000g$. The early rank budgets are

$$\begin{aligned} r &= 2K_1 + 2, & L_0 &= \lceil 100(D + 10) \rceil, & d_0 &= 4rL_0, \\ u_0 &= 10(K_1 + d_0 + 1), & K &\in \mathbb{N}, & K &\geq \frac{10(D + K_1 + u_0 + 10)}{\zeta}. \end{aligned} \quad (\text{A.2})$$

These are the bounds in (4.3) and (4.2); the construction takes K to be the least integer meeting the displayed lower bound. Only the first peeling is used. The auxiliary number u_0 is retained in the numerical definition of a conservative bound $K \geq K_1$; it does not count another operation in the present construction.

The remaining early tolerances in [Theorems 9.3](#) and [9.5](#) include

$$c_s = 10^{-12}(1 + r)^{-4}, \quad q_0 = 2^{-100r-10}, \quad \delta_{\text{cov}} = .0005. \quad (\text{A.3})$$

For each n , the moment order is the largest power of two $s \leq c_s N$; it is even for sufficiently large n , and $s \geq c_s N/2$. Choose $p_0 > 0$ sufficiently small in terms of r, D, c_s, q_0 and the fixed phase tolerance. Fix the transpose-injection error thresholds as well, with their total over all component

and endpoint roles below the required constant error, such as .001. These thresholds precede selector and channel sizes. For the bounded-tuple transpose tests in [Theorem 5.3](#), we may fix

$$q_{\text{test}} = d_0 + L_0 + 2r + 2K + 10.$$

This covers the tiny covers, fixed fresh probes, and bounded pin and marked-profile tests. The growing lists of length proportional to N in [Theorem 9.5](#) instead use an absolute reference-law count, as explained in its proof.

The exponential demands of the no-cover argument are simultaneously satisfiable in this order. Fix a phase-test threshold $\vartheta > 0$ smaller than the required fixed phase accuracy, and choose a constant $\Gamma > D + 2r + 1$ large enough to absorb the indicated density and bounded-rank tensor counts. With $L_\vartheta = \log_2(1/\vartheta)$, it suffices to arrange

$$\frac{c_s}{2} \left(\frac{q_0}{2} \log_2(1/p_0) - 2 - L_\vartheta \right) > \Gamma, \quad (\text{A.4})$$

$$\frac{c_s}{2} \left(\frac{hq_0}{4} - L_\vartheta \right) > \Gamma. \quad (\text{A.5})$$

The first determines a sufficiently small p_0 using only early quantities; the second is a later lower bound on h . Indeed the two moment terms are $2^{2s}p_0^{q_0s/2}$ and $2^{-hq_0s/4}$, and Markov at threshold ϑ costs 2^{sL_ϑ} . The comparison between raw and independent channel laws has a factor depending on h , but it is constant in n and does not alter the linear exponents.

For the bounded tiny-cover record use

$$R_0 = 2^{100(r+1)^2(d_0+|\mathcal{E}|+1)}, \quad \delta_{\text{ph}} = \frac{p_0\delta_{\text{cov}}}{64R_0}, \quad \delta_* = 10^{-6}\delta_{\text{ph}}/64.$$

The preparation argument supplies these early retained-mass bounds and $c_* = \delta_*^2/64 > 0$, as in [\(10.9\)](#). They measure retained unit mass and successful pair mass in the original pooled law with endpoint bound $M\pi$. Choose

$$0 < \eta_{\text{col}} < c_*/4, \quad 0 < a < \min\{1/100, \eta_{\text{col}}/800\}. \quad (\text{A.6})$$

A colour partition is refined into leaves with their actual weights; preparation is performed on their pooled law. Thus these margins are independent of both the number of colours and the smallest positive colour mass. Indeed original equal-colour pair mass is at most $\sum_c p_c^2 \leq \eta_{\text{col}}$; after restriction of mass δ_* its normalized mass is at most $\eta_{\text{col}}/\delta_*^2$. Either calculation gives strictly positive successful mass after the colour exclusion.

Now prescribe $0 < \beta \leq 1$, and choose $L > 10^6/\beta$, as in [\(16.6\)](#). Larger values of β follow by weakening the graph assertions. For the matrix obstruction one chooses β sufficiently small in terms of a , which is permitted here. Fix the additional bounded marginal factors and positive injection/phase tolerances required by the asymmetric test with this L . These later fixed tolerances may depend on L ; they do not change the preceding $\delta_*, c_*, \eta_{\text{col}}$. In particular the two preparation problems need not have the same success margin.

Next choose the row and selector parameters:

$$\begin{aligned} R_* &= 2(K + 20), & j_* &\geq 10(K + 1)(R_* + 20), \\ B_0 &= 2|\mathcal{E}|K(R_* + 14), & B_* &= 10(B_0 + 10000g + 1), \\ 2^{b-2j_*} &> 100g(B_* + 1), & p_* &= \sum_{j=0}^{j_*} \binom{b}{j}, \\ A &= 10(K + 1)p_*(g^2 + 5), & s_0 &= 4\lceil A \rceil, \quad J > 100(s_0 + K^2 + 1). \end{aligned} \quad (\text{A.7})$$

Integer inequalities are met by rounding upward when needed. This agrees with [Theorems 6.2](#) and [6.5](#). In particular j_* exceeds the interpolation degree required for $(K+1)(R_*+14)$ labels. The length b is chosen before A , and hence before s_0, J .

Now choose a sufficiently large integer r_0 , always setting

$$h = 1000r_0. \quad (\text{A.8})$$

The demands are finite in number and depend only on preceding constants:

1. the moment demand ([A.5](#));
2. fixed-cover and fresh-probe transpose-injection errors in [Theorem 9.3](#);
3. the unconditional transpose-injection estimate in [Theorem 9.5](#);
4. residual ranks and private-channel space in [Theorem 7.3](#);
5. the tester rank inequality in [Theorem 10.4](#).

Each is eventually satisfied by increasing r_0 . Quantitative checks appear below.

After fixing h , choose the thinning rate $\rho > 0$. It is enough to require

$$10^4 |\mathcal{E}| 2^{2h} \rho < \min\{\zeta/100, 10^{-6}, \epsilon_{\text{img}}, \epsilon_{\text{inj}}\}, \quad (\text{A.9})$$

where the positive constants ϵ_{img} and ϵ_{inj} are chosen below the reserved full-frame image and injection rate margins. Those margins are fixed before this choice. Put

$$\kappa_{\text{thin}} = 2|\mathcal{E}|(2^h - 1), \quad \tau_{\text{thin}} = 10|\mathcal{E}|2^h \rho.$$

The thinning lemma gives

$$\pi^2 \leq 2^{\tau_{\text{thin}} N} \mathbf{P}_0^2, \quad 3000g + \tau_{\text{thin}} < D - 1. \quad (\text{A.10})$$

Thus the distribution-test pair cap leaves a fixed positive reserve below the absolute cap $2^{DN} \mathbf{P}_0^2$, even after bounded normalizations and removal of exponentially light colours. The survival fraction is asymptotic to $2^{-\kappa_{\text{thin}} \rho N}$. Because $\kappa_{\text{thin}} \rho < .01$, the simultaneous additive concentration error $2^{-.1N}$ in the construction of the thinning sets is exponentially smaller than this fraction. Its use in conditional probabilities is therefore justified.

Finally choose a sufficiently large integer M_0 , also relative to $1/\rho$, and put

$$N = M_0 n, \quad m = 2^{C_0 g N}. \quad (\text{A.11})$$

The coefficient of every $O(n)$ count paid as a fraction of N has then been fixed. Choose M_0 to make all those fractions smaller than their prescribed slacks. Only afterward does n tend to infinity through integers with $n \geq 2r_0$. The resulting dependency order is

$$\begin{aligned} (C_0, g, D, M, k_{\text{max}}, \zeta, K_1) &\longrightarrow (r, L_0, d_0, u_0, K, \text{early tolerances}, p_0, \delta_*, c_*) \\ &\longrightarrow (\eta_{\text{col}}, a) \longrightarrow (\beta, L, \text{asymmetric tolerances}) \\ &\longrightarrow (R_*, j_*, B_0, B_*, b, p_*, A, s_0, J) \longrightarrow (r_0, h) \longrightarrow \rho \longrightarrow M_0 \longrightarrow n. \end{aligned}$$

Here b is only the selector length; β is the graph threshold. The fixed palette has size at most $\lceil 1/(2a) \rceil + 1$, and changes only bounded complexities and the sufficiently-large- n onset.

A.2 The channel demands precede later tests

In the frequent-output argument, a fixed collection of L_0 fresh probes succeeds on B-mass at least

$$q_* = (\delta_{\text{cov}}/2)(\delta_{\text{cov}}/4)^{L_0} > 0. \quad (\text{A.12})$$

The one-vertex density factor depends only on the early marked-law restrictions and p_0 . Transpose-injection failure on the fixed fresh vectors is at most that factor times $O(L_0)2^{L_0-h}$. It can be made smaller than $q_*/2$ by increasing h , without consulting a selector choice or a future common test. The later raw-event exponent is at most

$$-(.99 - .51)L_0N + DN + \text{small linear errors}.$$

The specified L_0 leaves a margin close to $(.48L_0 - D)N$. The small coefficient-count errors are paid by M_0 .

For [Theorem 9.5](#) only the unconditional accepting family is needed. The independent primal-span part uses the unchanged primal marginals. For the channel part, the estimate with $\ell = \lfloor N/(10(K+1)) \rfloor$ compares c^ℓ with $2^{-(h-2K)\ell+o(N)}$, where $c > 0$ is fixed by the required failure tolerance. A sufficient demand is

$$h > 2K + \log_2(1/c) + 30(K+1). \quad (\text{A.13})$$

It leaves a positive linear exponent margin. The thinning loss in [\(A.9\)](#) is chosen smaller than that margin; fixed marginal factors cost only constants. In particular this estimate does not require a table-dependent tolerance or a new pinning step.

For channel realization, the explicit bounds in [Equations \(7.5\) and \(7.10\) to \(7.12\)](#) are

$$\begin{aligned} B_{\text{lin}} &= 3K + 28, & R_{\text{res}} &= 14K + 140, \\ D_{\text{blk}} &= 2p_*|\mathcal{E}|(K + 14 + R_{\text{res}}), & D_{\text{quo}} &= 2p_*(1 + (g^2 + 3)D_{\text{blk}}) + 2K. \end{aligned}$$

The pure correction rank per component is at most $30r_0 + 28J|\mathcal{E}| + D_{\text{quo}}$, and the remaining channel pairing has rank at least $h - 2K - 4B_{\text{lin}}$. Thus [Equation \(7.14\)](#) asks precisely for

$$970r_0 > 28J|\mathcal{E}| + D_{\text{quo}} + 2K + 4B_{\text{lin}}. \quad (\text{A.14})$$

Every quantity on the right is independent of h, r_0, n and is fixed by parameters through J . The baseline first factors through bounded pin/key projections on primal inputs, then derivative corrections of bounded rank are chosen, and the small residual is compressed before the r_0 -sized pure terms are factored through private channels.

The tester inequality in [Theorem 10.4](#) is

$$\frac{g-1}{2} \left(g - \frac{4D}{g} \right) r_0 > 2(g-1)(h + 2JK_1) + 1. \quad (\text{A.15})$$

Substituting $D = 4000g$ and $h = 1000r_0$ gives

$$\frac{g-20000}{2} r_0 > 4JK_1 + \frac{1}{g-1}.$$

The coefficient on the left is positive for the specified g . This is therefore another compatible lower bound on r_0 .

A.3 Pin, atom, and tensor-rank ledger

Quantity	Bound and role
First pin dimension	K_1 , summed over components and signs.
Pin dimension used throughout	At most K_1 , hence at most the conservative budget K .
Tiny-cover dimension	At most $d_0 = 4rL_0$; its offset record has a bounded number of possibilities.
Effective space	$\dim C_i \leq K^2$; every member has total component rank at most K .
Attack atoms at one endpoint	At most 14, across its two lists.
Component rank in the annihilator argument	$2(K + 14) \leq R_* = 2(K + 20)$.
List length per cross edge	At most 7 on either side.
Atom positions in one unit's full query	At most $4 \cdot 7 = 28$.
General key-vector slots	$28 \cdot 2(g - 1) = k_{\max} = 56(g - 1)$.

Each tag atom occupies $g - 1$ star components and supplies one plus and one minus vector at each. These are vector counts, not scalar Gram counts. The label exclusions in [Theorem 6.3](#) give nominal independence modulo the pins, as proved in [Theorem 7.2](#), part (ii). The effective-space dimension is counted in projected pin bases: for a fixed endpoint $\sum_e \dim S_{i,e}^\pm \leq K$, so

$$\sum_e \dim(S_{i,e}^+ \otimes S_{i,e}^-) \leq \left(\sum_e \dim S_{i,e}^+ \right) \left(\sum_e \dim S_{i,e}^- \right) \leq K^2.$$

Thus an effective tensor basis is encoded in bounded projected coordinates, not as unrestricted matrices with $O(n^2)$ entries.

The unprojected representation of one gradient in [Theorem 7.3](#) has per-component rank at most

$$15r_0 + 14J|\mathcal{E}|.$$

Two endpoint contributions give $30r_0 + O(1)$. If A_+, A_- remove bounded pin/key subspaces, then

$$\text{rank}(M - A_+^T M A_-) \leq \text{rank}(I - A_+) + \text{rank}(I - A_-).$$

Hence this projection contributes only bounded residual rank, without multiplying the tester rank by a pin-dependent constant. The later blockwise compression and derivative corrections preserve that independence from r_0 .

A.4 Scalar records and the final dimension multiplier

Write

$$d_n = \dim \mathcal{B} = p_*(1 + (g^2 + 3)n). \quad (\text{A.16})$$

Then

$$\frac{d_n}{N} = \frac{p_*(g^2 + 3)}{M_0} + \frac{p_*}{M_0 n}.$$

This ratio, and the ratio after adding the bounded channel dimension, can be made arbitrarily small by choosing M_0 last. In particular the image estimate can use a fixed sufficiently small

$\varepsilon < \zeta/4$. For the tiny-cover calculation impose, in addition, that the total primal coefficient-count and image loss be smaller than $\rho/12$. The same choice leaves room for the injective frames and prescribed self-Gram values.

The collision records are linear in n . There are at most 56 point inputs across two units, costing at most

$$56(g^2 + 3)n + 56b$$

bits. A list of K nominal pin directions uses at most $2K(d_n + h)$ coefficient bits, plus bounded component, sign and rank indices. Once pin projections are fixed, a basis of C_i uses at most K^4 binary coefficients in its tensor-coordinate space; there are four endpoints. Tester summaries, numerical small tables, projection relationships and finite statuses have bounded lengths in n . The construction matrices E, L, R are fixed data and are not encoded afresh in each query record.

The total frozen dimension of a unit is at most $K + k_{\max}$, over all components and signs. There are $2(d_n + h)$ nominal columns of a given sign at each component. Recording all interactions of both units' columns with opposite frozen images therefore uses at most

$$4(d_n + h)(K + k_{\max})$$

scalar bits. The deterministic solution is selected from these data and the point parameters. No unrestricted $O(n^2)$ -entry matrix is appended. The number of record cells, including bounded nominal choices when needed, is consequently at most

$$2^{C_{\text{rec}}n + C_{\text{rec},0}}, \quad (\text{A.17})$$

for constants fixed before M_0 .

Only mixed primal-channel Gram entries are tested in the final target. For one cross Gram block and component, the two endpoint primal summands have total dimension $2d_n$ and the channel summands have total dimension $2h$ on either side. The two mixed products contain at most $8d_nh$ entries. There are two cross Gram blocks; hence at most

$$16|\mathcal{E}| h d_n \quad (\text{A.18})$$

bits are tested. Frozen or redundant entries only reduce this count. This is $O(n)$, with coefficient already fixed.

Choose M_0 to meet every earlier coefficient/codimension slack and, in particular, so that for sufficiently large n ,

$$\log_2(\text{record-cell count}) < .005N, \quad \text{tested scalar Gram bits} < .01N. \quad (\text{A.19})$$

The smaller image and pin-count slacks form a finite collection and can be met by the same final enlargement of M_0 .

The conditional point-mass estimate on a surviving cell is

$$\begin{aligned} \log_2 p_{\max}/N &\leq -(1 - 2\zeta)(k + t) + (k + .01) \\ &= -(1 - 2\zeta)t + .01 + 2\zeta k \leq -.95t \quad (t \geq 1, k \leq k_{\max}). \end{aligned}$$

Here $2\zeta k \leq .002$. A nonzero quotient character of rank t has mean at most

$$2^{tN/2} (2^{-.95tN} 2^{-.95tN})^{1/2} = 2^{-.45tN}.$$

This pays for fewer than $.01N$ scalar bits, leaving agreement probability at least $2^{-.02N}$. For each fixed parameter pair and key, pruning cells below $2^{-(k+.01)N}$ loses overlap at most

$$2|\mathcal{K}_n| 2^{C_{\text{rec}}n+C_{\text{rec},0}} 2^{-(k+.01)N},$$

which is exponentially small since $|\mathcal{K}_n| \leq 2^{kN}$ and (A.19) holds. This uses Fubini at a fixed parameter pair, without intersecting pruning decisions over all opposite parameters.

A.5 Probability scales and later analytic choices

The marginal distinctions used in the proof are summarized below. A bounded multiplier may depend on a fixed preparation restriction; it is independent of n .

Mechanism	Scale and permissible cost
Thinned reference law	Primal marginal exactly that of $\mathbf{P}_0$; full pair density at most $2^{\tau_{\text{thin}}N} \mathbf{P}_0^2$.
Original tested unit law	Joint cap $Q_N \pi^2$; both endpoint marginals at most $M\pi$, or $L\pi$ for the asymmetric test.
Prepared mixed law	Absolute joint cap within the reserved $2^{DN} \mathbf{P}_0^2$ budget; bounded marginals relative to π , hence bounded primal marginals relative to the original primal law.
Normalized leaves	A common $2^{C_{\text{leaf}}N} \mathbf{P}_0^2$ cap and fresh-image min-entropy with at most 2ζ slack; no bounded individual full-frame marginal is asserted.
Fixed-cost preparation restrictions	Positive mass fixed before h ; bounded logarithmic costs are absorbed by rate reserves.
Single-endpoint typicality	Error $2^{-\Omega(n)}$, tested on primal frames; arbitrary channel-dependent flags remain flags.
Paired oversampling and cross-rank failures	Error $2^{-\Omega(n^2)}$, surviving $2^{O(N)} = 2^{O(n)}$ joint inflation.
Tiny-cover compatible pairs	Positive mass fixed before h , using either the identically zero full-sum phase or the additional thinning gain.
Key equality	Factor $ \mathcal{K}_n ^{-1} \geq 2^{-kN}$, $k \leq k_{\text{max}}$.
Final four-hole probability	$2^{-(k_{\text{max}}+.03)N-o(N)}$, exceeding 2^{-100gN} for sufficiently large n .

For clarity, the extra gain from thinning is not obtained by subtracting the full-frame density loss from ρ . On a fixed tiny-cover record, the companion tuple of length t has primal point probability at most $2^{-tN+\rho N/12}$. If the full output is nonzero, one endpoint must have a nonzero channel combination in a prescribed nonzero translate of its allowed set. Conditional on its primal frame, Theorem 5.3 bounds that event by $2^{-\rho N/2}$. The point probability of companion plus nonzero full output is therefore at most $2^{-tN-\rho N/3}$, after absorbing fixed normalizations. The Walsh estimate pairs one such bound with a companion-only bound, giving at most

$$2^{(t_A+t_B)N/2} (2^{-t_A N+\rho N/12} 2^{-t_B N-\rho N/3})^{1/2} \leq 2^{-\rho N/12}.$$

There is no exponential union over these records: one of their boundedly many values was fixed at an early constant mass cost. If the full output is zero on the retained record, the lifted phase

vanishes for every ordered retained pair. Both cases supply constant compatibility mass. No further peeling or inverse-polynomial compatibility branch is used.

Fixed-cost restrictions are compatible with the leaf trimming in [Theorem 5.6](#). More generally, if a later analytical restriction has total mass $q_n = 2^{-o(N)}$, leaves on which its relative mass is below $2^{-\zeta N}$ contribute at most

$$2^{-\zeta N}/q_n = 2^{-(\zeta-o(1))N}$$

of the restricted law. On the other leaves division costs at most $2^{\zeta N}$, changing $(1-\zeta)tN$ to at worst $(1-2\zeta)tN$ for every positive rank t . This trimming preserves the old pins. Marginal hypotheses for later empirical common tests concern the primal projection; none requires replacing π by P_0 in a full-frame marginal inequality.

Partitions, entropy bounds, weak-product accuracies and finite transformation sample sizes are analytical choices made after the construction constants. They may depend on M_0 , a fixed positive test-cell mass, or a fixed accuracy. They do not change pin ranks, key-slot counts or the scalar record coefficient in [\(A.17\)](#). For each finite common-test stage these choices change only constants and the required onset in n ; a diagonal subsequence passes each fixed finite stage in turn. No rate uniform over all such stages is needed.

For example, unary positivity uses $\gamma = 2^{-K^2-3}$ and $b_{\text{int}} = 2^{-J+2s_0} < \gamma$. On a reference key cell of mass at least $\delta > 0$, the proof first obtains conditional input mass at least $\delta/2$, so it can use

$$u > \frac{2}{\delta(\gamma - b_{\text{int}})^2}$$

transformations. This changes the constant in $O_u(2^{-s_0 n})$, while $s_0 = 4\lceil A \rceil$ and the profile count 2^{An} remain fixed. Thus a smaller cell does not require a larger construction parameter J .

The final collision margin is

$$100g - [56(g-1) + .03] = 44g + 55.97 > 0.$$

This positive exponent gap absorbs any fixed positive subsequential overlap constant. A violating sequence for either distribution test would yield one limit model with positive successful overlap, which contradicts its asserted upper bound 2^{-100gN} . Uniformity over all limit models is unnecessary.

Finally, [Section 16](#) uses $Q_N = 2^{3000gN}$ and $m = 2^{1000gN}$. Entropy fingerprints have logarithmic count

$$O(N^3 2^{100gN}) = o(m).$$

For either fixed positive linear choice of k , the coloured exceptional-pair estimate is

$$m^{2k}(4/Q_N)^k = 2^{-(1000gN-2)k}.$$

The coloured position exception has mass less than $8/M$; at $k = \lceil m/100 \rceil$ its tail is at most $(800e/M)^k$. Fewer than $4/\eta_{\text{col}}$ exceptional colours, each of size at most $2am$, account for fewer than $m/100$ matching edges by [\(A.6\)](#). In the asymmetric transfer use $k = \lceil \beta m/100 \rceil$; vertex and point exceptional sets have mass below $1/L$, with tail at most $(100e/(\beta L))^k$. These bounds pay for the $\exp(o(m))$ terminal union and prove the graph properties at $m/20$ and $\beta m/10$, respectively.

B Compact measure constructions used in the proof

The spectral argument and the common signature model use compactness only for compact metric spaces, countable arrays, and bounded finite measures. We record constructions for these particular uses, including the jointly measurable densities in [Theorem 13.4](#).

Lemma B.1 (Compact arrays and finite measures). *Let X be a compact metric space and $C < \infty$. The positive Borel measures on X of mass at most C form a compact metrizable space for convergence against continuous functions. Countable products of compact metric spaces are compact metrizable. If ν_j converges weakly to ν and $F \subseteq X$ is closed, then $\limsup_j \nu_j(F) \leq \nu(F)$.*

Proof. Choose nested finite Borel partitions of X whose cell diameters tend to zero. Such partitions result by taking a finite cover by balls of arbitrarily small radius, making it disjoint by successive differences, and refining the previous partition. Given a sequence of measures, successive subsequences make every cell mass converge, as well as the total mass. The limiting cell masses are nonnegative and consistent under refinement.

Realize these masses by subdividing an interval of the limiting total length, first into intervals of the masses of the first partition, then recursively into intervals for its refinements. Outside the countable set of subdivision endpoints, each interval point specifies a nested sequence of cells. Representatives from those cells form a Cauchy sequence: all representatives after level j belong to one cell of diameter tending to zero. Compactness gives a limiting point of X . The representative step maps are measurable, so their limit is measurable. Push forward interval length by this limit map. Uniform continuity shows that the integral of each continuous function is the limit of its finite representative sums. The same sums approximate the integrals of the original measures uniformly. This proves weak subsequential compactness, including zero limiting mass.

Continuous functions on X have a countable uniformly dense family. For instance, finite nets from a countable dense subset of X , together with continuous distance cutoffs and rational values, give uniformly accurate piecewise weighted approximations to each continuous function. If (f_j) is such a family, convergence of the integrals of all f_j metrizes weak convergence on measures of bounded mass, by uniform approximation. The preceding subsequence argument therefore gives compactness in that metric. For a countable product use the metric $\sum_j 2^{-j} \min\{1, d_j\}$; successive coordinate subsequences and a diagonal choice prove its compactness.

Finally the continuous functions $f_k(x) = \max\{0, 1 - k d(x, F)\}$ decrease to $\mathbf{1}_F$. For every k , weak convergence gives $\limsup_j \nu_j(F) \leq \int f_k d\nu$. Bounded monotone approximation as $k \rightarrow \infty$ gives the stated closed-set inequality. $\square$

In particular, consistent masses of finite bit cylinders define a measure on the compact bit-sequence space: apply the interval subdivision construction directly to the successive bit prefixes. If the masses depend measurably on a stored record, all subdivision thresholds are measurable functions of that record. Using an additional uniform interval coordinate gives a jointly measurable sample. The same construction works for a compact metric target using the nested partitions above: evaluation of a measure on a Borel cell is measurable in the weak topology. For closed cells this follows from continuous distance cutoffs, and a monotone-class argument gives all Borel cells. Two independent interval coordinates sample independently from the two conditional measures stored in [Theorem 13.4](#). Thus the argument retains those conditional measures before taking limits; it does not interchange an unrecorded conditional distribution with a weak limit.

Lemma B.2 (Densities of dominated cylinder measures). *Let X be a compact bit-sequence space with Borel probability m , let θ have a probability law, and suppose ν_θ is a measurable family of measures satisfying $0 \leq \nu_\theta \leq m$. There is a jointly measurable $h(\theta, x) \in [0, 1]$ such that, for almost every θ , $\nu_\theta(A) = \int_A h(\theta, x) dm(x)$ for every Borel A . Conditional averages of an $L^2(m)$ function on increasing finite-prefix partitions converge in $L^2(m)$.*

Proof. Let $\mathcal{P}_k$ be the partition by the first k bits, and set $h_k(\theta, x) = \nu_\theta(C)/m(C)$ when $x \in C \in \mathcal{P}_k$

and $m(C) > 0$, with value zero on null cells. These functions are jointly measurable and lie in $[0, 1]$. Their successive differences are orthogonal in $L^2(d\Pr(\theta) dm(x))$, because the average of $h_{k+1} - h_k$ over each parent cell is zero for every θ . The squared norms of h_k are bounded by one. Hence (h_k) is Cauchy in L^2 , and has an L^2 limit h . One can see completeness here directly: choose a subsequence with summable L^2 norms of its successive differences; their L^1 norms are also summable, giving an almost-everywhere convergent sum with the required L^2 tail bound. A further almost-everywhere convergent subsequence makes h jointly measurable with values in $[0, 1]$.

For each cylinder C , all sufficiently late h_k integrate over C to $\nu_\theta(C)$. Convergence in the product L^2 space gives the same identity for h , first in L^1 of θ and thus almost surely. There are countably many cylinders, so these identities hold simultaneously outside one null set. They then extend to all Borel sets: the class of sets approximable in measure by the finite cylinder algebra is a sigma-algebra, and both measures are dominated by m .

Finite-cell averaging is an L^2 contraction by Cauchy–Schwarz. Cylinder-simple functions are dense in $L^2(m)$, by simple-function approximation and the same cylinder approximation of measurable sets. Approximating a function by such a simple function and using the contraction proves the final assertion. $\square$

Rescaling gives the same density construction for $\nu_\theta \leq Rm$. For countably many ordered measures, the versions can be chosen simultaneously with their almost-everywhere ordering; their increasing supremum represents the increasing supremum of the measures. This is the use in [Theorem 13.5](#). The integration identity for the sampled kernels holds first for rectangle indicators. The class of sets satisfying it is closed under complements and disjoint countable unions; hence it holds on the product sigma-algebra and then for nonnegative or bounded measurable integrands by simple-function approximation. These constructions justify the limiting integrations and projections used in [Section 13](#).

References

- [1] Kazuoki Azuma. Weighted sums of certain dependent random variables. *Tohoku Mathematical Journal, Second Series*, 19(3):357–367, 1967. doi: 10.2748/tmj/1178243286.
- [2] Xiaojun Chen, Yifan He, and Zaikun Zhang. Tight error bounds for the sign-constrained Stiefel manifold. *SIAM Journal on Optimization*, 35(1):302–329, 2025. doi: 10.1137/24M1659030.
- [3] Yves Colin de Verdière. Sur une hypothèse de transversalité d’Arnold. *Commentarii Mathematici Helvetici*, 63:184–193, 1988. URL <https://www-fourier.univ-grenoble-alpes.fr/~ycolver/All-Articles/88a.pdf>.
- [4] Yves Colin de Verdière. Sur un nouvel invariant des graphes et un critère de planarité. *Journal of Combinatorial Theory, Series B*, 50(1):11–21, 1990. doi: 10.1016/0095-8956(90)90093-F. URL [https://doi.org/10.1016/0095-8956\(90\)90093-F](https://doi.org/10.1016/0095-8956(90)90093-F).
- [5] Yves Colin de Verdière. *Spectres de graphes*, volume 4 of *Cours Spécialisés*. Société Mathématique de France, Paris, 1998. ISBN 2-85629-068-X. URL <https://smf.emath.fr/node/43207>.
- [6] Anders Cornect and Eduardo Martínez-Pedroza. Gromov’s approximating tree and the all-pairs bottleneck paths problem. *Involve*, 2025. URL <https://arxiv.org/abs/2408.05338v2>. To appear; arXiv:2408.05338v2.
- [7] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. John Wiley & Sons, second edition, 2006. ISBN 978-0-471-24195-9. doi: 10.1002/047174882X. URL <https://onlinelibrary.wiley.com/doi/book/10.1002/047174882X>.
- [8] Imre Csiszár. I -divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975. doi: 10.1214/aop/1176996454.
- [9] Gerard Debreu and I. N. Herstein. Nonnegative square matrices. *Econometrica*, 21(4):597–607, 1953. doi: 10.2307/1907925.
- [10] Miroslav Fiedler and Hans Schneider. Analytic functions of M -matrices and generalizations. *Linear and Multilinear Algebra*, 13(3):185–201, 1983. doi: 10.1080/03081088308817519.
- [11] L. R. Ford, Jr. and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956. doi: 10.4153/CJM-1956-045-5.
- [12] Alan Frieze and Ravi Kannan. Quick approximation to matrices and applications. *Combinatorica*, 19:175–220, 1999. doi: 10.1007/s004930050052. URL <https://link.springer.com/article/10.1007/s004930050052>.
- [13] M. Gromov. Hyperbolic groups. In S. M. Gersten, editor, *Essays in Group Theory*, volume 8 of *Mathematical Sciences Research Institute Publications*, pages 75–263. Springer, New York, 1987. doi: 10.1007/978-1-4613-9586-7_3.
- [14] Hugo Hadwiger. Über eine Klassifikation der Streckenkomplexe. *Vierteljahrsschrift der Naturforschenden Gesellschaft in Zürich*, 88:133–142, 1943. URL https://ngzh.ch/wp-content/uploads/2024/08/88_17.pdf.

- [15] Bo Jiang, Xiao Meng, Zaiwen Wen, and Xiaojun Chen. An exact penalty approach for optimization with nonnegative orthogonality constraints. *Mathematical Programming*, 198: 855–897, 2023. doi: 10.1007/s10107-022-01794-8.
- [16] Daniel J. Kleitman and Kenneth J. Winston. On the number of graphs without 4-cycles. *Discrete Mathematics*, 41(2):167–172, 1982. doi: 10.1016/0012-365X(82)90204-7.
- [17] Andrew Kotlov, László Lovász, and Santosh Vempala. The Colin de Verdière number and sphere representations of a graph. *Combinatorica*, 17(4):483–521, 1997. doi: 10.1007/BF01195002.
- [18] Monique Laurent and Bernard Mourrain. A generalized flat extension theorem for moment matrices. *Archiv der Mathematik*, 93(1):87–98, 2009. doi: 10.1007/s00013-009-0007-6.
- [19] László Lovász and Alexander Schrijver. A Borsuk theorem for antipodal links and a spectral characterization of linklessly embeddable graphs. *Proceedings of the American Mathematical Society*, 126(5):1275–1285, 1998. URL <https://homepages.cwi.nl/~lex/files/link.pdf>.
- [20] László Lovász and Alexander Schrijver. On the null space of a Colin de Verdière matrix. *Annales de l’Institut Fourier*, 49(3):1017–1026, 1999. doi: 10.5802/aif.1703. URL <https://www.numdam.org/item/10.5802/aif.1703.pdf>.
- [21] László Lovász and Alexander Schrijver. Nullspace embeddings for outerplanar graphs. In Martin Loeb, Jaroslav Nešetřil, and Robin Thomas, editors, *A Journey Through Discrete Mathematics*, pages 571–591. Springer, Cham, 2017. doi: 10.1007/978-3-319-44479-6_23. URL <https://ir.cwi.nl/pub/27485/27485.pdf>.
- [22] Colin McDiarmid. On the method of bounded differences. In J. Siemons, editor, *Surveys in Combinatorics, 1989*, volume 141 of *London Mathematical Society Lecture Note Series*, pages 148–188. Cambridge University Press, 1989. doi: 10.1017/CBO9781107359949.008.
- [23] OpenAI. A counterexample to Hadwiger’s conjecture. OpenAI Math Release preprint [OAI:A-counterexample-to-Hadwigers-conjecture-September-23-2026](#), 2026.
- [24] OpenAI. A linear list-coloring bound in terms of the Hadwiger number. OpenAI Math Release preprint [OAI:A-linear-list-coloring-bound-in-terms-of-the-Hadwiger-number-September-23-2026](#), 2026.
- [25] Bruce Reed and Paul Seymour. Fractional colouring and Hadwiger’s conjecture. *Journal of Combinatorial Theory, Series B*, 74(2):147–152, 1998. doi: 10.1006/jctb.1998.1835. URL <https://cgm.cs.mcgill.ca/~breed/SummerNSERC04/frachad.pdf>.
- [26] Wojciech Samotij. Counting independent sets in graphs. *European Journal of Combinatorics*, 48:5–18, 2015. doi: 10.1016/j.ejc.2015.02.005.
- [27] Hein van der Holst, László Lovász, and Alexander Schrijver. The Colin de Verdière graph parameter. In *Graph Theory and Combinatorial Biology*, volume 7 of *Bolyai Society Mathematical Studies*, pages 29–85. János Bolyai Mathematical Society, Budapest, 1999. URL <https://ir.cwi.nl/pub/1257/1257D.pdf>.